Marvell® Alaska® 88EA1517/88EA1512

Integrated 10/100/1000 Mbps Energy Efficient Ethernet Transceiver

Datasheet

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Alaska 88EA1517/88EA1512

Integrated 10/100/1000 Mbps

Energy Efficient Ethernet Transceiver

Datasheet

OVERVIEW

The Alaska® 88EA1512 device is a physical layer device containing a single 10/100/1000 Gigabit Ethernet transceiver. The transceiver implements the Ethernet physical layer portion of the 1000BASE-T, 100BASE-TX, and 10BASE-T standards. The Alaska® 88EA1517 device is a physical layer device containing a single 10/100 Ethernet transceiver. The transceiver implements the Ethernet physical layer portion of the 100BASE-TX and 10BASE-T standards. Both devices are manufactured using a standard digital CMOS process and contain all the active circuitry required to implement the physical layer functions to transmit and receive data on standard CAT 5 unshielded twisted pair.

Both devices support RGMII (Reduced pin count GMII), MII and 88EA1512 supports SGMII for direct connection to a MAC/Switch port. The SGMII can also be used on media/line side to connect to SFP modules that support 1000BASE-X, 100BASE-FX and SGMII. 88EA1512 also supports Copper/Fiber Auto-media applications with RGMII as the MAC interface. SGMII operates at 1.25 Gbps over a single differential pair thus reducing power and number of I/Os used on the MAC interface.

Both devices integrate MDI interface termination resistors into the PHY. This resistor integration simplifies board layout and reduces board cost by reducing the number of external components. The Marvell® calibrated resistor scheme achieves and exceeds the accuracy requirements of the IEEE 802.3 return loss specifications.

Both devices have an integrated switching voltage regulator to generate all required voltages. The device can run off a single 3.3V supply. The device supports 1.8V, 2.5V, and 3.3V LVCMOS I/O Standards.

Both devices support Synchronous Ethernet (SyncE) and Precise Timing Protocol (PTP) Time Stamping, which is based on IEEE1588 version 2 and IEEE802.1AS.

Both devices support IEEE 802.3az-2010 Energy Efficient Ethernet (EEE) and is IEEE 802.3az-2010 compliant.

The 88EA1517 device is specifically designed for automotive applications. It provides an MII or RGMII MAC interface and supports 100BASE-T by default. It also supports 10BASE-T. The 100BASE-T output is optimized to meet the demanding automotive EMI and EMC requirements and it meets AEC-Q100 Grade 2 requirements. This part is available in a small 40-Pin QFN package.

Both devices incorporate the Marvell Advanced Virtual Cable Tester® (VCT™) feature, which uses Time Domain Reflectometry (TDR) technology for the remote identification of potential cable malfunctions, thus reducing equipment returns and service calls. Using VCT, the 88EA1512 device detects and reports potential cabling issues such as pair swaps, pair polarity and excessive pair skew. Both devices will detect cable opens, shorts or any impedance mismatch in the cable and reporting accurately within one meter the distance to the fault.

The 88EA1512 uses advanced mixed-signal processing to perform equalization, echo and crosstalk cancellation, data recovery, and error correction at a gigabit per second data rate. The device achieves robust performance in noisy environments with very low power dissipation.

Features

- **88EA1512 is 10/100/1000BASE-T IEEE 802.3 compliant**
- **88EA1517 is 10/100BASE-T IEEE 802.3 compliant**
- **Multiple Operating Modes**
  - MII to Copper (88EA1517/88EA1512)
  - RGMII to Copper (88EA1517/88EA1512)
  - SGMII to Copper (88EA1512 devices only)
  - RGMII to Fiber/SGMII (88EA1512 device only)
  - RGMII to Copper/Fiber/SGMII with Auto-Media Detect (88EA1512 device only)
  - Copper to Fiber (1000BASE-X) (88EA1512 only)
- **Four RGMII timing modes including integrated delays - This eliminates the need for adding trace delays on the PCB**
- **88EA1512 only supports 1000BASE-X and 100BASE-FX on the Fiber interface along with SGMII**
- **Supports LVCMOS I/O Standards on the RGMII interface**
- **Supports Energy Efficient Ethernet (EEE) - IEEE 802.3az-2010 compliant**
  - EEE Buffering
  - Incorporates EEE buffering for seamless support of legacy MACs
- **Ultra Low Power**
- **Integrated MDI interface termination resistors that eliminate passive components**

- **Integrated Switching Voltage Regulators**
- **Supports Green Ethernet**
  - Active Power Save Mode
  - Energy Detect and Energy Detect+ low power modes
- **IEEE1588 version 2 Time Stamping**
- **Supports One-Step PTP**
- **88EA1512 only supports Synchronous Ethernet (SyncE) Clock Recovery**
- **Three loopback modes for diagnostics**
- **88EA1512 only supports “Downshift” mode for two-pair cable installations**
- **88EA1512 supports fully integrated digital adaptive equalizers, echo cancellers, and crosstalk cancellers**
- **Advanced digital baseline wander correction**
- **Automatic MDI/MDIX crossover at all speeds of operation**
- **Automatic polarity correction**
- **IEEE 802.3 compliant Auto-Negotiation**
- **Software programmable LED modes including LED testing**
- **MDC/MDIO Management Interface**
- **Two-Wire Serial Interface**
- **CRC checker, packet counter**
- **Packet generation**
- **Wake on Lan (WOL) event detection**
- **Advanced Virtual Cable Tester® (VCT™)**
- **Auto-Calibration for MAC Interface outputs**
- **Temperature Sensor**
- **Supports single 3.3V supply when using internal switching regulator**
- **I/O pads can be supplied with 1.8V, 2.5V, or 3.3V for all packages**
- **Automotive grade 2**
- **40-Pin QFN 6 mm x 6 mm Green package with EPAD (88EA1517), 56-Pin QFN 8 mm x 8 mm Green package with EPAD (88EA1512 device)**

Table 1: 88EA1517/88EA1512 Device Features

Features	88EA1517	88EA1512
MII to Copper	Yes	Yes
RGMII to Copper	Yes	Yes
SGMII to Copper	No	Yes
RGMII to Fiber/SGMII	No	Yes
RGMII to Copper/Fiber/SGMII with Auto-Media Detect	No	Yes
Copper to Fiber	No	Yes
1000BASE-T	No	Yes
I/O Voltage (VDDO)	3.3V/2.5V/1.8V	3.3V/2.5V/1.8V
IEEE 802.3az-2010 Energy Efficient Ethernet (EEE)	Yes	Yes
EEE Buffering	Yes	Yes
Synchronous Ethernet (SyncE)	No	Yes
Precise Timing Protocol (PTP)	Yes	Yes
Auto-Media Detect	No	Yes
Wake on LAN (WOL)	Yes	Yes
Package	40-pin QFN	56-pin QFN
Temperature Grade	Automotive	Automotive

Figure 1: RGMII/MII to Copper Device Application

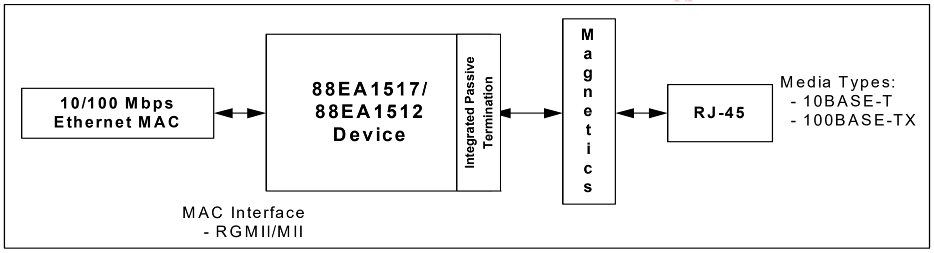


Figure 2: SGMII to Copper Application

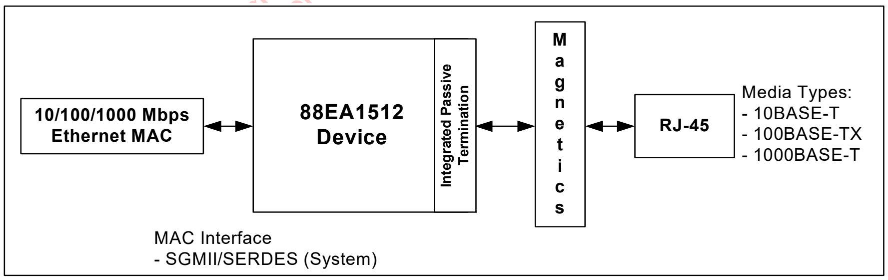


Figure 3: RGMII to Fiber/SGMII Application

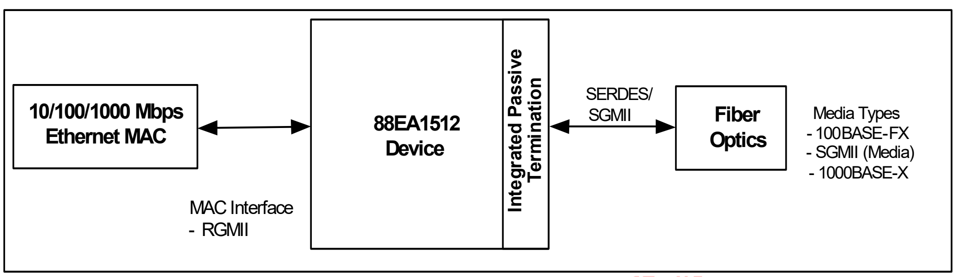
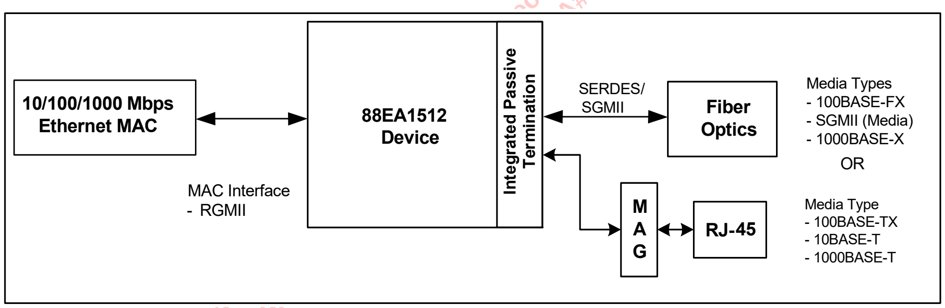


Figure 4: RGMII to Copper/Fiber/SGMII Auto-Media Application



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1.1 Pin Description

Table 2: Pin Type Definitions

Pin Type	Definition
H	Input with hysteresis
I/O	Input and output
I	Input only
O	Output only
PU	Internal pull-up
PD	Internal pull-down
D	Open drain output
Z	Tri-state output
mA	DC sink capability

1.1.1 88EA1517 40-Pin QFN Package Pinout

The 88EA1517 device is a 10/100BASE-T Ethernet transceiver.

Figure 5: 88EA1517 Device 40-Pin QFN Package (Top View)

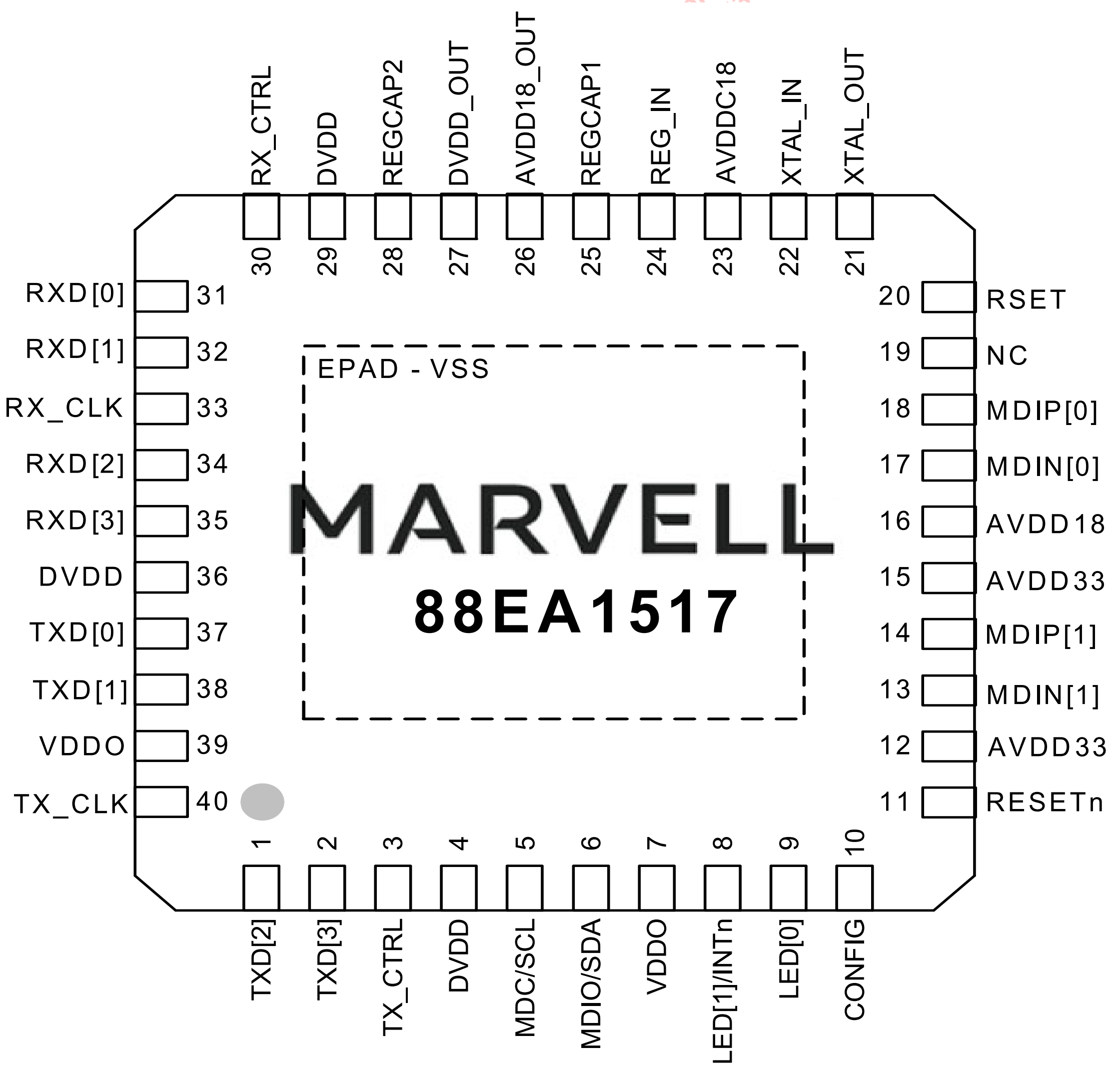


Table 3: Media Dependent Interface

40-QFN Pin #	Pin Name	Pin Type	Description
17 18	MDIN[0] MDIP[0]	I/O	Media Dependent Interface[0]. In 100BASE-TX and 10BASE-T modes in MDI configuration, MDIN/P[0] are used for the transmit pair. In MDIX configuration, MDIN/P[0] are used for the receive pair. Unused MDI pins must be left floating. The device contains an internal 100 ohm resistor between the MDIP/N[0] pins.
13 14	MDIN[1] MDIP[1]	I/O	Media Dependent Interface[1]. In 100BASE-TX and 10BASE-T modes in MDI configuration, MDIN/P[1] are used for the receive pair. In MDIX configuration, MDIN/P[1] are used for the transmit pair. Unused MDI pins must be left floating. The device contains an internal 100 ohm resistor between the MDIP/N[1] pins.

The RGMII/MII interface supports 10/100BASE-T modes of operation.

Table 4: RGMII/MII Interface

40-QFN Pin #	Pin Name	Pin Type	Description
40	TX_CLK	I/O	RGMII/MII Transmit Clock When RGMII is used, this pin is TXC and is therefore an input. It should be driven with a 25 MHz, or 2.5 MHz clock with ± 50 ppm tolerance depending on speed. When MII is used this pin is TX_CLK and therefore an output that drives out a 25 MHz or 2.5 MHz clock with ± 50 ppm tolerance depending on speed.
3	TX_CTRL	I	RGMII/MII Transmit Control. When RGMII is used, this pin is TX_CTL. This signal must be presented according to the RGMII timing mode requirements. When MII is used, this pin is TX_EN. This signal should be presented for sampling with the positive edge of TX_CLK.
2 1 38 37	TXD[3] TXD[2] TXD[1] TXD[0]	I	RGMII/MII Transmit Data. When RGMII is used, these pins are TD[3:0]. These signals must be presented according to the RGMII timing mode requirements. For 10/100BASE-T modes data is presented for sampling with the rising edge of TX_CLK only. When MII is used, these pins are TXD[3:0]. These signals should be presented for sampling with the positive edge of TX_CLK.
33	RX_CLK	O	RGMII/MII Receive Clock. When RGMII is used, this pin is RXC. It drives out a 25 MHz, or 2.5 MHz clock with ± 50 ppm tolerance derived from the received data stream depending on speed. When MII is used this pin is RX_CLK and drives out a 25 MHz, or 2.5 MHz clock with ± 50 ppm tolerance derived from the received data stream depending on speed.
30	RX_CTRL	O	RGMII/MII Receive Control When RGMII is used, this pin is RX_CTL. This signal will be presented according to the RGMII timing mode requirements. When MII is used, this pin is RX_DV. This signal will be presented for sampling with the positive edge of RX_CLK.
35 34 32 31	RXD[3] RXD[2] RXD[1] RXD[0]	O	RGMII/MII Receive Data. When RGMII is used, these pins are RD[3:0]. These signals will be presented according to the RGMII timing mode requirements. For 10/100BASE-T modes data is presented for sampling with the rising edge of RX_CLK only. When MII is used, these pins are TXD[3:0]. These signals are presented for sampling with the positive edge of RX_CLK.

Table 5: Management/TWSI Interface

40-QFN Pin #	Pin Name	Pin Type	Description
5	MDC/SCL	I	Management Interface/TWSI Clock When the Management Interface is used, this pin is MDC. A continuous clock stream is not required, however the maximum frequency allowed is 12 MHz. When TWSI is used this pin is SCL. The maximum frequency allowed is 400 KHz.
6	MDIO/SDA	I/O	Management Interface/TWSI Data When the Management Interface is used, this pin is MDIO and requires a pull-up resistor with a resistance from 1.5 kohm to 10 kohm. When TWSI is used this pin is SDA which is open drain and must be pulled up externally.

Table 6: LED Interface

40-QFN Pin #	Pin Name	Pin Type	Description
9	LED[0]	O	LED output.
8	LED[1]/INTn	I/O	LED output/PTP Event Request Input/PTP Trigger Generate Response Output. LED[1] can be configured as PTP Event Request Input or PTP Trigger Generate Response Output. Refer to 2.29 "Precise Timing Protocol (PTP) Time Stamping Support" on page 93 for further details. LED[1] pin also functions as an active low interrupt pin.

Table 7: Clock/Configuration/Reset/I/O

40-QFN Pin #	Pin Name	Pin Type	Description
10	CONFIG	I	Hardware Configuration.
22	XTAL_IN	I	Reference Clock. 25 MHz ± 50 ppm tolerance crystal reference or oscillator input. When XTAL_IN is driven directly from the oscillator or clock buffer, this pin should be AC coupled with a 0.1nF capacitor. **NOTE:** The XTAL_IN pin is not 2.5V/3.3V tolerant. Refer to 'Oscillator level shifting' application note to convert a 2.5V/3.3V clock source to 1.8V clock.

Table 7: Clock/Configuration/Reset/I/O (Continued)

40-QFN Pin #	Pin Name	Pin Type	Description
21	XTAL_OUT	O	Reference Clock. 25 MHz ± 50 ppm tolerance crystal reference. When the XTAL_OUT pin is not connected, it should be left floating.
11	RESETn	I	Hardware reset. Active low. 0 = Reset 1 = Normal operation

Table 8: Control and Reference

40-QFN Pin #	Pin Name	Pin Type	Description
20	RSET	I	Constant voltage reference. External 4.99 kohm 1% resistor connection to VSS is required for this pin.

Table 9: Power, Ground, and Internal Regulators

40-QFN Pin #	Pin Name	Pin Type	Description
23	AVDDC18	Power	Analog supply - 1.8V[1]. AVDDC18 can be supplied externally with 1.8V, or via the 1.8V internal regulator.
16	AVDD18	Power	Analog supply - 1.8V. AVDD18 can be supplied externally with 1.8V, or via the 1.8V internal regulator.
12 15	AVDD33	Power	Analog Supply - 3.3V.
24	REG_IN	Power	Analog Supply for the internal regulator – 3.3V. If the internal regulator is not used, this pin must be left floating– No connect. **NOTE:** For further details on pin connections, refer to the 2.35 "Regulators and Power Supplies" on page 125. **NOTE:** Ensure that this pin is left floating when the internal regulator is not used. Connecting this pin to either another power supply or ground will damage the device.
25 28	REGCAP1 REGCAP2		Capacitor terminal pins for the internal regulator. Connect a 220 nF ± 10% ceramic capacitor between REGCAP1 and REGCAP2 on the board and place it close to the device. If the internal regulator is not used, these pins must be left floating (no connect). Ensure that these pins are left floating when the internal regulator is not used. Connecting these two pins to either another power supply or ground will damage the device.
26	AVDD18_OUT	Power	Regulator output - 1.8V. If the internal regulator is used, this pin must be connected to the 1.8V power plane that connects to AVDD18 and AVDDC18. If the external supply is used, this pin must be left floating (no-connect).
27	DVDD_OUT	Power	Regulator output - 1.0V. If the internal regulator is used, this pin must be connected to the 1.0V power plane that connects to DVDD. If the external supply is used, this pin must be left floating (no-connect).
7 39	VDDO	Power	3.3V/2.5V/1.8V digital I/O supply[2]. VDDO must be supplied externally.
4 29 36	DVDD	Power	Digital core supply - 1.0V. DVDD can be supplied externally with 1.0V or via the 1.0V internal regulator.
EPAD	VSS	GND	Ground to device. The 40-pin QFN package has an exposed die pad (E-PAD) at its base. This EPAD must be soldered to VSS. Refer to the package mechanical drawings for the exact location and dimensions of the EPAD.

1. AVDDC18 supplies the XTAL_IN and XTAL_OUT pins.
2. VDDO supplies the MDC/SCL, MDIO/SDA, RESETn, LED[1:0], CONFIG, and the RGMII pins.

Table 10: No Connect

40-QFN Pin #	Pin Name	Pin Type	Description
19	NC	--	This pin must be left floating.

1.1.2 88EA1512 56-Pin QFN Package Pinout

The 88EA1512 device is a 10/100/1000BASE-T Gigabit Ethernet transceiver.

Figure 6: 88EA1512 Device 56-Pin QFN Package (Top View)

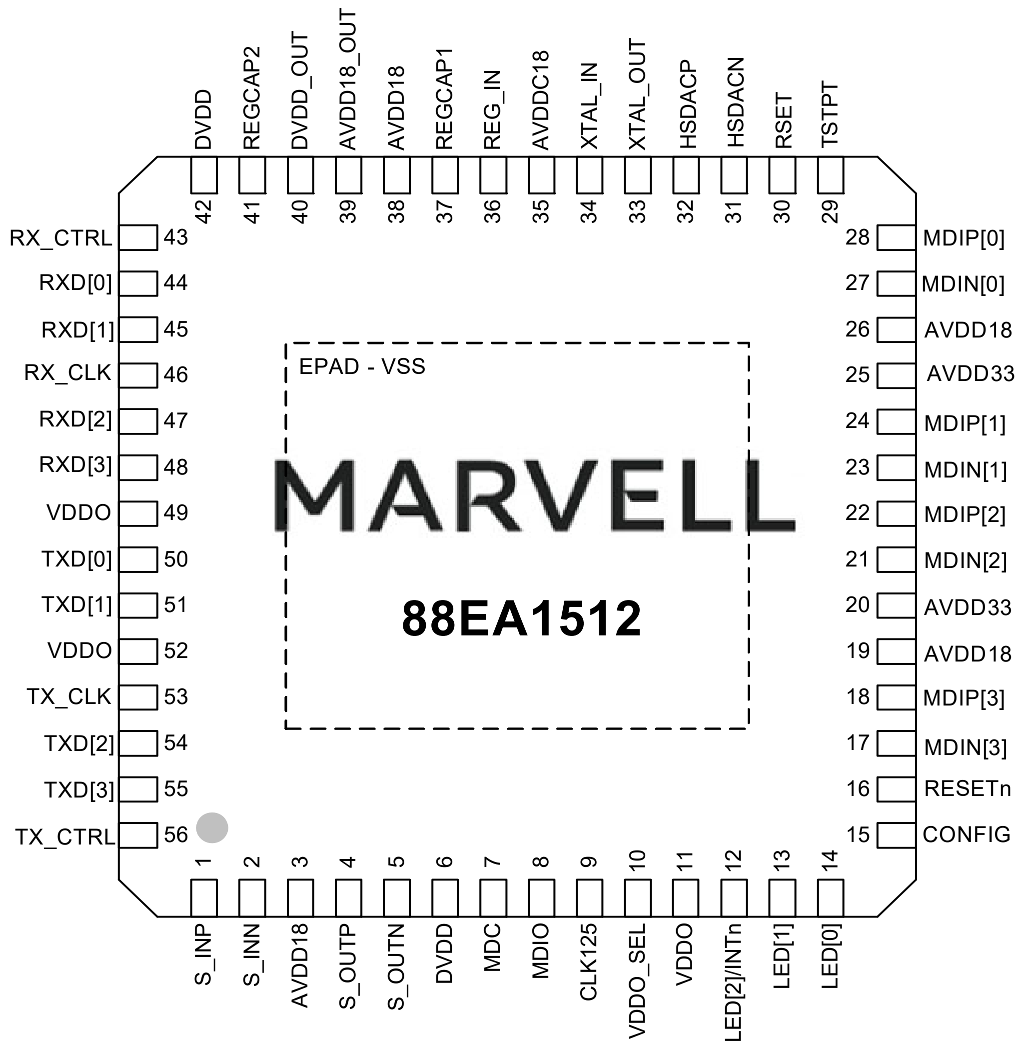


Table 11: Media Dependent Interface

56-QFN Pin #	Pin Name	Pin Type	Description
27 28	MDIN[0] MDIP[0]	I/O	Media Dependent Interface[0]. In 1000BASE-T mode in MDI configuration, MDIN/P[0] correspond to BI_DA±. In MDIX configuration, MDIN/P[0] correspond to BI_DB±. In 100BASE-TX and 10BASE-T modes in MDI configuration, MDIN/P[0] are used for the transmit pair. In MDIX configuration, MDIN/P[0] are used for the receive pair. **NOTE:** Unused MDI pins must be left floating. The device contains an internal 100 ohm resistor between the MDIP/N[0] pins.
23 24	MDIN[1] MDIP[1]	I/O	Media Dependent Interface[1]. In 1000BASE-T mode in MDI configuration, MDIN/P[1] correspond to BI_DB±. In MDIX configuration, MDIN/P[1] correspond to BI_DA±. In 100BASE-TX and 10BASE-T modes in MDI configuration, MDIN/P[1] are used for the receive pair. In MDIX configuration, MDIN/P[1] are used for the transmit pair. **NOTE:** Unused MDI pins must be left floating. The device contains an internal 100 ohm resistor between the MDIP/N[1] pins.
21 22	MDIN[2] MDIP[2]	I/O	Media Dependent Interface[2]. In 1000BASE-T mode in MDI configuration, MDIN/P[2] correspond to BI_DC±. In MDIX configuration, MDIN/P[2] corresponds to BI_DD±. In 100BASE-TX and 10BASE-T modes, MDIN/P[2] are not used. **NOTE:** Unused MDI pins must be left floating. The device contains an internal 100 ohm resistor between the MDIP/N[2] pins.
17 18	MDIN[3] MDIP[3]	I/O	Media Dependent Interface[3]. In 1000BASE-T mode in MDI configuration, MDIN/P[3] correspond to BI_DD±. In MDIX configuration, MDIN/P[3] correspond to BI_DC±. In 100BASE-TX and 10BASE-T modes, MDIN/P[3] are not used. **NOTE:** Unused MDI pins must be left floating. The device contains an internal 100 ohm resistor between the MDIP/N[3] pins.

Table 12: RGMII/MII Interface

56-QFN Pin #	Pin Name	Pin Type	Description
53	TX_CLK	I/O[1]	RGMII/MII Transmit Clock. When RGMII is used, this pin is TXC and is therefore an input. It should be driven with a 125 MHz, 25 MHz, or 2.5 MHz clock with ± 50 ppm tolerance depending on speed. When MII is used this pin is TX_CLK and therefore an output that drives out a 25 MHz, or 2.5 MHz clock with ± 50 ppm tolerance depending on speed.
56	TX_CTRL	I	RGMII/MII Transmit Control. When RGMII is used, This pin is TX_CTL. This signal must be presented according to the RGMII timing mode requirements. When MII is used, this pin is TX_EN. This signal should be presented for sampling with the positive edge of TX_CLK.
55 54 51 50	TXD[3] TXD[2] TXD[1] TXD[0]	I	RGMII/MII Transmit Data. When RGMII is used, these pins are TD[3:0]. These signals must be presented according to the RGMII timing mode requirements. For 1000BT, data is presented at a double data rate (both edges of TX_CLK are used to sample) whereas for 10/100BASE-T modes data is presented for sampling with the rising edge of TX_CLK only. When MII is used, these pins are TXD[3:0]. These signals should be presented for sampling with the positive edge of TX_CLK.
46	RX_CLK	O	RGMII/MII Receive Clock. When RGMII is used, this pin is RXC. It drives out a 125 MHz, 25 MHz, or 2.5 MHz clock with ± 50 ppm tolerance derived from the received data stream depending on speed. When MII is used this pin is RX_CLK and drives out a 25 MHz, or 2.5 MHz clock with ± 50 ppm tolerance derived from the received data stream depending on speed.
43	RX_CTRL	O	RGMII/MII Receive Control. When RGMII is used, This pin is RX_CTL. This signal will be presented according to the RGMII timing mode requirements. When MII is used, this pin is RX_DV. This signal will be presented for sampling with the positive edge of RX_CLK.
48 47 45 44	RXD[3] RXD[2] RXD[1] RXD[0]	O	RGMII/MII Receive Data. When RGMII is used, These pins are RD[3:0]. These signals will be presented according to the RGMII timing mode requirements. For 1000BT, data is presented at a double data rate (with respect to both edges of RX_CLK) whereas for 10/100BASE-T modes data is presented for sampling with the rising edge of RX_CLK only. When MII is used, these pins are RXD[3:0]. These signals are presented for sampling with the positive edge of RX_CLK.

1. This pin is an output pin on the MII Interface.

Table 13: Management/TWSI Interface

56-QFN Pin #	Pin Name	Pin Type	Description
7	MDC/SCL	I	Management Interface/TWSI Clock When the Management Interface is used, this pin is MDC. A continuous clock stream is not required for MDC, however the maximum frequency allowed is 12 MHz. When TWSI is used this pin is SCL.The maximum frequency allowed is 400 KHz.
8	MDIO/SDA	I/O	Management Interface/TWSI Data When the Management Interface is used, this pin is MDIO and requires a pull-up resistor with a resistance from 1.5 kohm to 10 kohm. When TWSI is used this pin is SDA which is open drain and must be pulled up externally.

Table 14: LED Interface

56-QFN Pin #	Pin Name	Pin Type	Description
14	LED[0]	O	LED output.
13	LED[1]	I/O	LED output/PTP Event Request Input/PTP Trigger Generate Response Output. LED[1] can be configured as PTP Event Request Input or PTP Trigger Generate Response Output. Refer to 2.29 "Precise Timing Protocol (PTP) Time Stamping Support" on page 93 for further details.
12	LED[2]/INTn	O, D	LED/Interrupt outputs. LED[2] pin also functions as an active low interrupt pin.

Table 15: Clock/Configuration/Reset/I/O

56-QFN Pin #	Pin Name	Pin Type	Description
15	CONFIG	I	Hardware Configuration.
9	CLK125	O	125 MHz Clock Output or SyncE Recovered Clock Output. Refer to 2.28 "CLK125" on page 92 for details.
34	XTAL_IN	I	Reference Clock. 25 MHz ± 50 ppm tolerance crystal reference or oscillator input. When XTAL_IN is driven directly from the oscillator or clock buffer, this pin should be AC coupled with a 0.1nF capacitor. **NOTE:** The XTAL_IN pin is not 2.5V/3.3V tolerant. Refer to 'Oscillator level shifting' application note to convert a 2.5V/3.3V clock source to 1.8V clock.

Table 15: Clock/Configuration/Reset/I/O (Continued)

56-QFN Pin #	Pin Name	Pin Type	Description
33	XTAL_OUT	O	Reference Clock. 25 MHz ± 50 ppm tolerance crystal reference. When the XTAL_OUT pin is not connected to a crystal, it should be left floating.
16	RESETn	I	Hardware reset. Active low. 0 = Reset 1 = Normal operation

Table 16: SGMII I/Os

56-QFN Pin #	Pin Name	Pin Type	Description
2 1	S_INN S_INP	I	SGMII Receive Data. 1.25 GBaud input - Positive and Negative.
5 4	S_OUTN S_OUTP	O	SGMII Transmit Data. 1.25 GBaud output - Positive and Negative.

Table 17: Control and Reference

56-QFN Pin #	Pin Name	Pin Type	Description
30	RSET	I	Constant voltage reference. External 4.99 kohm 1% resistor connection to VSS is required for this pin.

Table 18: Test

56-QFN Pin #	Pin Name	Pin Type	Description
31 32	HSDACN HSDACP	Analog O	Test Pins. These pins are used to bring out a differential TX_TCLK. Connect these pins with a 50 ohm termination resistor to VSS for IEEE testing. If IEEE testing are not important, these pins may be left floating.
29	TSTPT	O	DC Test Point. The TSTPT pin should be left floating.

Table 19: Power, Ground, and Internal Regulators

56-QFN Pin #	Pin Name	Pin Type	Description
35	AVDDC18	Power	Analog supply - 1.8V[1]. AVDDC18 can be supplied externally with 1.8V, or via the 1.8V internal regulator.
3 19 26 38	AVDD18	Power	Analog supply - 1.8V. AVDD18 can be supplied externally with 1.8V, or via the 1.8V internal regulator.
20 25	AVDD33	Power	Analog Supply - 3.3V.
36	REG_IN	Power	Analog Supply for the internal regulator – 3.3V. If the internal regulator is not used, this pin must be left floating– No connect. **NOTE:** For further details on pin connections, refer to the 2.35 "Regulators and Power Supplies" on page 125. **NOTE:** Ensure that this pin is left floating when the internal regulator is not used. Connecting this pin to either another power supply or ground will damage the device.
37 41	REGCAP1 REGCAP2		Capacitor terminal pins for the internal regulator. Connect a 220 nF ± 10% ceramic capacitor between REGCAP1 and REGCAP2 on the board and place it close to the device. If the internal regulator is not used, these pins must be left floating (no connect). **NOTE:** Ensure that these pins are left floating when the internal regulator is not used. Connecting these two pins to either another power supply or ground will damage the device.
39	AVDD18_OUT	Power	Regulator output - 1.8V. If the internal regulator is used, this pin must be connected to 1.8V power plane that connected to AVDD18 and AVDDC18. If the external supply is used, this pin must be left floating (no-connect).
40	DVDD_OUT	Power	Regulator output - 1.0V. If the internal regulator is used, this pin must be connected to 1.0V power plane that connected to DVDD. If the external supply is used, this pin must be left floating (no-connect).
11 49 52	VDDO	Power	3.3V or 2.5V or 1.8V digital I/O supply[2]. If VDDO is 2.5V or 3.3V it must be supplied externally. If VDDO is 1.8V, the 1.8V regulator output can be used to supply this or it may be supplied externally.
10	VDDO_SEL	Power	VDDO Voltage Control. For VDDO 2.5V/3.3V operation, VDDO_SEL must be tied to VSS. For VDDO 1.8V operation, VDDO_SEL must be tied to VDDO.
6 42	DVDD	Power	Digital core supply - 1.0V. DVDD can be supplied externally with 1.0V or via the 1.0V internal regulator.
EPAD	VSS	GND	Ground to device. The 56-pin QFN package has an exposed die pad (E-PAD) at its base. This EPAD must be soldered to VSS. Refer to the package mechanical drawings for the exact location and dimensions of the EPAD.

1. AVDDC18 supplies the XTAL_IN and XTAL_OUT pins.
2. VDDO supplies the MDC, MDIO, RESETn, LED[2:0], CONFIG, CLK125, and the RGMII pins.

1.2 Pin Assignment List

1.2.1 88EA1517 40-Pin QFN Pin Assignment List - Alphabetical by Signal Name

Pin #	Pin Name	Pin #	Pin Name
26	AVDD18_OUT	28	REGCAP2
16	AVDD18	24	REG_IN
23	AVDDC18	33	RX_CLK
12	AVDD33	30	RX_CTRL
15	AVDD33	11	RESETn
10	CONFIG	20	RSET
4	DVDD	31	RXD[0]
29	DVDD	32	RXD[1]
36	DVDD	34	RXD[2]
27	DVDD_OUT	35	RXD[3]
9	LED[0]	40	TX_CLK
8	LED[1]/INTn	3	TX_CTRL
5	MDC/SCL	37	TXD[0]
17	MDIN[0]	38	TXD[1]
13	MDIN[1]	1	TXD[2]
6	MDIO/SDA	2	TXD[3]
18	MDIP[0]	7	VDDO
14	MDIP[1]	39	VDDO
19	NC	22	XTAL_IN
25	REGCAP1	21	XTAL_OUT

1.2.2 88EA1512 56-Pin QFN Pin Assignment List - Alphabetical by Signal Name

Pin #	Pin Name	Pin #	Pin Name
39	AVDD18_OUT	37	REGCAP1
3	AVDD18	41	REGCAP2
19	AVDD18	36	REG_IN
26	AVDD18	46	RX_CLK
38	AVDD18	43	RX_CTRL
35	AVDDC18	16	RESETn
20	AVDD33	30	RSET
25	AVDD33	44	RXD[0]
9	CLK125	45	RXD[1]
15	CONFIG	47	RXD[2]
6	DVDD	48	RXD[3]
40	DVDD_OUT	2	S_INN
42	DVDD	1	S_INP
31	HSDACN	5	S_OUTN
32	HSDACP	4	S_OUTP
14	LED[0]	29	TSTPT
13	LED[1]	53	TX_CLK
12	LED[2]/INTn	56	TX_CTRL
7	MDC	50	TXD[0]
27	MDIN[0]	51	TXD[1]
23	MDIN[1]	54	TXD[2]
21	MDIN[2]	55	TXD[3]
17	MDIN[3]	11	VDDO
8	MDIO	49	VDDO
28	MDIP[0]	52	VDDO
24	MDIP[1]	10	VDDO_SEL
22	MDIP[2]	34	XTAL_IN
18	MDIP[3]	33	XTAL_OUT

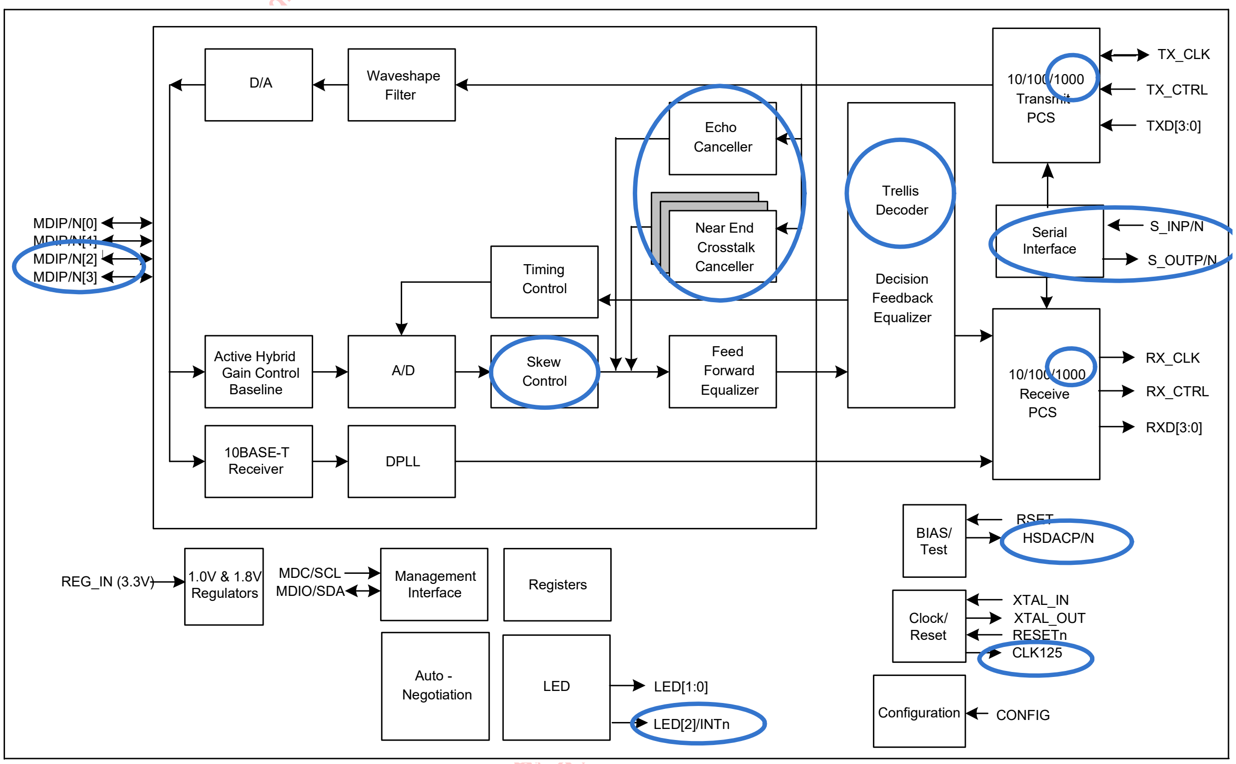
2 PHY Functional Specifications

The 88EA1512 is a single-port 10/100/1000 Gigabit Ethernet transceiver. The 88EA1517 is a single-port 10/100 Ethernet transceiver. Figure 7 shows the functional block diagram of the device. Items in Figure 7 that are circled in blue belong to 88EA1512 otherwise the items belong to both the devices.

Note

Refer to Overview for a list of features supported by the devices.

Figure 7: Device Functional Block Diagram



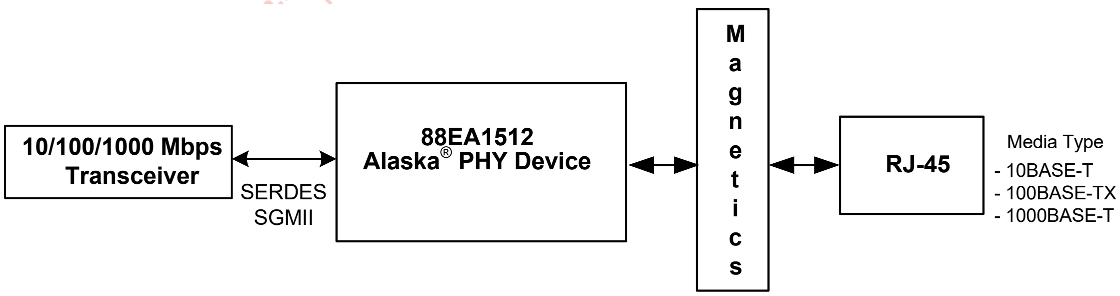
2.1 Modes of Operation and Major Interfaces

These devices collectively have four separate major electrical interfaces:

- MDI Interface to Copper Cable (88EA1517/88EA1512 devices)
- SERDES/SGMII Interface (88EA1512 device only)
- RGMII Interface (88EA1517/88EA1512 devices)
- MII Interface (88EA1517/88EA1512 devices)

The MDI Interface is always a media interface. The RGMII/MII interface is always a system interface. The SGMII interface can either be a System interface, or a media interface. (The System interface is also known as MAC interface. It is typically the connection between the PHY and the MAC or the system ASIC). The block diagrams describing the different applications of the 88EA1517/88EA1512 devices can be found below.

Figure 8: 88EA1512 SGMII/SERDES System to Copper Media Interface Example



The 88EA1512 device can be used in media conversion applications which require a conversion from 1000BASE-T to 1000BASE-X provided the 1000BASE-X auto negotiation is disabled on both the upstream device and 88EA1512. 88EA1512 device in this application must be configured to operate in SGMII to Copper(MODE[2:0]=001) for the conversion with the SGMII Auto-Negotiation disabled through Register 0_1.12=0.

When the SGMII interface is used as a system interface, the device implements the PHY SGMII Auto-Negotiation status (link, duplex, etc.) advertisements as specified in the Cisco SGMII specification. The System interface replicates the speed and duplex setting of the media interface

When the SGMII interface is used as a Media interface, the device implements the MAC SGMII Auto-Negotiation function, which monitors PHY status advertisements

For details of how SGMII Auto-Negotiation operates, see Section 2.10.3 as well as the Cisco SGMII specification 1.8.

Figure 9: 88EA1517 RGMII System to Copper Interface Example

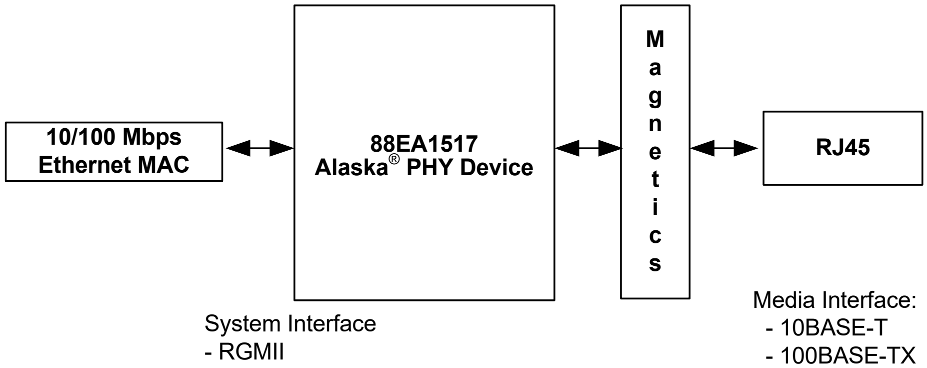


Figure 10: 88EA1517 MII System to Copper Interface Example

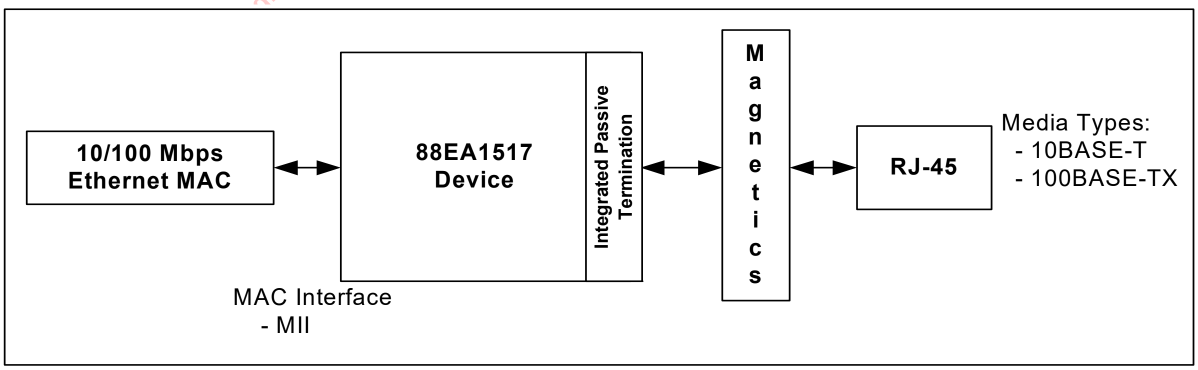


Figure 11: 88EA1512 RGMII System to SERDES Interface Example

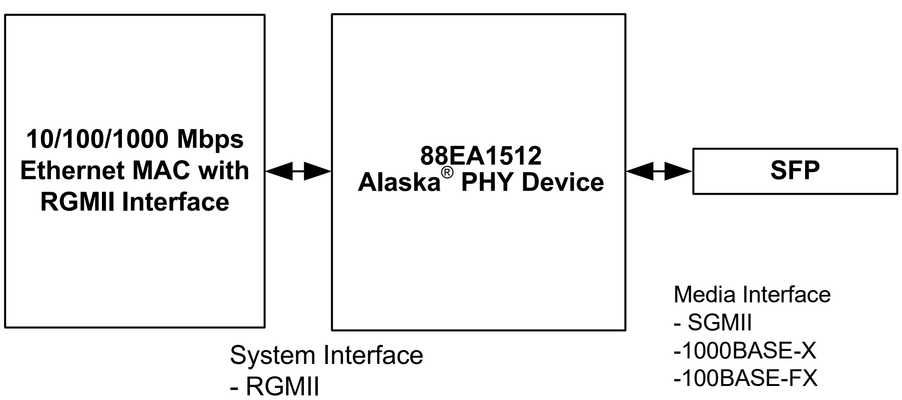
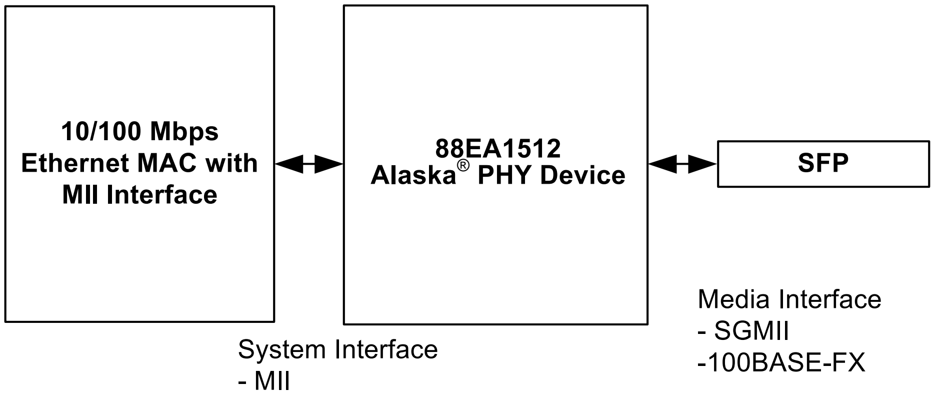


Figure 12: 88EA1512 MII System to SERDES Interface Example



The 88EA1512 device supports the modes of operation as shown in Table 20 and Table 21. On hardware reset the 88EA1512 device, by default is configured to RGMII to Auto-Media detect (Copper/1000BASE-X) if RGMII is the configured System mode and MII to copper if MII is the configured System mode. The System mode is either RGMII or MII depending on the Hardware Configuration of the RX_CTRL pin.

The behavior of the 1.25 GHz SERDES interface is selected by setting the MODE[2:0] register in 20_18.2:0. The SERDES can operate in 100BASE-FX, 1000BASE-X, SGMII (System), and SGMII (Media).

Table 20: MODE[2:0] Select (When System mode is RGMII)

MODE[2:0] Register 20_18.2:0	Description
000	RGMII (System mode) to Copper
001	SGMII (System mode) to Copper
010	RGMII (System mode) to 1000BASE-X
011	RGMII (System mode) to 100BASE-FX (when 20_18.6 = 0). RGMII (System mode) to Auto Media Detect Copper/100BASE-FX (when 20_18.6 = 1)
100	RGMII (System mode) to SGMII (Media mode).
101	Reserved.
110	RGMII (System mode) to Auto Media Detect Copper/SGMII (Media mode)
111	RGMII (System mode) to Auto Media Detect Copper/1000BASE-X

When the mode is set for Auto-Media Detect, Register 20_18.5:4 specifies which media is the preferred media. See Section Section 2.6 "Fiber/Copper Auto-Selection" on page 55 for more details.

Table 21: MODE[2:0] Select (When System mode is MII)

MODE[2:0] Register 20_18.2:0	Description
000	MII (System mode) to Copper
001	SGMII (System mode) to Copper
010	MII (System mode) to SGMII (Media mode)
011	MII (System mode) to 100BASE-FX
100	MII (System mode) to SGMII (Media mode)
101	Reserved
110	Reserved
111	Reserved

When link is up, two of the three interfaces pass packets back and forth. The unused interface is powered down.

When link is up, two of the three interfaces pass packets back and forth. The unused interface is powered down. There is no need to power down the unused interface via Registers 0_0.11 and 0_1.11. The unused interface will automatically power down when not needed.

The 88EA1517 device only supports RGMII or MII to copper mode for 10BASE-T and 100BASE-T operation. On hardware reset the 88EA1517 device, by default is configured to RGMII to copper if RGMII is the configured System mode and MII to copper if MII is the configured System mode. The System mode is either RGMII or MII depending on the Hardware Configuration of the RX_CTRL pin. Note that 20_18.2:0 = 000 is the only defined setting for 88EA1517.

Table 22: 88EA1517 and MII Interface Introduction Changes

Change Implemented	Applies to MII mode for 88EA1517	Applies to RGMII mode for 88EA1517	Applies to MII mode for 88EA1512
The MSB speed bit 0_0.6 is 0 always and a read of this bit always gives 0. A value of 1 programmed is ignored since Gigabit operation is not possible.	Yes	Yes	Yes
100BASE-T is selected by default. 0_0.13 defaults to 1 and may be programmed to 0 for 10BT.	Yes	Yes	Yes
Copper Auto-Negotiation is off by default (0_0.12 = 0). The user can write a 1 later and enable Auto-Negotiation if they want. This allows the 100BT link up to be fast.	Yes	Yes	No
Full-duplex bit 0_0.8 is 1 always and a read of this bit always gives 1. A value of 0 programmed is ignored since half-duplex operation is not possible.	Yes	No	Yes

Table 22: 88EA1517 and MII Interface Introduction Changes (Continued)

Change Implemented	Applies to MII mode for 88EA1517	Applies to RGMII mode for 88EA1517	Applies to MII mode for 88EA1512
4_0.7 and 4_0.5 are forced to 0 to make sure half-duplex is not advertised if Auto-Negotiation is enabled.	Yes	No	Yes
MDI or MDIX is forced depending on the MDI_MDIX config value. 16_0.5 defaults to this value with 16_0.6 = 0. If the user wants to program a different value later for 16_0.6:5 then this is allowed.	Yes	Yes	No
When EEE mode is enabled, the device is in Master mode (0_18.0 is forced to 1, is RO and only 1 may be read from it) where the EEE Buffer is enabled and LPI received are converted to regular idles on the MAC interface.	Yes	No	Yes
The default of 21_2.6 is 0 and only a value of 0 can be read (write of 1 is ignored)	Yes	Yes	Yes
The default of 21_2.13 is 1 and the user can write 0 if desired.	Yes	Yes	Yes
The default of 26_0.14 is 1 and only a value of 1 can be read (write of 0 is ignored).	Yes	Yes	Yes

2.2 Copper Media Interface

The copper interface consists of the MDIP/N[3:0] pins that connect to the physical media for 1000BASE-T, 100BASE-TX, and 10BASE-T modes of operation.

The device integrates MDI interface termination resistors. The IEEE 802.3 specification requires that both sides of a link have termination resistors to prevent reflections. Traditionally, these resistors and additional capacitors are placed on the board between a PHY device and the magnetics. The resistors have to be very accurate to meet the strict IEEE return loss requirements. Typically, ± 1% accuracy resistors are used on the board. These additional components between the PHY and the magnetics complicate board layout. Integrating the resistors has many advantages including component cost savings, better ICT yield, board reliability improvements, board area savings, improved layout, and signal integrity improvements.

2.2.1 Transmit Side Network Interface

2.2.1.1 Multi-mode TX Digital to Analog Converter

The device incorporates a multi-mode transmit DAC to generate filtered 4D PAM 5, MLT3, or Manchester coded symbols. The transmit DAC performs signal wave shaping to reduce EMI. The transmit DAC is designed for very low parasitic loading capacitances to improve the return loss requirement, which allows the use of low cost transformers.

2.2.1.2 Slew Rate Control and Waveshaping

In 1000BASE-T mode, partial response filtering and slew rate control are used to minimize high frequency EMI. In 100BASE-TX mode, slew rate control is used to minimize high frequency EMI. In 10BASE-T mode, the output waveform is pre-equalized via a digital filter.

2.2.2 Encoder

2.2.2.1 1000BASE-T

In 1000BASE-T mode, the transmit data bytes are scrambled to 9-bit symbols and encoded into 4D PAM5 symbols. Upon initialization, the initial scrambling seed is determined by the PHY address. This prevents multiple devices from outputting the same sequence during idle, which helps to reduce EMI.

2.2.2.2 100BASE-TX

In 100BASE-TX mode, the transmit data stream is 4B/5B encoded, serialized, and scrambled.

2.2.2.3 10BASE-T

In 10BASE-T mode, the transmit data is serialized and converted to Manchester encoding.

2.2.3 Receive Side Network Interface

2.2.3.1 Analog to Digital Converter

The device incorporates an advanced high speed ADC on each receive channel with greater resolution than the ADC used in the reference model of the IEEE 802.3ab standard committee. Higher resolution ADC results in better SNR, and therefore, lower error rates. Patented architectures and design techniques result in high differential and integral linearity, high power supply noise rejection, and low metastability error rate. The ADC samples the input signal at 125 MHz.

2.2.3.2 Active Hybrid

The device employs a sophisticated on-chip hybrid to substantially reduce the near-end echo, which is the super-imposed transmit signal on the receive signal. The hybrid minimizes the echo to reduce the precision requirement of the digital echo canceller. The on-chip hybrid allows both the transmitter and receiver to use the same transformer for coupling to the twisted pair cable, which reduces the cost of the overall system.

2.2.3.3 Echo Canceller

Residual echo not removed by the hybrid and echo due to patch cord impedance mismatch, patch panel discontinuity, and variations in cable impedance along the twisted pair cable result in drastic SNR degradation on the receive signal. The device employs a fully developed digital echo canceller to adjust for echo impairments from more than 100 meters of cable. The echo canceller is fully adaptive to compensate for the time varying nature of channel conditions.

2.2.3.4 NEXT Canceller

The 1000BASE-T physical layer uses all 4 pairs of wires to transmit data to reduce the baud rate requirement to only 125 MHz. This results in significant high frequency crosstalk between adjacent pairs of cable in the same bundle. The device employs 3 parallel NEXT cancellers on each receive channel to cancel any high frequency crosstalk induced by the adjacent 3 transmitters. A fully adaptive digital filter is used to compensate for the time varying nature of channel conditions.

2.2.3.5 Baseline Wander Canceller

Baseline wander is more problematic in the 1000BASE-T environment than in the traditional 100BASE-TX environment due to the DC baseline shift in both the transmit and receive signals. The device employs an advanced baseline wander cancellation circuit to automatically compensate for this DC shift. It minimizes the effect of DC baseline shift on the overall error rate.

2.2.3.6 Digital Adaptive Equalizer

The digital adaptive equalizer removes inter-symbol interference at the receiver. The digital adaptive equalizer takes unequalized signals from ADC output and uses a combination of feedforward equalizer (FFE) and decision feedback equalizer (DFE) for the best-optimized signal-to-noise (SNR) ratio.

2.2.3.7 Digital Phase Lock Loop

In 1000BASE-T mode, the slave transmitter must use the exact receive clock frequency it sees on the receive signal. Any slight long-term frequency phase jitter (frequency drift) on the receive signal must be tracked and duplicated by the slave transmitter; otherwise, the receivers of both the slave and master physical layer devices have difficulty canceling the echo and NEXT components. In the device, an advanced DPLL is used to recover and track the clock timing information from the receive signal. This DPLL has very low long-term phase jitter of its own, thereby maximizing the achievable SNR.

2.2.3.8 Link Monitor

The link monitor is responsible for determining if link is established with a link partner. In 10BASE-T mode, link monitor function is performed by detecting the presence of valid link pulses (NLPs) on the MDIP/N pins.

In 100BASE-TX and 1000BASE-T modes, link is established by scrambled idles.

If Force Link Good Register 16_0.10 is set high, the link is forced to be good and the link monitor is bypassed for 100BASE-TX and 10BASE-T modes. In the 1000BASE-T mode, Register 16_0.10 has no effect.

2.2.3.9 Signal Detection

In 1000BASE-T mode, signal detection is based on whether the local receiver has acquired lock to the incoming data stream.

In 100BASE-TX mode, the signal detection function is based on the receive signal energy detected on the MDIP/N pins that is continuously qualified by the squelch detect circuit, and the local receiver acquiring lock.

2.2.4 Decoder

2.2.4.1 1000BASE-T

In 1000BASE-T mode, the receive idle stream is analyzed so that the scrambler seed, the skew among the 4 pairs, the pair swap order, and the polarity of the pairs can be accounted for. Once calibrated, the 4D PAM 5 symbols are converted to 9-bit symbols that are then descrambled into 8-bit data values. If the descrambler loses lock for any reason, the link is brought down and calibration is restarted after the completion of Auto-Negotiation.

2.2.4.2 100BASE-TX

In 100BASE-TX mode, the receive data stream is recovered and converted to NRZ. The NRZ stream is descrambled and aligned to the symbol boundaries. The aligned data is then parallelized and 5B/4B decoded. The receiver does not attempt to decode the data stream unless the scrambler is locked. The descrambler "locks" to the *scrambler* state after detecting a sufficient number of consecutive idle code-groups. Once locked, the descrambler continuously monitors the data stream to make sure that it has not lost synchronization. The descrambler is always forced into the *unlocked* state when a link failure condition is detected, or when insufficient idle symbols are detected.

2.2.4.3 10BASE-T

In 10BASE-T mode, the recovered 10BASE-T signal is decoded from Manchester to NRZ, and then aligned. The alignment is necessary to insure that the start of frame delimiter (SFD) is aligned to the nibble boundary.

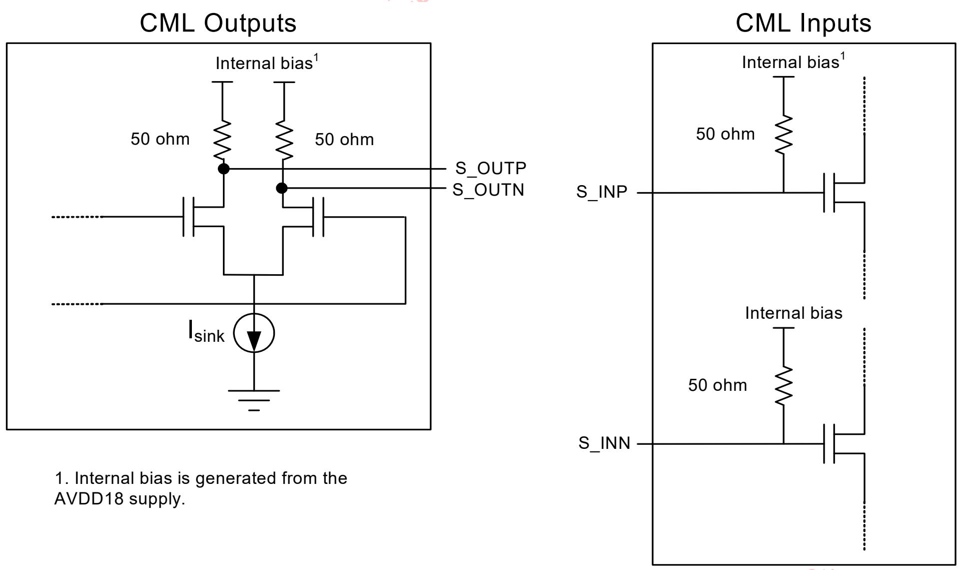
2.3 1.25 GHz SERDES Interface

The 1.25 GHz SERDES Interface, supported by the 88EA1512, can be configured as an SGMII to be hooked up to a MAC or as a 100BASE-FX/1000BASE-X/SGMII to be hooked up to the media.

2.3.1 Electrical Interface

The input and output buffers of the 1.25 GHz SERDES interface are internally terminated by 50 ohm impedance. No external terminations are required. The output swing can be adjusted by programming Register 26_1.2:0. The 1.25 GHz SERDES I/Os are Current Mode Logic (CML) buffers. CML I/Os can be used to connect to other components with PECL or LVDS I/Os. See the "Reference Design Schematics" and "Fiber Interface" application note for details.

Figure 13: CML I/Os



1. Internal bias is generated from the AVDD18 supply.

2.4 MAC Interfaces

2.4.1 SGMII Interface

The 88EA1512 device supports the SGMII specification revision 1.8, except for the carrier extension block that has to be carried out in software. This interface supports 10, 100 and 1000 Mbps modes of operation.

2.4.1.1 SGMII Speed and Link

When the SGMII MAC interface is used, the media interface can only be copper. The operational speed of the SGMII MAC interface is determined according to Table 23 media interface status and/or loopback mode.

Table 23: SGMII (System Interface) Operational Speed

Link Status or Media Interface Status	SGMII (MAC Interface) Speed
No Link	Determined by speed setting of Register 21_2.6,13
MAC Loopback	Determined by speed setting of Register 21_2.6, 13
1000BASE-T at 1000 Mbps	1000 Mbps
100BASE-TX at 100 Mbps	100 Mbps
10BASE-T at 10 Mbps	10 Mbps

Two registers are available to determine whether the SGMII achieved link and sync. Status Register 17_1.5 indicates that the SERDES locked onto the incoming KDKDKD… sequence. Register 17_1.10 indicates whether link is established on the SERDES. If SGMII Auto-Negotiation is disabled, Register 17_1.10 has the same meaning as Register 17_1.5. If SGMII Auto-Negotiation is enabled, then Register 17_1.10 indicates whether SGMII Auto-Negotiation successfully established link.

2.4.1.2 SGMII TRR Blocking

When the SGMII receives a packet with odd number of bytes, a single symbol of carrier extension will be passed on and transmitted onto 1000BASE-T. This carrier extension may cause problems with full-duplex MACs that incorrectly handle the carrier extension symbols. When Register 16_1.13 is set to 1, all carrier extend and carrier extend with error symbols received by the SGMII will be converted to idle symbols when operating in full-duplex. Carrier extend and carrier extend with error symbols will not be blocked when operating in half-duplex, or if Register 16_1.13 is set to 0. Note that symbol errors will continue to be propagated regardless of the setting of Register 16_1.13.

2.4.1.3 False SERDES Link Up Prevention

The SERDES interface can operate in 1000BASE-X mode where an unconnected optical receiver can sometimes send full swing noise into the PHY. This random noise will look like a real signal and falsely cause the 1000BASE-X PCS to link up.

A noise filtering state machine can be enabled to reduce the probability of false link up. When the state machine is enabled it will cause a small delay in link up time. 1000BASE-X noise filtering is enabled through the register below.

Table 24: Fiber Noise Filtering

Register	Function	Setting	Mode	HW Rst	SW Rst
26_1.14	1000BASE-X Noise Filtering	1 = Enable 0 = Disable	R/W	0	Retain

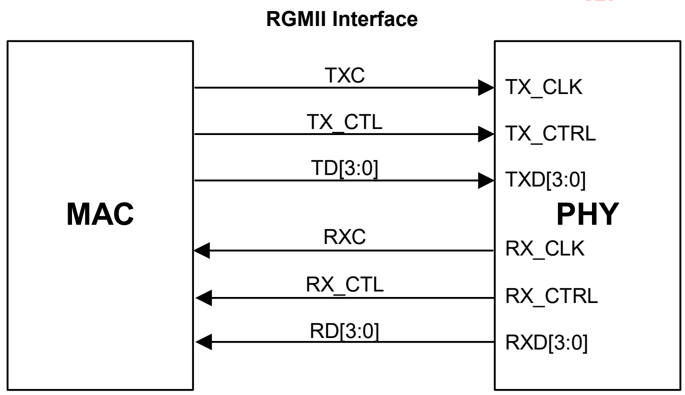
2.4.2 RGMII Interface

The 88EA1517 and 88EA1512 both support the RGMII specification (Version 1.2a, 9/22/2000, version 2.0, 04/2002, version 2.6,10/26/2009) where 88EA1517 only supports 100BASE-TX and 10BASE-T modes. Four RGMII timing modes, with different receive clock to data timing and transmit clock to data timing, can be programmed by setting 21_2.4 and 21_2.5 described in Register 21_2.5. See “RGMII Delay Timing for different RGMII timing modes” on page 289 for timing details.

Table 25: RGMII Signal Mapping

Device Pin Name	RGMII Spec Pin Name	Description
TX_CLK	TXC	125 MHz, 25 MHz, or 2.5 MHz transmit clock with ±50 ppm tolerance based on the selected speed.
TX_CTRL	TX_CTL	Transmit Control Signals. TX_EN is encoded on the rising edge of TX_CLK, TX_ER XORed with TX_EN is encoded on the falling edge of TX_CLK.
TXD[3:0]	TD[3:0]	Transmit Data. In 1000BASE-T mode, TXD[3:0] are synchronous with both edges of TX_CLK. In 100BASE-TX and 10BASE-T modes, TXD[3:0] are synchronous with the rising edge of TX_CLK.
RX_CLK	RXC	125 MHz, 25 MHz, or 2.5 MHz receive clock derived from the received data stream and based on the selected speed.
RX_CTRL	RX_CTL	Receive Control Signals. RX_DV is encoded on the rising edge of RX_CLK, RX_ER XORed with RX_DV is encoded on the falling edge of RX_CLK.
RXD[3:0]	RD[3:0]	Receive Data. In 1000BASE-T mode, RXD[3:0] are synchronous with both edges of RX_CLK. In 100BASE-TX and 10BASE-T modes, RXD[3:0] are synchronous with the rising edge of RX_CLK.

Figure 14: RGMII Signal Diagram



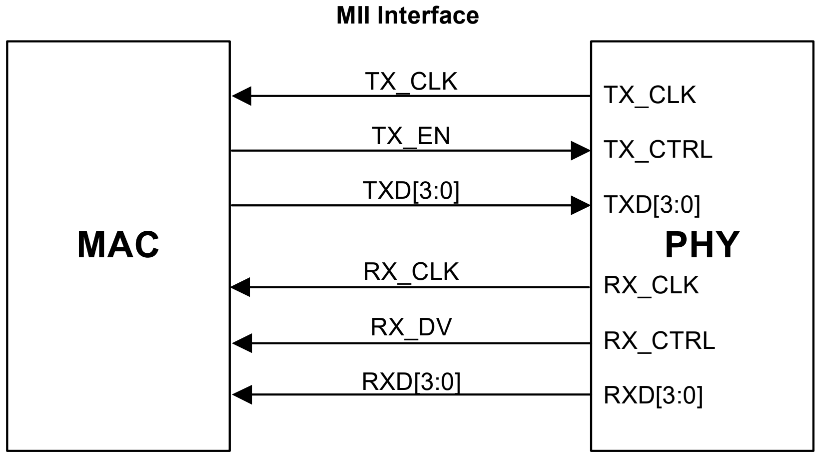
2.4.3 MII Interface

Table 26 indicates the signal mapping of the 88EA1517 and 88EA1512 devices to the MII interface. MII signaling to support 100BASE-TX and 10BASE-T modes is implemented by the pins of the MII interface. TX_ER, RX_ER, CRS and COL, as specified in IEEE802.3 Clause 22, are not implemented.

Table 26: MII Signal Mapping

Device Pin Name	MII Spec Pin Name	Description
TX_CLK	TX_CLK	25 MHz, or 2.5 MHz transmit clock based on the selected speed.
TX_CTRL	TX_EN	Transmit Enable Signal. This signal is synchronous with respect to the rising edge of TX_CLK.
TXD[3:0]	TXD[3:0]	Transmit Data. TXD[3:0] are synchronous with respect to the rising edge of TX_CLK.
RX_CLK	RX_CLK	25 MHz, or 2.5 MHz receive clock derived from the received data stream and based on the selected speed.
RX_CTRL	RX_DV	Receive Data Valid. This signal is synchronous with respect to rising edge of RX_CLK
RXD[3:0]	RXD[3:0]	Receive Data. In 100BASE-TX and 10BASE-T modes, RXD[3:0] are synchronous with respect to the rising edge of RX_CLK.

Figure 15: MII Signal Diagram



Note: During the transition from one speed to another, a dead time for a maximum duration of 1.5 clock cycles may occur on RX_CLK and TX_CLK to insure a glitch-free clock.

2.4.4 10/100 Mbps Functionality

This interface can be used to implement 10/100 Mbps Ethernet Media Independent Interface (MII) by reducing the clock rate to 25 MHz for 100 Mbps operation, and 2.5 MHz for 10 Mbps. The TX_CLK (TXC) signal is always sent out by the PHY, and the RX_CLK (RXC) signal is also generated by the PHY. The RGMII/MII supports 10 Mbps and 100 Mbps operation by reducing the clock-rate to 2.5 MHz and 25 MHZ respectively as shown in Table 25, RGMII Signal Mapping, on page 48.

During packet reception, RX_CLK may be stretched on either the positive or negative pulse to accommodate the transition from the free running clock to a data synchronous clock domain. When the speed of the PHY changes, a similar stretching of the positive or negative pulse is allowed. No glitching of the clocks is allowed during speed transitions.

The MAC must hold TX_CTRL (TX_CTL) low until the MAC has ensured that TX_CTRL (TX_CTL) is operating at the same speed as the PHY.

2.4.5 TX_ER and RX_ER Coding

See the RGMII Specifications for definitions of TX_ER, RX_ER, and in band status coding.

In RGMII mode, Register 21_2.3 is the register bit used to block carrier extension.

2.5 Loopback

The device implements various different loopback paths.

2.5.1 System Interface Loopback

The functionality, timing, and signal integrity of the System interface can be tested by placing the device in System interface loopback mode. This can be accomplished by setting Register 0_0.14 = 1, if copper is the selected media, or 0_1.14 = 1, if fiber is the selected media. In loopback mode, the data received from the MAC is not transmitted out on the media interface. Instead, the data is looped back and sent to the MAC. During loopback, media link will be lost and packets will not be received.

If loopback is enabled while Auto-Negotiating, FLP Auto-Negotiation codes will be transmitted onto the copper media. If loopback is enabled in forced 10BASE-T mode, 10BASE-T idle link pulses will be transmitted on the copper side. If loopback is enabled in forced 100BASE-T mode, 100BASE-T idles will be transmitted on the copper side.

The speed of the SGMII or RGMII/MII interface is determined by Register 21_2.6, 13 during loopback. 21_2.2:6,13 is 00 = 10 Mbps, 01 = 100 Mbps, 10 = 1000 Mbps.

Figure 16: MAC Interface Loopback Diagram - Copper Media Interface

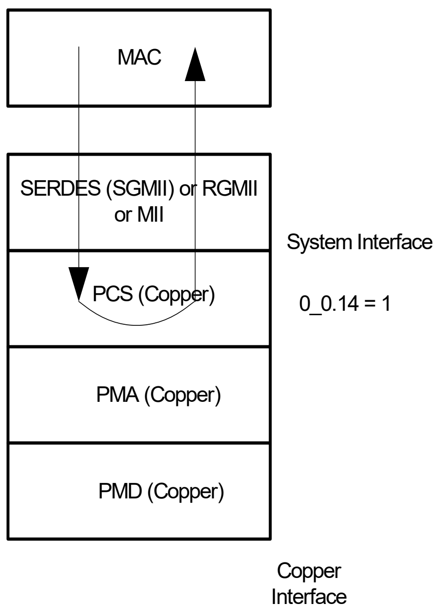
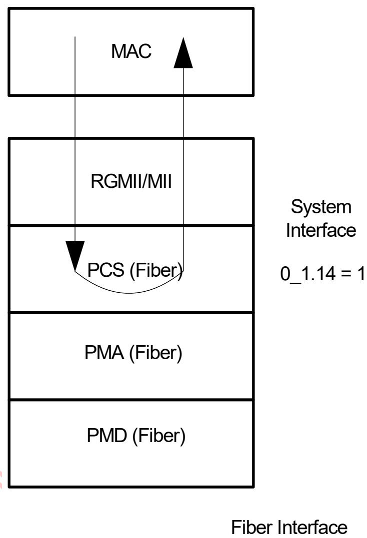


Figure 17: System Interface Loopback Diagram - Fiber Media Interface

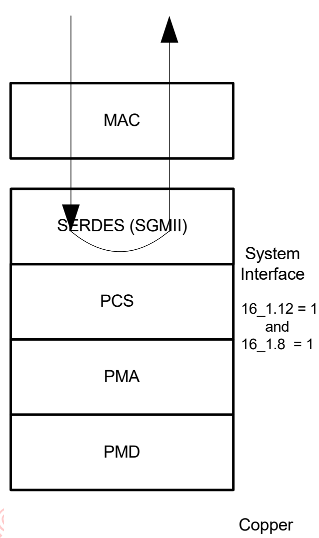


2.5.2 Synchronous SERDES Loopback

The 1.25 GHz SERDES can loop back the raw 10 bit symbol at the receiver back to the transmitter. In this mode of operation, the received data is assumed to be frequency locked with the transmit data output by the PHY. No frequency compensation is performed when the 10 bit symbol is looped back. This mode facilitates testing using non 8/10 symbols such as PRBS.

The 1.25 GHz SERDES synchronous loopback is enabled by setting Register 16_1.12 = 1 and 16_1.8 = 1.

Figure 18: Synchronous SERDES Loopback Diagram



2.5.3 Line Loopback

Line loopback allows a link partner to send frames into the device to test the transmit and receive data path. Frames from a link partner into the PHY, before reaching the MAC interface pins, are looped back and sent out on the line. They are also sent to the MAC. The packets received from the MAC are ignored during line loopback. Refer to Figure 19. This allows the link partner to receive its own frames.

Before enabling the line loopback feature, the PHY must first establish link to another PHY link partner. If Auto-Negotiation is enabled, both link partners should advertise the same speed and full-duplex. If Auto-Negotiation is disabled, both link partners need to be forced to the same speed and full-duplex. Once link is established, the line loopback mode can be enabled.

Register 21_2.14 = 1 enables the line loopback on the copper interface.

Register 16_1.12 = 1 and 16_1.8 = 0 enables the line loopback of the 1000BASE-X/SGMII media interface.

Figure 19: Copper Line Loopback Data Path

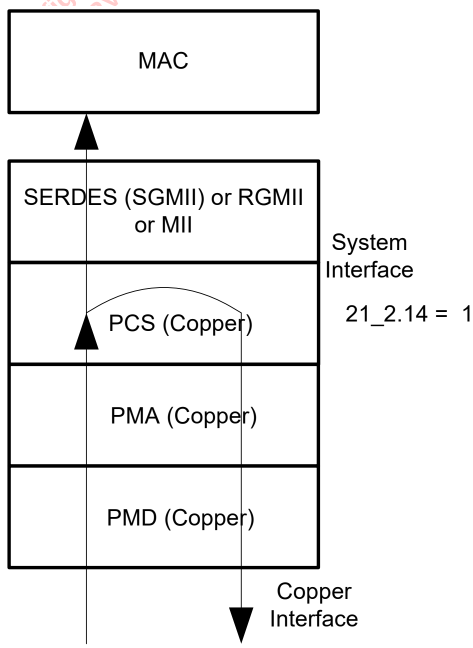
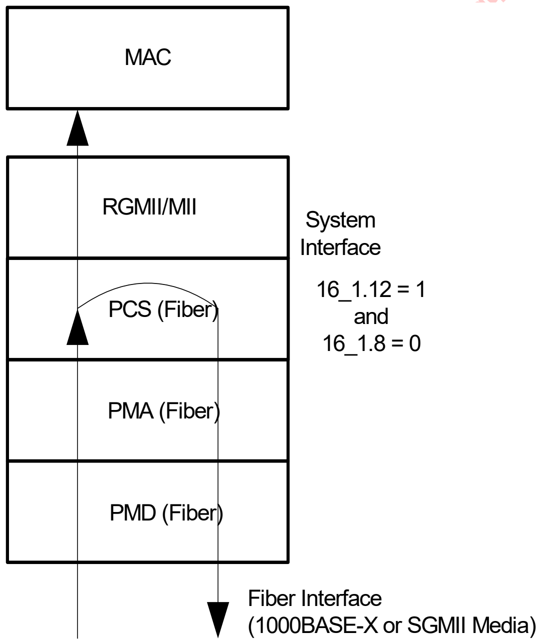


Figure 20: Fiber Line Loopback Data Path



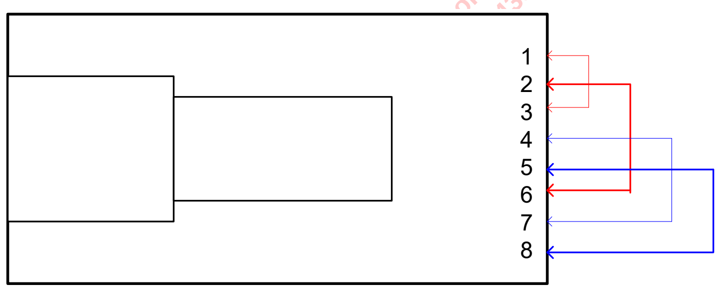
2.5.4 External Loopback

For production testing, an external loopback stub allows testing of the complete data path without the need of a link partner.

For 10BASE-T and 100BASE-TX modes, the loopback test requires no register writes. For 1000BASE-T mode, Register 18_6.3 must be set to 1 to enable the external loopback. All copper modes require an external loopback stub.

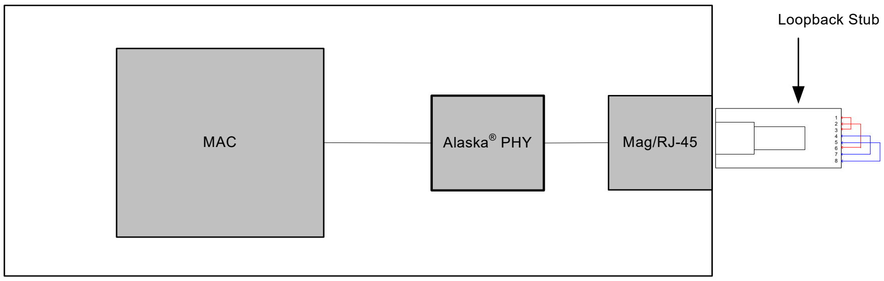
The loopback stub consists of a plastic RJ-45 header, connecting RJ-45 pair 1,2 to pair 3,6 and connecting pair 4,5 to pair 7,8, as seen in Figure 21.

Figure 21: Loopback Stub (Top View with Tab up)



The external loopback test setup requires the presence of a MAC that will originate the frames to be sent out through the PHY. Instead of a normal RJ-45 cable, the loopback stubs allows the PHY to self-link at 10/100/1000 Mbps. It also allows the actual external loopback. See Figure 22. The MAC should see the same packets it sent looped back to it.

Figure 22: Test Setup for 10/100/1000 Mbps Modes using an External Loopback Stub



2.6 Fiber/Copper Auto-Selection

The 88EA1512 has a patented feature to automatically detect and switch between fiber and copper cable connections. The auto-selection operates in one of three modes: Copper /1000BASE-X, Copper/100BASE-FX, and Copper/SGMII Media Interface.

Register 20_18.2:0 and Register 20_18.6 select the Fiber/Copper auto media modes of operation. See Table 27 for details.

Table 27: Fiber/Copper Auto-media Modes of Operation

Reg 20_18.2:0	Reg 20_18.6	Auto-media Modes of Operation
110	X	Copper/SGMII media
111	0	Copper/1000BASE-X
111	1	Copper/100BASE-FX
011	1	Copper/100BASE-FX
110	X	Copper/SGMII media
111	X	Copper/1000BASE-X

The device monitors the signals of the S_INP/N and the MDIP/N[3:0] lines. If a fiber optic cable is plugged in, the device will adjust itself to be in fiber mode. If an RJ-45 cable is plugged in, the device will adjust itself to be in copper mode. If both cables are connected then the first media to establish link, or the preferred media will be enabled. The media which is not enabled will be automatically turned off to save power. If the link on the first media is lost, then the inactive media will be powered up, and both media will once again start searching for link.

2.6.1 Preferred Media

The device can be programmed to give one media priority over the other. In other words if the non-preferred media establishes link first and subsequently energy is detected on the preferred media, the PHY will drop link on the non-preferred media for 4 seconds and give the preferred media a chance to establish link. Register 20_18.5:4 selects the preferred media.

- 00 = Link with the first media to establish link
- 01 = Prefer copper media
- 10 = Prefer fiber media

2.6.2 Definition of link in SGMII Media Interface in the context of auto media selection

In the conventional Copper/1000BASE-X definition of link, 1000BASE-X link is defined to be up when Auto-Negotiation is complete if 1000BASE-X Auto-Negotiation is turned on, or the acquisition of a comma if 1000BASE-X Auto-Negotiation is off. No link is defined to be up when the comma is not seen for some amount of time, or when Auto-Negotiation restarts.

In the Copper/SGMII Media Interface definition of link, the SGMII Media Interface link is up only if bit 15 of the SGMII Auto-Negotiation indicates that link is up. Completing Auto-Negotiation is not sufficient to bring the link up. With SGMII Auto-Negotiation turned off or in the link down case the link definition is identical to the 1000BASE-X case.

2.6.3 Notes on Determining which Media Linked Up

Since there are two sets of IEEE registers (one for copper and the other for fiber) the software needs to be aware of Register 22.7:0 so that the correct set of registers are selected. In general the sequence is as follows.

1. Set the Auto-Negotiation registers of the copper medium. (This step may not be necessary if the hardware configuration defaults are acceptable).
2. Set the Auto-Negotiation registers of the fiber medium. (This step may not be necessary if the hardware configuration defaults are acceptable).
3. Poll for link status.Go to step 4 if there is link.
4. Once there is link determine whether the link is copper or fiber medium.
5. Look at the Auto-Negotiation results for the medium that established link.
6. Poll for link status. If link status goes down then go back to step 3.

An example of a polling method procedure is as follows:

1. Write Register 22.7:0 to 0x00 to point to the copper medium.Write the appropriate Auto-Negotiation registers to advertise the desired capabilities.
2. Write Register 22.7:0 to 0x01 to point to the fiber medium. Write the appropriate Auto-Negotiation registers to advertise the desired capabilities.
3. If one medium is preferred over the other then write Register 22.7:0 to 0x02 to point to the MAC registers. Set Register 20_18.5:4 to the preferred media.
4. Write 0_0.15 and 0_1.15 to '1' to issue software reset. This causes the Auto-Negotiation settings to take effect.
5. Write Register 22.7:0 to 0x00 to point to the copper medium. Read the copper link status Register 1_0.2. Write Register 22.7:0 to 0x01 to point to the fiber medium. Read the fiber link status Register 1_1.2. Keep doing this until one of the link comes up. It should be clear which link goes up. When link is up go to the next step.
6. Once the link is up set Register 22.7:0 to 0x00 if the copper medium is up, or 0x01 if the fiber medium is up. Read the Auto-Negotiation registers for the Auto-Negotiation status if needed.
7. Poll Register 1_0.2 or 1_1.2 depending on copper or fiber link. If link goes down go back to step 5.

2.7 Synchronizing FIFO

The device has transmit synchronizing FIFOs to reconcile frequency differences between the clocks of the MAC interface and the media side. There are two FIFOs on the transmit paths of the copper and Fiber/SGMII as shown in Figure 23. The depth of the FIFOs can be independently programmed by programming register bits 16_1.15:14 and 16_2.15:14. See the "Alaska Ultra FAQs" for details on how to calculate required FIFO depth and the details of the different clocks used for transmit in each mode of operation.

The FIFO depths can be increased in length to support longer frames. The device can handle jumbo frame sizes up to 16 Kbytes with up to ± 100 PPM clock jitter. The deeper the FIFO depth, the higher the latency will be.

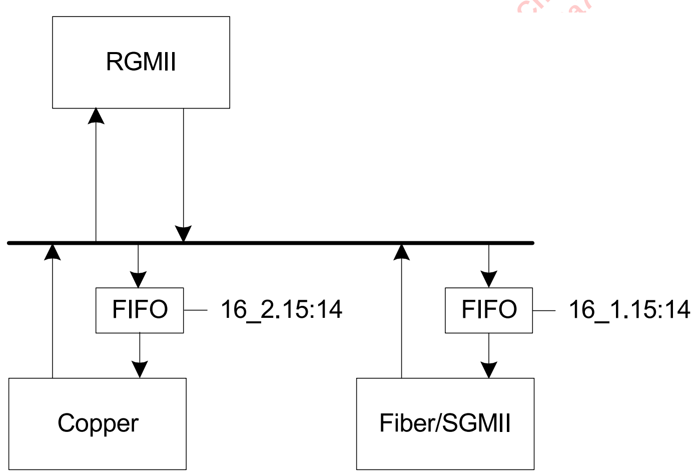
For the device, the status of the FIFO can be interrogated as in Table 28. Registers 19_2.3:2 are set depending on whether the copper transmit FIFO inserted or deleted idle symbols. Idles inserted or deleted will be flagged only if the interpacket gap is 24 bytes or less at the input of the FIFO. Inserted or deleted idles will be ignored if the inter packet gap is greater than 24 bytes.

The FIFO status bits can generate interrupts by setting the corresponding bits in Register 18_1 and 18_2.

Table 28: Device FIFO Status Bits

Register	Function	Setting
19_1.7	Fiber Transmit FIFO Over/Underflow	1 = Over/Underflow error 0 = No FIFO error
19_2.7	Copper Transmit FIFO Over/Underflow	1 = Over/Underflow error 0 = No FIFO error
19_2.3	Copper Transmit FIFO Idle Inserted	1 = Idle inserted 0 = No idle inserted
19_2.2	Copper Transmit FIFO Idle Deleted	1 = Idle deleted 0 = No idle deleted

Figure 23: FIFO Locations



There are two new FIFO modes- a low latency PPM FIFO mode and a Phase FIFO mode.

The default TX FIFO depth setting is 01 and the register bits used to program the copper TX FIFO depth are 16_2.15:14

Table 29: MAC Specific Control Register 1, Page 2, Register 16

Bits	Field	Mode	HW Rst	SW Rst	Description
15:14	Transmit FIFO Depth	R/W	01	Retain	00 = ± 16 Bits 01 = ± 24 Bits 10 = ± 32 Bits 11 = ± 40 Bits See 20_0.14 description. This bit enables a TX FIFO low latency mode- packets size is max 1518 bytes. 20_0.14 has priority over 16_2.15:14.

However, this setting uses a larger FIFO depth and bigger latency. To take advantage of the lowest latency for the TX path when there is PPM program 20_0.14 = 1 which enables a low latency PPM FIFO setting. The register bit description is shown below:

Table 30: Copper Specific Control Register 2, Page 0, Register 20

Bits	Field	Mode	HW Rst	SW Rst	Description
14	TX FIFO Low Latency Enable	R/W	0	Retain	1 = Enable TX FIFO low latency mode- packets size is max 1518 bytes 0 = 16_2.15:14 control the FIFO depth

Note that the packet size for this setting cannot exceed 1518 bytes. If packets need to be sent with sizes greater than 1518 bytes and up to 10K-byte jumbo packets then program 16_2.15:14 = 00 (and 20_0.14 = 0). The TX latency will not be the minimum, but it will still be lower than the latency associated with the default depth setting.

There is also a Phase FIFO mode which allows for lower latency assuming that the FIFO read and write clocks are synchronous. To enable the TX FIFO to be a phase FIFO, program 20_0.13 = 1:

Table 31: Copper Specific Control Register 2, Page 0, Register 20

Bits	Field	Mode	HW Rst	SW Rst	Description
0.20.13	Phase FIFO Enable	R/W	0	Update	1 = Enable Phase FIFO for TX FIFO 0 = TX FIFO handles ppm difference

2.8 Resets

In addition to the hardware reset pin (RESETn) there are several software reset bits as summarized in Table 32.

The copper, fiber, and RGMII circuits are reset per port via Register 0_0.15 and 0_1.15 respectively. A reset in one circuit does not directly affect another circuit.

Register 20_18.15 resets the mode control, port power management, and generator and checkers.

All the reset registers described so far self clear. However, Register 20_6.9 is not self clearing. It resets the PTP circuit when it is not used. To re-enable the PTP circuit Register 20_6.9 must be manually cleared. When Register 20_6.9 is set to 1, registers in banks 8, 9, 10 and 11 are not accessible.

Table 32: Reset Control Bits

Reset Register	Register Effect	Functional Block
0_0.15	Software Reset for Registers in page 0, 2, 3, 5, 7	Copper
0_1.15	Software Reset for Registers in page 1	Fiber/SGMII
20_18.15	Software Reset for Registers in page 6 and 18	Generator/Checker/Mode
20_6.9	Power Down/Reset for Registers in page 8, 9, 12, 14. This register does not self clear. Registers in page 8, 9, 12, and 14 are not accessible if this bit is in reset.	TAI

2.9 Power Management

The device supports several advanced power management modes that conserve power.

2.9.1 Low Power Modes

Four low power modes are supported in the device.

- IEEE 22.2.4.1.5 compliant power down
- Energy Detect (Mode 1)
- Energy Detect+^{TM} (Mode 2)
- IEEE 802.3az-2010 Energy Efficient Ethernet (EEE) compliant low power modes

IEEE 22.2.4.1.5 power down compliance allows for the PHY to be placed in a low-power consumption state by register control.

Energy Detect (Mode 1) allows the device to wake up when energy is detected on the wire.

Energy Detect+^{TM} (Mode 2) is identical to Mode 1 with the additional capability to wake up a link partner. In Mode 2, the 10BASE-T link pulses are sent once every second while listening for energy on the line.

Energy Efficient ethernet allows the PHY to go into a Low Power Idle (LPI) mode to save power.

Details of each mode are described below.

2.9.1.1 IEEE Power Down Mode

The standard IEEE power down mode is entered by setting Register 0_0.11 or 0_1.11. In this mode, the PHY does not respond to any System interface (i.e., RGMII/SGMII) signals except the MDC/MDIO. It also does not respond to any activity on the copper or fiber media.

In this power down mode, the PHY cannot wake up on its own by detecting activity on the media. It can only wake up by setting Registers 0_0.11 and 16_0.2 = 0 for copper and 0_1.11 = 0 for Fiber.

Note that Register 0_0.11 or 16_0.2 may be set to 1 to power down the copper media.

As shown in Table 33, each power down control independently powers down its respective circuits. In general, it is not necessary to power down an unused interface. The PHY will automatically power down any unused circuit. For example when auto-media detect is turned on, the unused interface will automatically power down.

The automatic PHY power management can be overridden by setting the power down control bits. These bits have priority over the PHY power management in that the circuit can not be powered up by the power management when its associated power down bit is set to 1. When a circuit is power back up by setting the bit to 0, a software reset is also automatically sent to the corresponding circuit.

Table 33: Power Down Control Bits

Reset Register	Register Effect
0_0.11	Copper Power Down
16_0.2	Copper Power Down
0_1.11	Fiber/SGMII Power Down

2.9.1.2 Copper Energy Detect Modes

The device can be placed in energy detect power down modes by selecting either of the two energy detect modes. Both modes enable the PHY to wake up on its own by detecting activity on the CAT 5 cable. The status of the energy detect is reported in Register 17_0.4 and the energy detect changes are reported in Register 19_0.4. The energy detect modes only apply to the copper media. The energy detect modes will not work while Fiber/Copper Auto Select (2.5 "Fiber/Copper Auto-Selection" on page 45) is enabled. Normal 10/100/1000 Mbps operation can be entered by turning off energy detect mode by setting Register 16_0.9:8 to 0x.

Energy Detect (Mode 1)

Energy Detect (Mode 1) is entered by setting Register 16_0.9:8 to 10.

In Mode 1, only the signal detection circuitry and serial management interface are active. How the PHY wakes up depends on the register setting of 16_0.7. If 16_0.7 is set to 1, the PHY can wake up by writing 16_0.4 to 1. If 16_0.7 is set to 0, the PHY can automatically wake up when energy is detected on the wire. When the PHY wakes up, it starts to Auto-Negotiate sending FLPs for 5 seconds. If at the end 0f 5 seconds the Auto-Negotiation is not completed, then the PHY stops sending FLPs and goes back to monitoring receive energy. If Auto-Negotiation is completed, then the PHY goes into normal 10/100/1000 Mbps operation. If during normal operation the link is lost, the PHY will re-start Auto-Negotiation. If no energy is detected after 5 seconds, the PHY goes back to monitoring receive energy.

Energy Detect +^{TM} (Mode 2)

Energy Detect (Mode 2) is entered by setting Register 16_0.9:8 to 11.

In Mode 2, the PHY sends out a single 10 Mbps NLP (Normal Link Pulse) every one second. Except for this difference, Mode 2 is identical to Mode 1 operation. If the device is in Mode 1, it cannot wake up a connected device; therefore, the connected device must be transmitting NLPs, or either device must be woken up through register access. If the device is in Mode 2, then it can wake a connected device.

Power State Upon Exiting Power Down

When the PHY exits power down (Register 0_0.11 or 16_0.2) the active state will depend on whether the energy detect function is enabled (Register 16_0.9:8 = 1x). If the energy detect function is enabled, the PHY will transition to the energy detect state first and will wake up only if there is a signal on the wire.

Table 34: Power State after Exiting Power Down

Register 0_0.11	Register 16_0.2	Register 16_0.9:8	Behavior
1	x	xx	IEEE Power down
0	1	xx	Power down with signal detect
1 to 0	1	xx	Transition to power down with signal detect state
1 to 0	0	00	Transition to power up
0	1 to 0	00	Transition to power up
1 to 0	0	1x	Transition to energy detect state
0	1 to 0	1x	Transition to energy detect state

2.9.1.3 Energy Efficient Ethernet (EEE) Low Power Modes

The 88EA1517/88EA1512 device supports IEEE 802.3az-2010 Energy Efficient Ethernet (EEE). The device supports EEE on the following Media Interface:

- 10BASE-Te
- 100BASE-TX EEE
- 1000BASE-T EEE

See Section 2.15 "Energy Efficient Ethernet (EEE)" on page 74 for further details.

2.9.2 RGMII/SGMII MAC Interface Power Down

In some applications, the MAC interface must run continuously regardless of the state of the media interface. Additional power will be required to keep the MAC interface running during low power states.

If absolute minimal power consumption is required during network interface power down mode or in the Energy Detect modes, then Register 16_2.3 or 16_1.3 should be set to 0 to allow the MAC interface to power down.

In general 16_2.3 is used when the network interface is copper and 16_1.3 is used when the network interface is fiber. Note that for these settings to take effect a software reset must be issued.

In the auto media detect modes (MODE = 110 and 111) both the fiber side and copper side has to indicate power down before the RGMII port can be powered down.

2.10 Auto-Negotiation

The device supports three types of Auto-Negotiation.

- 10/100/1000BASE-T Copper Auto-Negotiation. (IEEE 802.3 Clauses 28 and 40)
- 1000BASE-X Fiber Auto-Negotiation (IEEE 802.3 Clause 37)
- SGMII Auto-Negotiation (Cisco specification)
- EEE Auto-Negotiation. For further details, refer to Section 2.15.3 "Energy Efficient Ethernet Auto-Negotiation" on page 76.

Auto-Negotiation provides a mechanism for transferring information from the local station to the link partner to establish speed, duplex, and Master/Slave preference (in the case of Copper Auto-Negotiation) during a link session.

Auto-Negotiation is initiated upon any of the following conditions:

- Power up reset
- Hardware reset
- Software reset (Register 0_0.15 or 0_1.15)
- Restart Auto-Negotiation (Register 0_0.9 or 0_1.9)
- Transition from power down to power up (Register 0_0.11 or 0_1.11)
- The link goes down

The following sections describe each of the Auto-Negotiation modes in detail.

2.10.1 10/100/1000BASE-T Auto-Negotiation

The 10/100/1000BASE-T Auto-Negotiation (AN) is based on Clause 28 and 40 of the IEEE 802.3 specification. It is used to negotiate speed, duplex, and flow control over CAT5 UTP cable. Once Auto-Negotiation is initiated, the device determines whether or not the remote device has Auto-Negotiation capability. If so, the device and the remote device negotiate the speed and duplex with which to operate.

If the remote device does not have Auto-Negotiation capability, the device uses the parallel detect function to determine the speed of the remote device for 100BASE-TX and 10BASE-T modes. If link is established based on the parallel detect function, then it is required to establish link at half-duplex mode only. Refer to IEEE 802.3 clauses 28 and 40 for a full description of Auto-Negotiation.

After hardware reset, 10/100/1000BASE-T Auto-Negotiation can be enabled and disabled via Register 0_0.12. Auto MDI/MDIX and Auto-Negotiation may be disabled and enabled independently. When Auto-Negotiation is disabled, the speed and duplex can be set via Registers 0_0.13, 0_0.6, and 0_0.8 respectively. When Auto-Negotiation is enabled the abilities that are advertised can be changed via Registers 4_0 and 9_0.

Changes to Registers 0_0.12, 0_0.13, 0_0.6 and 0_0.8 do not take effect unless one of the following takes place:

- Software reset (Registers 0_0.15)
- Restart Auto-Negotiation (Register 0_0.9)
- Transition from power down to power up (Register 0_0.11)
- The copper link goes down

To enable or disable Auto-Negotiation, Register 0_0.12 should be changed simultaneously with either Register 0_0.15 or 0_0.9. For example, to disable Auto-Negotiation and force 10BASE-T half-duplex mode, Register 0_0 should be written with 0x8000.

Registers 4_0 and 9_0 are internally latched once every time the Auto-Negotiation enters the Ability Detect state in the arbitration state machine. Hence, a write into Register 4_0 or 9_0 has no effect once the device begins to transmit Fast Link Pulses (FLPs).This guarantees that sequences of FLPs transmitted are consistent with one another.

Register 7_0 is treated in a similar way as Registers 4_0 and 9_0 during additional next page exchanges.

If 1000BASE-T mode is advertised, then the device automatically sends the appropriate next pages to advertise the capability and negotiate Master/Slave mode of operation. If the user does not wish to transmit additional next pages, then the next page bit (Register 4_0.15) can be set to zero, and the user needs to take no further action.

If next pages in addition to the ones required for 1000BASE-T are needed, then the user can set Register 4_0.15 to one, and send and receive additional next pages via Registers 7_0 and 8_0, respectively. The device stores the previous results from Register 8 in internal registers, so that new next pages can overwrite Register 8_0.

Note that 1000BASE-T next page exchanges are automatically handled by the device without user intervention, regardless of whether or not additional next pages are sent.

Once the device completes Auto-Negotiation, it updates the various status in Registers 1_0, 5_0, 6_0, and 10_0. Speed, duplex, page received, and Auto-Negotiation completed status are also available in Registers 17_0 and 19_0.

Refer to Register 17_0 and 19_0.

2.10.2 1000BASE-X Auto-Negotiation

1000BASE-X Auto-Negotiation is defined in Clause 37 of the IEEE 802.3 specification. It is used to Auto-Negotiate duplex and flow control over fiber cable. Registers 0_1, 4_1, 5_1, 6_1, and 15_1 are used to enable AN, advertise capabilities, determine link partner's capabilities, show AN status, and show the duplex mode of operation respectively.

Register 22.7:0 must be set to one to view the fiber Auto-Negotiation Registers.

The device supports Next Page option for 1000BASE-X Auto-Negotiation. Register 7_1 of the fiber pages is used to transmit Next Pages, and Register 8_1 of the fiber pages is used to store the received Next Pages. The Next Page exchange occurs with software intervention. The user must set Register 4_1.15 to enable fiber Next Page exchange. Each Next Page received in the registers should be read before a new Next Page to be transmitted is loaded in Register 7_1.

If the PHY enables 1000BASE-X Auto-Negotiation and the link partner does not, the link cannot link up. The device implements an Auto-Negotiation bypass mode. See Section 2.10.3.2 for more details.

2.10.3 SGMII Auto-Negotiation

SGMII is a de-facto standard designed by Cisco. SGMII uses 1000BASE-X coding to send data as well as Auto-Negotiation information between the PHY and the MAC. However, the contents of the SGMII Auto-Negotiation are different than the 1000BASE-X Auto-Negotiation. See the “Cisco SGMII Specification” and the “MAC Interfaces and Auto-Negotiation” application note for further details.

The device supports SGMII interface with and without Auto-Negotiation. Auto-Negotiation can be enabled or disabled by writing to Register 0_1.12 followed by a soft reset. If SGMII Auto-Negotiation is disabled, the MAC interface link, speed, and duplex status (determined by the media side) cannot be conveyed to the MAC from the PHY. The user must program the MAC with this information in some other way (e.g., by reading PHY registers for link, speed, and duplex status). However, the operational speed of the SGMII will follow the speed of the media (See Table 23 on page 47) regardless of whether the Auto-Negotiation is enabled or disabled.

In case of RGMII to SGMII mode of operation, the SGMII interface behaves as if it were the SGMII on the MAC side of the interface. When Auto-Negotiation is enabled, the SGMII Auto-Negotiation information like the speed, duplex, and link received from the PHY is used for determining the mode of operation. The RGMII interface will be adjusted accordingly when the SGMII Auto-Negotiation is completed.

2.10.3.1 Flow control Enhancement to SGMII Auto-Negotiation

During standard SGMII Auto-Negotiation the PHY passes the link, speed, and duplex information to the MAC. The flow control information is not communicated. Typically, the MAC will have to read the registers of the PHY to find out the flow control capability of the link partner. The device has added in-line flow control information by using some of the reserved bits of the Auto-Negotiation base page as defined by the SGMII specification. This feature is optional. The user can select the standard SGMII Auto-Negotiation or the enhanced mode as described in Table 36. Table 35 shows the bit definitions for the enhanced mode.

Bit 9 corresponds to Register 17_1.3 and bit 8 corresponds to Register 17_1.2.

Register 16_1.7 will enable this feature (Table 36). 0 = set bits 9:7 always to 000, 1 = set bits 9:7 according to Table 35. The default is disabled.

Table 35: Enhanced SGMII PHY Status

TX_CONFIG_REG[15:0]			
Bit Number	**Reg 16_1.7:6 = ‘00’**	**Reg 16_1.7:6 = ‘01’**	**Reg 16_1.7:6 = ‘10’**
15	1 = Link Up 0 = Link Down	1 = Link Up 0 = Link Down	1 = Link Up 0 = Link Down
14	Acknowledge	Acknowledge	Acknowledge
13	0 = Reserved	0 = Reserved	0 = Reserved
12	1 = Full-duplex 0 = Half-duplex	1 = Full-duplex 0 = Half-duplex	1 = Full-duplex 0 = Half-duplex
11:10	00 = 10 Mbps 01 = 100 Mbps 10 = 1000 Mbps 11 = Reserved	00 = 10 Mbps 01 = 100 Mbps 10 = 1000 Mbps 11 = Reserved	00 = 10 Mbps 01 = 100 Mbps 10 = 1000 Mbps 11 = Reserved
9	1 = EEE supported 0 = EEE not supported	1 = Transmit pause enabled 0 = Disabled	1 = EEE supported 0 = EEE not supported

Table 35: Enhanced SGMII PHY Status (Continued)

TX_CONFIG_REG[15:0]			
Bit Number	**Reg 16_1.7:6 = ‘00’**	**Reg 16_1.7:6 = ‘01’**	**Reg 16_1.7:6 = ‘10’**
8	1 = EEE clock stop supported 0 = EEE clock stop not supported	1 = Received pause enabled 0 = Disabled	1 = EEE clock stop supported 0 = EEE clock stop not supported
7	0 = Reserved	0 = 10/100/1000BASE-T 1 = 100BASE-FX/1000BASE-X	0 = Reserved
6	0 = Reserved	1 = EEE supported 0 = EEE not supported	0 = Reserved
5	0 = Reserved	1 = EEE clock stop supported 0 = EEE clock stop not supported	0 = Reserved
4	0 = Reserved	0 = Reserved	0 = Reserved
3	0 = Reserved	0 = Reserved	1 = Transmit pause enabled 0 = Disabled
2	0 = Reserved	0 = Reserved	1 = Received pause enabled 0 = Disabled
1	0 = Reserved	0 = Reserved	0 = 10/100/1000BASE-T 1 = 100BASE-FX/1000BASE-X
0	Always 1	Always 1	Always 1

Table 36: MAC Specific Control Register 1

Register	Function	Setting	Mode	HW Rst	SW Rst
16_1.7:6	Enhanced SGMII	00 = Do not pass flow control bits through SGMII Auto-Negotiation. SGMII (System mode) Registers 4_1.9:7 are always 000. EEE bits are passed per IEEE. 01 = Pass flow control bits through SGMII Auto-Negotiation using 4_1.9:7. EEE bits are passed in alternate locations. 10 = Pass flow control bits through SGMII Auto-Negotiation using bits other than 4_1.9:7. EEE bits use 9:8. 11 = Reserved	R/W	0x0	Update

2.10.3.2 Serial Interface Auto-Negotiation Bypass Mode

If the MAC or the PHY implements the Auto-Negotiation function and the other does not, two-way communication is not possible unless Auto-Negotiation is manually disabled and both sides are configured to work in the same operational modes. To solve this problem, the device implements the SGMII Interface Auto-Negotiation Bypass Mode. When entering the state “Ability_Detect”, a bypass timer begins to count down from an initial value of approximately 200 ms. If the device receives idles during the 200 ms, the device will interpret that the other side is “alive” but cannot send configuration codes to perform Auto-Negotiation. After 200 ms, the state machine will move to a new state called “Bypass_Link_Up” in which the device assumes a link-up status and the operational mode is set to the value listed under the “Comments” column of Table 37. Refer to the Section 2.1 for further details.

Table 37: SGMII Auto-Negotiation modes

Reg. 0_1.12	Reg. 26_1.6	Comments
0	X	No Auto-Negotiation. User responsible for determining speed, link, and duplex status by reading PHY registers.
1	0	Normal SGMII Auto-Negotiation. Speed, link, and duplex status automatically communicated to the MAC during Auto-Negotiation.
1	1	MAC Auto-Negotiation enabled. Normal operation.
		MAC Auto-Negotiation disabled. After 200 ms the PHY will disable Auto-Negotiation and link based on idles.

2.11 Downshift Feature

Without the downshift feature enabled, connecting between two Gigabit link partners requires a four-pair RJ-45 cable to establish 10, 100, or 1000 Mbps link. However, there are existing cables that have only two-pairs, which are used to connect 10 Mbps and 100 Mbps Ethernet PHYs. With the availability of only pairs 1, 2 and 3,6, Gigabit link partners can Auto-Negotiate to 1000 Mbps, but fail to link. The Gigabit PHY will repeatedly go through the Auto-Negotiation but fail 1000 Mbps link and never try to link at 10 Mbps or 100 Mbps.

With the Marvell® downshift feature enabled, the device is able to Auto-Negotiate with another Gigabit link partner using cable pairs 1,2 and 3,6 to downshift and link at 10 Mbps or 100 Mbps, whichever is the next highest advertised speed common between the two Gigabit PHYs.

In the case of a three pair cable (additional pair 4,5 or 7,8 - but not both) the same downshift function for two-pair cables applies.

By default, the downshift feature is turned off. Refer to Register 16_0.14:11 which describe how to enable this feature and how to control the downshift algorithm parameters.

To enable the downshift feature, the following registers must be set:

- Register 16_0.11 = 1 - enables downshift
- Register 16_0.14:12 - sets the number of link attempts before downshifting

2.12 Fast 1000BASE-T Link Down Indication

Per the IEEE 802.3 Clause 40 standard, a 1000BASE-T PHY is required to wait 750 milliseconds or more to report that link is down after detecting a problem with the link. For Metro Ethernet applications, a Fast Failover in 50 ms is specified, which cannot be met if the PHY follows the 750 ms wait time. This delay can be reduced by intentionally violating the IEEE standard by setting Register 26_0.9 to 1.

The delay at which link down is to be reported can be selected by setting Register 26_0.11:10. 00 = 0ms, 01 = 10 ± 2 ms, 10 = 20 ± 2ms, 11 = 40 ± 2ms.

2.13 Advanced Virtual Cable Tester®

The device Advanced Virtual Cable Tester feature uses Time Domain Reflectometry (TDR) to determine the quality of the cables, shorts, cable impedance mismatch, bad connectors, termination mismatch, and bad magnetics. The device transmits a signal of known amplitude (+1V) down each of the four pairs of an attached cable. It will conduct the cable diagnostic test on each pair, testing the MDI_0_0P/N, MDI_0_1P/N, MDI_0_2P/N, and MDI_0_3P/N pairs sequentially. The transmitted signal will continue down the cable until it reflects off of a cable imperfection.

The Advanced VCT™ has 4 modes of operation that is programmable via Register 23_5.7:6. The first mode returns the peak with the maximum amplitude that is above a certain threshold. The second mode returns the first peak detected that is above a certain threshold. The third mode measures the systematic offset at the receiver. The fourth mode measures the amplitude seen at a certain specified distance.

The VCT test is initiated by setting Register 23_5.15 to 1. This bit will self clear when the test is completed. Register 23_5.14 will be set to a 1 indicating that the TDR results in the registers are valid.

Each point in the VCT reflected waveform is sampled multiple times and averaged. The number of samples to take is programmable via Register 23_5.10:8.

Each time the VCT test is enable, the results seen on the four receive channels are reported in Registers 16_5, 17_5, 18_5, and 19_5. Register 23_5.13:11 selects which channel transmits the test pulse.

When Register 23_5.13:11 is set to 000 the same channel reflection is recorded. In other words, channel 0 transmits a pulse and the reflection seen on channel 0 receiver is reported. Channel 1 transmits a pulse and the reflection seen on channel 1 receiver is reported. The same for channel 2 and channel 3.

When Register 23_5.13:11 is set to 100 all 4 receive channels report the reflection seen by a pulse transmitted by channel 0.

When Register 23_5.13:11 is set to 101 all 4 receive channels report the reflection seen by a pulse transmitted by channel 1.

When Register 23_5.13:11 is set to 110 all 4 receive channels report the reflection seen by a pulse transmitted by channel 2.

When Register 23_5.13:11 is set to 111 all 4 receive channels report the reflection seen by a pulse transmitted by channel 3.

Hence, if only the reflection seen on the same channel is desired the VCT test should be run with 23_5.13:11 = 000. If all same channel and cross channel combinations are desired then the VCT test must be run 4 times with 23_5.13:11 set to 100, 101, 110, and 111 for the 4 runs. Registers 16_5, 17_5, 18_5, and 19_5 should be read and stored between each run.

2.13.1 Maximum Peak

When Register 23_5.7:6 is set to 00, the maximum peak above a certain threshold is reported. Pulses are sent out and recorded according to the setting of Register 23_5.13:11.

There are 10 threshold settings for same channel reflections and are specified by Registers 26_5.6:0, 26_5.14:8, 27_5.6:0, 27_5.14:8, and 28_5.6:0 for positive reflections and 26_7.6:0, 26_7.14:8, 27_7.6:0, 27_7.14:8, and 28_7.6:0 for negative reflections.

These settings correspond to the amplitude threshold the reflected signal has to exceed before it is counted. Any reflected signal below this threshold level is ignored. The threshold settings are based on cable length with the breakpoints at 10 m, 50 m, 110 m, and 140 m.

There are 4 threshold settings for cross-channel specified by Registers 25_5.6:0 and 25_5.14:8 for positive reflections and 25_7.6:0 and 25_7.14:8 for negative reflections. The threshold settings are based on cable length with the breakpoints at 10 m.

The default values are targeted to 85 ohm to 115 ohms. However these threshold settings should be calibrated for the desired impedance setting on the target system.

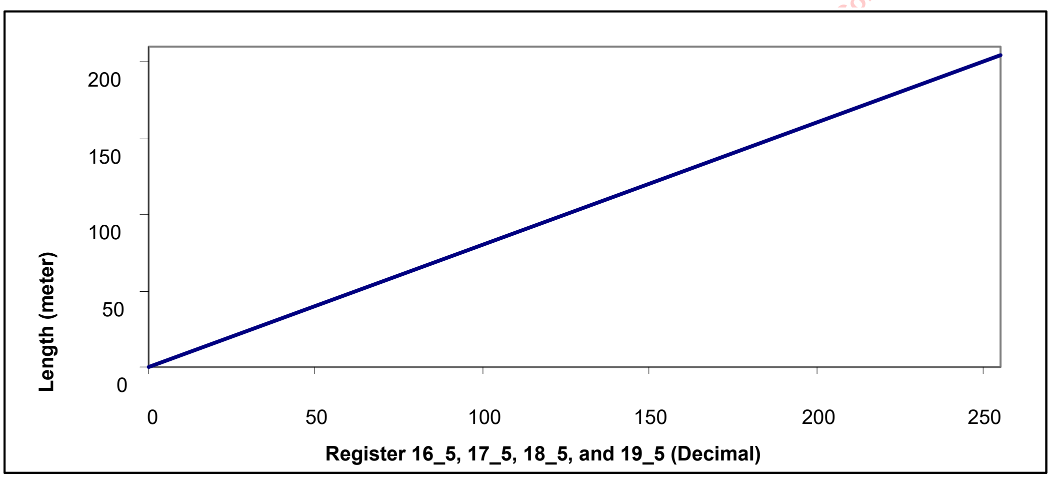
The results are stored in Registers 16_5, 17_5, 18_5, and 19_5. Bits 7:0 report the distance of the peak. The distance can be converted to using the trend line in Figure 24. Bits 14:8 report the reflected amplitude. Bit 15 reports whether the reflected amplitude was positive or negative. When bits 15:8 return a value of 0x80, it means there was no peak detected above the threshold. If bits 15:8 return a value of 0x00 then the test failed.

Register 28_5.7 controls the exact distance that is reported. When set to 0 the distance where the amplitude falls to 50% of the peak amplitude is reported. When set to 1 the distance where the peak amplitude actually occurs is reported. In either case, the magnitude of the maximum amplitude is reported in bits 14:8.

In the maximum peak mode, Register 24_5.7:0 is used to set the starting distance of the sweep. Normally this value should be set to 0. If this value is set to a non-zero value, any reflection below the starting distance is ignored. Note that 24_5.8 is ignored.

Note that the maximum peak only measures up to about 200 meters of cable.

Figure 24: Cable Fault Distance Trend Line



2.13.2 First Peak

When Register 23_5.7:6 is set to 01, the first peak above a certain threshold is reported. The first peak operates in exactly the same way as the maximum peak except that there has to be some qualification as to what constitutes a peak since the first peak is not necessarily the maximum peak. The first peak is defined as the maximum amplitude seen before the reflected amplitude drops by some value below this peak. This hysteresis value is defined by Register 23_5.5:0.

For example, in Figure 25, if Pa is greater than the hysteresis value in 23_5.5:0 and Va is above the threshold value, then Va and Da are reported since it is the first peak that is above the threshold.

If Pa is less than the hysteresis value in 23_5.5:0, then Va and Da are not reported as the first peak. Vb and Db will be reported as the first peak if Pb is greater than the hysteresis value in 23_5.5:0 and Vb is above the threshold value.

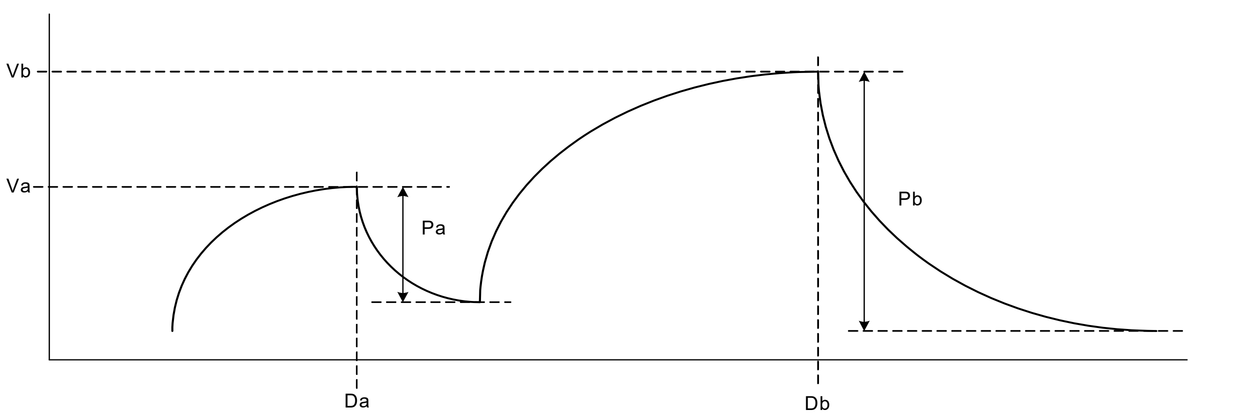
If Pa is greater than the hysteresis value in 23_5.5:0 but Va is below the threshold value then Va and Da are not reported as the first peak. Vb and Db will be reported as the first peak if Pb is greater than the hysteresis value in 23_5.5:0 and Vb is above the threshold value.

Register 28_5.7 controls the exact distance that is reported. When set to 0 the distance where the amplitude falls below the peak amplitude minus the hysteresis level as defined in Register 23_5.5:0 is reported. When set to 1 the distance where the peak amplitude actually occurs is reported. In either case, the magnitude of the maximum amplitude of the first peak is reported in bits 14:8.

In the first peak mode Register 24_5.7:0 is used to set the starting distance of the sweep. Normally, this value should be set to 0. If this value is set to a non-zero value, any reflection below the starting distance is ignored. This may be useful to ignore reflections at the transformer that are reported as the first peak. Note that 24_5.8 is ignored.

Note that the maximum peak only measures up to about 200 meters of cable.

Figure 25: First Peak Example



2.13.3 Offset

The offset reports the offset seen at the receiver. This is a debug mode. Bits 7:0 of Registers 16_5, 17_5, 18_5, and 19_5 have no meaning. When bits 15:8 return a value of 0x80 it means there is zero offset. If bit 15:8 returns a value of 0x00 then the test failed.

Note that in the maximum peak, first peak, and sample point modes, the systematic offset is automatically subtracted from the results.

2.13.4 Sample Point

When Register 23_5.7:6 is set to 11, the amplitude of the reflected pulse at a particular distance on the cable is reported. Unlike the maximum peak and first peak modes which only measures up to about 200 meters of cable, the sample point mode can measure up to 400 meters of cable.

The sample point returns the amplitude of the reflected pulse at a particular distance on the cable. The distance is set by Register 24_5.8:0. The threshold Registers 25_5, 26_5, 27_5, 28_5.6:0, 25_7, 26_7, 27_7, and 28_7.6:0 are ignored.

Bits 7:0 of Registers 16_5, 17_5, 18_5, and 19_5 return the same value as 24_5.7:0. Note that Register 24_5.8 is not returned. Bits 14:8 return the amplitude, and bit 15 the sign of the amplitude. If the test failed bits 15:8 will return 00000000 (zero amplitude will always return as 10000000).

By programming Register 24_5.8:0 from 0x000 to 0x1FF and running the sample point test at each distance it is possible to reconstruct the reflected amplitude. Note that since the threshold is ignored, it is possible that some small reflections in the same channel measurements will be reported at short cable lengths when there are none. This is because the analog hybrid does not 100% cancel out the transmitted signal.

2.13.5 Pulse Amplitude and Pulse Width

The transmitted pulse amplitude and pulse width can be adjusted via Registers 28_5.9:8 and 28_5.11:10 respectively. They should normally be set to full amplitude and full pulse width.

2.13.6 Drop Link

When Register 28_5.12 is set to 0 the circuit will wait 1.5 seconds to break the link before starting VCT™. When set to 1 this delay is bypassed.

2.13.7 VCT™ with Link Up

The following status requires the PHY to link up with a link partner.

- Register 20_5 reports the pair skew of each pair of wires relative to each other.
- Register 21_5.3:0 reports the polarity of each pair of wires.
- Register 21_5.5:4 reports the crossover status
- Register 20_5 and 21_5 are not valid unless Register 21_5.6 is set to 1.

2.13.8 Alternate VCT Control

The registers in page 7 provides an alternate means to control VCT. When using page 7 to control the VCT, the VCT results in Registers 16_5, 17_5, 18_5, and 19_5 will be altered and are not meaningful. Register 21_7.14 must be set to 0 when controlling VCT using registers in page 5.

VCT will run when any one of the following three events occur:

- Register 21_7.12 transitions from 0 to 1.
- Register 21_7.15 transitions from 0 to 1.
- Link transitions from link up to link down and Register 21_7.14 = 1.

Note that if the VCT circuit is busy when any of the three events occur, the event is ignored.

If Register 21_7.15 triggers VCT then the 1.5s break link prior to measurement is bypassed.

If any of the other 2 events triggers VCT then the 1.5s break link prior to measurement is not bypassed.

If Register 21_7.13 = 1 then only same pair is checked using the first peak method.

If Register 21_7.13 = 0 then same pair and cross pair are checked using the first peak method.

Register 21_7.11 indicates whether the cable testing is completed or not. Once completed the results are presented in Registers 16_7, 17_7, 18_7, 19_7, and 20_7. It may be possible for multiple faults to occur on the same pair but only one fault can be reported. The priority in reporting in descending order is as follows.

- Test failed (busy).
- Fault at the shortest distance from the PHY (open, short, cross pair short).
- In case of a tie in distance, open and short has higher priority than cross pair short.
- Cable good

When a pair is reported as good, the value in its corresponding cable length register is invalid.

Register 21_7.10 selects how the cable length is to be presented – in meters or centimeters. Note that Register 21_7.10 should not be change in the middle of a VCT test. Once the testing is complete and cable length values written into Registers 16_7, 17_7, 18_7, and 19_7, the values will not change when Register 21_7.10 changes. Changes to 21_7.10 will only take effect the next time VCT is started.

Table 38: Alternate VCT™ Control

Register	Function	Setting	Mode
16_7.15:0	Pair 0 Cable Length	Length to fault in meters or centimeters based on Register 21_7.10.	RO
17_7.15:0	Pair 1 Cable Length	Length to fault in meters or centimeters based on Register 21_7.10.	RO
18_7.15:0	Pair 2 Cable Length	Length to fault in meters or centimeters based on Register 21_7.10.	RO
19_7.15:0	Pair 3 Cable Length	Length to fault in meters or centimeters based on Register 21_7.10.	RO
20_7.15:12	Pair 3 Fault Code	0000 = Invalid 0001 = Pair Ok 0010 = Pair Open 0011 = Same Pair Short 0100 = Cross Pair Short 1001 = Pair Busy else = Reserved	RO

Table 38: Alternate VCT™ Control (Continued)

Register	Function	Setting	Mode
20_7.11:8	Pair 2 Fault Code	0000 = Invalid 0001 = Pair Ok 0010 = Pair Open 0011 = Same Pair Short 0100 = Cross Pair Short 1001 = Pair Busy else = Reserved	RO
20_7.7:4	Pair 1 Fault Code	0000 = Invalid 0001 = Pair Ok 0010 = Pair Open 0011 = Same Pair Short 0100 = Cross Pair Short 1001 = Pair Busy else = Reserved	RO
20_7.3:0	Pair 0 Fault Code	0000 = Invalid 0001 = Pair Ok 0010 = Pair Open 0011 = Same Pair Short 0100 = Cross Pair Short 1001 = Pair Busy else = Reserved	RO
21_7.15	Run Immediately	0 = No Action 1 = Run VCT Now	R/W, SC
21_7.14	Run at each Auto-Negotiation Cycle	0 = Do No Run At Auto-Negotiation Cycle 1 = Run At Auto-Negotiation Cycle	R/W
21_7:13	Disable Cross Pair Check	0 = Enable Cross Pair Check 1 = Disable Cross Pair Check	R/W
21_7.12	Run After Break Link	0 = No Action 1 = Run VCT After Breaking Link	R/W, SC
21_7.11	Cable Diagnostics Status	0 = Complete 1 = In Progress	RO
21_7.10	Cable Length Unit	0 = Centimeters 1 = Meters	R/W

2.14 Data Terminal Equipment (DTE) Detect

The device supports the Data Terminal Equipment (DTE) power function. The DTE power function is used to detect if a link partner requires power supplied by the POE PSE device.

The DTE power function can be enabled by writing to Register 26_0.8. When DTE is enabled, the device will first monitor for any activity transmitted by the link partner. If the link partner is active, then the link partner has power and power from the POE PSE device is not required. If there is no activity coming from the link partner, DTE power engages, and special pulses are sent to detect if the link partner requires DTE power. If the link partner has a low pass filter (or similar fixture) installed, the link partner will be detected as requiring DTE power.

The DTE power status register (Register 17_0.2) immediately comes up as soon as link partner is detected as a device requiring DTE power. Register 19_0.2 is a stray bit that reports the DTE power status has changed states.

If a link partner that requires DTE power is unplugged, the DTE power status (Register 17_0.2) will drop after a user controlled delay (default is 20 seconds - Register 26_0.7:4) to avoid DTE power status register drop during the link partner powering up (for most applications), since the low pass filter (or similar fixture) is removed during power up. If DTE power status drop is desired to be reported immediately, write Register 26_0.7:4 to 4'b0000.

A detailed description of the register bits used for DTE power detection for the device are shown in Table 39.

Table 39: Registers for DTE Power

Register	Description
26_0.8 - Enable power over Ethernet detection	1 = Enable DTE detect 0 = Disable DTE detect A soft reset is required to enable this feature HW reset: 0 SW reset: Update
17_0.2 - Power over Ethernet detection status	1 = Need power 0 = Do not need power HW reset: 0 SW reset: 0
19_0.2 - Power over Ethernet detection state changed	1 = Changed 0 = No change HW reset: 0 SW reset: 0
26_0.7:4 - DTE detect status drop	Once the PHY no longer detects that the link partner filter, the PHY will wait a period of time before clearing the power over Ethernet detection status bit (17_0.2). The wait time is 5 seconds multiplied by the value of these bits. Example: (5 * 0x4 = 20 seconds) Default at HW reset: 0x4 At SW reset: retain

2.15 Energy Efficient Ethernet (EEE)

2.15.1 Energy Efficient Ethernet Low Power Modes

The 100BASE-TX EEE capability is enabled through XMDIO Register bit 7.60.1, by default, this bit is set to 0. The 1000BASE-T EEE capability is enabled through XMDIO Register bit 7.60.2, by default, this bit is set to 0. For both 100BASE-TX and 1000BASE-T, link partners negotiate the EEE capability through the auto-negotiation process. If the auto-negotiation result indicates that the EEE mode is supported, the MAC sub-layer can request entering and exiting EEE low power mode by asserting and de-asserting Low Power Idle (LPI) on the RGMII and SGMII interfaces. Passing LPI on the MII interface is not supported.

When LPI is asserted on RGMI or SGMII interfaces by the MAC sub-layer, local PHY transmitter signals a sleep request to its link partner to indicate that the local transmitter is entering LPI mode. The 100BASE-TX transmitter will enter LPI mode after signaling a sleep indication to its link partner. The 1000BASE-T transmitter will enter LPI mode after it signals a sleep indication to its link partner and receives a sleep indication signal from its link partner's transmitter. When the PHY transmitter is in LPI mode, it powers off part of its circuits and keeps the media interface quiet, then powers up the transmitter periodically to send the refresh signals for the link partner's receiver to update adaptive coefficients and timing circuits with. The PHY receiver enters LPI mode after receiving a sleep indication from the link partner's transmitter. When the receiver is in LPI mode, it powers off part of the receiver circuits, asserts LPI on the RGMII or SGMII interface, and updates adaptive coefficients and timing circuits upon receiving refresh signals from the link partner.

This sleep-refresh cycle continues until the MAC sub-layer de-asserts LPI on the RGMII or SGMII interface to request the PHY to resume normal operational mode. The PHY transmitter sends wake signals and then after a pre-defined period of time, the PHY enters normal operational mode.

10BASE-T EEE is enabled by MDIO Register bit 20_0.7. When 10BASE-T EEE is enabled, the transmit waveform shape is changed on the media interface to reduce the power consumption.

The 1000BASE-X PCS is modified to transmit low power idle code groups /LI1/ and /LI2/ ordered sets on the media interface when LPI is asserted on the RGMII interface by MAC sub-layer, and asserts LPI on the RGMII interface when /LI1/ or /LI2/ ordered sets are received on the media interface.

The RGMII interface is modified to receive LPI on the media transmit direction and pass LPI on the media receive direction. SGMII PCS is modified in a similar way as 1000BASE-X to receive and assert /LI1/ and /LI2/.The 88EA1517/88EA1512 devices support two EEE modes:

a) Legacy or Master mode

b) EEE aware or Slave mode

when RGMII or SGMII is the MAC interface.

When MII is the MAC interface only the Legacy or Master mode is supported.

2.15.1.1 Master (Legacy) mode

The legacy mode of operation is used in systems where the external MAC is not EEE capable. The MAC and the system are completely transparent to the EEE operation. Entering and exiting out of EEE operation or Low Power Idle (LPI) mode is completely managed and controlled by the PHY alone. Once EEE capabilities are exchanged with an EEE compliant link partner, the PHY waits for an enter timer to expire. This enter timer tracks the amount of time without a data transaction between the MAC and the PHY. Once the enter timer expires the PHY begins to transmit the low power idles to save power. Once the low power mode is disabled, either because of a wake signal received from the link partner or data received from the MAC, the PHY waits for an exit timer to expire before it resumes normal operation. The exit timer tracks the transition from the low power idle mode to normal operation.

The Legacy mode of operation uses the EEE buffer for storing the data from the MAC before it exits out of the EEE LPI. Refer to Section 2.15.2 for further details.

The EEE buffer has an enter timer and an exit timer. The enter timer and exit timer settings are available in Register 1_18 and 2_18. By default, it assumes IEEE timer values. When the EEE buffer is empty, an enter timer will be initiated. Before the enter timer expires, if the EEE buffer receives any packet from the legacy MAC the enter timer will be restarted. But if the buffer remains empty until the expiration of the enter timer, then the device enters the LPI mode and begins transmitting LPI. As long as the buffer remains empty, the device continues to transmit LPI.

Once the EEE buffer begins receiving packets from the MAC, an exit timer is initiated and the device sends a WAKE signal to the link partner indicating that data will be transmitted shortly. Alternatively, the device can also transition out of the LPI mode if it receives a WAKE from its link partner. When the exit timer expires, the device transitions out of LPI mode to resume normal transmission.

Note

When using MII mode as the System mode and EEE mode is enabled, only EEE Master Mode may be used. Reg. 0_18.0 is 1 always where the EEE Buffer is enabled and LPI received is converted to regular idles at the MII receive interface.

2.15.1.2 Slave (EEE Aware) mode

By default the PHY is configured in EEE Slave mode (Reg 0_18.0 = 0) if the MII interface is not used. In the EEE Slave mode, the external MAC connected to the system interface must be EEE-capable and must meet the IEEE 802.3az-2010 Clause 78 requirements. In this mode, the EEE buffering is disabled and EEE enter/exit timers settings are ignored. The MAC is expected to handle all time requirements. The MAC is expected to buffer the packets while the PHY transitions from the LPI mode to normal mode. The MAC is also responsible for initiating the low power mode by sending the LPI idles (to initiate LPI mode) or normal idles (to exit the LPI mode) to the PHY. The PHY will pass the LPI idles from the system interface to the media interface and vice versa.

2.15.1.3 10BASE-Te

The 88EA1517/88EA1512 devices support 10BASE-Te. When 10BASE-Te is enabled power consumption is reduced by changing the waveform shape of the TX output on the media interface. 10BASE-Te is enabled by Register bit 20_0.7. 10BASE-Te capability is not part of 10/100/1000BASE-T Auto-Negotiation.

2.15.2 Energy Efficient Ethernet (EEE) Buffering for Master mode

The 88EA1517/88EA1512 devices have a built-in buffer to be used when the devices transition into Master mode EEE operation. This buffer is used to store data from a non -EEE compliant legacy MAC, when the PHY media interface is exiting the Low Power Idle (LPI) mode. The buffer basically ensures that no packets are lost by the PHY during the transition from LPI mode to normal mode of operation. Once the media interface exits the LPI mode, the data in the buffer is sent out through the media interface. The EEE buffer may be enabled by setting 0_18.0 = 1.

When the PHY in LPI mode detects data coming from the MAC interface, the LPI idles will be stopped on the media interface. This triggers the PHY to initiate a WAKE to the link partner to indicate that the data will be transmitted shortly. During this time an exit timer will enable the buffer to store the incoming data from the MAC until the media interface has completely come out of the sleep state. The exit timer values are stored in Register 1_18.

The 1000BASE-X, SGMII PCS are modified to support low power idle code groups /LI1/ and /LI2/ ordered sets. The SGMII Auto-Negotiation configuration code groups contain additional two bits to communicate EEE status through in-band SGMII Auto-Negotiation (See Table 35, Enhanced SGMII PHY Status, on page 64).

Note

The slave and the Master mode EEE operation must not be confused with the Master and Slave mode operation of 1000BASE-T

2.15.3 Energy Efficient Ethernet Auto-Negotiation

In order for the EEE to work both the local PHY and its link partner needs to support EEE. The EEE capabilities are exchanged through standard 10/100/1000BASE-T Auto-Negotiation (IEEE 802.3 Clauses 28 and 40). These capabilities are exchanged through the Next Page exchanges. The Auto-Negotiation process is used only for the exchange of speed and duplex information and not any timer information with regards to the sleep, refresh and wake cycles. 10BASE-Te capability is not part of the 10/100/1000BASE-T Auto-Negotiation.

The device implements Energy Efficient Ethernet (EEE) functions based on IEEE 802.3az-2010. The device supports EEE on the following System and Media Interfaces:

- 10BASE-Te
- 100BASE-TX
- 1000BASE-T

The EEE registers are defined in Clause 22 Page 18 registers and Clause 45 XMDIO space. Refer to "PHY MDIO Register Description" on page 129 for details.

2.16 CRC Error Counter and Frame Counter

The CRC counter and packet counters, normally found in MACs, are available in the device. The error counter and packet counter features are enabled through register writes and each counter is stored in eight register bits.

Register 18_18.2:0 controls which path the CRC checker and packet counter is counting.

If Register 18_18.2:0 is set to 010 then the Copper receive path is checked.

If Register 18_18.2:0 is set to 100 then the SGMII input path is checked.

If Register 18_18.2:0 is set to 110 then the RGMII/MII input path is checked.

2.16.1 Enabling The CRC Error Counter and Packet Counter

To enable the counters to count, set Register 18_18.2:0 to a non-zero value.

To disable the counters, set Register 18_18.2:0 to 000.

To read the CRC counter and packet counter, read Register 17_18.

17_18.15:8 (Frame count is stored in these bits)

17_18.7:0 (CRC error count is stored in these bits)

The CRC counter and packet counter do not clear on a read command. To clear the counters, write Register 18_18.4 = 1. The Register 18_18.4 is a self-clear bit. Disabling the counters by writing Register 18_18.2:0 to 000 will also reset the counters.

2.17 Packet Generator

The device contains a very simple packet generator. Register 16_18.7:5 lists the device Packet Generator register details.

When Register 16_18.7:5 is set to 010 packets are generated on the copper transmit path.

When Register 16_18.7:5 is set to 100 packets are generated on the SGMII transmit path.

When Register 16_18.7:5 is set to 110 packets are generated on the RGMII/MII transmit path.

Once enabled, fixed length packets of 64 or 1518 bytes (including CRC) will be transmitted separated by 12 bytes of IPG.The preamble length will be 8 bytes. The payload of the packet is either a fixed 5A, A5, 5A, A5 pattern or a pseudo random pattern. A correct IEEE CRC is appended to the end of the packet. An error packet can also be generated.

The registers are as follows:

16_18.7:5 Packet generation enable. 000 = Normal operation, Else = Enable internal packet generator.

16_18.2 Payload type. 0 = Pseudo random, 1 = Fixed 5A, A5, 5A, A5,...

16_18.1 Packet length. 0 = 64 bytes, 1 = 1518 bytes

16_18.0 Error packet. 0 = Good CRC, 1 = Symbol error and corrupt CRC.

16_18.15:8 Packet Burst Size. 0x00 = Continuous, 0x01 to 0xFF = Burst 1 to 255 packets.

If Register 16_18.15:8 is set to a non-zero value, then Register 16_18.7:5 will self clear once the required number of packets are generated. Note that if Register 16_18.7:5 is manually set to 0 while packets are still bursting, the bursting will cease immediately once the current active packet finishes transmitting. The value in Register 16_18.15:8 should not be changed while 16_18.7:5 is set to a non-zero value.

2.18 RX_ER Byte Capture

Whenever there is an RX_ER in the internal GMII interface the PHY will capture 4 bytes before the RX_ER is asserted. Once the bytes preceding the RX_ER assertion are captured into the registers, they will not be over written by new errors and they will only be cleared after the registers are read.

The capture register for the copper path is Register 20_2. The capture register for the fiber path is Register 26_18. The one for the RGMII path is Register 20_4. The description below applies to the copper/fiber path and the RGMII paths but refers to Register 26_18 only. Register 20_4 should be used when accessing the RGMII path and 20_2 should be used when accessing the copper path.

Once an error event is captured Register 26_18.15 is set to 1 indicating that the capture data is valid. No further errors are captured until all captured registers are read. Register 26_18.13:12 is set to 00. Register 26_18.9:0 outputs the byte that is the earliest received. Once Register 26_18 is read Register 26_18.13:12 increments and Register 26_18.9.0 is updated with the next earliest byte. The register is incremented and byte updated until the fourth read occurs. After the fourth read to Register 26_18 is completed, Register 26_18.15 is set to 0 and the error capturing resumes four RX_CLK cycles after the final read is completed. The 4 RX_CLK cycle delay is required to insure that the register has 4 valid bytes loaded prior to being frozen. Note that a side effect of doing this is the RX_ER may be high in the captured bytes.

Table 40: Error Byte Capture

Register	Function	Setting
26_18.15	Capture Data Valid	1 = Bits 14:0 Valid 0 = Bits 14:0 Invalid
26_18.13:12	Byte Number	00 = 4 bytes before RX_ER asserted 01 = 3 bytes before RX_ER asserted 10 = 2 bytes before RX_ER asserted 11 = 1 byte before RX_ER asserted The byte number increments after every read when Register 26_18.15 is set to 1.
26_18.9	RX_ER	RX Error. Normally this bit will be low since the capture is triggered by RX_ER being high. However, it is possible to see an RX_ER high when the capture is re-enabled after reading the fourth byte and there happens to be a long sequence of RX_ER when the capture restarts.
26_18.8	RX_DV	RX Data Valid
26_18.7:0	RXD[7:0]	RX Data

2.19 1.25G PRBS Generator and Checker

A PRBS generator and checker are available for use on the 1.25G SERDES. PRBS7,PRBS23, and PRBS31 are supported.

A 32-bit checker is implemented. Note that the reads are atomic. A read to the LSB will update the MSB register. The counters only clear when Register 23_1.4 is set to 1. This bit self clears.

The checker and generator polarity can be inverted by setting Registers 23_1.7 and 23_1.6 respectively.

Register 23_1.5 controls whether the checker has to lock before counting commences.

Table 41: 1.25 GHz SERDES PRBS Registers

Register	Function	Setting
23_1.7	Invert Checker Polarity	0 = Invert 1 = Normal
23_1.6	Invert Generator Polarity	0 = Invert 1 = Normal
23_1.5	PRBS Lock	0 = Counter Free Runs 1 = Do not start counting until PRBS locks first
23_1.4	Clear Counter	0 = Normal 1 = Clear Counter

Table 41: 1.25 GHz SERDES PRBS Registers

Register	Function	Setting
23_1.3:2	Pattern Select	00 = PRBS 7 01 = PRBS 23 10 = PRBS 31 11 = Generate 1010101010...pattern
23_1.1	PRBS Checker Enable	0 = Disable 1 = Enable
23_1.0	PRBS Generator Enable	0 = Disable 1 = Enable
24_1.15:0	PRBS Error Count LSB	A read to this register freezes Register 25_1. Cleared only when Register 23_1.4 is set to 1.
25_1.15:0	PRBS Error Count MSB	This register does not update unless Register 24_1 is read first. Cleared only when Register 23_1.4 is set to 1.

2.20 MDI/MDIX Crossover

The device automatically determines whether or not it needs to cross over between pairs as shown in Table 42 so that an external crossover cable is not required. If the device interoperates with a device that cannot automatically correct for crossover, the device makes the necessary adjustment prior to commencing Auto-Negotiation. If the device interoperates with a device that implements MDI/MDIX crossover, a random algorithm as described in IEEE 802.3 clause 40.4.4 determines which device performs the crossover.

When the device interoperates with legacy 10BASE-T devices that do not implement Auto-Negotiation, the device follows the same algorithm as described above since link pulses are present. However, when interoperating with legacy 100BASE-TX devices that do not implement Auto-Negotiation (i.e. link pulses are not present), the device uses signal detect to determine whether or not to crossover.

The auto MDI/MDIX crossover function can be disabled via Register 16_0.6:5.

The pin mapping in MDI and MDIX modes is shown in Table 42.

Table 42: Media Dependent Interface Pin Mapping

Pin	MDI			MDIX		
	1000BASE-T	100BASE-TX	10BASE-T	1000BASE-T	100BASE-TX	10BASE-T
MDIP/N[0]	BI_DA±	TX±	TX±	BI_DB±	RX±	RX±
MDIP/N[1]	BI_DB±	RX±	RX±	BI_DA±	TX±	TX±
MDIP/N[2]	BI_DC±	unused	unused	BI_DD±	unused	unused
MDIP/N[3]	BI_DD±	unused	unused	BI_DC±	unused	unused

Note: Table 42 assumes no crossover on PCB.

The MDI/MDIX status is indicated by Register 17_0.6. This bit indicates whether the receive pairs (3,6) and (1,2) are crossed over. In 1000BASE-T operation, the device can correct for crossover between pairs (4,5) and (7,8) as shown in Table 42. However, this is not indicated by Register 17_0.6.

If 1000BASE-T link is established, pairs (1,2) and (3,6) crossover is reported in Register 21_5.4, and pairs (4,5) and (7,8) crossover is reported in Register 21_5.5.

For the 88EA1517, the MDI or MDIX mode is configured via the HW configuration pin - RXD[3] pin. So by default the device is either good MDI or MDIX depending on the value of the RXD[3] pin. The auto-MDI/MDIX crossover function can be enabled via register 16_0.6:5.

2.21 Unidirectional Transmit

IEEE 802.3ah requires OAM support with unidirectional transmit capability. Unidirectional transmit allows a PHY to transmit data when the PHY does not have link due to potential issues on the receive path. 802.3ah formally requires two bits for this capability. Register 0.5 enables this capability, and 1.7 advertises this ability. This ability only applies to 10BASE-T, 100BASE-TX or 1000BASE-X. It doesn't apply to 1000BASE-T since 1000BASE-T requires a MASTER/SLAVE relationship and training with both link partners participating, which requires that link exists for any data transmit.

The device can support transmit of packets when there is no link by using Register bit 16_0.10 = 1 (Force Copper Link Good) and 16_1.10 = 1(Force Fiber Link Good). This is not the official bit specified by the IEEE 802.3ah but serves the same function.

2.22 Polarity Correction

The device automatically corrects polarity errors on the receive pairs in 1000BASE-T and 10BASE-T modes. In 100BASE-TX mode, the polarity does not matter.

In 1000BASE-T mode, receive polarity errors are automatically corrected based on the sequence of idle symbols. Once the descrambler is locked, the polarity is also locked on all pairs. The polarity becomes unlocked only when the receiver loses lock.

In 10BASE-T mode, polarity errors are corrected based on the detection of validly spaced link pulses. The detection begins during the MDI crossover detection phase and locks when the 10BASE-T link is up. The polarity becomes unlocked when link is down.

The polarity correction status is indicated by Register 17_0.1. This bit indicates whether the receive pair (3,6) is polarity reversed in MDI mode of operation. In MDIX mode of operation, the receive pair is (1,2) and Register 17_0.1 indicates whether this pair is polarity reversed. Although all pairs are corrected for receive polarity reversal, Register 17_0.1 only indicates polarity reversal on the pairs described above.

If 1000BASE-T link is established Register 21_5.3:0 reports the polarity on all 4 pairs.

Polarity correction can be disabled by Register write 16_0.1 = 1. Polarity will then be forced in normal 10BASE-T mode.

2.23 FLP Exchange Complete with No Link

Sometimes when link does not come up, it is difficult to determine whether the failure is due to the Auto-Negotiation Fast Link Pulse (FLP) not completing or from the 10/100/1000BASE-T link not being able to come up.

Register 19_0.3 is a sticky bit that gets set to 1 whenever the FLP exchange is completed but the link cannot be established for some reason. Once the bit is set, it can be cleared only by reading the register.

This bit will not be set if the FLP exchange is not completed, or if link is established.

2.24 Duplex Mismatch Indicator

When operating in half-duplex mode collisions should occur within the first 512 bit times. Collisions that are detected after this point can indicate an incorrect environment (too many repeaters in the system, too long cable) or it can indicate that the link partner thinks the link is a full-duplex link.

Registers 23_6.7:0, 23_6.15:8, 24_6.7:0, and 24_6.15:8 are 8 bit counters that count late collisions.

They will increment only when the PHY is in half-duplex mode and only applies to the copper interface. Each counter increments when a late collision is detected in a certain window as shown in Table 43. The four late collision counters will increment based on when the late collision starts. The counters clear on read. If the counter reaches FF it will not roll over.

Table 43: Late Collision Registers

Register	Function	Setting	Mode
23_6.15:8	Late Collision 97-128 bytes	This counter increments by 1 when the PHY is in half-duplex and a start of packet is received while the 96th to 128th bytes of the packet are transmitted. The measurement is done at the internal GMII interface. The counter will not roll over and will clear on read.	RO, SC
23_6.7:0	Late Collision 65-96 bytes	This counter increments by 1 when the PHY is in half-duplex and a start of packet is received while the 65th to 96th bytes of the packet are transmitted. The measurement is done at the internal GMII interface. The counter will not roll over and will clear on read.	RO, SC
24_6.15:8	Late Collision >192 bytes	This counter increments by 1 when the PHY is in half-duplex and a start of packet is received after 192 bytes of the packet are transmitted. The measurement is done at the internal GMII interface. The counter will not roll over and will clear on read.	RO, SC
24_6.7:0	Late Collision 129-192 bytes	This counter increments by 1 when the PHY is in half-duplex and a start of packet is received while the 129th to 192nd bytes of the packet are transmitted. The measurement is done at the internal GMII interface. The counter will not roll over and will clear on read.	RO, SC
25_6.12:8	Late Collision Window Adjust	Number of bytes to advance in late collision window. 0 = start at 64th byte, 1 = start at 63rd byte, etc.	R/W

The real point of measurement for late collision should be done at the MAC and not at the PHY. In order to compensate for additional latency between the PHY and the MAC Register 25_6.12:8 is used to move the window earlier. For example, if Register 25_6.12.8 is set to 2 then the first window is 63 to 94 bytes, the second window is 95 to 129 bytes, etc. It is up to the user to program this register correctly since it is system dependent.

2.25 Link Disconnect Counter

The link disconnect counter increments every time the link transitions from up to down. This applies regardless of whether the link is a copper link or fiber link.

Register 25_18.7:0 is used as the counter. It clears on read and will not roll over when it reaches 0xFF.

2.26 Two-wire Serial Interface

The 88EA1517/88EA1512 device supports a Two-wire Serial Interface (TWSI). The TWSI operates with a serial data line (SDA) and a serial clock line (SCL). The bus interface is enabled through the hardware configuration pin RXD[0] and, during data transfer, the MDC/MDIO pins are utilized as the bus interface connections. For the TWSI device address, in the 88EA1512 device the lower 4 bits (PHYADR[3:0]), are latched during hardware reset, and the device address bits ([6:4]) are fixed at '100'. In the 88EA1517 device, the lower 3 bits (PHYADR[2:0]), are latched during hardware reset, and the device address bits ([6:3]) are fixed at '1000'. The SDA is a bi-directional line, while the SCL line is not. SDA requires a 1.5 kohm pull-up resistor. The 88EA1517/88EA1512 device operates as the Slave port of the bus interface, and all references to Slave refer to the 88EA1517/88EA1512 device.

The 88EA1517/88EA1512 device will be available for read/write operations 5 ms after hardware reset.

Table 44 indicates the pin mapping of the 88EA1517/88EA1512 device to the TWSI.

Table 44: 88EA1517/88EA1512 device to Two-Wire Serial Interface Signal Mapping

88EA1517/88EA1512 Device Pins	100/400 Kbps Mode	Description
MDIO	SDA	Serial data line
MDC	SCL	Serial clock line

The 88EA1517/88EA1512 device TWSI features are:

- 7-bit device address/8-bit data transfers
- 100 Kbps mode
- 400 Kbps mode

Multiple devices using the TWSI can share and lump up the MDC and MDIO lines, and are pulled up with a resistor ranging from 4.5 kohm to 10 kohm.

The 88EA1517/88EA1512 device will be available for read/write operations 5 ms after hardware reset.

2.26.1 Bus Operation

The Master generates one clock pulse for each data bit transferred. The high or low state of the data line can only change when the clock signal on the SCL line is low. A high to low transition on the SDA line while SCL is high defines a Start. A low to high transition on the SDA line while the SCL is high defines a Stop. Start (S), Repeated Start (Sr), and Stop (P) conditions are always generated by the Master. Acknowledge (A) and Not Acknowledge ($\overline{A}$) can be generated by either the Slave or Master.

The Master continuously monitors for Start and Stop conditions. Whenever a Stop is detected, the 88EA1517/88EA1512 device goes into standby mode, and the current operation is canceled. The Slave recovers from this error condition, and waits for the next transfer to begin.

Data transfer with Acknowledge is always obligatory. The receiver must pull down the SDA line during the Acknowledge clock pulse so that it remains stable low during the high period of this clock pulse.

If the Slave does not Acknowledge the device address, but some time later in the transfer cannot receive any more data bytes, the Master must abort the transfer. This is indicated by the Slave generating the Not Acknowledge on the first byte to follow. The Slave device then leaves the data line high, and the Master must generate a Stop or a Repeated Start condition. When the Slave is transmitting data on the bus and the Master responds with a Not Acknowledge, the Slave must receive a Stop or a Repeated Start condition. If neither is received, it is an error condition. The Slave recovers from this error condition and waits for the next transfer to begin.

The bus interface is also active in power down mode. (See Section 2.9, Power Management, on page 59 for power down mode details.) Whenever the Slave is addressed by the Master, the 88EA1517/88EA1512 device comes out of power down mode if it is in power down mode. Read and Write operations are detailed in the following section.

2.26.2 Read and Write Operations

Write operations require an 8-bit Slave address followed by the register address and Acknowledgement. The Slave address is 7 bits long followed by an eighth bit. The first bit of a data transfer is the most significant bit (MSb). The eighth bit is the least significant bit (LSb), which is a direction bit (R/W) - Read = 1, Write = 0). Read operations are executed the same way as Write operations with the exception that the R/W bit is set to one.

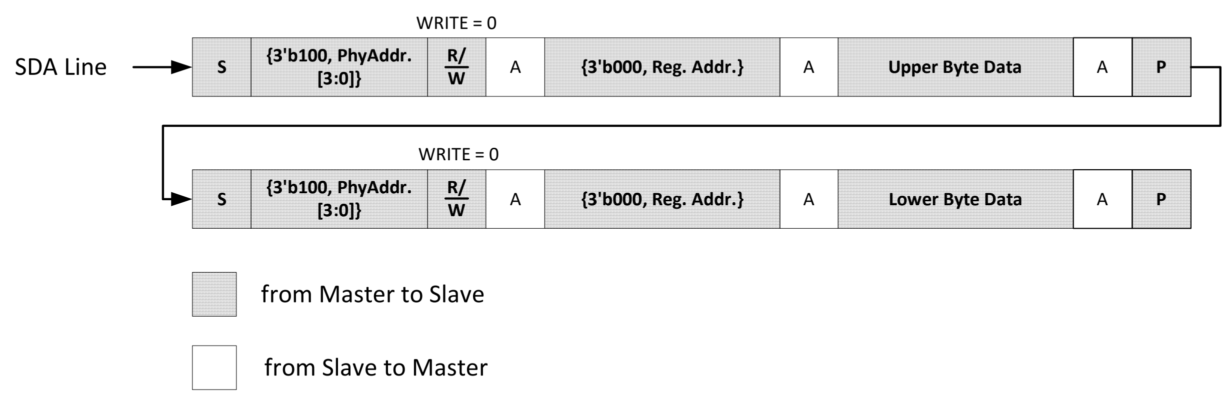
A complete byte write operation includes the upper and lower bytes. The upper byte is written first, followed by the lower byte.

The register address counter maintains the last address accessed during the last read or write operation, incremented by one. The address remains valid between operations as long as chip power is maintained. The register limit is 32 registers. Once the counter reaches the lower byte of Register 31, it rolls over to the upper byte of Register 0.

2.26.2.1 Random Write

For random writes, both the upper byte and lower byte must be written to the Slave before the data is written to the addressed register.

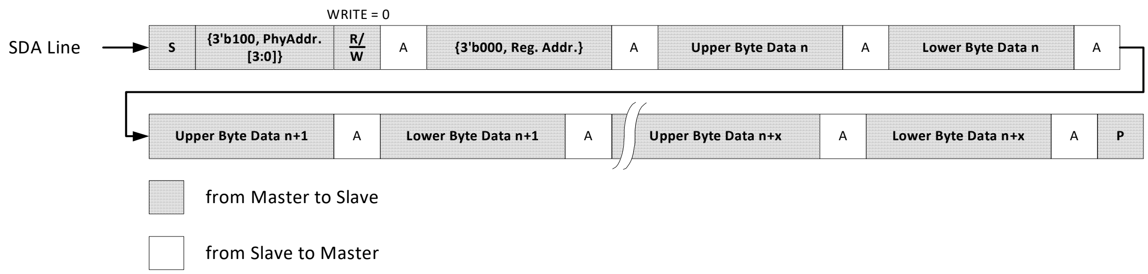
Figure 26: Random Write Operation



2.26.2.2 Sequential Write

A sequential write is started by a random address write. After the register address is received by the Slave, the Slave responds with an Acknowledge. The Slave generates an Acknowledge as long as the Master does not generate a Stop. In sequential write, only the even transfer of bytes is accepted by the 88EA1517/88EA1512 device. If the last byte is odd, it is held internally by the Slave, but is not written to the Slave register.

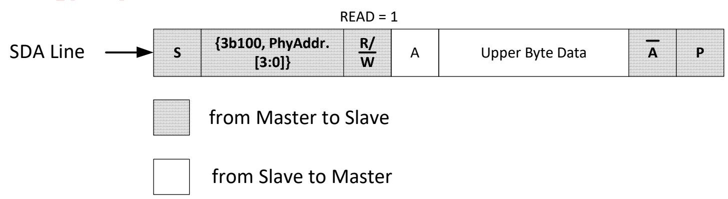
Figure 27: Sequential Write Operation



2.26.2.3 Current Address Read

A current address read is used to read the data at the current register address. A Start begins a current address read and resets the Slave to synchronize with the Master.

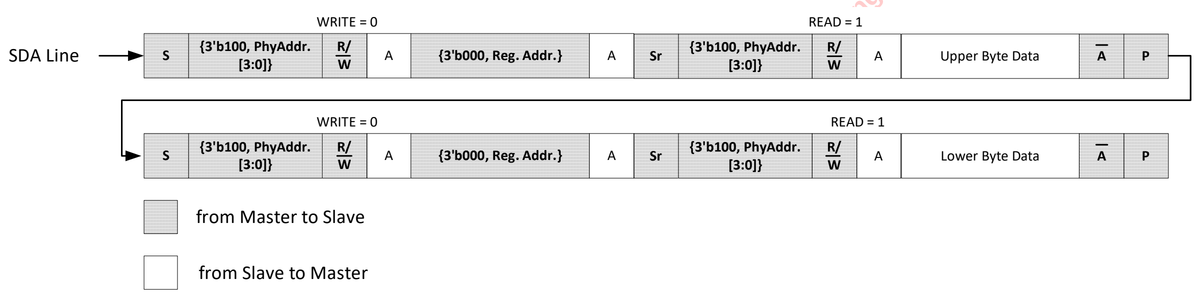
Figure 28: Current Address Read



2.26.2.4 Random Read

A random read is used to access any particular register. A "dummy" byte write is required to load in the data word address. When the device address and the data word address are Acknowledged, another Start condition must be generated. Figure 29 is an example of a random read.

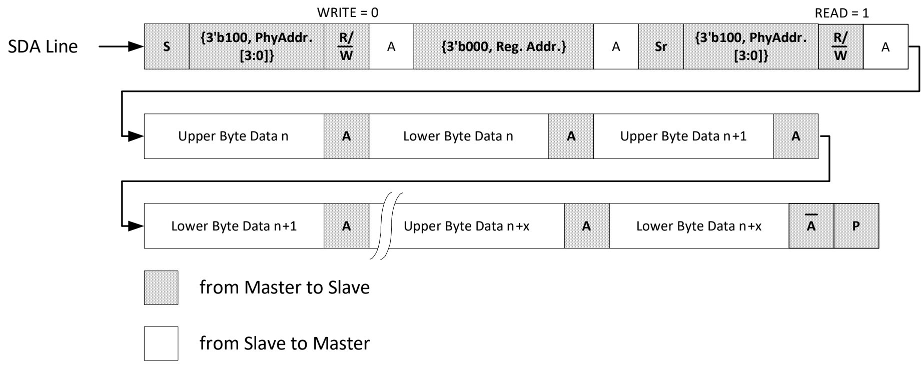
Figure 29: Random Read Operation



2.26.2.5 Sequential Read

A sequential read is used to read particular registers in their address order. A sequential read is initiated by either a current or random address read. After the Master receives the register data, it responds with an Acknowledge. As long as the Slave receives an Acknowledge, register data continues to be read incrementally. The sequential read operation is stopped when the Master does not respond with a zero, but does generate a Stop condition. Figure 30 is an example of a sequential read started by a random read.

Figure 30: Sequential Read Operation



2.27 LED

The LED[2:0] pins can be used to drive LED pins. Registers 16_3, 17_3, and 18_3 control the operation of the LED pins. LED[1:0] are used to configure the PHY per Section •, MDC/MDIO register writes (managed applications)88EA1512 Hardware Configuration, on page 117. After the configuration is completed, LED[1:0] will operate per the setting in 16_3.7:0.

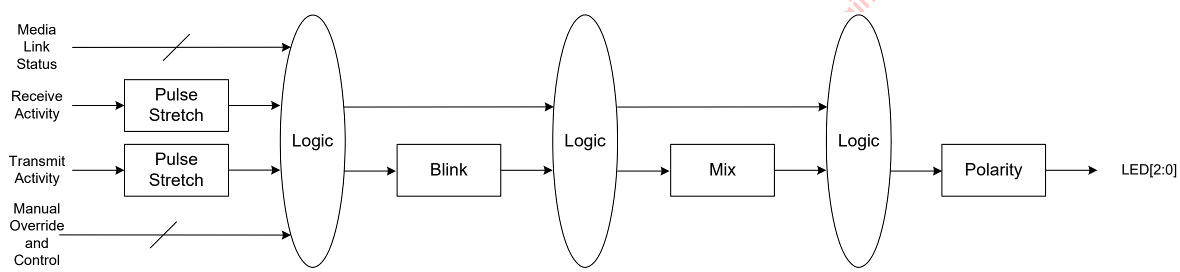
In general, 16_3.11:8 control the LED[2] pin, 16_3.7:4 control the LED[1] pin, and 16_3.3:0 control the LED[0] pin. These are referred to as single LED modes.

However, there are some LED modes where LED[1:0] operate as a unit. These are entered when 16_3.3:2 are set to 11. These are referred to as dual LED modes. In dual LED modes, Register 16_3.7:4 have no meaning when 16_3.3:2 are set to 11.

For 88EA1517, only LED[1:0] pins and associated functionality for LED[1] and LED[0] are available.

Figure 31 shows the general chaining of function for the LEDs. The various functions are described in the following sections.

Figure 31: LED Chain



2.27.1 LED Polarity

There are a variety of ways to hook up the LEDs. Some examples are shown in Figure 32. In order to make things more flexible Registers 17_3.5:4, 17_3.3:2, and 17_3.1:0 specify the output polarity for the LED[2:0]. The lower bit of each pair specifies the on (active) state of the LED, either high or low. The upper bit of each pair specifies whether the off state of the LED should be driven to the opposite level of the on state or Hi-Z.

Figure 32: Various LED Hookup Configurations

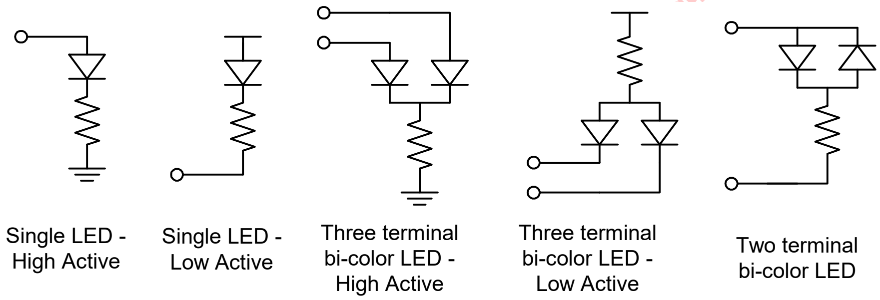


Table 45: LED Polarity

Register	Pin	Definition
17_3.5:4	LED[2] Polarity	00 = On - drive LED[2] low, Off - drive LED[2] high 01 = On - drive LED[2] high, Off - drive LED[2] low 10 = On - drive LED[2] low, Off - tristate LED[2] 11 = On - drive LED[2] high, Off - tristate LED[2]
17_3.3:2	LED[1] Polarity	00 = On - drive LED[1] low, Off - drive LED[1] high 01 = On - drive LED[1] high, Off - drive LED[1] low 10 = On - drive LED[1] low, Off - tristate LED[1] 11 = On - drive LED[1] high, Off - tristate LED[1]
17_3.1:0	LED[0] Polarity	00 = On - drive LED[0] low, Off - drive LED[0] high 01 = On - drive LED[0] high, Off - drive LED[0] low 10 = On - drive LED[0] low, Off - tristate LED[0] 11 = On - drive LED[0] high, Off - tristate LED[0]

2.27.2 Pulse Stretching and Blinking

Register 18_3.14:12 specify the pulse stretching duration of a particular activity. Only the transmit activity, receive activity, and (transmit or receive) activity are stretched. All other statuses are not stretched since they are static in nature and no stretching is required.

Some status will require blinking instead of a solid on. Register 18_3.10:8 specify the blink rate. Note that the pulse stretching is applied first and the blinking will reflect the duration of the stretched pulse.

The stretched/blinked output will then be mixed if needed (Section 2.27.3) and then inverted/Hi-Z according to the polarity described in section (Section 2.27.1).

Table 46: Pulse Stretching and Blinking

Register	Pin	Definition
18_3.14:12	Pulse stretch duration	000 = No pulse stretching 001 = 21 ms to 42 ms 010 = 42 ms to 84 ms 011 = 84 ms to 170 ms 100 = 170 ms to 340 ms 101 = 340 ms to 670 ms 110 = 670 ms to 1.3s 111 = 1.3s to 2.7s

Table 46: Pulse Stretching and Blinking

Register	Pin	Definition
18_3.10:8	Blink Rate	000 = 42 ms 001 = 84 ms 010 = 170 ms 011 = 340 ms 100 = 670 ms 101 to 111 = Reserved

2.27.3 Bi-Color LED Mixing

In the dual LED modes the mixing function allows the 2 colors of the LED to be mixed to form a third color. This is useful since the PHY is tri-speed and the three colors each represent one of the speeds. Register 17_3.15:12 control the amount to mix in the LED[1] pin. Register 17_3.11:8 control the amount to mix in the LED[0] pin. The mixing is determined by the percentage of time the LED is on during the active state. The percentage is selectable in 12.5% increments.

Note that there are two types of bi-color LEDs. There is the three terminal type and the 2 terminal type. For example, the third and fourth LED block from the left in Figure 32 illustrates three terminal types, and the one on the far right is the two terminal type. In the three terminal type both of the LEDs can be turned on at the same time. Hence the sum of the percentage specified by 17_3.15:12 and 17_3.11:8 can exceed 100%. However, in the two terminal type the sum should never exceed 100% since only one LED can be turned on at any given time.

The mixing only applies when Register 16_3.3:0 are set to 11xx. There is no mixing in single LED modes.

Table 47: Bi-Color LED Mixing

Register	Function	Definition
17_3.15:12	LED[1] mix percentage	When using 2 terminal bi-color LEDs the mixing percentage should not be set greater than 50%. 0000 = 0% 0001 = 12.5% . . 0111 = 87.5% 1000 = 100% 1001 to 1111 = Reserved
17_3.11:8	LED[0] mix percentage	When using 2 terminal bi-color LEDs the mixing percentage should not be set greater than 50%. 0000 = 0% 0001 = 12.5%, . . . 0111 = 87.5% 1000 = 100% 1001 to 1111 = Reserved

2.27.4 Modes of Operation

The LED pins relay some modes of the PHY so that these modes can be displayed by the LEDs. Most of the single LED modes are self-explanatory from the register map of Register 16_3. The non-obvious ones are covered in this section.

Table 48: Modes of Operation

Register	Pin	Definition
16_3.11:8	LED[2] Control	0000 = On - Link, Off - No Link 0001 = On - Link, Blink - Activity, Off - No Link 0010 = On- Full Duplex, Blink- Collision, Off- Half Duplex 0011 = On - Activity, Off - No Activity 0100 = Blink - Activity, Off - No Activity 0101 = On - Transmit, Off - No Transmit 0110 = On - 10/1000 Mbps Link, Off - Else 0111 = On - 10 Mbps Link, Off - Else 1000 = Force Off 1001 = Force On 1010 = Force Hi-Z 1011 = Force Blink 11xx = Reserved
16_3.7:4	LED[1] Control	If 16_3.3:2 is set to 11 then 16_3.7:4 has no effect 0000 = On- Receive, Off- No Receive 0001 = On - Link, Blink - Activity, Off - No Link 0010 = On - Link, Blink - Receive, Off - No Link 0011 = On - Activity, Off - No Activity 0100 = Blink - Activity, Off - No Activity 0101 = On- 100 Mbps Link/ Fiber Link 0110 = On - 100/1000 Mbps Link, Off - Else 0111 = On - 100 Mbps Link, Off - Else 1000 = Force Off 1001 = Force On 1010 = Force Hi-Z 1011 = Force Blink 11xx = Reserved
16_3.3:0	LED[0] Control	0000 = On - Link, Off - No Link 0001 = On - Link, Blink - Activity, Off - No Link 0010 = 3 blinks - 1000 Mbps 2 blinks - 100 Mbps 1 blink - 10 Mbps 0 blink - No Link 0011 = On - Activity, Off - No Activity 0100 = Blink - Activity, Off - No Activity 0101 = On - Transmit, Off - No Transmit 0110 = On - Copper Link, Off - Else 0111 = On - 1000 Mbps Link, Off - Else 1000 = Force Off 1001 = Force On 1010 = Force Hi-Z 1011 = Force Blink 1100 = MODE 1 (Dual LED mode) 1101 = MODE 2 (Dual LED mode) 1110 = MODE 3 (Dual LED mode) 1111 = MODE 4 (Dual LED mode)

2.27.4.1 Compound LED Modes

Compound LED modes are defined in Table 49.

Table 49: Compound LED Status

Compound Mode	Description
Activity	Transmit Activity OR Receive Activity
Link	10BASE-T link OR 100BASE-TX Link OR 1000BASE-T Link

2.27.4.2 Speed Blink

When 16_3.3:0 is set to 0010 the LED[0] pin takes on the following behavior.

LED[0] outputs the sequence shown in Table 50 depending on the status of the link. The sequence consists of 8 segments. If a 1000 Mbps link is established the LED[0] outputs 3 pulses, 100 Mbps 2 pulses, 10 Mbps 1 pulse, and no link 0 pulses. The sequence repeats over and over again indefinitely.

The odd numbered segment pulse duration is specified in 18_3.1:0. The even numbered pulse duration is specified in 18_3.3:2.

Table 50: Speed Blinking Sequence

Segment	10 Mbps	100 Mbps	1000 Mbps	No Link	Duration
1	On	On	On	Off	18_3.1:0
2	Off	Off	Off	Off	18_3.3:2
3	Off	On	On	Off	18_3.1:0
4	Off	Off	Off	Off	18_3.3:2
5	Off	Off	On	Off	18_3.1:0
6	Off	Off	Off	Off	18_3.3:2
7	Off	Off	Off	Off	18_3.1:0
8	Off	Off	Off	Off	18_3.3:2

Table 51: Speed Blink

Register	Pin	Definition
18_3.3:2	Pulse Period for even segments	00 = 84 ms 01 = 170 ms 10 = 340 ms 11 = 670 ms
18_3.1:0	Pulse Period for odd segments	00 = 84 ms 01 = 170 ms 10 = 340 ms 11 = 670 ms

2.27.4.3 Manual Override

When 16_3.11:10, 16_3.7:6, and 16_3.3:2 are set to 10 the LED[2:0] are manually forced. Registers 16_3.9:8, 16_3.5:4, and 16_3.1:0 then select whether the LEDs are to be on, off, Hi-Z, or blink.

If bi-color LEDs are used, the manual override will select only one of the 2 colors. In order to get the third color by mixing MODE 1 and MODE 2 should be used (Section 2.27.4.4).

2.27.4.4 MODE 1, MODE 2, MODE 3, MODE 4

MODE 1 to 4 are dual LED modes. These are used to mix to a third color using bi-color LEDs.

When 16_3.3:0 is set to 11xx then one of the 4 modes are enabled.

MODE 1 – Solid mixed color. The mixing is discussed in Section 2.27.3.

MODE 2 – Blinking mixed color. The mixing is discussed in Section 2.27.3. The blinking is discussed in section Section 2.27.2.

MODE 3 – Behavior according to Table 52.

MODE 4 – Behavior according to Table 53.

Note that MODE 4 is the same as MODE 3 except the 10 Mbps and 100 Mbps are reversed.

Table 52: MODE 3 Behavior

Status	LED[1]	LED[0]
1000 Mbps Link - No Activity	Off	Solid On
1000 Mbps Link - Activity	Off	Blink
100 Mbps Link - No Activity	Solid Mix	Solid Mix
100 Mbps Link - Activity	Blink Mix	Blink Mix
10 Mbps Link - No Activity	Solid On	Off
10 Mbps Link - Activity	Blink	Off
No link	Off	Off

Table 53: MODE 4 Behavior

Status	LED[1]	LED[0]
1000 Mbps Link - No Activity	Off	Solid On
1000 Mbps Link - Activity	Off	Blink
100 Mbps Link - No Activity	Solid On	Off
100 Mbps Link - Activity	Blink	Off
10 Mbps Link - No Activity	Solid Mix	Solid Mix
10 Mbps Link - Activity	Blink Mix	Blink Mix
No link	Off	Off

2.28 CLK125

The CLK125 output supplies a 125 MHz clock that can be used to clock other digital logic. The CLK125 should not be used as an input to devices that require a higher quality clock.

Note

Note that the 88EA1517 does not support the CLK125 output and associated functionality.

Register 16_2.1 is used to disable/enable the CLK125 output pin, where Register 16_2.1 = 1 disables the CLK125 output pin by tristating it and 16_2.1 = 0 enables the current selected clock to go out. The default value of 16_2.1 is 0 on power on reset or hardware reset. The CLK125 output will not glitch when going in or out of tristate.

The CLK125 can be in one of two states: hardware reset or normal operation. When transitioning between states, the CLK125 will not glitch.

Hardware Reset State

When hardware reset is asserted the CLK125 outputs a 25 MHz copy of the XTAL_IN clock.

Normal Operation

Two sources (an internally generated 125 MHz clock or SyncE clock) can be selected to drive the CLK125 output. Register 16_2.0 = 1 will output the SyncE clock and Register 16_2.0 = 0 will output internally generated 125 MHz clock. The default value of 16_2.0 is 0 for power on reset or hardware reset.

When Register 16_2.0 = 0 (an internally generated 125 MHz clock is selected), the CLK125 outputs a 125 MHz clock if Register 16_2.2 is set to 0. The CLK125 will continue to toggle when the PHY enters the power down state (Register 0_0.11 = 1) or in energy detect state. When Register 16_2.2 is set to 1, the CLK125 signal is held low. The default value of 16_2.2 is 0 on power on reset or hardware reset.

2.28.1 Synchronous Ethernet Recovered Clock

When Register 16_2.0 = 1 (SyncE clock is selected), the CLK125 outputs either a 125 MHz or 25 MHz clock that is based on the 125 MHz recovered clock on the copper receive path when linked up in1000BASE-T or 100BASE-TX. If a 25 MHz clock is selected, the 125 MHz recovered clock is internally divided by 5.

Register 16_2.11 selects whether the CLK125 outputs 25 MHz based on the XTAL clock or drives LOW when the link is down or when the copper receiver is linked to 10BASE-T. The default value of 16_2.11 is 0 on power on reset or hardware reset.

- 0 = CLK125 outputs 25 MHz XTAL clock during link down or 10BASE-T
- 1 = CLK125 drives LOW during link down or 10BASE-T

Setting Register 16_2.10 to 1 allows the recovered clock to go out on the CLK125. When Register 16_2.10 is set to 0, the recovered clock is held low. The default value of 16_2.10 is 1 on power on reset or hardware reset.

Register 16_2.12 selects whether the CLK125 outputs 25 MHz or 125 MHz(0 = 25 MHz, 1 = 125 MHz). The default value of 16_2.12 is 0 on power on reset or hardware reset.

Synchronous Ethernet recovered clock support is provided when the device is operating in 1000BASE-X or 100BASE-FX on the line interface. The CLK125 pin is used to output the recovered clock. Refer to Section 2.28, CLK125, on page 92 for configuring the CLK125 pin as recovered clock output. When the device is configured in SLAVE mode of operation, the CLK125 will output the recovered clock from the line interface. If the device is configured in 1000BASE-T MASTER mode of operation, the CLK125 will output the clock based on the XTAL.

2.29 Precise Timing Protocol (PTP) Time Stamping Support

Precise Time Protocol (PTP) is used by IEEE specifications to determine the time of day for systems across a network. The IEEE specifications are IEEE 802.1AS, IEEE 1588 version 1, and IEEE 1588 version 2. The PTP protocol is typically used in audio video bridging (PTP) applications, or industrial and test automation applications.

The fundamental concept is to be able to time stamp the PTP frames with high precision as close to the physical wires as possible. As such, doing the time stamping in the PHY increases the accuracy compared to doing it in the MAC or higher layers since the MAC interface FIFOs can add up to ±2 bytes of uncertainty.

The PTP core in the device consists of two sub-cores, namely the Packet Time Stamping and the Time Application Interface (TAI). The time stamping core supports time stamping of frame formats as defined in IEEE 802.1AS, IEEE 1588v1, and IEEE1588v2 frames.

2.29.1 PTP Control

To support the PTP Time Stamping function, the device has four pins that are global to the entire PHY.

- PTP clock input pin (The CONFIG pin is used for this purpose.)
- PTP Event Request input pin (The LED[1] pin is used for this purpose)
- PTP Trigger Generate output pin (The LED[1][1] pin is used for this purpose)
- Interrupt Pin (The LED[2] pin is used for this purpose)

2.29.1.1 PTP Clock Input

The PTP clock input is an option to use an external clock for the Time of Day counter instead of the free running internal clock. Register 20_6.8 is used to select the clock source. After configuration is completed and the external clock source is enabled (Register 20_6.8 = 1), the CONFIG pin is used as the external 125 MHz reference clock input.

Note

Because LED[1] pin is also used for PTP Event Request input, the PTP Trigger Generate output feature can never be used at the same time.

2.29.1.2 PTP Event Request

The PTP Event Request input pin can be configured to capture an external event (referred to as EventReq) and record the time at which the event occurred using the PTP Global Time Register (PTP Global Time Register - Page 12, Registers 14 and 15). Users must program 20_6.7 to 1 to enable this function. The definition of an external event is a low to high transition or a high to low transition on the LED[1] pin. Page 12, Register 0, bit 13 (Event Phase) selects which transition (rising or falling) is used. The event time is captured in EventCapRegister (Event Capture Register, Page 12, Register 10 and 11). This field is validated by EventCapValid bit (TAI Global Configuration Register 9 - Page 12, Register 9).

1. Refer to Section 2.27, LED, on page 86 for further details on LED[1] pin.

2.29.1.3 PTP Trigger Generate

The PTP Trigger Generate output pin is used to output an external signal (referred to as TrigGenResp) when the internal Time of Day counter matches a value programmed into a PHY register. When there is a match this output will go from low to high or high to low based on Page 12, Register 0, bit 12 (TrigPhase). The LED[1] pin is used for this function. It is selected by setting Register 20_6.6 = 1. The trigger output can also be a pulse or a clock depending on what is programmed in Page 12, Register 0, bit 1 (TrigMode).

2.29.1.4 PTP Control Register

The PTP circuit timestamps packets as it passes through the PHY. The register control to this function can be accessed via pages 8, 9, 12, and 14. The PTP circuit can be powered down when it is not used via Register 20_6.9. When Register 20_6.9 is set to 1 the registers in pages 8, 9, 12, and 14 are not accessible since the entire circuit is powered down.

By default Register 20_6.9 is set to 1. Register 20_6.9 must be set to 0 to enable PTP.

Table 54: PTP Control Register

Register	Function	Setting
20_6.9	PTP Power Down	1 = Power Down 0 = Power Up
20_6.8	PTP Reference Clock Source	1 = Use 125 MHz clock supplied to the CONFIG pin 0 = Use internal 125 MHz clock
20_6.7	PTP Input Source	1 = Use LED[1] pin for PTP Event Request input 0 = Force input to 0
20_6.6	PTP Output Source	1 = Use LED[1] pin for PTP Trigger Generate Output 0 = Use LED[1] for non-PTP functions.
20_6.6:0	Reserved	Reserved

2.29.2 Packet Time Stamping

2.29.2.1 Time Stamping without Hardware Acceleration

The device supports two sets of hardware arrival time stamp registers to be able to capture two different PTP event messages' time stamp before the CPU reads the time stamp registers out of the device. For every incoming PTP message type either PTPArr0Time (PTP Arrival 0 Time Registers - Page 8, Registers 9 and 10) or PTPArr1Time (PTP Arrival 1 Time Registers - Page 8, Registers 13 and 14) registers can be chosen by configuring TSArrPtr (PTP Global Configuration Register 2 - Page 14, Register 2). The SequenceID from the PTP Common header is captured as part of Arrival 0, Arrival 1 and/or Departure time stamp register sets so software can correlate the collected time stamps with the received or transmitted PTP event message. Hardware can be enabled to generate an interrupt upon capturing the time stamp information by writing a 0x1 to the interrupt enable register bits PTPArrIntEn (PTP Port Configuration Register 2 - Page 8, Register 2) for incoming PTP event messages or PTPDepIntEn (PTP Port Configuration Register 2 - Page 8, Register 2) for outgoing PTP event messages. In addition to generating an interrupt on the interrupt pin, an interrupt status (PTPArr0IntStatus, PTPArr1IntStatus, and PTPDepIntStatus) gets generated which indicates if there were to be an error related to the time stamp register. The interrupt status gets set to 0x1 when the time stamp counter gets overwritten before the previous time stamp registers have been read out. The interrupt status gets set to a 0x2, when a time stamp could not be captured for a PTP event message because DisTSOverwrite (PTP Port Configuration Register - Page 8, Register 0) is set to a 0x1.

Given that the device registers are accessed in units of 16-bits, to retain the entire 32-bit time stamp and the associated error messages and the SequenceID for a given PTP frame, the hardware treats the Arrival 0 block of registers (Page 8, Registers 8 through 11), Arrival 1 block registers (Page 8, Registers 12 through 15) and Departure block registers (Page 9, Registers 0 through 3) as a group and atomic operations are supported for these block of registers.

It is important to note that the device does not alter the contents of any PTP packet in either the ingress or egress directions. Time stamping at the device level does not involve adding a time stamp to a packet or changing its CRC. It also does not involve the device looking at time stamps that are embedded within a packet. The PHY does not add any additional latency when the PTP function is enabled. The PHY only identifies that a packet is a PTP frame and records the enter and exit time. The PHY does this by parsing the packet for certain fields as described in later sections. If the packet is identified as such a PTP packet, then the device loads the value of an internal “time of day counter” to a register showing the time for the first byte or SFD of the packet. The device then can inform the CPU or the PTP higher level firmware/software that such an event happened by activating an interrupt pin. The CPU can then read the relevant registers to find out if the event was in the RX or TX directions and the value of the “time of day counter” when the event happened.

Using the above time information and following the PTP protocol, the CPU or higher level entity can then determine the offset in time of day between a grand master clock and the SLAVE node, as well as the frequency difference between the Grand Master Clock and the SLAVE clock in case they are not frequency locked. These calculations are above the PHY level. The PHY provides full flexibility by allowing its time of day counter to be adjusted based on the CPU’s calculations, as well as a totally new time of day value to be entered. The CPU can also use the time information provided by the PHY to create PTP packets that inform the link partners or the Grand Master when the packets had arrived or left the port in question.

The maximum jitter associated with capturing the time stamps collected by the logic is one TSClkPer (TAI Global Configuration Register 1 - Page 12, Register 1). Note that there are inherent delay variation introduced in PHY layer pipelines both in receive and transmit direction which add to the overall jitter of the time stamps collected by the hardware and frequency/phase computations done in PTP protocol software.

For achieving higher accuracies in terms of PTP, it is recommended that an external clock device is used to adjust the time stamping clock with the frequency/phase offset information computed in PTP protocol software. The frequency and/or phase adjusted clock can in turn be fed back into the device to be used by the time stamping logic.

2.29.2.2 Time Stamping with Hardware Acceleration

Hardware acceleration is available and can be turned on to offload the CPU from processing time stamp information.

Without the hardware acceleration, PTP frames are detected and the time stamp information is extracted and placed in registers so there is no alteration of any frames. A CPU or a higher level entity will need to access the relevant registers to obtain the time stamp information.

With the hardware acceleration, time stamp information is inserted directly into the PTP frame before it is transmitted to the CPU. Therefore, the CPU can obtain the time stamp information once the frame is received and does not need to access registers. In this case, PTP frames are being modified and additional latency is introduced due to the frame buffering and time stamp insertion.

Frames Involved

Hardware acceleration is achieved by modifying the seven frames mentioned earlier based on the mode of operation. Table 55 lists the fields of each of these frames.

Table 55: List of Frame's Fields

# Octet	Sync	Follow_Up	Delay_Req	Delay_Resp	Pdelay_Req	Pdelay_Resp	Pdelay_Resp_ Follow_Up
6	Dest. Addr	Dest. Addr	Dest. Addr	Dest. Addr	Dest. Addr	Dest. Addr	Dest. Addr
6	Source Addr	Source Addr	Source Addr	Source Addr	Source Addr	Source Addr	Source Addr
2	EtherType	EtherType	EtherType	EtherType	EtherType	EtherType	EtherType
	UDP Portion	UDP Portion	UDP Portion	UDP Portion	UDP Portion	UDP Portion	UDP Portion
1	TransSpec, MessageType	TransSpec, MessageType	TransSpec, MessageType	TransSpec, MessageType	TransSpec, MessageType	TransSpec, MessageType	TransSpec, MessageType
1 2	Rsvd, VersionPTP MessageLength	Rsvd, VersionPTP MessageLength	Rsvd, VersionPTP MessageLength	Rsvd, VersionPTP MessageLength	Rsvd, VersionPTP MessageLength	Rsvd, VersionPTP MessageLength	Rsvd, VersionPTP MessageLength
1	DomainNumber	DomainNumber	DomainNumber	DomainNumber	DomainNumber	DomainNumber	DomainNumber
1	Rsvd	Rsvd	Rsvd	Rsvd	Rsvd	Rsvd	Rsvd
2	FlagField	FlagField	FlagField	FlagField	FlagField	FlagField	FlagField
8	CorrectionField	CorrectionField	CorrectionField	CorrectionField	CorrectionField	CorrectionField	CorrectionField
4	Rsvd	Rsvd	Rsvd	Rsvd	Rsvd	Rsvd	Rsvd
10	SourcePortID	SourcePortID	SourcePortID	SourcePortID	SourcePortID	SourcePortID	SourcePortID
2	SeqID	SeqID	SeqID	SeqID	SeqID	SeqID	SeqID
1	ControlField	ControlField	ControlField	ControlField	ControlField	ControlField	ControlField
1	LogMsgInterval	LogMsgInterval	LogMsgInterval	LogMsgInterval	LogMsgInterval	LogMsgInterval	LogMsgInterval
10	OriginTS - 1588 Rsvd – 802.1AS	Precise OriginTS	OriginTS	ReceiveTS	OriginTS -1588 Rsvd –802.1AS	ReqReceiptTS	Response OriginTS
10		TLV – 802.1AS		ReqPortID	Rsvd	ReqPortID	ReqPortID
22	-		-	-	-	-	-

Data Receive Path

To accelerate the frames in hardware the egress port needs information from the ingress port. The frame's ingress timestamp value is needed at the egress port to calculate the frame's residence time in the switch. When event frames ingress the switch, the hardware needs to embed the arrival time into the frame (in the 4 bytes RSVD location). In addition to the arrival time, in some cases, the correction field needs to be updated with the mean path delay and the delay asymmetry values at the ingress port.

Table 56: Receive Path Frame Modifications

Receive path
Fields highlighted in
Gray – Decode these fields in hardware (to determine if it is a supported PTP frame)
Blue – Modify these fields of the frame (as needed) in hardware
Green – Capture these fields of the frame in hardware to use for comparisons of associated frames

# Octet	Sync	Follow_Up	Delay_Req	Delay_Resp	Pdelay_Req	Pdelay_Resp	Pdelay_Resp_ Follow_Up
6	Dest. Addr	Dest. Addr	Dest. Addr	Dest. Addr	Dest. Addr	Dest. Addr	Dest. Addr
6	Source Addr	Source Addr	Source Addr	Source Addr	Source Addr	Source Addr	Source Addr
2	EtherType	EtherType	EtherType	EtherType	EtherType	EtherType	EtherType
	UDP Portion	UDP Portion	UDP Portion	UDP Portion	UDP Portion	UDP Portion	UDP Portion
1	TransSpec, MessageType	TransSpec, MessageType	TransSpec, MessageType	TransSpec, MessageType	TransSpec, MessageType	TransSpec, MessageType	TransSpec, MessageType
1	Rsvd, VersionPTP	Rsvd, VersionPTP	Rsvd, VersionPTP	Rsvd, VersionPTP	Rsvd, VersionPTP	Rsvd, VersionPTP	Rsvd, VersionPTP
2	MessageLength	MessageLength	MessageLength	MessageLength	MessageLength	MessageLength	MessageLength
1	DomainNumber	DomainNumber	DomainNumber	DomainNumber	DomainNumber	DomainNumber	DomainNumber
1	Rsvd	Rsvd	Rsvd	Rsvd	Rsvd	Rsvd	Rsvd
2	FlagField	FlagField	FlagField	FlagField	FlagField	FlagField	FlagField
8	CorrectionField	CorrectionField	CorrectionField	CorrectionField	CorrectionField	CorrectionField	CorrectionField
4	Rsvd	Rsvd	Rsvd	Rsvd	Rsvd	Rsvd	Rsvd
10	SourcePortID	SourcePortID	SourcePortID	SourcePortID	SourcePortID	SourcePortID	SourcePortID
2	SeqID	SeqID	SeqID	SeqID	SeqID	SeqID	SeqID
1	ControlField	ControlField	ControlField	ControlField	ControlField	ControlField	ControlField
1	LogMsgInterval	LogMsgInterval	LogMsgInterval	LogMsgInterval	LogMsgInterval	LogMsgInterval	LogMsgInterval
10	OriginTS - 1588 Rsvd – 802.1AS	Precise OriginTS	OriginTS	ReceiveTS	OriginTS -1588 Rsvd –802.1AS	ReqReceiptTS	Response OriginTS
10		TLV – 802.1AS		ReqPortID	Rsvd	ReqPortID	ReqPortID
22	-		-	-	-	-	-

Data Transmit Path

Hardware acceleration involves modifying some of the frame's fields at the egress port. The required information is extracted from the frames and used to update them before transmitting the frames. The ingress time value should be extracted from the reserved location and the field zeroed out. Correction field will be updated on top of any modifications made at the ingress port.

Table 57: Transmit Path Frame Modification

Transmit path
Fields highlighted in
Gray – Decode these fields in hardware (to determine if it is a supported PTP frame)
Blue – Modify these fields of the frame (as needed) in hardware
Green – Capture these fields of the frame in hardware to use for comparisons of associated frames
Red – Hardware can't modify these fields. The modification has to be done by software (after checking for associated frames)

# Octet	Sync	Follow_Up	Delay_Req	Delay_Resp	Pdelay_Req	Pdelay_Resp	Pdelay_Resp_ Follow_Up
6	Dest. Addr	Dest. Addr	Dest. Addr	Dest. Addr	Dest. Addr	Dest. Addr	Dest. Addr
6	Source Addr	Source Addr	Source Addr	Source Addr	Source Addr	Source Addr	Source Addr
2	EtherType	EtherType	EtherType	EtherType	EtherType	EtherType	EtherType
	UDP Portion	UDP Portion	UDP Portion	UDP Portion	UDP Portion	UDP Portion	UDP Portion
1	TransSpec, MessageType	TransSpec, MessageType	TransSpec, MessageType	TransSpec, MessageType	TransSpec, MessageType	TransSpec, MessageType	TransSpec, MessageType
1	Rsvd, VersionPTP	Rsvd, VersionPTP	Rsvd, VersionPTP	Rsvd, VersionPTP	Rsvd, VersionPTP	Rsvd, VersionPTP	Rsvd, VersionPTP
2	MessageLength	Message Length	MessageLength	MessageLength	MessageLength	MessageLength	MessageLength
1	DomainNumber	DomainNumber	DomainNumber	DomainNumber	DomainNumber	DomainNumber	DomainNumber
1	Rsvd	Rsvd	Rsvd	Rsvd	Rsvd	Rsvd	Rsvd
2	FlagField	FlagField	FlagField	FlagField	FlagField	FlagField	FlagField
8	CorrectionField	CorrectionField	CorrectionField	CorrectionField	CorrectionField	CorrectionField	CorrectionField
4	Rsvd	Rsvd	Rsvd	Rsvd	Rsvd	Rsvd	Rsvd
10	SourcePortID	SourcePortID	SourcePortID	SourcePortID	SourcePortID	SourcePortID	SourcePortID
2	SeqID	SeqID	SeqID	SeqID	SeqID	SeqID	SeqID
1	ControlField	ControlField	ControlField	ControlField	ControlField	ControlField	ControlField
1	LogMsgInterval	LogMsgInterval	LogMsgInterval	LogMsgInterval	LogMsgInterval	LogMsgInterval	LogMsgInterval
10	OriginTS - 1588 Rsvd – 802.1AS	Precise OriginTS	OriginTS	ReceiveTS	OriginTS -1588 Rsvd – 802.1AS	ReqReceiptTS	Response OriginTS
10		TLV – 802.1AS		ReqPortID	Rsvd	ReqPortID	ReqPortID
22	-		-	-	-	-	-

Frame Ingress Path Modification

When frame ingresses a port, decoding its header determines if it is a PTP frame. If hardware acceleration is enabled (HA bit Page 8, Register 2) then IEEE 1588/802.1AS layer 2 frames of know time domains (PTP Global - Page 15, Register 2) and version number (PTP Global - Page 14, Register 7) are accelerated in hardware. The event frame's ingress time (IntPTPTime Register PTP Global - Page 14, Register 7) is placed in the 4 bytes of reserved space in the frame (bytes 17-20). The correction field of the frames is updated with MeanPathDelay and IngressDelayAsymmetry values when ever applicable based on the mode of operation as described on "Equations defining frame field values" on page 103.

When a frame contains unknown domain number, unknown message type or unsupported version number, it can't be accelerated by hardware. In such cases, the event frame's arrival time (domain specific time or the hardware timer value) is embedded into the frame itself based on ArrTSMode register value (PTP Port- Page 8, Register 2) so that CPU has the ingress time information. If the ArrTSMode value is zero, the frame's arrival time is placed in the status registers (PTP Port - Page 8 Register 8 to Register 15, Page 9 Register 0).

A latency of about 8 byte times is introduced in the data path to achieve the hardware acceleration. Switching between the bypass mode (no acceleration) and hardware acceleration mode should be made only when the port is idle to avoid corrupting the frames.Frame Transmit Path Modification

When a frame egresses a port, decoding its header determines if it is a PTP frame. If hardware acceleration is enabled (HA bit Page 8, Register 2) then IEEE 1588/802.1AS layer 2 frames of know time domains (Page 15, Register 2) and version number (Page 14, Register 7, index 0x0) are accelerated in hardware. These frames could be coming from an ingress port (hardware accelerated at ingress) or from the CPU port (CPU needs to place the needed info into the frames).

The event frame's ingress time (IntPTPTime Register Page 14, Register 7) is extracted from the 4 bytes of reserved space in the frame (bytes 17-20) for use in further calculations and the reserved bytes zeroed out (unless KeepRxData bit of action vectors is set). The OriginTS field and the Correction field of the frames are updated when ever applicable based on the mode of operation as described on "Equations defining frame field values" on page 103.

A latency of about 12 byte times is introduced in the data path to achieve the hardware acceleration. Switching between the bypass mode (no hardware acceleration) and hardware acceleration mode should be made only when the port is idle to avoid corrupting the frames.

Sync – Follow_Up Frames

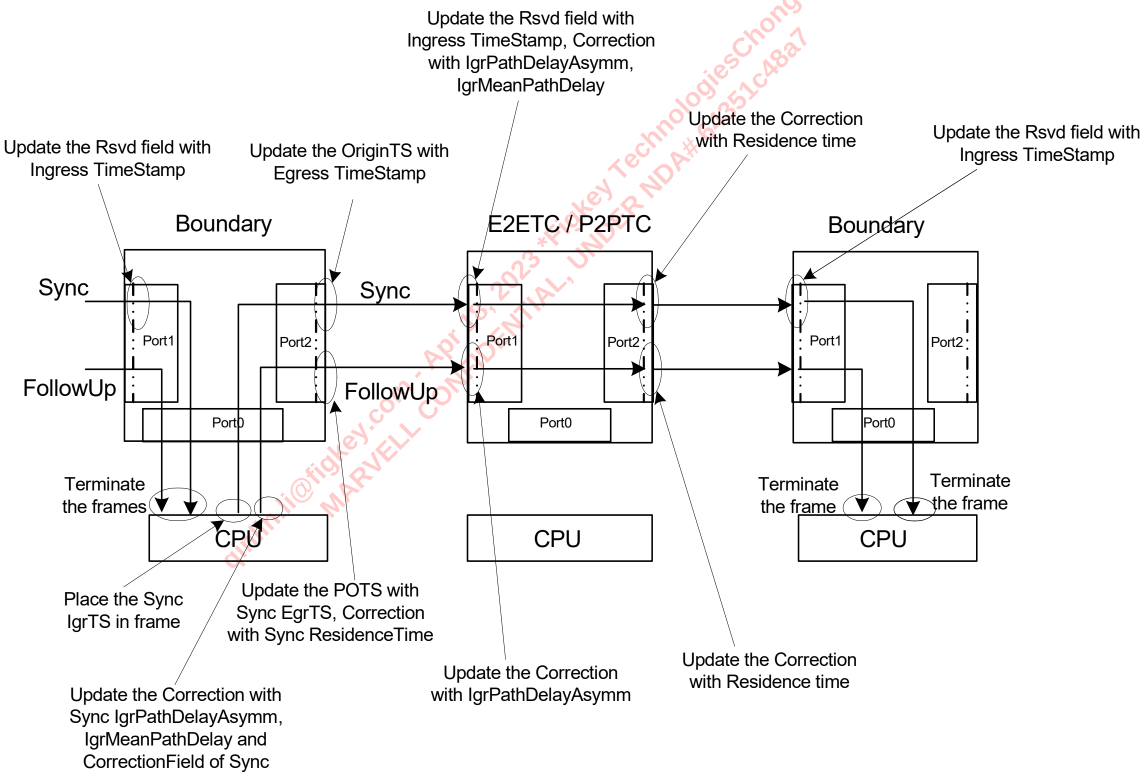
Sync and FollowUp frames travel in the same direction - from master to the slave. These frames contain master's timing information. A FollowUp frame and Sync frame are said to be associated frames if their SourcePortID and the SeqID match. A FollowUp frame should be modified only after checking its SourcePortID and SeqID against the previously received Sync frame to ensure that the associated frames are modified correctly. The fields to be updated in the FollowUp frame are ahead of the SourcePortID and the SeqID fields. The 'compare and then modify' method results in a long latency. Thus, 'modify and then compare' method is used to avoid the latency, FollowUp frames are modified on the fly even before checking their SourcePortID and SeqID. After such modifications, when the SourcePortID and SeqID fields are reached in the frame, they are compared with that of the previous Sync frame. If they do not match then the FCS value of the FollowUp frame is purposefully corrupted before transmitting the frame.

At ingress, the hardware inserts the receive time into the Sync frames. The CorrectionField of the Sync and FollowUp frames is updated with MeanPathDelay and IngressDelayAsymmetry values when ever applicable based on the mode of operation.

At egress port, the Sync frame's ingress time is extracted from the reserved bytes (bytes 17-20 of PTP header) and the field is zeroed out before being transmitted (unless KeepRxData bit of action vectors is set). The Sync frame's OriginTS and CorrectionField are modified based on the equations matrix. The residence time of Sync frame is saved to be used in updating the CorrectionField of FollowUp frame when applicable. The FollowUp frame's preciseOriginTS and CorrectionField are modified on the egress path based on the equations matrix.

Figure 33 lists all the possible modifications needed on Sync and FollowUp frames. However, it should be noted that the changes described in the figure do not all occur at the same time. Based on the mode the device is in i.e. one-step, two-step, IEEE1588 etc only some of the changes will be made at a time.

Figure 33: Sync-FollowUp Frame's Hardware Acceleration Path



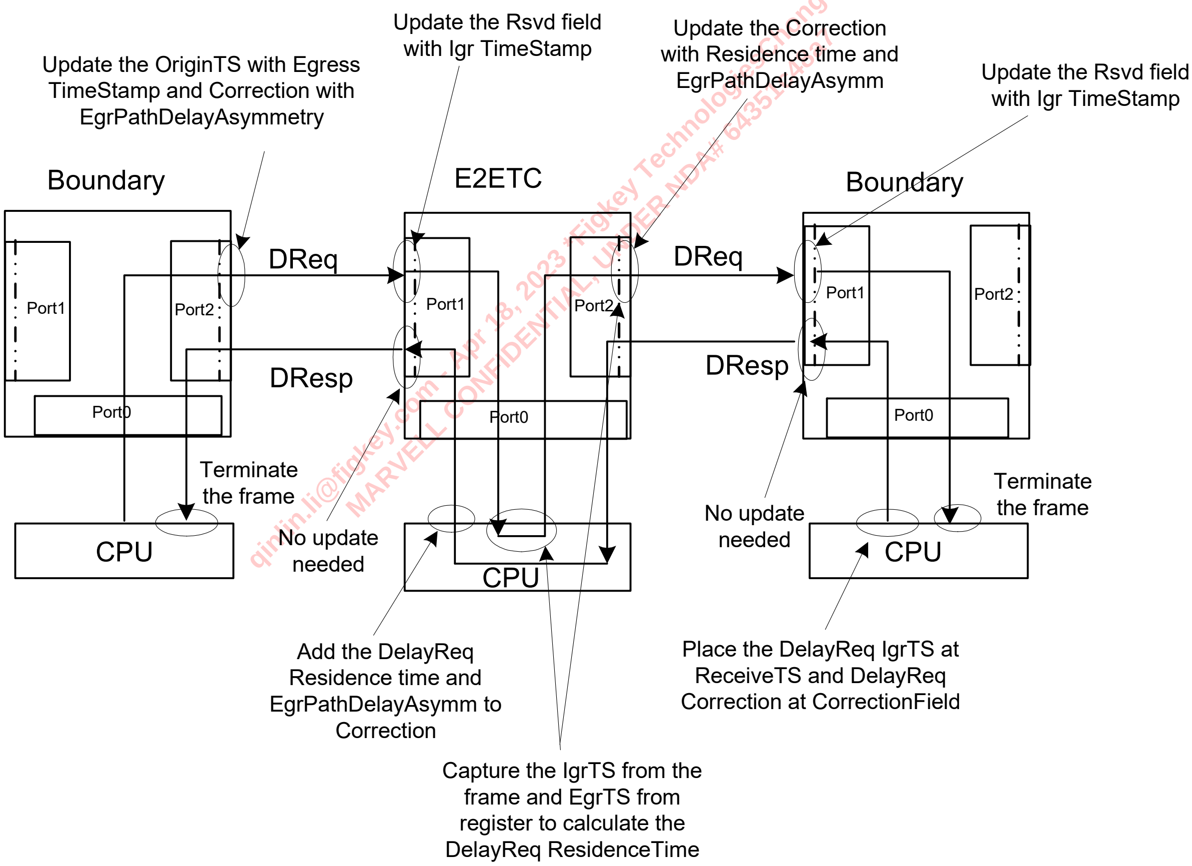
Delay_Request – Delay_Response Frames

DelayReq and DelayResp frames travel in opposite directions. DelayReq frame is sent by the slave and DelayResp frame is sent by the master port. At ingress, the hardware inserts the receive time into the DelayReq frames before sending them to the CPU. At egress port, the DelayReq frame's ingress time is extracted from the reserved bytes (bytes 17-20 of PTP header) and the field is zeroed out before being transmitted (unless KeepRxData bit of action vectors is set). The DelayReq frame's OriginTS and CorrectionField are modified based on the equations matrix. DelayReq frame's egress time is placed into the status registers so that the CPU can extract this information and use it to update the DelayResp frame.

A DelayReq frame and DelayResp frame are said to be associated frames if their SourcePortID and the SeqID match. A DelayResp frame should be modified with information from previously received DelayReq only after matching its SourcePortID and SeqID to ensure that they are associated frames. However, the hardware at egress port of DelayResp frame doesn't have the SourcePortId or SeqID values of the received DelayReq frame (opposite direction). Thus CPU needs to check SourcePortId and SeqId of the frames and add in the correct DelayReq frame's information (residence time, CorrectionField value and EgrPathDelayAsymm) to the DelayResp frames before transmission. The hardware doesn't modify/accelerate the DelayResp frames at egress.

Figure 34 lists all the possible modifications needed on DelayReq and DelayResp frames. However, it should be noted that the changes described in the figure do not all occur at the same time. Based on the mode the device is in i.e. one-step, two-step, IEEE1588 etc only some of the changes will be made at a time.

Figure 34: DelayReq-DelayResp Frame's Hardware Acceleration Path



PDelayRequest, PDelayResponse and PDelayResponseFollowUp frames

PDelayReq frame is sent by the initiator to the responder. At ingress, the hardware inserts the receive time into the PDelayReq frames before sending them to the CPU. At egress port, the PDelayReq frame's ingress time is extracted from the reserved bytes (bytes 17-20 of PTP header) and the field is zeroed out before being transmitted (unless KeepRxData bit of action vectors is set). The PDelayReq frame's OriginTS and CorrectionField are modified based on the equations matrix. PDelayReq frame's egress time is placed into the status registers so that the CPU can extract this information and use it to update the PDelayResp/ PDelayRespFollowUp frame.

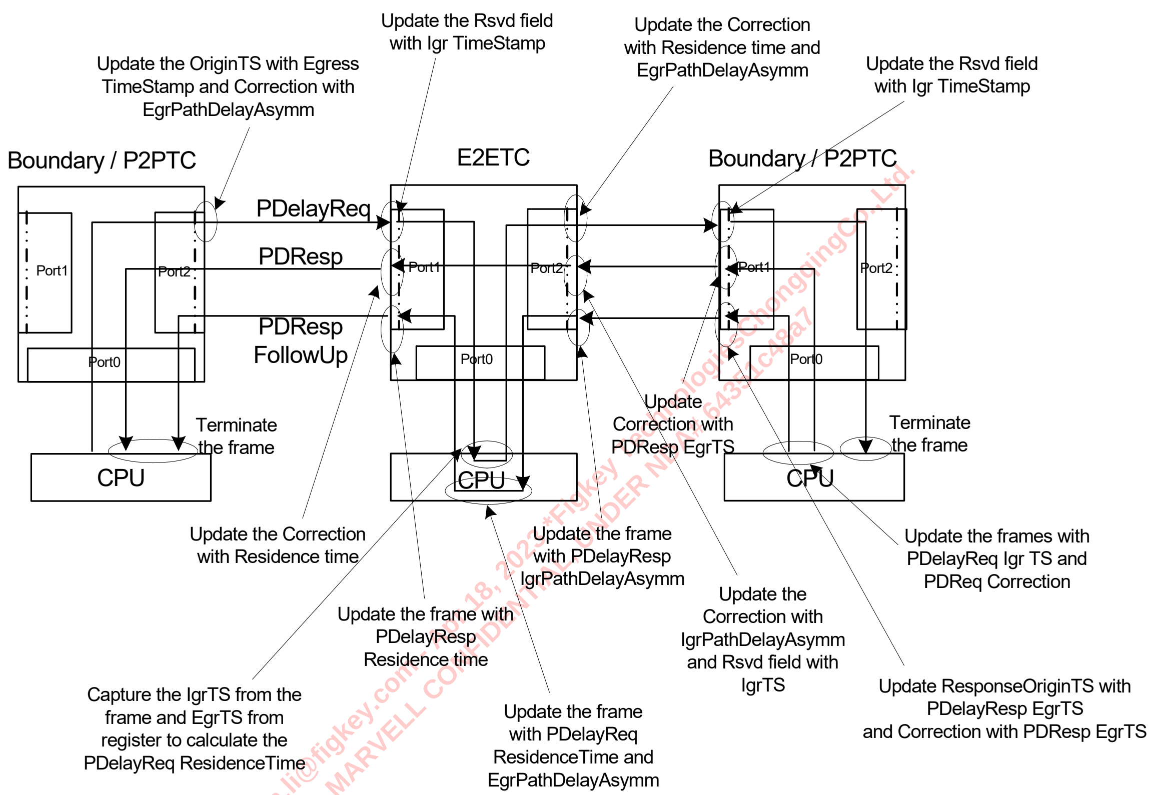
The PDelayResp frame is sent by the responder to the initiator i.e. in opposite direction than the PDelayReq frame. At ingress, the hardware inserts the receive time into the PDelayResp frames. A PDelayReq frame and PDelayResp frame are associated frames if the SourcePortID and SeqID of PDelayReq frame matches with the ReqPortID and the SeqID of PDelayResp frame. A PDelayResp frame should be modified only after checking its ReqPortID and SeqID against the SourcePortId and SeqID of previously received PDelayReq frame to ensure that the associated frames are modified correctly. However, the hardware at egress port of PDelayResp frame doesn't have the SourcePortId or SeqID values of the received PDelayReq frame (opposite direction). Thus CPU needs to match the frames and add in the correct PDelayReq frame's information to the PDelayResp frames before transmission. The CorrectionField value of PDelayReq needs to be added to the CorrectionField of PDelayResp fames when applicable. CPU should place the

PDelayReq frame's IgrTS in the 4 reserved bytes (bytes 17-20 of PTP header) of PDelayResp frames when applicable. At egress, the hardware extracts this information and updates the PDelayResp frame's ReqReceiptTS and CorrectionField based on the equation's matrix.

PDelayResp and PDelayRespFollowUp frames travel in the same direction, from the responder to the initiator. A PDelayRespFollowUp frame and PDelayResp frame are associated with each other if their ReqPortID and the SeqID match. A PDelayRespFollowUp frame should be modified only after checking its ReqPortID and SeqID against the previously received PDelayResp frame to ensure that the associated frames are modified correctly. The fields to be updated in the PDRespFollowUp frame are ahead of the ReqPortID and the SeqID fields. The 'compare and then modify' method results in a long latency. Thus, 'modify and then compare' method is used to avoid the latency, PDelayRespFollowUp frames are modified on the fly even before checking their ReqPortID and SeqID. After such modifications, when the ReqPortID and SeqID fields are reached in the frame, they are compared with that of the previous PDelayResp frame. If they do not match then the FCS value of the PDelayRespFollowUp frame is purposefully corrupted before transmitting the frame. Similar to the PDelayResp frame's case, hardware depends on CPU to make PDelayReq related timing information changes to PDelayRespFollowUp whenever applicable.

Figure 35 lists all the possible modifications needed on PDelayReq, PDelayResp and PDelayRespFollowUp frames. However, it should be noted that the changes described in the figure do not all occur at the same time. Based on the mode the device is in i.e. one-step, two-step, IEEE1588 etc only some of the changes will be made at a time.

Figure 35: PDelayReq, PDelayResp, PDelayRespFollowUp Frame's Hardware Acceleration Path



Equations defining frame field values

The tables below describe the equations that govern the frame field values based on various configurations supported.

Table 58: Sync Frame Equations Implemented

1. The boxes filled in red depict the special cases where the device works differently than described in IEEE spec. The text describes the device's behavior in those cases.
2. The text highlighted in light grey is the ingress path function that is not performed in hardware. The frame is sent to CPU and it should take care of this portion.
3. The text highlighted in green is the information that CPU is required to supply the hardware in the egress path. That portion needs to be placed in the frame by the software.
4. Rest of the portion is done in hardware (combination of the ingress path and egress path)

			Ordinary / Boundary	Peer-To-Peer Transparent Clock	End-To-End Transparent Clock
Sync	**One Step**	**1588**	**OriginTS** = SyncEgrTS **CorrectionField** = Rx's CorrectionField (0) **CorrectionField(Rx)** = CorrField + IgrPathDelayAsym	**OriginTS**=Rx's OriginTS **Correction**= Rx's CorrectionField + IgrMeanPathDelay + ResidenceTime + IgrPathDelayAsym	**OriginTS**=Rx's OriginTS **Correction**= Rx's CorrectionField + ResidenceTime + IgrPathDelayAsym
	Two Step w/ Rx TwoStepFlag = False	**1588**	NA	**OriginTS**=Rx's OriginTS **Correction** = Rx's CorrectionField *i.e frame not modified*	**OriginTS**=Rx's OriginTS **Correction** = Rx's CorrectionField *i.e frame not modified*
	Two Step w/ Rx TwoStepFlag = True	**1588**	**OriginTS**= Rx's OriginTS (0 or estimate SyncEgrTS) OR SyncEgrTS (Program action reg bit) **CorrectionField**= Rx's CorrectionField (0) **CorrectionField(Rx)** = CorrField + IgrPathDelayAsym	**OriginTS**=Rx's OriginTS **Correction** = Rx's CorrectionField *i.e frame not modified*	**OriginTS**=Rx's OriginTS **Correction** = Rx's CorrectionField *i.e frame not modified*
		802.1 AS	**OriginTS** - NA **CorrectionField**= Rx's CorrectionField (0) *i.e frame not modified*	NA	NA

Table 59: FollowUp Frame Equations Implemented

			Ordinary / Boundary	Peer-To-Peer Transparent Clock	End-To-End Transparent Clock
FolloWUp	**One Step**	**1588**	NA	**PreciseOriginTS**=Rx's PreciseOriginTS **Correction** = Rx's CorrectionField *i.e frame not modified*	**PreciseOriginTS**= Rx's PreciseOriginTS **Correction** = Rx's CorrectionField *i.e frame not modified*
	Two Step w/ Rx TwoStepFlag = False	**1588**	NA	**PreciseOriginTS**=Rx's PreciseOriginTS **Correction** = Rx's CorrectionField *i.e frame not modified*	**PreciseOriginTS**= Rx's PreciseOriginTS **Correction** = Rx's CorrectionField *i.e frame not modified*
	Two Step w/ Rx TwoStepFlag = True	**1588**	**PreciseOriginTS**= SyncEgrTS **Correction** = Rx's CorrectionField (0 – CorrectionField Sync)	**PreciseOriginTS**= Rx's PreciseOriginTS **Correction** = Rx's CorrectionField + ResidenceTime + IgrMeanPathDelay + IgrPathDelayAsym	**PreciseOriginTS**= Rx's PreciseOriginTS **Correction** = Rx's CorrectionField + ResidenceTime + IgrPathDelayAsym
		802.1 AS	**PreciseOriginTS**= (a) If GrandMaster: SyncEgrTS (b) If not the GrandMaster: Rx's PreciseOriginTS **Correction** = (a) If GrandMaster: Rx's CorrectionField (0) (b) If not the GrandMaster: Rx's CorrectionField + ResidenceTime + PropagationDelay (MeanPathDelay) + IgrPathDelayAsym	NA	NA

Table 60: DelayReq-Resp Frame's Equations Implemented

			Ordinary / Boundary	Peer-To-Peer Transparent Clock	End-To-End Transparent Clock
Delay Req	**One Step**	**1588**	**OriginTS** = Rx's OriginTS (0 /estimate DelayReqEgrTS) OR DelayRespEgrTS (Program action reg) **Correction** = Rx's CorrectionField (0) - EgrPathDelayAsym	Discard	**OriginTS** =Rx's OriginTS **Correction** = Rx's CorrectionField + ResidenceTime - EgrPathDelayAsym
	Two Step w/ Rx TwoStepFlag = False	**1588**	**OriginTS** = Rx's OriginTS (0 /estimate DelayReqEgrTS) OR DelayRespEgrTS (Program action reg) **Correction** = Rx's CorrectionField (0) - EgrPathDelayAsym	Discard	**OriginTS** = Rx's OriginTS **Correction** = Rx's CorrectionField *i.e frame not modified*
	Two Step w/ Rx TwoStepFlag = True	**1588**	**OriginTS** = Rx's OriginTS (0 /estimate DelayReqEgrTS) OR DelayRespEgrTS (Program action reg) **Correction** = Rx's CorrectionField (0) - EgrPathDelayAsym	Discard	**OriginTS** = Rx's OriginTS **Correction** = Rx's CorrectionField *i.e frame not modified*
		802.1 AS	N/A	N/A	N/A
Delay Resp	**One Step**	**1588**	**ReceiveTS** = DelayReqIgrTS **Correction** = 0 + CorrectionField DelayReq *i.e frame not modified*	Discard	**ReceiveTS** = Rx's ReceiveTS **Correction** = Rx's CorrectionField *i.e frame not modified*
	Two Step w/ Rx TwoStepFlag =False	**1588**	**ReceiveTS** = DelayReqIgrTS **Correction** = 0 + CorrectionField DelayReq *i.e frame not modified*	Discard	**ReceiveTS** = Rx's ReceiveTS **Correction** = Rx's CorrectionField + ResidenceTime DelayReq – EgrPathDelayAsym of DelayReq *i.e frame not modified*
	Two Step w/ Rx TwoStepFlag = True	**1588**	**ReceiveTS** = DelayReqIgrTS **Correction** = 0 + CorrectionField DelayReq *i.e frame not modified*	Discard	**ReceiveTS** = Rx's ReceiveTS **Correction** = Rx's CorrectionField + ResidenceTime DelayReq – EgrPathDelayAsym of DelayReq *i.e frame not modified*
		802.1 AS	N/A	N/A	N/A

Table 61: PDelayReq Frame Equations Implemented

			Ordinary / Boundary	Peer-To-Peer Transparent Clock	End-To-End Transparent Clock
Pdelay Req	**One Step**	**1588**	**OriginTS** = Rx's OriginTS (0/ estimate PdelayReqEgrTS) OR PdelayReqEgrTS (Program action reg) **Correction** = Rx's CorrectionField (0) - EgrPathDelayAsym	**OriginTS** = Rx's OriginTS (0/ estimate PdelayReqEgrTS) OR PdelayReqEgrTS (Program action reg) **Correction** = Rx's CorrectionField (0) - EgrPathDelayAsym	**OriginTS** = Rx's OriginTS **Correction** = Rx's CorrectionField + Residence Time PdelayReq – EgrPathDelayAsym
	Two Step w/ Rx TwoStepFlag =False	**1588**	**OriginTS** = Rx's OriginTS (0/ estimate PdelayReqEgrTS) OR PdelayReqEgrTS (Program action reg) **Correction** = Rx's CorrectionField (0) - EgrPathDelayAsym	**OriginTS** = Rx's OriginTS (0/ estimate PdelayReqEgrTS) OR PdelayReqEgrTS (Program action reg) **Correction** = Rx's CorrectionField (0) - EgrPathDelayAsym	**OriginTS** = Rx's OriginTS **Correction** = Rx's CorrectionField *i.e frame not modified*
	Two Step w/ Rx TwoStepFlag = True	**1588**	**OriginTS** = Rx's OriginTS (0/ estimate PdelayReqEgrTS) OR PdelayReqEgrTS (Program action reg) **Correction** = Rx's CorrectionField (0) - EgrPathDelayAsym	**OriginTS** = Rx's OriginTS (0/ estimate PdelayReqEgrTS) OR PdelayReqEgrTS (Program action reg) **Correction** = Rx's CorrectionField (0) - EgrPathDelayAsym	**OriginTS** = Rx's OriginTS **Correction** = Rx's CorrectionField *i.e frame not modified*
		802.1AS	**OriginTS** = NA **Correction** = Rx's CorrectionField (0) *i.e frame not modified*	NA	NA

Table 62: PDelayResp frame equations implemented

			Ordinary / Boundary	Peer-To-Peer Transparent Clock	End-To-End Transparent Clock
Pdelay Resp	**One Step**	**1588**	**ReqReceiptTS** = Rx's ReqReceiptTS (0) **Correction** = CorrectionField of PdelayReq + (PDRespEgrTS – PDReqIgrTS) **Rsvd** = PDReqIgrTS **CorrectionField(Rx)** = CorrField + IgrPathDelayAsym	**ReqReceiptTS** = Rx's ReqReceiptTS (0) **Correction** = CorrectionField of PdelayReq + (PDRespEgrTS – PDReqIgrTS) **Rsvd** = PDReqIgrTS **CorrectionField(Rx)** = CorrField + IgrPathDelayAsym	**ReqReceiptTS** = Rx's ReqReceiptTS **Correction** = Rx's CorrectionField + ResidenceTime Pdelay_Resp + IgrPathDelayAsym
	Two Step w/ Rx TwoStepFlag = False	**1588**	NA	NA	**ReqReceiptTS** = Rx's ReqReceiptTS **Correction** = Rx's CorrectionField *i.e frame not modified*
	Two Step w/ Rx TwoStepFlag = True	**1588**	**ReqReceiptTS** = (a) Rx's ReqReceiptTS (0) OR (b) ReqReceiptTS (PDelayReqIgrTS) **Correction** = (a) Rx's CorrectionField (0) OR (b) Rx's CorrectionField (0) **CorrectionField(Rx)** = CorrField + IgrPathDelayAsym	**ReqReceiptTS** = (a) Rx's ReqReceiptTS (0) OR (b) ReqReceiptTS (PDelayReqIgrTS) **Correction** = (a) Rx's CorrectionField (0) OR (b) Rx's CorrectionField (0) **CorrectionField(Rx)** = CorrField + IgrPathDelayAsym	**ReqReceiptTS** = Rx's ReqReceiptTS **Correction** = Rx's CorrectionField *i.e frame not modified*
		802.1 AS	**ReqReceiptTS** = Rx's ReqReceiptTS (PDelayReqIgrTS) **Correction** = Rx's CorrectionField (0) *i.e frame not modified*	NA	NA

Table 63: PDelayRespFollowUp frame equations implemented

			Ordinary / Boundary	Peer-To-Peer Transparent Clock	End-To-End Transparent Clock
PdelayRespFolloWUp	**One Step**	**1588**	NA	NA	**ResponseOriginTS** = Rx's ResponseOriginTS **Correction** = Rx's CorrectionField *i.e frame not modified*
	Two Step w/ Rx TwoStepFlag = False	**1588**	NA	NA	**ResponseOriginTS** = Rx's ResponseOriginTS **Correction** = Rx's CorrectionField *i.e frame not modified*
	Two Step w/ Rx TwoStepFlag = True	**1588**	**ResponseOriginTS** = (a) Rx's ResponseOriginTS (0) OR (b) PdelayRespEgrTS **Correction** = Correction Field PdelayReq + (a) PdelayRespEgrTS-PdelayReqIgrTS OR (b) 0 **Rsvd** = (a) PdelayReqIgrTS	**ResponseOriginTS** = (a) Rx's ResponseOriginTS (0) OR (b) PdelayRespEgrTS **Correction** = Correction Field PdelayReq + (a) PdelayRespEgrTS-PdelayReqIgrTS OR (b) 0 **Rsvd** = (a) PdelayReqIgrTS	**ResponseOriginTS** = Rx's ResponseOriginTS **Correction** = Rx's Correction field + Residence Time of PdelayReq + Residence Time of PdelayResp + IgrPathDelayAsym of PDelayResp – EgrPathDelayAsym of PDelayReq
		802.1 AS	**ResponseOriginTS** = PdelayRespEgrTS **Correction** = Rx's Correction field (0)	NA	NA

The PdelayResp and PdelayRespFollowUp frames follow either scheme (a) or (b) based on the register selection.

2.29.3 Time Application Interface (TAI)

The Precise Timing Protocol provides both frequency and time-of-day with respect to the PTP Grand Master for the entire PTP network. In a given endpoint device (media talker or a media listener), once the PTP Grand Master aware clock and time are available, it needs to be transported over to the rest of the subsystem without loss of accuracy. For example, if a Digital Video Recorder is the end device, the network clock and time needs to be transported to the Video SoC and/or Storage SoC and the host processor seamlessly. This ensures that when the media is played out of the DVR, the network time aware presentation time of the content is required to be carried through.

The TAI “Timing Interface Block” supports features required for the above purpose. This block utilizes two signals to offer various services. One signal is called EventRequest input signal and the second is called TriggerGenerate output signal.

Using the above signals there are several functions that this block supports:

1. An event pulse capture function.
2. Multiple event counter function.
3. A trigger pulse generate function with pulse width control.
4. A trigger clock generate function with digital clock compensation.
5. A multi-ptp device time sync function.

2.29.3.1 Event pulse capture interface

In many IEEE 1588 applications like industrial automation etc. it is important to precisely capture the time at which a particular event has happened. The event is defined by a low to high or high to low transition on an external signal called EventRequest. The event time is captured in EventCapRegister (Page 12, Register 10 and 11). This field is validated by EventCapValid bit (TAI Global Configuration Register 9 - Page 12, Register 9).

The captured event time register needs to be read out by software and valid bit cleared before the hardware captures another event. If there were to be two back to back events before the software read the results of the first event an error indication is set in EventCapErr (TAI Global Configuration Register 9 - Page 12, Register 9). If the user chooses that the hardware rather overwrite the Event capture register then it can be configured by setting a 0x1 to EventCapOv (TAI Global Configuration Register 0 - Page 12, Register 0).

Once an event has been captured the software can optionally (EventCapIntEn, Page 12, Register 0) be interrupted and an EventInt (Page 12, Register 9) bit is also set.

The maximum jitter associated with capturing the EventRequest signal pulse is one TSClkPer (TAI Global Configuration, Register 1 - Page 12, Register 1). The minimum pulse width of the EventRequest signal needs to be 1.5 times the TSClkPer (TAI Global Configuration, Register 1 - Page 12, Register 1). In order for the hardware logic to detect distinct events on the EventRequest signal, the minimum gap between two events needs to be 150 ns plus 5 times TSClkPer amount.

2.29.3.2 Multiple Event Counter Function

Similar to the Event Pulse capture interface described above, if multiple events need to be captured for an application to detect how many times a particular event is happening on the EventRequest input signal, EventCtrStart (TAI Global Configuration Register - Page 12, Register 0) needs to be set to a 0x1 and EventCapOv (TAI Global Configuration Register - Page 12, Register 0) needs to set to a 0x1.

The Multiple Event Counter function is capable of capturing up to 255 events in EventCapCtr (TAI Global Configuration Register 9 - Page 12, Register 9).

The maximum jitter associated with capturing the EventRequest signal pulse is one TSClkPer (TAI Global Configuration Register 1 - Page 12, Register 1). The minimum pulse width of the EventRequest signal needs to be 1.5 times the TSClkPer (TAI Global Configuration Register 1 -

Page 12, Register 1). In order for the hardware logic to detect distinct events on the EventRequest signal, the minimum gap between two events needs to be 150 ns plus 5 times TSClkPer amount.

Note

In the multiple event counter mode, the EventCapRegister (Page 12, Registers 10 and 11) indicate the time stamp value for the last captured event register.

2.29.3.3 Trigger Pulse Generate Function

In many PTP applications, the time of day computed in PTP needs to be distributed in some form to the rest of the node. One commonly used method is to generate a pulse whenever the PTP Global Time matches certain configured value. The pulse gets output on an TrigGenResp output signal.

The above function can be achieved by:

a) Configuring the TrigMode (TAI Global Configuration Register 0 - Page 12, Register 0) to 0x1 and

b) Configuring the time amount when the pulse needs to be generated in TrigGenAmt (TAI Global Configuration Register 2 - Page 12, Register 2 and 3) and

c) Configuring the TrigGenReq (TAI Global Configuration Register 0 - Page 12, Register 0) to a 0x1

The PTP Global Timer gets compared to the TrigGenAmt and upon a match a pulse signal gets generated on the TrigGenResp output signal.

Optionally after generating the TrigGenResp pulse the CPU can be notified by setting the TrigGenIntEn (TAI Global Configuration Register 0, Page 12, Register 0) and along with the pulse output, TrigGenInt (TAI Global Configuration Register 8, Page 12, Register 8) bit gets set. Upon receiving the interrupt, it is the CPUs job to clear the interrupt bit.

The pulse width of the output signal can be controlled by PulseWidth (TAI Global Configuration Register 5 - Page 12, Register 5). Do not set the PulseWidth to a zero value.

2.29.3.4 Trigger Clock Generate Function

Similar to the trigger pulse generation function described above, the same set of registers can be used to generate a periodic clock. The value specified in TrigGenAmt (TAI Global Configuration Register 2 - Page 12, Register 2 and 3) is used to generate the base period of the clock output. For this functional mode the TrigMode (TAI Global Configuration Register 0 - Page 12, Register 0) needs to be set a 0x0 and TrigGenReq (TAI Global Configuration Register 0 - Page 12, Register 0) needs to be set to a 0x1.

The output clock can be compensated by configuring the field TrigClkComp (TAI Global Configuration Register 4 - Page 12, Register 4) and TrigClkCompSubps (TAI Global Configuration Register 5 - Page 12, Register 5). This field specifies the remainder amount for the clock that is being generated with the period specified by the TrigGenAmt. The TrigClkComp amount gets constantly accumulated and when this accumulated amount exceeds the value specified in TSClkPer, a TSClkPer gets added to the output clock momentarily to compensate for the remainder accumulated over time.

A start time for the clock may be chosen (TAI Global Register, Page 13, Register 0 and Register 1). Also a compensation direction may be chosen by indicating the amount to add or subtract in Page 12, Register 4, bit 5 (TrigComp Dir)

Note

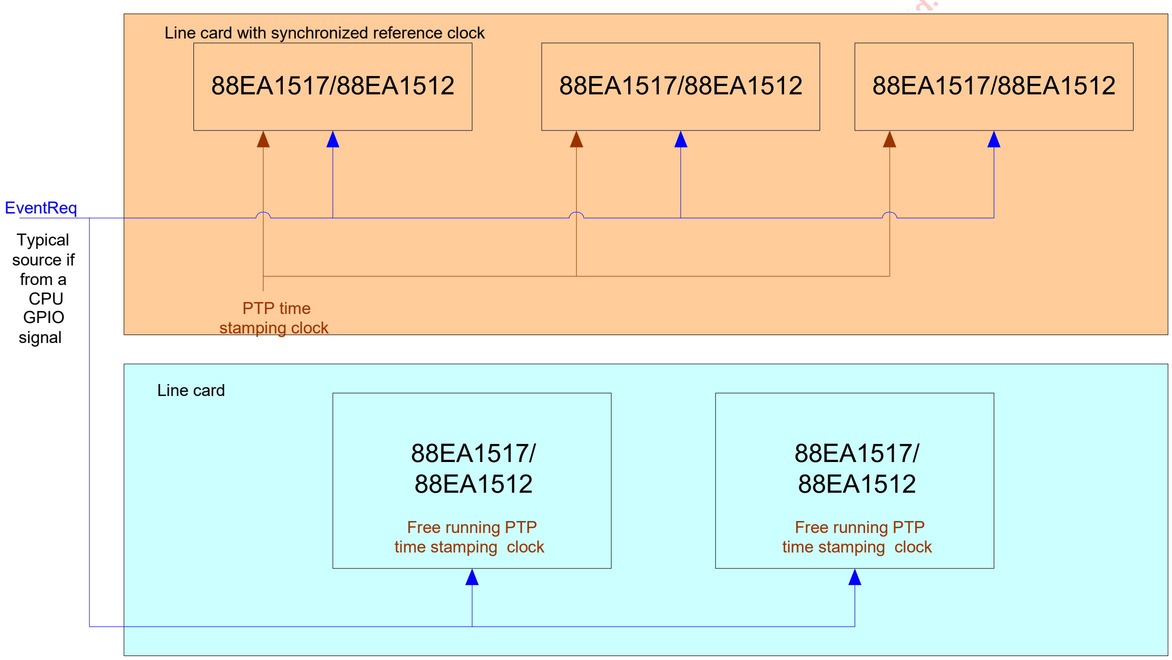
The TrigGenAmt should be set to no less than 2 times the TSCLKPer amount.

2.29.3.5 Multi-PTP device time sync Function

When PTP is enabled on devices on enterprise, service provider line cards and/or chassis, it is important for various PTP capable devices on the system to have the common notion of PTP Global Time. Typically a central processing card is present in these chassis which runs the PTP software for the entire chassis and all the data line cards send the PTP frames and the time stamp information to the central PTP software entity. Given that there can be so many ports on each line card and each of the line cards may be plugged into the chassis at different times, the PTP global Time counter value in each of the PTP devices on the line cards may not be synchronized. Thus the hardware needs to provide a sure shot way for all the PTP capable devices across various line cards regardless of when the line cards have been powered on, to be synchronized to the same PTP Global Time so that the PTP protocol software doesn't have to remember the offsets with respect to the PTP Grand Master and also with respect to each of these devices.

The background for this feature is that the PTP nodes derive the time of day via PTP message exchange. The derived time of day consists of 64 bits of seconds and 32 bits of nanoseconds. The PTP slave nodes are expected to derive the frequency and phase offset information with respect to the PTP Grand Master node. The derived offset information is used to periodically adjust various device PTP global timer values. For multi PTP (Marvell devices only) capable devices the goal is to synchronize all the nodes with a common 32-bit nanoseconds field and also adjust the nanoseconds field with the computed PTP Grand Master offset information.

Figure 36: Multiple devices across multiple line cards connected by an EventReq input signal



This feature is enabled by setting a 0x1 to MultiPTPSyncMode (TAI Global Configuration Register 0 - Page 12, Register 0). Note that once this bit is enabled, the functions EventRequest and TriggerGen are disabled.

When MultiPTPSyncMode is 0x1, a low to high or high to low transition on the EventRequest signal triggers transfer of TrigGenAmt to the PTP Global Timer register. At the timer of the low to high or high to low transition for further software time correlation, the EventCapTime register is also updated with the value of the PTP Global Time before it got overwritten.

Note that even though the above schema ensures that the PTP Global Time register value is synchronized, the following are the possible sources of jitter associated from a system level (which can be avoided):

a) The reference clock sources for various PTP devices that support the multi-sync EventReq interface on the same line card and/or across line cards may not be synchronized.

b) The added delays in the clock path between various PTP devices that support the multi-sync EventReq interface on the same line card and/or across line cards.

c) Inherent operating system related jitter from the point a command is issued to when it actually gets executed in hardware. This is applicable for EventReq pulse generation from a GPIO as well. This tends to be more predictable with real time operating systems.

One guaranteed method to avoid the above mentioned jitter factors a and b is by using a flip-flop with the EventReq as the data input and PTP time stamping clock as the clock input into the flop and the flop output gets distributed with matched delays across various PTP devices that support the multi-sync EventReq. One method to reduce the operating system associated uncertainties is to choose an operating system in which the scheduler tasks are predictable down to 1's of nanoseconds accuracy.

2.29.4 ReadPlus Command

The PTP Global Time Registers are used as 32-bit global timer value that is running off of the free running PHY clock. A Read from the PTP Global Time Registers must be done with ReadPlus command. A Read directly to the PTP Global Time Registers without using ReadPlus command will return 0. The PTP Global Time Registers value can be loaded directly by writing back-to-back to the PTP Global Time Register Byte 3 & 2 – Page 12, Register 15 first followed by PTP Global Time Register Byte 1 & 0 – Page 12, Register 14.

To read from PTP Global Time Register:

1. Write to Reg 22 = 0xE (Page 14)
2. Write to Reg 14 = 0x8E0E (ReadPlus Command from PTP Global Time Register)
3. Read from Reg 15 (PTP Global Time Registers Bit[15:0])
4. Read from Reg 15 (PTP Global Time Registers Bit[31:16])

Also the new TOD-related registers are all only ReadPlus (Page 15, Register 0 to Register 14)

To read from PTP TOD related Register:

1. Write to Reg 22 = 0xE (Page 14)
2. Write to Reg 14 = 0x8F10 (ReadPlus Command from TOD load point register)
3. Read from Reg 15 (PTP TOD load point Registers Bit[15:0])
4. Read from Reg 15 (PTP TOD load point Registers Bit[31:16])

Note

These registers need to use the ReadPlus command otherwise the readback value is not correct
Page 15, Register 0 to Register 14
Page 12, Register 14 to Register 15
Page 12, Register 2 to Register 3
Page 12, Register 10 to Register 11
Page 13, Register 0 to Register 1

2.30 Interrupt

When Register 18_3.7 is set to 1, LED[1] outputs the interrupt for 88EA1517 and the LED[2] outputs the interrupt for the 88EA1512. Register 18_3.11 selects the polarity of the interrupt signal when it is active, where 18_3.11 = 1 means it is active low and 18_3.11 = 0 means it is active high.

Registers 18_0 and 18_2 are the Interrupt Enable registers for the copper media.

Registers 19_0 and 19_2 are the Interrupt Status registers for the copper media.

Register 18_1 is the Interrupt Enable register and 19_1 is the Interrupt Status register for the fiber media.

There are force bits and polarity bits for fiber and copper media See Table 64 and Table 65.

Table 64: Copper

Register	Function
18_3.15	Force Interrupt
18_3.11	Set Polarity

Table 65: Fiber

Register	Function
26_1.15	Force Interrupt
16_1.2	Set Polarity

For Auto-Media Detect (AMD) modes, copper bits should be used. If the active media is fiber then the bits in Table 65 should be used. If the active media is copper then the bits in Table 64 should be used.

2.31 Automatic and Manual Impedance Calibration

2.31.1 MAC Interface Calibration Circuit

Auto-calibration is available for the MAC interface I/Os. The PHY runs the automatic calibration circuit with a 47.3 ohm impedance target by default after hardware reset. Other impedance targets are available by changing the impedance target and restarting the auto calibration through register writes. Individual NMOS and PMOS output transistors can be controlled.

Manual NMOS and PMOS settings are available if the automatic calibration is not desired. If the PCB traces are different from 47.3 ohms, the output impedance of the MAC interface I/O buffers can be programmed to match the trace impedance. Users can adjust the NMOS and PMOS driver output strengths to perfectly match the transmission line impedance and eliminate reflections completely.

2.31.2 MAC Interface Calibration Register Definitions

Table 66: RGMII Output Impedance Calibration Override
Page 2, Register 24

Bits	Field	Mode	HW Rst	SW Rst	Description
15	Restart Calibration	R/W, SC	0	Retain	Calibration will start once bit 15 is set to 1. 0 = Normal, 1 = Restart
14	Calibration Complete	RO	0	Retain	Calibration is done once bit 14 becomes 1. 0 = Not done, 1 = Done
13	VDDO Level	RW	See Descr.	Retain	VDDO level- must be programmed to indicate the VDDO supply voltage used when 1.8V is not used. The bit mapping is: 0 = 3.3V 1 = 2.5V If the CONFIG pin input values bit 1:0 are: 00, then VDDO Level = 3.3V 11, then VDDO Level = 3.3V 10, then VDDO Level = 2.5V 01, then VDDO Level = 2.5V Note: 3.3V is assumed initially until this value is changed.
12	1.8V VDDO Used	R/O	See Descr	Retain	This bit indicates whether VDDO = 1.8V is used or not. 1 = VDDO = 1.8V 0 = VDDO = 2.5V or 3.3V
11:8	PMOS Value	R/W	See Descr	Retain	0000 = all fingers off 1111 = all fingers on The automatic calibrated values are stored here after calibration completes. Once 24_2.6 is set to 1 the new calibration value is written into the I/O pad. The automatic calibrated value is lost.
7	Reserved	RW	0	Retain	0

Table 66: RGMII Output Impedance Calibration Override (Continued)
Page 2, Register 24

Bits	Field	Mode	HW Rst	SW Rst	Description
6	Force PMOS/NMOS	R/W	0x0	Retain	1 = force value from 24_2.11:8 to PMOS, and 24_2.3:0 to NMOS. (Used for manual settings)
5:4	Reserved	R/O	0x0	Retain	Reserved.
3:0	NMOS value	R/W	See Descr	Retain	0000 = All fingers off 1111 = All fingers on The automatic calibrated values are stored here after calibration completes. Once 24_2.6 is set to 1 the new calibration value is written into the I/O pad. The automatic calibrated value is lost.

Table 67: RGMII Output Impedance Target
Page 2, Register 25

Bits	Field	Mode	HW Rst	SW Rst	Description
2:0	Calibration target	RW	0x3	Retain	000 = 78.8 Ohm 001 = 64.5 Ohm 010 = 54.6 Ohm 011 = 47.3 Ohm 100 = 41.7 Ohm 101 = 37.3 Ohm 110 = 33.8 Ohm 111 = 30.9 Ohm

2.31.3 Changing Auto Calibration Targets

The PHY runs the automatic calibration circuit with a 47.3 ohm impedance target by default after hardware reset. Other impedance targets are available by changing the impedance target and restarting the auto calibration through register writes.

To change the auto calibration targets:

Write to Register 25_2.2:0 with the target impedance and then

Write to Register 24_2 = 0x8000 (Restarts the auto-calibration with the new target).

2.31.4 Manual Settings to The Calibration Registers

To use manual calibration, write to the following registers:

Write to Register 24_2.11:8 = b'PPPP and Register 24_2.3:0 = b'NNNN adjusts the PMOS and NMOS fingers accordingly.

Where PPPP is the 4 bit value for the PMOS strength.

Where NNNN is the 4 bit value for the NMOS strength.

PPPP or NNNN will depend on the PCB used. The '1111' value enables all the fingers for maximum drive strength and for minimum impedance. The '0000' value turns all fingers off for minimum drive strength and for maximum impedance. For assessment of the auto-calibration required on a particular PCB, the RGMII pins at the destination can be monitored and depending on the signal integrity on the PCB the auto-calibration values can be changed accordingly. For example, if the automatic calibration has a 47.3 ohm target, and the RGMII trace impedance on the board is 60 ohms, then by monitoring the RX_CLK pin at the destination, reflections can be noticed. This is shown in Figure 37. Through manual calibration the reflections can be eliminated as shown in Figure 38.

Figure 37: Signal Reflections, using the 50 ohm setting, 60 ohm line

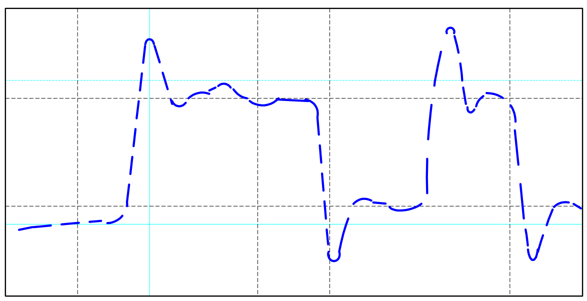
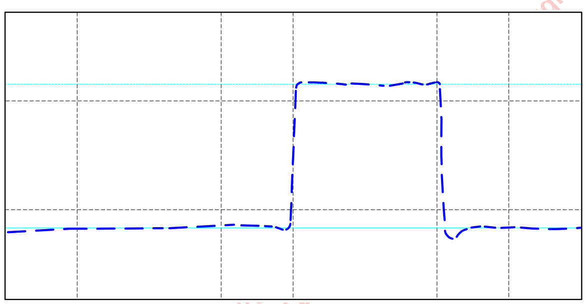


Figure 38: Clean signal after manual calibration for the 60 ohm



2.32 Configuring the 88EA1517/88EA1512 Device

The device can be configured two ways:

- Hardware configuration strap options (unmanaged applications)
- MDC/MDIO register writes (managed applications)88EA1512 Hardware Configuration

After the deassertion of RESETn the device will be hardware configured.

The device is configured through the CONFIG, RXD[3:0], and RX_CTRL pins as shown in Table 68.

Table 68: 88EA1512 Configuration Pin Mapping

Pin	Configuration	Sampled Bit Definition	Register Affected
CONFIG	PHYAD[0]	PHYAD bit 0	None
	VDDO_LEVEL	VDDO level at power up 1 = 2.5V 0 = 3.3V 3.3V is assumed until this bit is initialized.	24_2.13
RXD[0]	MDIO_TWSI	1 = MDIO 0 = TWSI	None
RXD[1]	PHYAD[1][1]	1 = PHYAD[1] is 0 0 = PHYAD[1] is 1	None
RXD[2]	PHYAD[2]	1 = PHYAD[2] is 0 0 = PHYAD[2] is 1	None
RXD[3]	PHYAD[3]	1 = PHYAD[3] is 0 0 = PHYAD[3] is 1	None
RX_CTRL	RGMII_MII	1 = RGMII 0 = MII	20_18.3 (Read Only)

1. PHYAD[4] = 0.

The VDDO_LEVEL configuration bit can be overwritten by software. PHYAD[3:0], RGMII_MII and MDIO_TWSI cannot be overwritten.

The CONFIG pin is used to configure 2 bits. The 2-bit value is set depending on what is connected to the CONFIG pin soon after the deassertion of hardware reset. The 2-bit mapping is shown in Table 69.

Table 69: Two-Bit Mapping

Pin	Bit 1,0
VSS	00
LED[0]	01
LED[1]	10
LED[2]	Unused
VDDO	11

The 2 bits for the CONFIG pin are mapped as shown in Table 70.

Table 70: Configuration Mapping

Pin	CONFIG Bit1	CONFIG Bit 0	Value Assignment
CONFIG	0	0	PHYAD[0] = 0 VDDO_LEVEL[1] = 3.3V
CONFIG	1	1	PHYAD[0] = 1 VDDO_LEVEL[1] = 3.3V
CONFIG	1	0	PHYAD[0] = 0 VDDO_LEVEL[1] = 2.5V
CONFIG	0	1	PHYAD[0] = 1 VDDO_LEVEL[1] = 2.5V

1. 88EA1512: Both VDDO_LEVEL and VDDO_SEL pin need to be set for 2.5/3.3V. As for 1.8V, only VDDO_SEL needs to be set and VDDO_LEVEL is ignored.

The RXD[3:0] and RX_CTRL pins are also used for configuration. Each of these are described below. For the RXD[3:0] and RX_CTRL pins, an internal pull-up resistor is enabled by default, so a HIGH value is sampled, If a LOW value needs to be sampled, then a pull-down resistor (4.7 kohm) must be added to the pin externally.

The RX_CTRL pin is used to configure the system interface in RGMII or MII mode. The configuration value is set depending on what is connected to the RX_CTRL pin after the deassertion of hardware reset. The mapping is shown in Table 71.

Table 71: RX_CTRL Mapping for Configuration

RX_CTRL Pin	Value	Configuration Definition
Unconnected	1	RGMII
4.7 kohm pull-down resistor	0	MII

The RXD[0] pin is used to configure the Management Interface pins in MDIO or Two-wire Serial Interface (TWSI). The configuration value is set depending on which pin is connected to the RXD[0] pin soon after the deassertion of hardware reset. The mapping is shown in Table 72.

Table 72: RXD[0] Mapping for Configuration

RXD[0] Pin	Value	Configuration Definition
Unconnected	1	MDIO
4.7 kohm pull-down resistor	0	TWSI

The RXD[1], RXD[2] and RXD[3] pins are used to configure PHYAD[1], PHYAD[2] and PHYAD[3] respectively. In order to configure a “0” for a PHYAD config bit, leave the pin that configures it floating. In order to configure a “1” for a PHYAD config bit, add a 4.7 kohm resistor to the pin that configures it. The mapping in shown in Table 73.

Table 73: RXD[3:1] Mapping Configuration

RXD[3] pin	RXD[2] pin	RXD[1] Pin	PHYAD[3:1] value
Unconnected	Unconnected	Unconnected	000
Unconnected	Unconnected	4.7 kohm pull-down resistor	001
Unconnected	4.7 kohm pull-down resistor	Unconnected	010
Unconnected	4.7 kohm pull-down resistor	4.7 kohm pull-down resistor	011
4.7 kohm pull-down resistor	Unconnected	Unconnected	100
4.7 kohm pull-down resistor	Unconnected	4.7 kohm pull-down resistor	101
4.7 kohm pull-down resistor	4.7 kohm pull-down resistor	Unconnected	110
4.7 kohm pull-down resistor	4.7 kohm pull-down resistor	4.7 kohm pull-down resistor	111

Table 74 shows possible values for the PHY Address Bits in the second column and the Configuration Pin settings needed to get each value in the first column.

Table 74: RXD[3:1] Mapping Configuration

{RXD[3:1],CONFIG[0]}	PHYADR[3:0]
1110	0000
1111	0001
1100	0010
1101	0011
1010	0100
1011	0101
1000	0110
1001	0111
0110	1000
0111	1001
0100	1010
0101	1011
0010	1100
0011	1101
0000	1110
0001	1111

Note that RXD[3:1] are pulled up when not connected and so 1s are sampled for these.

However the backward compatible value (which is the same as no 4.7 kohm pull-downs added) for the PHYADR[3:1] is 000. To resolve this the configuration values are inverted internally. So all RXD[3:1] values assigned must be the inverse of what is needed.

2.32.1 88EA1517 Hardware Configuration

After the deassertion of RESETn the device will be hardware configured.

The device is configured through the CONFIG, RXD[3:0], and RX_CTRL pins as shown in Table 68.

Table 75: 88EA1517 Configuration Pin Mapping

Pin	Configuration	Sampled Bit Definition	Register Affected
CONFIG	PHYAD[0]	PHYAD bit 0	None
	VDDO_LEVEL	VDDO level at power up 1 = 2.5V 0 = 3.3V 3.3V is assumed until this bit is initialized.	24_2.13
RXD[0]	MDIO_TWSI	1 = MDIO 0 = TWSI	None
RXD[1]	PHYAD[1][1]	1 = PHYAD[1] is 0 0 = PHYAD[1] is 1	None
RXD[2]	PHYAD[2]	1 = PHYAD[2] is 0 0 = PHYAD[2] is 1	None
RXD[3]	MDI_MDIX	1 = MDI 0 = MDIX	16_0.5
RX_CTRL	RGMII_MII	1 = RGMII 0 = MII	20_18.3 (Read Only)

1. PHYAD[4:3] = 0.

The VDDO_LEVEL configuration bit and the MDI_MDIX configuration bit can be overwritten by software. PHYAD[2:0], RGMII_MII and MDIO_TWSI cannot be overwritten.

The CONFIG pin is used to configure 2 bits. The 2-bit value is set depending on what is connected to the CONFIG pin soon after the deassertion of hardware reset. The 2-bit mapping is shown in Table 69.

Table 76: Two-Bit Mapping

Pin	Bit 1,0
VSS	00
LED[0]	01
LED[1]	10
LED[2]	Unused
VDDO	11

The 2 bits for the CONFIG pin are mapped as shown in Table 70.

Table 77: Configuration Mapping

Pin	CONFIG Bit1	CONFIG Bit 0	Value Assignment
CONFIG	0	0	PHYAD[0] = 0 VDDO_LEVEL[1] = 3.3V
CONFIG	1	1	PHYAD[0] = 1 VDDO_LEVEL[1] = 3.3V
CONFIG	1	0	PHYAD[0] = 0 VDDO_LEVEL[1] = 2.5V
CONFIG	0	1	PHYAD[0] = 1 VDDO_LEVEL[1] = 2.5V

1. VDDO level needs to be set here.

The RXD[3:0] and RX_CTRL pins are also used for configuration. Each of these are described below. For the RXD[3:0] and RX_CTRL pins, an internal pull-up resistor is enabled by default, so a HIGH value is sampled. If a LOW value needs to be sampled, then a pull-down resistor (4.7 kohm) must be added to the pin externally.

The RX_CTRL pin is used to configure the system interface in RGMII or MII mode. The configuration value is set depending on what is connected to the RX_CTRL pin after the deassertion of hardware reset. The mapping is shown in Table 71.

Table 78: RX_CTRL Mapping for Configuration

RX_CTRL Pin	Value	Configuration Definition
Unconnected	1	RGMII
4.7 kohm pull-down resistor	0	MII

The RXD[0] pin is used to configure the Management Interface pins in MDIO or Two-wire Serial Interface (TWSI). The configuration value is set depending on what is connected to the RXD[0] pin soon after the deassertion of hardware reset. The mapping is shown in Table 72.

Table 79: RXD[0] Mapping for Configuration

RXD[0] Pin	Value	Configuration Definition
Unconnected	1	MDIO
4.7 kohm pull-down resistor	0	TWSI

The RXD[1] and RXD[2] pins are used to configure PHYAD[1] and PHYAD[2] respectively. In order to configure a “0” for a PHYAD config bit, leave the pin that configures it floating. In order to configure a “1” for a PHYAD config bit, add a 4.7 kohm resistor to the pin that configures it. The mapping in shown in Table 73.

Table 80: RXD[2:1] Mapping Configuration

RXD[2] pin	RXD[1] Pin	PHYAD[2:1] value
Unconnected	Unconnected	00
Unconnected	4.7 kohm pull-down resistor	01

Table 80: RXD[2:1] Mapping Configuration (Continued)

RXD[2] pin	RXD[1] Pin	PHYAD[2:1] value
4.7 kohm pull-down resistor	Unconnected	10
4.7 kohm pull-down resistor	4.7 kohm pull-down resistor	11

The RXD[3] pin is used to configure MDI/MDIX for 88EA1517 package only. For the 88EA1512, this configuration bit does not do anything. The configuration value is set depending on what is connected to the RXD[3] pin soon after the deassertion of hardware reset. The mapping is shown in Table 81.

Table 81: RXD[3] Mapping Configuration

RXD[3] pin	Value	Configuration Definition
Unconnected	1	MDI
4.7 kohm pull-down resistor	0	MDIX

Table 82 shows possible values for the PHY Address Bits in the second column and the Configuration Pin settings needed to get each value in the first column.

Table 82: RXD[2:1] Mapping Configuration

{RXD[2:1],CONFIG[0]}	PHYADR[2:0]
110	000
111	001
100	010
101	011
010	100
011	101
000	110
001	111

Note that RXD[2:1] are pulled up when not connected and so 1s are sampled for these.

However the backward compatible value (which is the same as no 4.7 kohm pull-downs added) for the PHYADR[2:1] is 00. To resolve this the configuration values are inverted internally. So all RXD[2:1] values assigned must be the inverse of what is needed.

2.32.2 Software Configuration - Management Interface

The management interface provides access to the internal registers via the MDC and MDIO pins and is compliant with IEEE 802.3u Clause 22 and Clause 45 MDIO protocol. MDC is the management data clock input and, it can run from DC to a maximum rate of 12 MHz. At high MDIO fanouts the maximum rate may be decreased depending on the output loading. MDIO is the management data input/output and is a bi-directional signal that runs synchronously to MDC.

The MDIO pin requires a pull-up resistor in a range from 1.5 kohm to 10 kohm that pulls the MDIO high during the idle and turnaround phases of read and write operations.

In the 88EA1512 device, bits[3:0] of the PHY Address are configured during the hardware reset sequence, and PHY Address bit 4 is set to 0 internally. In the 88EA1517 device, bits[2:0] of the PHY

Address are configured during the hardware reset sequence, and PHY Address bits 4:3 are set to 0 internally. Refer to Section •, MDC/MDIO register writes (managed applications)88EA1512 Hardware Configuration, on page 117 and Section 2.32.1, 88EA1517 Hardware Configuration, on page 120 for more information on how to configure this.

Typical read and write operations on the management interface are shown in Figure 39 and Figure 40. All the required serial management registers are implemented as well as several optional registers. A description of the registers can be found in Section 3, General Register Description, on page 129.

Figure 39: Typical MDC/MDIO Read Operation

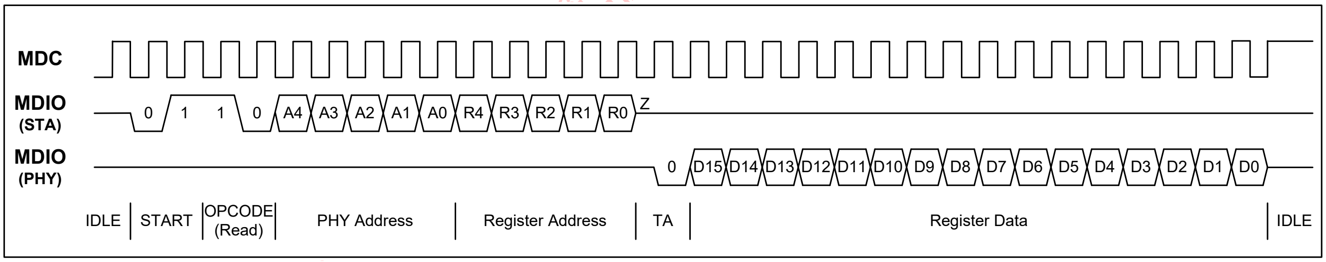


Figure 40: Typical MDC/MDIO Write Operation

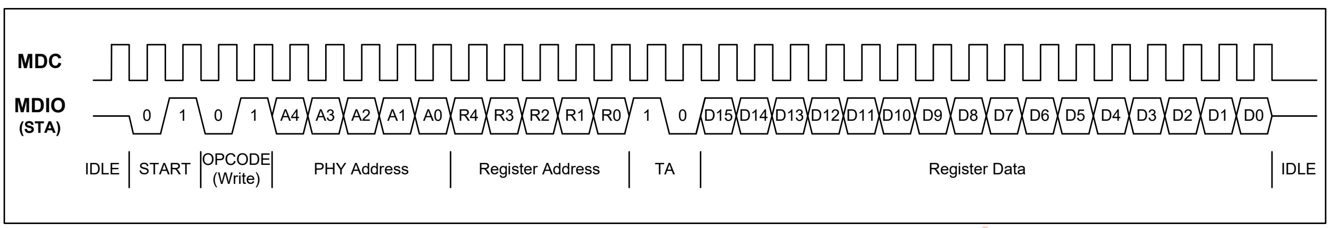


Table 83 is an example of a read operation.

Table 83: Serial Management Interface Protocol

32-Bit Preamble	Start of Frame	OpCode Read = 10 Write = 01	5-Bit PHY Device Address	5-Bit PHY Register Address (MSB)	2-Bit Turn around Read = z0 Write = 10	16-Bit Data Field	Idle
11111111	01	10	00000	00000	z0	0001001100000000	11111111

2.32.2.1 Preamble Suppression

The device is permanently programmed for preamble suppression. A minimum of one idle bit is required between operations.

2.33 Jumbo Packet Support

The device supports jumbo packets up to 16Kbytes on all data paths.

2.34 Temperature Sensor

The device features an internal temperature sensor. The sensor reports the die temperature and is updated approximately once per second. The temperature is obtained by reading the value in Register 26_6:4:0 and performing conversion functions as described in Table 84.

An interrupt can be generated when the temperature exceeds a certain threshold.

Register 26_6.6 is set high whenever the temperature is greater than or equal to the value programmed in Register 26_6.12:8. Register 26_6.6 remains high until read.

Register 26_6.7 controls whether the interrupt pin is asserted when Register 26_6.6 is high.

Table 84: Temperature Sensor

Register	Function	Setting	Mode	HW Rst	SW Rst
26_6.12:8	Temperature Threshold	Temperature in C = 5 x 26_6.4:0 - 25 i.e., for 100C the value is 11001	R/W	11001	Retain
26_6.7	Temperature Sensor Interrupt Enable	1= Interrupt Enable 0 = Interrupt Enable	R/W	0	Retain
26_6.6	Temperature Sensor Interrupt	1 = Temperature Reached Threshold 0 = Temperature Below Threshold	RO, LH	0	0
26_6.4:0	Temperature Sensor	Temperature in C = 5 x 26_6.4:0 - 25 i.e., for 100C the value is 11001	RO	xxxxx	xxxxx

2.35 Regulators and Power Supplies

The 88EA1517/88EA1512 devices have built-in switch-cap regulators to support single rail operation from a 3.3V source. These internal regulators generate 1.8V and 1.0V. The integrated regulators greatly reduce the PCB BOM cost. If regulators are not used then an external 1.8V and 1.0V supply are needed. Table 85 and Table 86 lists the valid combinations of regulator usage.

The VDDO supply can operate at 1.8V/2.5V/3.3V supplies.

Note

- If VDDO is tied to either 1.8V or 2.5V, then the I/Os are not 3.3V tolerant.
- AVDDC18 is tied to 1.8V, so the XTAL_IN pin is not 2.5V/3.3V tolerant.

Table 85: Power Supply Options - Integrated Switching Regulator (REG_IN)

Functional Description	AVDD33	AVDDC18/ AVDD18	DVDD	Setup
Supply Source	3.3V	1.8V from Internal Regulator	1.0V from Internal Regulator	Single 3.3V external supply Internal regulator enabled

Table 86: Power Supply Options - External Supplies

Functional Description	AVDD33	AVDDC18/ AVDD18	DVDD	Setup
Supply Source	3.3V	1.8V External	1.0V from External	3.3V, 1.8V, 1.0V external supplies Internal regulator disabled.

Note

When internal regulator option is preferred, both 1.0V and 1.8V regulators must be used. Supplying 1.0V internally and 1.8V externally (or vice versa) is not supported.

2.35.1 AVDD18

AVDD18 is used as the 1.8V analog supply. AVDD18 can be supplied externally with 1.8V, or via the 1.8V regulator.

2.35.2 AVDDC18

AVDDC18 is used as a 1.8V analog supply for XTAL_IN/OUT pins. AVDDC18 can be supplied externally with 1.8V, or via the 1.8V regulator.

2.35.3 AVDD33

AVDD33 is used as a 3.3V analog supply.

2.35.4 DVDD

DVDD is used as the 1.0V digital supply. DVDD can be supplied externally with 1.0V, or via the internal switching 1.0V regulator.

2.35.5 REG_IN

REG_IN is used as the 3.3V supply to the internal regulator that generates the 1.8V for AVDD18 and AVDDC18 and 1.0V for DVDD. If the 1.8V or 1.0V regulators are not used, REG_IN must be left floating in addition to leaving REGCAP1 and REGCAP2 floating.

2.35.6 AVDD18_OUT

AVDD18_OUT is the internal regulator 1.8V output. This must be connected to 1.8V power plane that connects to AVDD18 and AVDDC18. If an external supply is used to supply AVDD18 and AVDDC18, AVDD18_OUT must be left floating.

2.35.7 DVDD_OUT

DVDD_OUT is the internal regulator 1.0V output. When internal regulator is used, DVDD_OUT must be connected to the DVDD plane. If an external supply is used to supply DVDD, DVDD_OUT must be left floating.

2.35.8 VDDO

VDDO supplies all digital I/O pins which use LVCMOS I/O standards. The supported voltages are 1.8V, 2.5V or 3.3V for 88EA1517. 88EA1512 supports 2.5V/3.3V if VDDO_SEL is tied to VSS and 1.8V if VDDO_SEL is tied to VDDO which is 1.8V. For VDDO 1.8V operation, the power can be supplied by the internal regulator.

2.35.9 Power Supply Sequencing

On power-up, no special power supply sequencing is required.

2.36 Wake on Lan (WOL) Event Detection

The device supports the detection and reporting of Wake On LAN (WOL) events. There are 3 types of events:

- Wake Up Frame Event
- Magic Packet Event
- Link Change Event

WOL related registers are grouped in 16_17 to 29_17. See Registers 16_17 to 29_17 for further details.

Register 19_0.7 indicates whether any of the events have been detected or not. If 19_0.7 = 1, then at least one of the 3 types of events have been detected. Setting 18_0.7 to 1 will enable an interrupt to be generated which can be sent out the LED[2] pin. See LED 19_0.7 for further details.

2.36.1 Wake Up Frame Event

There is an on-chip SRAM that can store up to eight 128-byte frame patterns to be matched to. Each pattern can be individually enabled, including it in the matching. The matching itself is enabled by setting 16_17.15 to 1 and 16_17.7:0 are used to enable which patterns are used in the matching. If any of the frame patterns enabled for matching are present in frames received by the PHY, then a Wake Up Frame Event has happened and 17_17.15 will assert to 1. Which patterns were detected in the frames received is shown by 17_17.7:0. These status bits may be cleared by setting 16_17.12 to 1.

The length of each frame pattern is programmable using 18_17 to 21_17. The maximum length of each pattern is 128 bytes. A received frame cannot be considered matching to a frame pattern if that received frame has a shorter length than what is specified by 18_17 to 21_17 even if all of the bytes in the received frame match the frame pattern up to the received frame's length. A received frame is considered matching to a frame pattern if that received frame has an equal or longer length than what is specified in addition to all bytes match up.

Each byte in the frame pattern has an associated compare bit, written to the SRAM in 27_17.8. When that compare bit is 1, the associated byte (written to the SRAM in 27_17.7:0) is compared to the corresponding byte in a received packet, otherwise when the compare bit is 0, it is not compared. As long as all bytes in a received frame that are enabled for comparison to the corresponding frame pattern bytes from Start of Frame (SOF) to the pattern length match, a Wake Up Frame Event will be indicated. Note that each compare bit and byte to be written to the SRAM must be programmed in 27_17 first then 26_17.10 must be set 1 in order for this data to actually be written into the SRAM.

The SRAM can be read on a per byte basis by programming the address in 26_17 and reading the data retrieved in 28_17.8:0 after 26_17.11 is set to 1. The data structure in the SRAM is shown in Table 87. For further details, refer to the 88E1310 WOL application notes.

Table 87: SRAM Data Structure

Bits	71:64	63:56	55:48	47:40	39:32	31:24	23:16	15:8	7:0
Row add 0x00	64 = Enable pat 0 B0 65 = Enable pat 1 B0 . . 71 = Enable pat 7 B0	pat 7 byte 0	pat 6 byte 0	pat 5 byte 0	pat 4 byte 0	pat 3 byte 0	pat 2 byte 0	pat 1 byte 0	pat 0 byte 0
Row add 0x01	64 = Enable pat 0 B1 65 = Enable pat 1 B1 . . 71 = Enable pat 7 B1	pat 7 byte 1	pat 6 byte 1	pat 5 byte 1	pat 4 byte 1	pat 3 byte 1	pat 2 byte 1	pat 1 byte 1	pat 0 byte 1
...	...		.	.	.	.	.	.	.
Row add 0x7f	64 = Enable pat 0 B127 65 = Enable pat 1 B127 . . 71 = Enable pat 7 B127	pat 7 byte 127	pat 6 byte 127	pat 5 byte 127	pat 4 byte 127	pat 3 byte 127	pat 2 byte 127	pat 1 byte 127	pat 0 byte 127

Note that when writing or reading the SRAM using register accesses, the SRAM packet matching logic must be disabled by setting 16_17.15 = 0 otherwise internal memory accesses done for this may conflict with register memory accesses.

2.36.2 Magic Packet Event

If any of the frames received by the device contain 6 bytes of FF followed by 16 iterations of the destination address programmed in Register 23_17, 24_17 and 25_17, then such a frame is considered to be a Magic Packet and a Magic Packet Event has happened. Magic Packet detection can be enabled/disabled by 16_17.14. 17_17.14 will be set to 1 when a Magic Packet Event has happened and can be cleared by setting 16_17.12 to 1.

2.36.3 Link Change Event

If the link status changes from link down to up, then a Link Change Event has happened. This detection can be enabled/disabled by 16_17.13. 17_17.13 will be set to 1 when a Link Change Event has happened and can be cleared by setting 16_17.12 to 1.

3 General Register Description

The 88EA1517/88EA1512 device's General Registers are accessible using the MDC and MDIO pins and support the IEEE Serial Management Interface (SMI – Clause 22) used for PHY devices.

3.1 Register Types

The General Registers in the 88EA1517/88EA1512 device are made up of one or more fields. The way in which each of these fields operate is defined by the field's Type. The function of each Type is described below.

Table 88: Register Types

Type	Description
LH	Register field with latching high function. If status is high, then the register is set to one and remains set until a read operation is performed through the management interface or a reset occurs.
LL	Register field with latching low function. If status is low, then the register is cleared to zero and remains zero until a read operation is performed through the management interface or a reset occurs.
RES	Reserved. All reserved bits are read as zero unless otherwise noted.
Retain	The register value is retained after software reset is executed.
RO	Read only.
ROC	Read only clear. After read, register field is cleared.
RW	Read and Write with initial value indicated.
RWC	Read/Write clear on read. All bits are readable and writable. After reset or after the register field is read, register field is cleared to zero.
SC	Self-Clear. Writing a one to this register causes the desired function to be immediately executed, then the register field is automatically cleared to zero when the function is complete.
Update	Value written to the register field doesn't take effect until soft reset is executed.
WO	Write only. Reads from this type of register field return undefined data.
NR	Non-Rollover Register

3.2 PHY MDIO Register Description

The device supports both Clause 22 MDIO register access protocol and Clause 45 XMDIO register access protocol. The device also supports Clause 22 MDIO access to registers in Clause 45 XMDIO register space using Page 0 registers 13 and 14.

Table 89: Register Map

Register Name	Register Address	Table and Page
Copper Control Register	Page 0, Register 0	Table 90, p. 134
Copper Status Register	Page 0, Register 1	Table 91, p. 136
PHY Identifier 1	Page 0, Register 2	Table 92, p. 137
PHY Identifier 2	Page 0, Register 3	Table 93, p. 137
Copper Auto-Negotiation Advertisement Register	Page 0, Register 4	Table 94, p. 138
Copper Link Partner Ability Register - Base Page	Page 0, Register 5	Table 95, p. 141
Copper Auto-Negotiation Expansion Register	Page 0, Register 6	Table 96, p. 142
Copper Next Page Transmit Register	Page 0, Register 7	Table 97, p. 143
Copper Link Partner Next Page Register	Page 0, Register 8	Table 98, p. 143
1000BASE-T Control Register	Page 0, Register 9	Table 99, p. 144
1000BASE-T Status Register	Page 0, Register 10	Table 100, p. 145
XMDIO MMD control Register	Page 0, Register 13	Table 101, p. 146
XMDIO MMD address data Register	Page 0, Register 14	Table 102, p. 146
Extended Status Register	Page 0, Register 15	Table 103, p. 147
Copper Specific Control Register 1	Page 0, Register 16	Table 104, p. 147
Copper Specific Status Register 1	Page 0, Register 17	Table 105, p. 149
Copper Specific Interrupt Enable Register	Page 0, Register 18	Table 106, p. 150
Copper Interrupt Status Register	Page 0, Register 19	Table 107, p. 151
Copper Specific Control Register 2	Page 0, Register 20	Table 108, p. 152
Copper Specific Receive Error Counter Register	Page 0, Register 21	Table 109, p. 153
Page Address	Page Any, Register 22	Table 110, p. 153
Global Interrupt Status	Page 0, Register 23	Table 111, p. 154
Copper Specific Control Register 3	Page 0, Register 26	Table 112, p. 154
Fiber Control Register	Page 1, Register 0	Table 113, p. 155
Fiber Status Register	Page 1, Register 1	Table 114, p. 157
PHY Identifier	Page 1, Register 2	Table 115, p. 158
PHY Identifier	Page 1, Register 3	Table 116, p. 159
Fiber Auto-Negotiation Advertisement Register - 1000BASE-X Mode (Register 16_1.1:0 = 01)	Page 1, Register 4	Table 117, p. 159
Fiber Auto-Negotiation Advertisement Register - SGMII (System mode) (Register 16_1.1:0 = 10)	Page 1, Register 4	Table 118, p. 160
Fiber Auto-Negotiation Advertisement Register - SGMII (Media mode) (Register 16_1.1:0 = 11)	Page 1, Register 4	Table 119, p. 161
Fiber Link Partner Ability Register - 1000BASE-X Mode (Register 16_1.1:0 = 01)	Page 1, Register 5	Table 120, p. 161
Fiber Link Partner Ability Register - SGMII (System mode) (Register 16_1.1:0 = 10)	Page 1, Register 5	Table 121, p. 163
Fiber Link Partner Ability Register - SGMII (Media mode) (Register 16_1.1:0 = 11)	Page 1, Register 5	Table 122, p. 163
Fiber Auto-Negotiation Expansion Register	Page 1, Register 6	Table 123, p. 164

Table 89: Register Map

Register Name	Register Address	Table and Page
Fiber Next Page Transmit Register	Page 1, Register 7	Table 124, p. 165
Fiber Link Partner Next Page Register	Page 1, Register 8	Table 125, p. 165
Extended Status Register	Page 1, Register 15	Table 103, p. 147
Fiber Specific Control Register 1	Page 1, Register 16	Table 127, p. 166
Fiber Specific Status Register	Page 1, Register 17	Table 128, p. 167
Fiber Interrupt Enable Register	Page 1, Register 18	Table 129, p. 169
Fiber Interrupt Status Register	Page 1, Register 19	Table 130, p. 169
Fiber Receive Error Counter Register	Page 1, Register 21	Table 131, p. 170
PRBS Control	Page 1, Register 23	Table 132, p. 170
PRBS Error Counter LSB	Page 1, Register 24	Table 133, p. 171
PRBS Error Counter MSB	Page 1, Register 25	Table 134, p. 171
Fiber Specific Control Register 2	Page 1, Register 26	Table 135, p. 171
MAC Specific Control Register 1	Page 2, Register 16	Table 136, p. 172
MAC Specific Interrupt Enable Register	Page 2, Register 18	Table 137, p. 173
MAC Specific Status Register	Page 2, Register 19	Table 138, p. 174
Copper RX_ER Byte Capture	Page 2, Register 20	Table 139, p. 174
MAC Specific Control Register 2	Page 2, Register 21	Table 140, p. 175
RGMII Output Impedance Calibration Override	Page 2, Register 24	Table 141, p. 176
RGMII Output Impedance Target	Page 2, Register 25	Table 142, p. 177
LED[2:0] Function Control Register	Page 3, Register 16	Table 144, p. 178
LED[2:0] Polarity Control Register	Page 3, Register 17	Table 145, p. 179
LED Timer Control Register	Page 3, Register 18	Table 146, p. 181
RGMII RX_ER Byte Capture	Page 4, Register 20	Table 148, p. 183
Advanced VCT TX to MDI[0] Rx Coupling	Page 5, Register 16	Table 149, p. 184
Advanced VCT TX to MDI[1] Rx Coupling	Page 5, Register 17	Table 150, p. 185
Advanced VCT TX to MDI[2] Rx Coupling	Page 5, Register 18	Table 151, p. 186
Advanced VCT TX to MDI[3] Rx Coupling	Page 5, Register 19	Table 152, p. 187
1000BASE-T Pair Skew Register	Page 5, Register 20	Table 153, p. 188
1000BASE-T Pair Swap and Polarity	Page 5, Register 21	Table 154, p. 188
Advance VCT Control	Page 5, Register 23	Table 155, p. 189
Advanced VCT Sample Point Distance	Page 5, Register 24	Table 156, p. 189
Advanced VCT Cross Pair Positive Threshold	Page 5, Register 25	Table 157, p. 190
Advanced VCT Same Pair Impedance Positive Threshold 0 and 1	Page 5, Register 26	Table 158, p. 190
Advanced VCT Same Pair Impedance Positive Threshold 2 and 3	Page 5, Register 27	Table 159, p. 190
Advanced VCT Same Pair Impedance Positive Threshold 4 and Transmit Pulse Control	Page 5, Register 28	Table 160, p. 191
Copper Port Packet Generation	Page 6, Register 16	Table 161, p. 192
Copper Port CRC Counters	Page 6, Register 17	Table 162, p. 193

Table 89: Register Map

Register Name	Register Address	Table and Page
Checker Control	Page 6, Register 18	Table 163, p. 193
Copper Port Packet Generation	Page 6, Register 19	Table 164, p. 193
General Control Register	Page 6, Register 20	Table 165, p. 194
Late Collision Counters 1 & 2	Page 6, Register 23	Table 166, p. 194
Late Collision Counters 3 & 4	Page 6, Register 24	Table 167, p. 195
Late Collision Window Adjust/Link Disconnect	Page 6, Register 25	Table 168, p. 195
Misc Test	Page 6, Register 26	Table 169, p. 195
Misc Test: Temperature Sensor Alternative Reading	Page 6, Register 27	Table 170, p. 196
PHY Cable Diagnostics Pair 0 Length	Page 7, Register 16	Table 171, p. 196
PHY Cable Diagnostics Pair 1 Length	Page 7, Register 17	Table 172, p. 197
PHY Cable Diagnostics Pair 2 Length	Page 7, Register 18	Table 173, p. 197
PHY Cable Diagnostics Pair 3 Length	Page 7, Register 19	Table 174, p. 197
PHY Cable Diagnostics Results	Page 7, Register 20	Table 175, p. 197
PHY Cable Diagnostics Control	Page 7, Register 21	Table 176, p. 198
Advanced VCT Cross Pair Negative Threshold	Page 7, Register 25	Table 177, p. 199
Advanced VCT Same Pair Impedance Negative Threshold 0 and 1	Page 7, Register 26	Table 178, p. 199
Advanced VCT Same Pair Impedance Negative Threshold 2 and 3	Page 7, Register 27	Table 179, p. 199
Advanced VCT Same Pair Impedance Negative Threshold 4	Page 7, Register 28	Table 180, p. 200
WOL Control	Page 17, Register 16	Table 181, p. 200
WOL Status	Page 17, Register 17	Table 182, p. 201
SRAM Packets 7/6 Length	Page 17, Register 18	Table 183, p. 203
SRAM Packets 5/4 Length	Page 17, Register 19	Table 184, p. 203
SRAM Packets 3/2 Length	Page 17, Register 20	Table 185, p. 203
SRAM Packets 1/0 Length	Page 17, Register 21	Table 186, p. 204
Magic Packet Destination Address Word 2	Page 17, Register 23	Table 187, p. 204
Magic Packet Destination Address Word 1	Page 17, Register 24	Table 188, p. 204
Magic Packet Destination Address Word 0	Page 17, Register 25	Table 189, p. 204
SRAM Byte Address Control	Page 17, Register 26	Table 190, p. 205
SRAM Byte Data Control	Page 17, Register 27	Table 191, p. 205
SRAM Read Control	Page 17, Register 28	Table 192, p. 206
EEE Buffer Control Register 1	Page 18, Register 0	Table 193, p. 206
EEE Buffer Control Register 2	Page 18, Register 1	Table 194, p. 207
EEE Buffer Control Register 3	Page 18, Register 2	Table 195, p. 207
Packet Generation	Page 18, Register 16	Table 197, p. 207
CRC Counters	Page 18, Register 17	Table 198, p. 208
Checker Control	Page 18, Register 18	Table 199, p. 209
Packet Generation	Page 18, Register 19	Table 200, p. 209

Table 89: Register Map

Register Name	Register Address	Table and Page
General Control Register 1	Page 18, Register 20	Table 201, p. 209
Link Disconnect count	Page 18, Register 25	Table 202, p. 211
SERDES RX_ER Byte Capture	Page 18, Register 26	Table 203, p. 211

Table 90: Copper Control Register
Page 0, Register 0

Bits	Field	Mode	HW Rst	SW Rst	Description
15	Copper Reset	R/W, SC	0x0	SC	Copper Software Reset. Affects Pages 0, 2, 3, 5, and 7. Writing a 1 to this bit causes the PHY state machines to be reset. When the reset operation is done, this bit is cleared to 0 automatically. The reset occurs immediately. 1 = PHY reset 0 = Normal operation
14	Loopback	R/W	0	0	When loopback is activated, the data from the MAC presented to the PHY is looped back inside the PHY and then sent back to the MAC. Link is broken when loopback is enabled. Loopback speed is determined by Registers 21_2.{6,13}. 1 = Enable Loopback 0 = Disable Loopback
13	Speed Select (LSB)	R/W	See Desc	Update	Changes to this bit are disruptive to the normal operation; therefore, any changes to these registers must be followed by a software reset to take effect. A write to this register bit does not take effect until any one of the following also occurs: Software reset is asserted (Register 0_0.15) Restart Auto-Negotiation is asserted (Register 0_0.9) Power down (Register 0_0.11, 16_0.2) transitions from power down to normal operation This bit defaults to 1 for 88EA1517 and when MII mode is used for either package, otherwise it defaults to 0. Bit 6, 13 11 = Reserved 10 = 1000 Mbps 01 = 100 Mbps 00 = 10 Mbps
12	Auto-Negotiation Enable	R/W	1	Update	Changes to this bit are disruptive to the normal operation. A write to this register bit does not take effect until any one of the following occurs: Software reset is asserted (Register 0_0.15) Restart Auto-Negotiation is asserted (Register 0_0.9) Power down (Register 0_0.11, 16_0.2) transitions from power down to normal operation If Register 0_0.12 is set to 0 and speed is manually forced to 1000 Mbps in Registers 0.13 and 0.6, then Auto-Negotiation will still be enabled and only 1000BASE-T full-duplex is advertised if register 0_0.8 is set to 1, and 1000BASE-T half-duplex is advertised if 0.8 is set to 0. Registers 4.8:5 and 9.9:8 are ignored. Auto-Negotiation is mandatory per IEEE for proper operation in 1000BASE-T. This bit defaults to 0 for 88EA1517, otherwise it defaults to 1. 1 = Enable Auto-Negotiation Process 0 = Disable Auto-Negotiation Process

Table 90: Copper Control Register
Page 0, Register 0

Bits	Field	Mode	HW Rst	SW Rst	Description
11	Power Down	R/W	0x0	Retain	Power down is controlled via register 0.11 and 16_0.2. Both bits must be set to 0 before the PHY will transition from power down to normal operation. When the port is switched from power down to normal operation, software reset and restart Auto-Negotiation are performed even when bits Reset (0_15) and Restart Auto-Negotiation (0.9) are not set by the user. IEEE power down shuts down the chip except for the RGMII interface if 16_2.3 is set to 1. If 16_2.3 is set to 0, then the RGMII interface also shuts down. 1 = Power down 0 = Normal operation
10	Isolate	R/W	0x0	0x0	1 = Isolate 0 = Normal Operation
9	Restart Copper Auto-Negotiation	R/W, SC	0x0	SC	Auto-Negotiation automatically restarts after hardware or software reset regardless of whether or not the restart bit (0_0.9) is set. 1 = Restart Auto-Negotiation Process 0 = Normal operation
8	Copper Duplex Mode	R/W	See Desc	Update	Changes to this bit are disruptive to the normal operation; therefore, any changes to these registers must be followed by a software reset to take effect. A write to this register bit does not take effect until any one of the following also occurs: Software reset is asserted (Register 0_0.15) Restart Auto-Negotiation is asserted (Register 0_0.9) Power down (Register 0_0.11, 16_0.2) transitions from power down to normal operation This bit is a 1 always (a write of 0 has no effect) when MII mode is used, otherwise this bit defaults to 1 and may be written to 0. 1 = Full-duplex 0 = Half-Duplex
7	Collision Test	RO	0x0	0x0	This bit has no effect.

Table 90: Copper Control Register
Page 0, Register 0

Bits	Field	Mode	HW Rst	SW Rst	Description
6	Speed Selection (MSB)	R/W	See Descr	Update	Changes to this bit are disruptive to the normal operation; therefore, any changes to these registers must be followed by a software reset to take effect. A write to this register bit does not take effect until any one of the following occurs: Software reset is asserted (Register 0_0.15) Restart Auto-Negotiation is asserted (Register 0_0.9) Power down (Register 0_0.11, 16_0.2) transitions from power down to normal operation This bit is a 0 always (a write of 1 has no effect) for 88EA1517 and when MII mode is used for either package, otherwise this bit defaults to 1 and may be written to 0. bit 6, 13 11 = Reserved 10 = 1000 Mbps 01 = 100 Mbps 00 = 10 Mbps
5:0	Reserved	RO	Always 000000	Always 000000	Will always be 0.

Table 91: Copper Status Register
Page 0, Register 1

Bits	Field	Mode	HW Rst	SW Rst	Description
15	100BASE-T4	RO	Always 0	Always 0	100BASE-T4. This protocol is not available. 0 = PHY not able to perform 100BASE-T4
14	100BASE-X Full-Duplex	RO	Always 1	Always 1	1 = PHY able to perform full-duplex 100BASE-X
13	100BASE-X Half-Duplex	RO	Always 1	Always 1	1 = PHY able to perform half-duplex 100BASE-X
12	10 Mbps Full-Duplex	RO	Always 1	Always 1	1 = PHY able to perform full-duplex 10BASE-T
11	10 Mbps Half-Duplex	RO	Always 1	Always 1	1 = PHY able to perform half-duplex 10BASE-T
10	100BASE-T2 Full-Duplex	RO	Always 0	Always 0	This protocol is not available. 0 = PHY not able to perform full-duplex
9	100BASE-T2 Half-Duplex	RO	Always 0	Always 0	This protocol is not available. 0 = PHY not able to perform half-duplex
8	Extended Status	RO	Always 1	Always 1	1 = Extended status information in Register 15
7	Reserved	RO	Always 0	Always 0	Must always be 0.
6	MF Preamble Suppression	RO	Always 1	Always 1	1 = PHY accepts management frames with preamble suppressed

Table 91: Copper Status Register
Page 0, Register 1

Bits	Field	Mode	HW Rst	SW Rst	Description
5	Copper Auto-Negotiation Complete	RO	0x0	0x0	1 = Auto-Negotiation process complete 0 = Auto-Negotiation process not complete
4	Copper Remote Fault	RO,LH	0x0	0x0	1 = Remote fault condition detected 0 = Remote fault condition not detected
3	Auto-Negotiation Ability	RO	Always 1	Always 1	1 = PHY able to perform Auto-Negotiation
2	Copper Link Status	RO,LL	0x0	0x0	This register bit indicates that link was down since the last read. For the current link status, either read this register back-to-back or read Register 17_0.10 Link Real Time. 1 = Link is up 0 = Link is down
1	Jabber Detect	RO,LH	0x0	0x0	1 = Jabber condition detected 0 = Jabber condition not detected
0	Extended Capability	RO	Always 1	Always 1	1 = Extended register capabilities

Table 92: PHY Identifier 1
Page 0, Register 2

Bits	Field	Mode	HW Rst	SW Rst	Description
15:0	Organizationally Unique Identifier Bit 3:18	RO	0x0141	0x0141	Marvell® OUI is 0x005043 0000 0000 0101 0000 0100 0011 ^ ^ bit 1.................................bit 24 Register 2.[15:0] show bits 3 to 18 of the OUI. 0000000101000001 ^ ^ bit 3..................bit18

Table 93: PHY Identifier 2
Page 0, Register 3

Bits	Field	Mode	HW Rst	SW Rst	Description
15:10	OUI Lsb	RO	Always 000011	Always 000011	Organizationally Unique Identifier bits 19:24 00 0011 ^.........^ bit 19...bit24

Table 93: PHY Identifier 2
Page 0, Register 3

Bits	Field	Mode	HW Rst	SW Rst	Description
9:4	Model Number	RO	Always 011101	Always 011101	Model Number 011101
3:0	Revision Number	RO	See Descr	See Descr	Rev Number. Contact Marvell® FAEs for information on the device revision number.

Table 94: Copper Auto-Negotiation Advertisement Register
Page 0, Register 4

Bits	Field	Mode	HW Rst	SW Rst	Description
15	Next Page	R/W	0x0	Update	A write to this register bit does not take effect until any one of the following occurs: Software reset is asserted (Register 0_0.15) Restart Auto-Negotiation is asserted (Register 0_0.9) Power down (Register 0_0.11, 16_0.2) transitions from power down to normal operation Copper link goes down. If 1000BASE-T is advertised then the required next pages are automatically transmitted. Register 4.15 should be set to 0 if no additional next pages are needed. 1 = Advertise 0 = Not advertised
14	Ack	RO	Always 0	Always 0	Must be 0.
13	Remote Fault	R/W	0x0	Update	A write to this register bit does not take effect until any one of the following occurs: Software reset is asserted (Register 0_0.15) Restart Auto-Negotiation is asserted (Register 0_0.9) Power down (Register 0_0.11, 16_0.2) transitions from power down to normal operation Copper link goes down. 1 = Set Remote Fault bit 0 = Do not set Remote Fault bit
12	Reserved	R/W	0x0	Update	A write to this register bit does not take effect until any one of the following occurs: Software reset is asserted (Register 0_0.15) Restart Auto-Negotiation is asserted (Register 0_0.9) Power down (Register 0_0.11, 16_0.2) transitions from power down to normal operation Copper link goes down Reserved bit is R/W to allow for forward compatibility with future IEEE standards.

Table 94: Copper Auto-Negotiation Advertisement Register
Page 0, Register 4

Bits	Field	Mode	HW Rst	SW Rst	Description
11	Asymmetric Pause	R/W	0x0	Update	A write to this register bit does not take effect until any one of the following occurs: Software reset is asserted (Register 0.15) Restart Auto-Negotiation is asserted (Register 0.9) Power down (Register 0.11, 16_0.2) transitions from power down to normal operation Copper link goes down. 1 = Asymmetric Pause 0 = No asymmetric Pause
10	Pause	R/W	0x0	Update	A write to this register bit does not take effect until any one of the following occurs: Software reset is asserted (Register 0.15) Restart Auto-Negotiation is asserted (Register 0.9) Power down (Register 0.11, 16_0.2) transitions from power down to normal operation Copper link goes down. 1 = MAC PAUSE implemented 0 = MAC PAUSE not implemented
9	100BASE-T4	R/W	0x0	Retain	0 = Not capable of 100BASE-T4
8	100BASE-TX Full-Duplex	R/W	0x1	Update	A write to this register bit does not take effect until any one of the following occurs: Software reset is asserted (Register 0.15) Restart Auto-Negotiation is asserted (Register 0.9) Power down (Register 0.11, 16_0.2) transitions from power down to normal operation Copper link goes down. If register 0.12 is set to 0 and speed is manually forced to 1000 Mbps in Registers 0.13 and 0.6, then Auto-Negotiation will still be enabled and only 1000BASE-T full-duplex is advertised if register 0.8 is set to 1, and 1000BASE-T half-duplex is advertised if 0.8 set to 0. Registers 4.8:5 and 9.9:8 are ignored. Auto-Negotiation is mandatory per IEEE for proper operation in 1000BASE-T. 1 = Advertise 0 = Not advertised

Table 94: Copper Auto-Negotiation Advertisement Register
Page 0, Register 4

Bits	Field	Mode	HW Rst	SW Rst	Description
7	100BASE-TX Half-Duplex	R/W	See Descr	Update	A write to this register bit does not take effect until any one of the following occurs: Software reset is asserted (Register 0.15) Restart Auto-Negotiation is asserted (Register 0.9) Power down (Register 0.11, 16_0.2) transitions from power down to normal operation Copper link goes down. If register 0.12 is set to 0 and speed is manually forced to 1000 Mbps in Registers 0.13 and 0.6, then Auto-Negotiation will still be enabled and only 1000BASE-T full-duplex is advertised if register 0.8 is set to 1, and 1000BASE-T half-duplex is advertised if 0.8 set to 0. Registers 4.8:5 and 9.9:8 are ignored. Auto-Negotiation is mandatory per IEEE for proper operation in 1000BASE-T. This bit is a 0 always (a write of 1 has no effect) when MII mode is used, otherwise this bit defaults to 1 and it may be written to 0. 1 = Advertise 0 = Not advertised
6	10BASE-TX Full-Duplex	R/W	0x1	Update	A write to this register bit does not take effect until any one of the following occurs: Software reset is asserted (Register 0.15) Restart Auto-Negotiation is asserted (Register 0.9) Power down (Register 0.11, 16_0.2) transitions from power down to normal operation Copper link goes down. If register 0.12 is set to 0 and speed is manually forced to 1000 Mbps in Registers 0.13 and 0.6, then Auto-Negotiation will still be enabled and only 1000BASE-T full-duplex is advertised if register 0.8 is set to 1, and 1000BASE-T half-duplex is advertised if 0.8 set to 0. Registers 4.8:5 and 9.9:8 are ignored. Auto-Negotiation is mandatory per IEEE for proper operation in 1000BASE-T. 1 = Advertise 0 = Not advertised

Table 94: Copper Auto-Negotiation Advertisement Register
Page 0, Register 4

Bits	Field	Mode	HW Rst	SW Rst	Description
5	10BASE-TX Half-Duplex	R/W	See Descr	Update	A write to this register bit does not take effect until any one of the following occurs: Software reset is asserted (Register 0.15) Restart Auto-Negotiation is asserted (Register 0.9) Power down (Register 0.11, 16_0.2) transitions from power down to normal operation Copper link goes down. If register 0.12 is set to 0 and speed is manually forced to 1000 Mbps in Registers 0.13 and 0.6, then Auto-Negotiation will still be enabled and only 1000BASE-T full-duplex is advertised if register 0.8 is set to 1, and 1000BASE-T half-duplex is advertised if 0.8 set to 0. Registers 4.8:5 and 9.9:8 are ignored. Auto-Negotiation is mandatory per IEEE for proper operation in 1000BASE-T. This bit is a 0 always (a write of 1 has no effect) when MII mode is used, otherwise this bit defaults to 1 and it may be written to 0. 1 = Advertise 0 = Not advertised
4:0	Selector Field	R/W	0x01	Retain	Selector Field mode 00001 = 802.3

Table 95: Copper Link Partner Ability Register - Base Page
Page 0, Register 5

Bits	Field	Mode	HW Rst	SW Rst	Description
15	Next Page	RO	0x0	0x0	Received Code Word Bit 15 1 = Link partner capable of next page 0 = Link partner not capable of next page
14	Acknowledge	RO	0x0	0x0	Acknowledge Received Code Word Bit 14 1 = Link partner received link code word 0 = Link partner does not have Next Page ability
13	Remote Fault	RO	0x0	0x0	Remote Fault Received Code Word Bit 13 1 = Link partner detected remote fault 0 = Link partner has not detected remote fault
12	Technology Ability Field	RO	0x0	0x0	Received Code Word Bit 12
11	Asymmetric Pause	RO	0x0	0x0	Received Code Word Bit 11 1 = Link partner requests asymmetric pause 0 = Link partner does not request asymmetric pause

Table 95: Copper Link Partner Ability Register - Base Page
Page 0, Register 5

Bits	Field	Mode	HW Rst	SW Rst	Description
10	Pause Capable	RO	0x0	0x0	Received Code Word Bit 10 1 = Link partner is capable of pause operation 0 = Link partner is not capable of pause operation
9	100BASE-T4 Capability	RO	0x0	0x0	Received Code Word Bit 9 1 = Link partner is 100BASE-T4 capable 0 = Link partner is not 100BASE-T4 capable
8	100BASE-TX Full-Duplex Capability	RO	0x0	0x0	Received Code Word Bit 8 1 = Link partner is 100BASE-TX full-duplex capable 0 = Link partner is not 100BASE-TX full-duplex capable
7	100BASE-TX Half-Duplex Capability	RO	0x0	0x0	Received Code Word Bit 7 1 = Link partner is 100BASE-TX half-duplex capable 0 = Link partner is not 100BASE-TX half-duplex capable
6	10BASE-T Full-Duplex Capability	RO	0x0	0x0	Received Code Word Bit 6 1 = Link partner is 10BASE-T full-duplex capable 0 = Link partner is not 10BASE-T full-duplex capable
5	10BASE-T Half-Duplex Capability	RO	0x0	0x0	Received Code Word Bit 5 1 = Link partner is 10BASE-T half-duplex capable 0 = Link partner is not 10BASE-T half-duplex capable
4:0	Selector Field	RO	0x00	0x00	Selector Field Received Code Word Bit 4:0

Table 96: Copper Auto-Negotiation Expansion Register
Page 0, Register 6

Bits	Field	Mode	HW Rst	SW Rst	Description
15:5	Reserved	RO	0x000	0x000	Reserved. Must be 00000000000.
4	Parallel Detection Fault	RO,LH	0x0	0x0	Register 6_0.4 is not valid until the Auto-Negotiation complete bit (Reg 1_0.5) indicates completed. 1 = A fault has been detected via the Parallel Detection function 0 = A fault has not been detected via the Parallel Detection function
3	Link Partner Next page Able	RO	0x0	0x0	Register 6_0.3 is not valid until the Auto-Negotiation complete bit (Reg 1_0.5) indicates completed. 1 = Link Partner is Next Page able 0 = Link Partner is not Next Page able

Table 96: Copper Auto-Negotiation Expansion Register
Page 0, Register 6

Bits	Field	Mode	HW Rst	SW Rst	Description
2	Local Next Page Able	RO	0x1	0x1	Register 6_0.2 is not valid until the Auto-Negotiation complete bit (Reg 1_0.5) indicates completed. 1 = Local Device is Next Page able 0 = Local Device is not Next Page able
1	Page Received	RO, LH	0x0	0x0	Register 6_0.1 is not valid until the Auto-Negotiation complete bit (Reg 1_0.5) indicates completed. 1 = A New Page has been received 0 = A New Page has not been received
0	Link Partner Auto-Negotiation Able	RO	0x0	0x0	Register 6_0.0 is not valid until the Auto-Negotiation complete bit (Reg 1_0.5) indicates completed. 1 = Link Partner is Auto-Negotiation able 0 = Link Partner is not Auto-Negotiation able

Table 97: Copper Next Page Transmit Register
Page 0, Register 7

Bits	Field	Mode	HW Rst	SW Rst	Description
15	Next Page	R/W	0x0	0x0	A write to register 7_0 implicitly sets a variable in the Auto-Negotiation state machine indicating that the next page has been loaded. Link fail will clear Reg 7_0. Transmit Code Word Bit 15
14	Reserved	RO	0x0	0x0	Transmit Code Word Bit 14
13	Message Page Mode	R/W	0x1	0x1	Transmit Code Word Bit 13
12	Acknowledge2	R/W	0x0	0x0	Transmit Code Word Bit 12
11	Toggle	RO	0x0	0x0	Transmit Code Word Bit 11
10:0	Message/ Unformatted Field	R/W	0x001	0x001	Transmit Code Word Bit 10:0

Table 98: Copper Link Partner Next Page Register
Page 0, Register 8

Bits	Field	Mode	HW Rst	SW Rst	Description
15	Next Page	RO	0x0	0x0	Received Code Word Bit 15
14	Acknowledge	RO	0x0	0x0	Received Code Word Bit 14
13	Message Page	RO	0x0	0x0	Received Code Word Bit 13

Table 98: Copper Link Partner Next Page Register
Page 0, Register 8

Bits	Field	Mode	HW Rst	SW Rst	Description
12	Acknowledge2	RO	0x0	0x0	Received Code Word Bit 12
11	Toggle	RO	0x0	0x0	Received Code Word Bit 11
10:0	Message/ Unformatted Field	RO	0x000	0x000	Received Code Word Bit 10:0

Table 99: 1000BASE-T Control Register
Page 0, Register 9

Bits	Field	Mode	HW Rst	SW Rst	Description
15:13	Test Mode	R/W	000	Retain	TX_TCLK can be brought out to the HSDACP/N pins for jitter testing in test modes 2 and 3 by setting 26_6.15 = 1. After exiting the test mode, hardware reset or software reset (Register 0_0.15) should be issued to ensure normal operation. A restart of Auto-Negotiation will clear these bits. These bits are always 000 for 88EA1517 and when MII mode is used for either package, otherwise they may be programmed as follows: 000 = Normal Mode 001 = Test Mode 1 - Transmit Waveform Test 010 = Test Mode 2 - Transmit Jitter Test (MASTER mode) 011 = Test Mode 3 - Transmit Jitter Test (SLAVE mode) 100 = Test Mode 4 - Transmit Distortion Test 101, 110, 111 = Reserved
12	MASTER/SLAVE Manual Configu-ration Enable	R/W	0x0	Update	A write to this register bit does not take effect until any of the following also occurs: Software reset is asserted (Register 0_0.15) Restart Auto-Negotiation is asserted (Register 0_0.9) Power down (Register 0_0.11, 16_0.2) transitions from power down to normal operation Copper link goes down. 1 = Manual MASTER/SLAVE configuration 0 = Automatic MASTER/SLAVE configuration
11	MASTER/SLAVE Configuration Value	R/W	0x0	Update	A write to this register bit does not take effect until any of the following also occurs: Software reset is asserted (Register 0.15) Restart Auto-Negotiation is asserted (Register 0.9) Power down (Register 0.11, 16_0.2) transitions from power down to normal operation Copper link goes down. 1 = Manual configure as MASTER 0 = Manual configure as SLAVE

Table 99: 1000BASE-T Control Register
Page 0, Register 9

Bits	Field	Mode	HW Rst	SW Rst	Description
10	Port Type	R/W	0x0	Update	A write to this register bit does not take effect until any of the following also occurs: Software reset is asserted (Register 0_0.15) Restart Auto-Negotiation is asserted (Register 0_0.9) Power down (Register 0_0.11, 16_0.2) transitions from power down to normal operation Copper link goes down. Register 9_0.10 is ignored if Register 9_0.12 is equal to 1. 1 = Prefer multi-port device (MASTER) 0 = Prefer single port device (SLAVE)
9	1000BASE-T Full-Duplex	R/W	See Desc	Update	A write to this register bit does not take effect until any of the following also occurs: Software reset is asserted (Register 0_0.15) Restart Auto-Negotiation is asserted (Register 0_0.9) Power down (Register 0_0.11, 16_0.2) transitions from power down to normal operation link goes down. This bit is always 0 for 88EA1517 and when MII mode is used for either package, otherwise it defaults to 1. 1 = Advertise 0 = Not advertised
8	1000BASE-T Half-Duplex	R/W	See Desc	Update	A write to this register bit does not take effect until any of the following also occurs: Software reset is asserted (Register 0.15) Restart Auto-Negotiation is asserted (Register 0.9) Power down (Register 0.11, 16_0.2) transitions from power down to normal operation Copper link goes down. This bit is always 0 for 88EA1517 and when MII mode is used for either package, otherwise it defaults to 1. 1 = Advertise 0 = Not advertised
7:0	Reserved	R/W	0x00	Retain	0

Table 100: 1000BASE-T Status Register
Page 0, Register 10

Bits	Field	Mode	HW Rst	SW Rst	Description
15	MASTER/SLAVE Configuration Fault	RO,LH	0x0	0x0	This register bit will clear on read. 1 = MASTER/SLAVE configuration fault detected 0 = No MASTER/SLAVE configuration fault detected
14	MASTER/SLAVE Configuration Resolution	RO	0x0	0x0	1 = Local PHY configuration resolved to MASTER 0 = Local PHY configuration resolved to SLAVE
13	Local Receiver Status	RO	0x0	0x0	1 = Local Receiver OK 0 = Local Receiver is Not OK

Table 100: 1000BASE-T Status Register
Page 0, Register 10

Bits	Field	Mode	HW Rst	SW Rst	Description
12	Remote Receiver Status	RO	0x0	0x0	1 = Remote Receiver OK 0 = Remote Receiver Not OK
11	Link Partner 1000BASE-T Full-Duplex Capability	RO	0x0	0x0	1 = Link Partner is capable of 1000BASE-T full-duplex 0 = Link Partner is not capable of 1000BASE-T full-duplex
10	Link Partner 1000BASE-T Half-Duplex Capability	RO	0x0	0x0	1 = Link Partner is capable of 1000BASE-T half-duplex 0 = Link Partner is not capable of 1000BASE-T half-duplex
9:8	Reserved	RO	0x0	0x0	Reserved
7:0	Idle Error Count	RO, SC	0x00	0x00	MSB of Idle Error Counter These register bits report the idle error count since the last time this register was read. The counter pegs at 11111111 and will not roll over.

Table 101: XMDIO MMD control Register
Page 0, Register 13

Bits	Field	Mode	HW Rst	SW Rst	Description
15:14	Function	R/W	0x0	0x0	15.14 00 = Address 01 = Data, no post increment 10 = Data, post increment on reads and writes 11 = Data, post increment on writes only
13:5	Reserved	RO	0x000	0x000	Reserved
4:0	DEVAD	R/W	0x00	0x00	Device address

Table 102: XMDIO MMD address data Register
Page 0, Register 14

Bits	Field	Mode	HW Rst	SW Rst	Description
15:0	Address Data	R/W	0x0000	0x0000	If 13.15:14 = 00, this register is the MMD DEVAD's address register. Otherwise, it is the MMD DEVAD's data register as indicated by the contents of its address register

Table 103: Extended Status Register
Page 0, Register 15

Bits	Field	Mode	HW Rst	SW Rst	Description
15	1000BASE-X Full-Duplex	RO	Always 0	Always 0	0 = Not 1000BASE-X full-duplex capable
14	1000BASE-X Half-Duplex	RO	Always 0	Always 0	0 = Not 1000BASE-X half-duplex capable
13	1000BASE-T Full-Duplex	RO	Always 1	Always 1	1 = 1000BASE-T full-duplex capable
12	1000BASE-T Half-Duplex	RO	Always 1	Always 1	1 = 1000BASE-T half-duplex capable
11:0	Reserved	RO	0x000	0x000	000000000000

Table 104: Copper Specific Control Register 1
Page 0, Register 16

Bits	Field	Mode	HW Rst	SW Rst	Description
15	Disable Link Pulses	R/W	0x0	0x0	1 = Disable Link Pulse 0 = Enable Link Pulse
14:12	Downshift counter	R/W	0x3	Update	Changes to these bits are disruptive to the normal operation; therefore, any changes to these registers must be followed by software reset to take effect. 1x, 2x,...8x is the number of times the PHY attempts to establish Gigabit link before the PHY downshifts to the next highest speed. 000 = 1x 100 = 5x 001 = 2x 101 = 6x 010 = 3x 110 = 7x 011 = 4x 111 = 8x
11	Downshift Enable	R/W	0x0	Update	Changes to these bits are disruptive to the normal operation; therefore, any changes to these registers must be followed by software reset to take effect. 1 = Enable downshift. 0 = Disable downshift.
10	Force Copper Link Good	R/W	0x0	Retain	If link is forced to be good, the link state machine is bypassed and the link is always up. In 1000BASE-T mode this has no effect. 1 = Force link good 0 = Normal operation

Table 104: Copper Specific Control Register 1
Page 0, Register 16

Bits	Field	Mode	HW Rst	SW Rst	Description
9:7	Energy Detect	R/W	See Desc	Update	0xx = Off 100 = sense only on receiver (Energy Detect), auto wake-up, 101 = sense only on receiver (Energy Detect), SW wake-up 110 = sense and periodically transmit NLP (Energy Detect+TM), auto wake-up 111 = sense and periodically transmit NLP (Energy Detect+TM), SW wake-up
6:5	MDI Crossover Mode	R/W	11	Update	Changes to these bits are disruptive to the normal operation; therefore, any changes to these registers must be followed by a software reset to take effect. Bit 6 defaults to 0 and Bit 5 defaults to MDI_MDIX (the value sampled from RXD[3] during hardware configuration) for 88EA1517, otherwise bits 6 and 5 are both 1 by default 00 = Manual MDI configuration 01 = Manual MDIX configuration 10 = Reserved 11 = Enable automatic crossover for all modes
4	Energy Detect wake up control	R/W or RO, SC	0x0	0	This bit controls how the device wakes up from Energy detect state. If 16_0.7 = 1 (SW wake-up), this register bit is R/W. SW write of 1 to this bit will wake up the PHY. Then the bit will self clear after the device wakes up. 1 = Sleep 0 = Active If 16_0.7 = 0, this register bit is RO and reflects wake up status. The device will wake up from energy detect state automatically based on the energy detected from line.
3	Copper Transmitter Disable	R/W	0x0	Retain	1 = Transmitter Disable 0 = Transmitter Enable
2	Power Down	R/W	0x0	Retain	Power down is controlled via register 0_0.11 and 16_0.2. Both bits must be set to 0 before the PHY will transition from power down to normal operation. When the port is switched from power down to normal operation, software reset and restart Auto-Negotiation are performed even when bits Reset (0_0.15) and Restart Auto-Negotiation (0_0.9) are not set by the user. 1 = Power down 0 = Normal operation
1	Polarity Reversal Disable	R/W	0x0	Retain	If polarity is disabled, then the polarity is forced to be normal in 10BASE-T. 1 = Polarity Reversal Disabled 0 = Polarity Reversal Enabled The detected polarity status is shown in Register 17_0.1, or in 1000BASE-T mode, 21_5.3:0.
0	Disable Jabber	R/W	0x0	Retain	Jabber has effect only in 10BASE-T half-duplex mode. 1 = Disable jabber function 0 = Enable jabber function

Table 105: Copper Specific Status Register 1
Page 0, Register 17

Bits	Field	Mode	HW Rst	SW Rst	Description
15:14	Speed	RO	See Desc	Retain	These status bits are valid only after resolved bit 17_0.11 = 1. The resolved bit is set when Auto-Negotiation is completed or Auto-Negotiation is disabled. 11 = Reserved 10 = 1000 Mbps 01 = 100 Mbps 00 = 10 Mbps
13	Duplex	RO	0x0	Retain	This status bit is valid only after resolved bit 17_0.11 = 1. The resolved bit is set when Auto-Negotiation is completed or Auto-Negotiation is disabled. 1 = Full-duplex 0 = Half-duplex
12	Page Received	RO, LH	0x0	0x0	1 = Page received 0 = Page not received
11	Speed and Duplex Resolved	RO	0x0	0x0	When Auto-Negotiation is not enabled 17_0.11 = 1. 1 = Resolved 0 = Not resolved
10	Copper Link (real time)	RO	0x0	0x0	1 = Link up 0 = Link down
9	Transmit Pause Enabled	RO	0x0	0x0	This is a reflection of the MAC pause resolution. This bit is for information purposes and is not used by the device. This status bit is valid only after resolved bit 17_0.11 = 1. The resolved bit is set when Auto-Negotiation is completed or Auto-Negotiation is disabled. 1 = Transmit pause enabled 0 = Transmit pause disable
8	Receive Pause Enabled	RO	0x0	0x0	This is a reflection of the MAC pause resolution. This bit is for information purposes and is not used by the device. This status bit is valid only after resolved bit 17_0.11 = 1. The resolved bit is set when Auto-Negotiation is completed or Auto-Negotiation is disabled. 1 = Receive pause enabled 0 = Receive pause disabled
7	Reserved	RO	0x0	0x0	Reserved.
6	MDI Crossover Status	RO	See Desc	Retain	This status bit is valid only after resolved bit 17_0.11 = 1. The resolved bit is set when Auto-Negotiation is completed or Auto-Negotiation is disabled. This bit is 0 or 1 depending on what is written to 16.6:5 in manual configuration mode. Register 16.6:5 are updated with software reset. 1 = MDIX 0 = MDI
5	Downshift Status	RO	0x0	0x0	1 = Downshift 0 = No Downshift

Table 105: Copper Specific Status Register 1 (Continued)
Page 0, Register 17

Bits	Field	Mode	HW Rst	SW Rst	Description
4	Copper Energy Detect Status	RO	0x0	0x0	1 = Sleep 0 = Active
3	Global Link Status	RO	0x0	0x0	1 = Copper link is up 0 = Copper link is down
2	DTE power status	RO	0x0	0x0	1 = Link partner needs DTE power 0 = Link partner does not need DTE power
1	Polarity (real time)	RO	0x0	0x0	1 = Reversed 0 = Normal Polarity reversal can be disabled by writing to Register 16_0.1. In 1000BASE-T mode, polarity of all pairs are shown in Register 21_5.3:0.
0	Jabber (real time)	RO	0x0	0x0	1 = Jabber 0 = No jabber

Table 106: Copper Specific Interrupt Enable Register
Page 0, Register 18

Bits	Field	Mode	HW Rst	SW Rst	Description
15	Auto-Negotiation Error Interrupt Enable	R/W	0x0	Retain	1 = Interrupt enable 0 = Interrupt disable
14	Speed Changed Interrupt Enable	R/W	0x0	Retain	1 = Interrupt enable 0 = Interrupt disable
13	Duplex Changed Interrupt Enable	R/W	0x0	Retain	1 = Interrupt enable 0 = Interrupt disable
12	Page Received Interrupt Enable	R/W	0x0	Retain	1 = Interrupt enable 0 = Interrupt disable
11	Auto-Negotiation Completed Interrupt Enable	R/W	0x0	Retain	1 = Interrupt enable 0 = Interrupt disable
10	Link Status Changed Interrupt Enable	R/W	0x0	Retain	1 = Interrupt enable 0 = Interrupt disable
9	Symbol Error Interrupt Enable	R/W	0x0	Retain	1 = Interrupt enable 0 = Interrupt disable
8	False Carrier Interrupt Enable	R/W	0x0	Retain	1 = Interrupt enable 0 = Interrupt disable
7	WOL Event Interrupt Enable	R/W	0x0	Retain	1 = Interrupt enable 0 = Interrupt disable

Table 106: Copper Specific Interrupt Enable Register (Continued)
Page 0, Register 18

Bits	Field	Mode	HW Rst	SW Rst	Description
6	MDI Crossover Changed Interrupt Enable	R/W	0x0	Retain	1 = Interrupt enable 0 = Interrupt disable
5	Downshift Interrupt Enable	R/W	0x0	Retain	1 = Interrupt enable 0 = Interrupt disable
4	Copper Energy Detect Interrupt Enable	R/W	0x0	Retain	1 = Interrupt enable 0 = Interrupt disable
3	FLP Exchange Complete but no Link Interrupt Enable	R/W	0x0	Retain	1 = Interrupt enable 0 = Interrupt disable
2	DTE power detection status changed interrupt enable	R/W	0x0	Retain	1 = Interrupt enable 0 = Interrupt disable
1	Polarity Changed Interrupt Enable	R/W	0x0	Retain	1 = Interrupt enable 0 = Interrupt disable
0	Jabber Interrupt Enable	R/W	0x0	Retain	1 = Interrupt enable 0 = Interrupt disable

Table 107: Copper Interrupt Status Register
Page 0, Register 19

Bits	Field	Mode	HW Rst	SW Rst	Description
15	Copper Auto-Negotiation Error	RO,LH	0x0	0x0	An error is said to occur if MASTER/SLAVE does not resolve, parallel detect fault, no common HCD, or link does not come up after negotiation is completed. 1 = Auto-Negotiation Error 0 = No Auto-Negotiation Error
14	Copper Speed Changed	RO,LH	0x0	0x0	1 = Speed changed 0 = Speed not changed
13	Copper Duplex Changed	RO,LH	0x0	0x0	1 = Duplex changed 0 = Duplex not changed
12	Copper Page Received	RO,LH	0x0	0x0	1 = Page received 0 = Page not received
11	Copper Auto-Negotiation Completed	RO,LH	0x0	0x0	1 = Auto-Negotiation completed 0 = Auto-Negotiation not completed
10	Copper Link Status Changed	RO,LH	0x0	0x0	1 = Link status changed 0 = Link status not changed

Table 107: Copper Interrupt Status Register (Continued)
Page 0, Register 19

Bits	Field	Mode	HW Rst	SW Rst	Description
9	Copper Symbol Error	RO,LH	0x0	0x0	1 = Symbol error 0 = No symbol error
8	Copper False Carrier	RO,LH	0x0	0x0	1 = False carrier 0 = No false carrier
7	WOL Event Happened	RO, LH	0x0	0x0	1 = WOL event happened 0 = WOL event did not happen
6	MDI Crossover Changed	RO,LH	0x0	0x0	1 = Crossover changed 0 = Crossover not changed
5	Downshift Inter-rupt	RO,LH	0x0	0x0	1 = Downshift detected 0 = No down shift
4	Copper Energy Detect Changed	RO,LH	0x0	0x0	1 = Energy Detect state changed 0 = No Energy Detect state change detected
3	FLP Exchange Complete but no Link	RO,LH	0x0	0x0	1 = FLP Exchange Completed but Link Not Established 0 = No Event Detected
2	DTE power detection status changed interrupt	RO,LH	0x0	0x0	1 = DTE power detection status changed 0 = No DTE power detection status change detected
1	Polarity Changed	RO,LH	0x0	0x0	1 = Polarity Changed 0 = Polarity not changed
0	Jabber	RO,LH	0x0	0x0	1 = Jabber 0 = No jabber

Table 108: Copper Specific Control Register 2
Page 0, Register 20

Bits	Field	Mode	HW Rst	SW Rst	Description
15	Transmitter Always On Enable	R/W	0	Retain	1 = Transmitter is always on 0 = Transmitter turns off in between FLP burst transmissions during Auto-Negotiation to save power.
14	TX FIFO Low Latency Enable	R/W	0	Retain	1 = Enable TX FIFO low latency mode- packet size is max 1518 bytes 0 = 16_2.15:14 control the FIFO depth
13	Phase FIFO Enable	R/W	0	Update	1 = Enable Phase FIFO for TX FIFO 0 = TX FIFO handles PPM difference
12:9	Reserved	R/W	0	Retain	Reserved
8	10BASE-T Low Power Mode Enable	R/W	0	Retain	Enables 10BT Low Power Mode when the EEE Buffer is used. 1 = Enable 10BT Low Power Mode 0 = Disable 10BT Low Power Mode

Table 108: Copper Specific Control Register 2 (Continued)
Page 0, Register 20

Bits	Field	Mode	HW Rst	SW Rst	Description
7	10BASE-Te Enable	R/W	0	Retain	Enables 10BT EEE Low Power Mode. 0 = Disable 10BASE-Te 1 = Enable 10BASE-Te
6	Break Link On Insufficient IPG	R/W	0x0	Retain	0 = Break link on insufficient IPGs in 10BASE-T and 100BASE-TX 1 = Do not break link on insufficient IPGs in 10BASE-T and 100BASE-TX
5	100 BASE-T Transmitter Clock Source	R/W	0x1	Update	1 = Local Clock 0 = Recovered Clock
4	Accelerate 100BASE-T Link Up	R/W	0x0	Retain	0 = No Acceleration 1 = Accelerate
3	Reverse MDIP/ N[3] Transmit Polarity	R/W	0x0	Retain	0 = Normal Transmit Polarity 1 = Reverse Transmit Polarity
2	Reverse MDIP/ N[2] Transmit Polarity	R/W	0x0	Retain	0 = Normal Transmit Polarity 1 = Reverse Transmit Polarity
1	Reverse MDIP/ N[1] Transmit Polarity	R/W	0x0	Retain	0 = Normal Transmit Polarity 1 = Reverse Transmit Polarity
0	Reverse MDIP/ N[0] Transmit Polarity	R/W	0x0	Retain	0 = Normal Transmit Polarity 1 = Reverse Transmit Polarity

Table 109: Copper Specific Receive Error Counter Register
Page 0, Register 21

Bits	Field	Mode	HW Rst	SW Rst	Description
15:0	Receive Error Count	RO, LH	0x0000	Retain	Counter will peg at 0xFFFF and will not roll over. Both False carrier and symbol errors are reported.

Table 110: Page Address
Page Any, Register 22

Bits	Field	Mode	HW Rst	SW Rst	Description
15:8	Reserved	RO	0x00	0x00	All 0's
7:0	Page select for registers 0 to 28	R/W	0x00	Retain	Page Number

Table 111: Global Interrupt Status
Page 0, Register 23

Bits	Field	Mode	HW Rst	SW Rst	Description
15:1	Reserved	RO	0x0000	0x0000	Reserved.
0	Interrupt	RO	0x0	0x0	1 = Interrupt active on port X 0 = No interrupt active on port X

Table 112: Copper Specific Control Register 3
Page 0, Register 26

Bits	Field	Mode	HW Rst	SW Rst	Description
15	Reserved	R/W	0	Retain	Reserved.
14	Disable 1000BASE-T	R/W	See Desc	Retain	When set to disabled, 1000BASE-T will not be advertised even if registers 9.9 or 9.8 are set to 1. A write to this register bit does not take effect until any one of the following occurs: Software reset is asserted (Register 0.15) Restart Auto-Negotiation is asserted (Register 0.9) Power down (Register 0.11, 16_0.2) transitions from power down to normal operation Copper link goes down. This bit is a 1 always (a write of 0 has no effect) for 88EA1517 and when MII mode is used for either package, otherwise this bit defaults to 0 and may be written to 1. 1 = Disable 1000BASE-T Advertisement 0 = Enable 1000BASE-T Advertisement
13	Reverse Auto-Neg	R/W	0x0	Retain	A write to this register bit does not take effect until any one of the following occurs: Software reset is asserted (Register 0.15) Restart Auto-Negotiation is asserted (Register 0.9) Power down (Register 0.11, 16_0.2) transitions from power down to normal operation Copper link goes down. 1 = Reverse Auto-Negotiation 0 = Normal Auto-Negotiation
12	Disable 100BASE-T	R/W	0	Retain	When set to disabled, 100BASE-TX will not be advertised even if registers 4_0.8 or 4_0.7 are set 1. A write to this register bit does not take effect until any one of the following occurs: Software reset is asserted (Register 0_0.15) Restart Auto-Negotiation is asserted (Register 0_0.9) Power down (Register 0_0.11, 16_0.2) transitions from power down to normal operation Copper link goes down. 1 = Disable 100BASE-TX Advertisement 0 = Enable 100BASE-TX Advertisement

Table 112: Copper Specific Control Register 3 (Continued)
Page 0, Register 26

Bits	Field	Mode	HW Rst	SW Rst	Description
11:10	Gigabit Link Down Delay	R/W	0x0	Retain	This register only have effect if register 26_0.9 is set to 1. 00 = 0ms 01 = 10 ± 2 ms 10 = 20 ± 2 ms 11 = 40 ± 2 ms
9	Speed Up Gigabit Link Down Time	R/W	0x0	Retain	1 = Enable faster gigabit link down 0 = Use IEEE gigabit link down
8	DTE detect enable	R/W	0x0	Update	1 = Enable DTE detection 0 = Disable DTE detection
7:4	DTE detect status drop hysteresis	R/W	0x4	Retain	0000: Report immediately 0001: Report 5s after DTE power status drop .... 1111: Report 75s after DTE power status drop
3:2	100 MB test select	R/W	0x0	Retain	0x = Normal Operation 10 = Select 112 ns sequence 11 = Select 16 ns sequence
1	10 BT polarity force	R/W	0x0	Retain	1 = Force negative polarity for Receive only 0 = Normal Operation
0	Force 1G Mode	R/W	See Desc	Retain	This bit allows for bypassing auto-negotiation and allowing 1G link up much sooner. 1 = Force 1G link up, bypassing auto-negotiation 0 = Have 1G link happen with auto-negotiation

Table 113: Fiber Control Register
Page 1, Register 0

Bits	Field	Mode	HW Rst	SW Rst	Description
15	Fiber Reset	R/W	0x0	SC	Fiber Software Reset. Affects page 1. Writing a 1 to this bit causes the PHY state machines to be reset. When the reset operation is done, this bit is cleared to 0 automatically. The reset occurs immediately. 1 = PHY reset 0 = Normal operation
14	Loopback	R/W	0x0	0x0	When loopback is activated, the transmitter data presented on TXD of the internal bus is looped back to RXD of the internal bus. Link is broken when loopback is enabled. Loopback speed is determined by the mode the device is in. 1000BASE-X - loopback is always in 1000Mbps. 100BASE-FX - loopback is always in 100Mbps. 1 = Enable Loopback 0 = Disable Loopback

Table 113: Fiber Control Register (Continued)
Page 1, Register 0

Bits	Field	Mode	HW Rst	SW Rst	Description
13	Speed Select (LSB)	RO, R/W	0x0	Retain	If register 16_1.1:0 (MODE[1:0]) = 00 then this bit is always 1. If register 16_1.1:0 (MODE[1:0]) = 01 then this bit is always 0. If register 16_1.1:0 (MODE[1:0]) = 10 then this bit is 1 when the PHY is at 100Mb/s, else it is 0. If register 16_1.1:0 (MODE[1:0]) = 11 then this bit is R/W. bit 6,13 10 = 1000 Mbps 01 = 100 Mbps 00 = 10 Mbps
12	Auto-Negotiation Enable	R/W	See Descr	Retain	If the value of this bit is changed, the link will be broken and Auto-Negotiation Restarted This bit has no effect when in 100BASE-FX mode When this bit gets set/reset, Auto-negotiation is restarted (bit 0_1.9 is set to 1). On hardware reset this bit takes on the value of S_ANEG 1 = Enable Auto-Negotiation Process 0 = Disable Auto-Negotiation Process
11	Power Down	R/W	0x0	0x0	When the port is switched from power down to normal operation, software reset and restart Auto-Negotiation are performed even when bits Reset (0_1.15) and Restart Auto-Negotiation (0_1.9) are not set by the user. On hardware reset, bit 0_1.11 1 = Power down 0 = Normal operation
10	Isolate	RO	0x0	0x0	This function is not supported
9	Restart Fiber Auto-Negotiation	R/W, SC	0x0	SC	Auto-Negotiation automatically restarts after hardware, software reset (0_1.15) or change in Auto-Negotiation enable (0_1.12) regardless of whether or not the restart bit (0_1.9) is set. The bit is set when Auto-negotiation is Enabled or Dis-abled in 0_1.12 1 = Restart Auto-Negotiation Process 0 = Normal operation
8	Duplex Mode	R/W	0x1	Retain	Writing this bit has no effect unless one of the following events occur: Software reset is asserted (Register 0_1.15) Restart Auto-Negotiation is asserted (Register 0_1.9) Auto-Negotiation Enable changes (Register 0_1.12) Power down (Register 0_1.11) transitions from power down to normal operation 1 = Full-duplex 0 = Half-duplex
7	Collision Test	RO	0x0	0x0	This bit has no effect.

Table 113: Fiber Control Register (Continued)
Page 1, Register 0

Bits	Field	Mode	HW Rst	SW Rst	Description
6	Speed Selection (MSB)	RO, R/W	0x1	Retain	If register 16_1.1:0 (MODE[1:0]) = 00 then this bit is always 0. If register 16_1.1:0 (MODE[1:0]) = 01 then this bit is always 1. If register 16_1.1:0 (MODE[1:0]) = 10 then this bit is 1 when the PHY is at 1000Mb/s, else it is 0. If register 16_1.1:0 (MODE[1:0]) = 11 then this bit is R/W. bit 6,13 10 = 1000 Mbps 01 = 100 Mbps 00 = 10 Mbps
5:0	Reserved	RO	Always 000000	Always 000000	Always 0.

Table 114: Fiber Status Register
Page 1, Register 1

Bits	Field	Mode	HW Rst	SW Rst	Description
15	100BASE-T4	RO	Always 0	Always 0	100BASE-T4. This protocol is not available. 0 = PHY not able to perform 100BASE-T4
14	100BASE-X Full-Duplex	RO	See Descr	See Descr	If register 16_1.1:0 (MODE[1:0]) = 00 then this bit is 1, else this bit is 0. bit 6,13 1 = PHY able to perform full duplex 100BASE-X 0 = PHY not able to perform full duplex 100BASE-X
13	100BASE-X Half-Duplex	RO	See Descr	See Descr	If register 16_1.1:0 (MODE[1:0]) = 00 then this bit is 1, else this bit is 0. bit 6,13 1 = PHY able to perform half-duplex 100BASE-X 0 = PHY not able to perform half-duplex 100BASE-X
12	10 Mbps Full-Duplex	RO	Always 0	Always 0	0 = PHY not able to perform full-duplex 10BASE-T
11	10 Mbps Half-Duplex	RO	Always 0	Always 0	0 = PHY not able to perform half-duplex 10BASE-T
10	100BASE-T2 Full-Duplex	RO	Always 0	Always 0	This protocol is not available. 0 = PHY not able to perform full-duplex
9	100BASE-T2 Half-Duplex	RO	Always 0	Always 0	This protocol is not available. 0 = PHY not able to perform half-duplex
8	Extended Status	RO	Always 1	Always 1	1 = Extended status information in Register 15
7	Reserved	RO	Always 0	Always 0	Must always be 0.

Table 114: Fiber Status Register (Continued)
Page 1, Register 1

Bits	Field	Mode	HW Rst	SW Rst	Description
6	MF Preamble Suppression	RO	Always 1	Always 1	1 = PHY accepts management frames with preamble suppressed
5	Fiber Auto-Negotiation Complete	RO	0x0	0x0	1 = Auto-Negotiation process complete 0 = Auto-Negotiation process not complete Bit is not set when link is up due of Fiber Auto-negotiation Bypass or if Auto-negotiation is disabled.
4	Fiber Remote Fault	RO,LH	0x0	0x0	1 = Remote fault condition detected 0 = Remote fault condition not detected This bit is always 0 in SGMII modes.
3	Auto-Negotiation Ability	RO	See Descr	See Descr	If register 16_1.1:0 (MODE[1:0]) = 00 then this bit is 0, else this bit is 1. bit 6,13 1 = PHY able to perform Auto-Negotiation 0 = PHY not able to perform Auto-Negotiation
2	Fiber Link Status	RO,LL	0x0	0x0	This register bit indicates when link was lost since the last read. For the current link status, either read this register back-to-back or read Register 17_1.10 Link Real Time. 1 = Link is up 0 = Link is down
1	Reserved	RO,LH	Always 0	Always 0	Must be 0
0	Extended Capability	RO	Always 1	Always 1	1 = Extended register capabilities

Table 115: PHY Identifier
Page 1, Register 2

Bits	Field	Mode	HW Rst	SW Rst	Description
15:0	Organizationally Unique Identifier Bit 3:18	RO	0x0141	0x0141	Marvell® OUI is 0x005043 0000 0000 0101 0000 0100 0011 ^ ^ bit 1...................................bit 24 Register 2.[15:0] show bits 3 to 18 of the OUI. 0000000101000001 ^ ^ bit 3..................bit18

Table 116: PHY Identifier
Page 1, Register 3

Bits	Field	Mode	HW Rst	SW Rst	Description
15:10	OUI Lsb	RO	Always 000011	Always 000011	Organizationally Unique Identifier bits 19:24 000011 ^.........^ bit 19...bit24
9:4	Model Number	RO	Always 011101	Always 011101	Model Number 011101
3:0	Revision Number	RO	Always 0000	0x0	Rev Number = 0000

Table 117: Fiber Auto-Negotiation Advertisement Register - 1000BASE-X Mode (Register 16_1.1:0 = 01)
Page 1, Register 4

Bits	Field	Mode	HW Rst	SW Rst	Description
15	Next Page	R/W	0x0	Retain	A write to this register bit does not take effect until any one of the following occurs: Software reset is asserted (Register 0_1.15) Restart Auto-Negotiation is asserted (Register 0_1.9) Power down (Register 0_1.11) transitions from power down to normal operation Link goes down 1 = Advertise 0 = Not advertised
14	Reserved	RO	Always 0	Always 0	Reserved
13:12	Remote Fault 2/ Remote Fault 1	R/W	0x0	Retain	A write to this register bit does not take effect until any one of the following also occurs: Software reset is asserted (Register 0_1.15) Re-start Auto-Negotiation is asserted (Register 0_1.9) Power down (Register 0_1.11) transitions from power down to normal operation Link goes down Device has no ability to detect remote fault. 00 = No error, link OK (default) 01 = Link Failure 10 = Offline 11 = Auto-Negotiation Error
11:9	Reserved	RO	Always 000	Always 000	Reserved

Table 117: Fiber Auto-Negotiation Advertisement Register - 1000BASE-X Mode (Register 16_1.1:0 = 01) (Continued)
Page 1, Register 4

Bits	Field	Mode	HW Rst	SW Rst	Description
8:7	Pause	R/W	See Descr.	Retain	A write to this register bit does not take effect until any one of the following also occurs: Software reset is asserted (Register 0_1.15) Re-start Auto-Negotiation is asserted (Register 0_1.9) Power down (Register 0_1.11) transitions from power down to normal operation Link goes down Upon hardware reset both bits takes on the value of ENA_PAUSE. 00 = No PAUSE 01 = Symmetric PAUSE 10 = Asymmetric PAUSE toward link partner 11 = Both Symmetric PAUSE and Asymmetric PAUSE toward local device.
6	1000BASE-X Half-Duplex	R/W	See Descr.	Retain	A write to this register bit does not take effect until any one of the following also occurs: Software reset is asserted (Register 0_1.15) Re-start Auto-Negotiation is asserted (Register 0_1.9) Power down (Register 0_1.11) transitions from power down to normal operation Link goes down Upon hardware reset this bit takes on the value of C_ANEG[0]. 1 = Advertise 0 = Not advertised
5	1000BASE-X Full-Duplex	R/W	0x1	Retain	A write to this register bit does not take effect until any one of the following also occurs: Software reset is asserted (Register 0_1.15) Re-start Auto-Negotiation is asserted (Register 0_1.9) Power down (Register 0_1.11) transitions from power down to normal operation Link goes down 1 = Advertise 0 = Not advertised
4:0	Reserved	R/W	0x00	0x00	Reserved

Table 118: Fiber Auto-Negotiation Advertisement Register - SGMII (System mode) (Register 16_1.1:0 = 10)
Page 1, Register 4

Bits	Field	Mode	HW Rst	SW Rst	Description
15	Link Status	RO	0x0	0x0	0 = Link is Not up on the Copper Interface 1 = Link is up on the Copper Interface
14	Reserved	RO	Always 0	Always 0	Reserved
13	Reserved	RO	Always 0	Always 0	Reserved
12	Duplex Status	RO	0x0	0x0	0 = Interface Resolved to Half-duplex 1 = Interface Resolved to Full-duplex

Table 118: Fiber Auto-Negotiation Advertisement Register - SGMII (System mode) (Register 16_1.1:0 = 10) (Continued)
Page 1, Register 4

Bits	Field	Mode	HW Rst	SW Rst	Description
11:10	Speed[1:0]	RO	0x0	0x0	00 = Interface speed is 10 Mbps 01 = Interface speed is 100 Mbps 10 = Interface speed is 1000 Mbps 11 = Reserved
9	Transmit Pause	RO	0x0	0x0	Note that if register 16_1.7 is set to 0 then this bit is always forced to 0. 0 = Disabled, 1 = Enabled
8	Receive Pause	RO	0x0	0x0	Note that if register 16_1.7 is set to 0 then this bit is always forced to 0. 0 = Disabled, 1 = Enabled
7	Fiber/Copper	RO	0x0	0x0	Note that if register 16_1.7 is set to 0 then this bit is always forced to 0. 0 = Copper media, 1 = Fiber media
6:0	Reserved	RO	Always 0000001	Always 0000001	Reserved

Table 119: Fiber Auto-Negotiation Advertisement Register - SGMII (Media mode) (Register 16_1.1:0 = 11)
Page 1, Register 4

Bits	Field	Mode	HW Rst	SW Rst	Description
15:0	Reserved	RO	Always 0x0001	Always 0x0001	Reserved

Table 120: Fiber Link Partner Ability Register - 1000BASE-X Mode (Register 16_1.1:0 = 01)
Page 1, Register 5

Bits	Field	Mode	HW Rst	SW Rst	Description
15	Next Page	RO	0x0	0x0	Register bit is cleared when link goes down and loaded when a base page is received Received Code Word Bit 15 1 = Link partner capable of next page 0 = Link partner not capable of next page
14	Acknowledge	RO	0x0	0x0	Register bit is cleared when link goes down and loaded when a base page is received Acknowledge Received Code Word Bit 14 1 = Link partner received link code word 0 = Link partner has not received link code word

Table 120: Fiber Link Partner Ability Register - 1000BASE-X Mode (Register 16_1.1:0 = 01) Page 1, Register 5

Bits	Field	Mode	HW Rst	SW Rst	Description
13:12	Remote Fault 2/ Remote Fault 1	RO	0x0	0x0	Register bit is cleared when link goes down and loaded when a base page is received Received Code Word Bit 13:12 00 = No error, link OK (default) 01 = Link Failure 10 = Offline 11 = Auto-Negotiation Error
11:9	Reserved	RO	0x0	0x0	Register bit is cleared when link goes down and loaded when a base page is received Received Code Word Bit 11:9
8:7	Asymmetric Pause	RO	0x0	0x0	Register bit is cleared when link goes down and loaded when a base page is received Received Code Word Bit 8:7 00 = No PAUSE 01 = Symmetric PAUSE 10 = Asymmetric PAUSE toward link partner 11 = Both Symmetric PAUSE and Asymmetric PAUSE toward local device.
6	1000BASE-X Half-Duplex	RO	0x0	0x0	Register bit is cleared when link goes down and loaded when a base page is received Received Code Word bit 6 1 = Link partner capable of 1000BASE-X half-duplex. 0 = Link partner not capable of 1000BASE-X half-duplex.
5	1000BASE-X Full-Duplex	RO	0x0	0x0	Register bit is cleared when link goes down and loaded when a base page is received Received Code Word bit 5 1 = Link partner capable of 1000BASE-X full-duplex. 0 = Link partner not capable of 1000BASE-X full-duplex.
4:0	Reserved	RO	0x00	0x00	Register bit is cleared when link goes down and loaded when a base page is received Received Code Word Bits 4:0 Must be 0

Table 121: Fiber Link Partner Ability Register - SGMII (System mode) (Register 16_1.1:0 = 10)
Page 1, Register 5

Bits	Field	Mode	HW Rst	SW Rst	Description
15	Reserved	RO	0x0	0x0	Must be 0
14	Acknowledge	RO	0x0	0x0	Acknowledge Register bit is cleared when link goes down and loaded when a base page is received Received Code Word Bit 14 1 = Link partner received link code word 0 = Link partner has not received link code word
13:0	Reserved	RO	0x0000	0x0000	Received Code Word Bits 13:0 Must receive 00_0000_0000_0001 per SGMII spec

Table 122: Fiber Link Partner Ability Register - SGMII (Media mode) (Register 16_1.1:0 = 11)
Page 1, Register 5

Bits	Field	Mode	HW Rst	SW Rst	Description
15	Link	RO	0x0	0x0	Register bit is cleared when link goes down and loaded when a base page is received Received Code Word Bit 15 1 = Copper Link is up on the link partner 0 = Copper Link is not up on the link partner
14	Acknowledge	RO	0x0	0x0	Register bit is cleared when link goes down and loaded when a base page is received Acknowledge Received Code Word Bit 14 1 = Link partner received link code word 0 = Link partner has not received link code word
13	Reserved	RO	0x0	0x0	Register bit is cleared when link goes down and loaded when a base page is received Received Code Word Bit 13 Must be 0
12	Duplex Status	RO	0x0	0x0	Register bit is cleared when link goes down and loaded when a base page is received Received Code Word Bit 12 1 = Copper Interface on the link Partner is capable of Full Duplex 0 = Copper Interface on the link partner is capable of Half Duplex
11:10	Speed Status	RO	0x0	0x0	Register bits are cleared when link goes down and loaded when a base page is received Received Code Word Bit 11:10 00 = 10 Mbps 01 = 100 Mbps 10 = 1000 Mbps 11 = reserved

Table 122: Fiber Link Partner Ability Register - SGMII (Media mode) (Register 16_1.1:0 = 11)
Page 1, Register 5

Bits	Field	Mode	HW Rst	SW Rst	Description
9	Transmit Pause Status	RO	0x0	0x0	This bit is non-zero only if the link partner supports enhanced SGMII auto negotiation. Received Code Word Bit 9 0 = Disabled, 1 = Enabled
8	Receive Pause Status	RO	0x0	0x0	This bit is non-zero only if the link partner supports enhanced SGMII auto negotiation. Received Code Word Bit 8 0 = Disabled, 1 = Enabled
7	Fiber/Copper Status	RO	0x0	0x0	This bit is non-zero only if the link partner supports enhanced SGMII auto negotiation. Received Code Word Bit 7 0 = Copper media, 1 = Fiber media
6:0	Reserved	RO	0x00	0x00	Register bits are cleared when link goes down and loaded when a base page is received Received Code Word Bits 6:0 Must be 0000001

Table 123: Fiber Auto-Negotiation Expansion Register
Page 1, Register 6

Bits	Field	Mode	HW Rst	SW Rst	Description
15:4	Reserved	RO	0x000	0x000	Reserved. Must be 00000000000.
3	Link Partner Next page Able	RO	0x0	0x0	SGMII and 100BASE-FX modes this bit is always 0. In 1000BASE-X mode register 6_1.3 is set when a base page is received and the received link control word has bit 15 set to 1. The bit is cleared when link goes down. 1 = Link Partner is Next Page able 0 = Link Partner is not Next Page able
2	Local Next Page Able	RO	Always 1	Always 1	1 = Local Device is Next Page able
1	Page Received	RO, LH	0x0	0x0	Register 6_1.1 is set when a valid page is received. 1 = A New Page has been received 0 = A New Page has not been received
0	Link Partner Auto-Negotiation Able	RO	0x0	0x0	This bit is set when there is sync status, the fiber receiver has received 3 non-zero matching valid configuration code groups and Auto-negotiation is enabled in register 0_1.12 1 = Link Partner is Auto-Negotiation able 0 = Link Partner is not Auto-Negotiation able

Table 124: Fiber Next Page Transmit Register
Page 1, Register 7

Bits	Field	Mode	HW Rst	SW Rst	Description
15	Next Page	R/W	0x0	0x0	A write to register 7_1 implicitly sets a variable in the Auto-Negotiation state machine indicating that the next page has been loaded. Register 7_1 only has effect in the 1000BASE-X mode. Transmit Code Word Bit 15
14	Reserved	RO	0x0	0x0	Transmit Code Word Bit 14
13	Message Page Mode	R/W	0x1	0x1	Transmit Code Word Bit 13
12	Acknowledge2	R/W	0x0	0x0	Transmit Code Word Bit 12
11	Toggle	RO	0x0	0x0	Transmit Code Word Bit 11. This bit is internally set to the opposite value each time a page is received
10:0	Message/ Unformatted Field	R/W	0x001	0x001	Transmit Code Word Bit 10:0

Table 125: Fiber Link Partner Next Page Register
Page 1, Register 8

Bits	Field	Mode	HW Rst	SW Rst	Description
15	Next Page	RO	0x0	0x0	Register 8_1 only has effect in the 1000BASE-X mode. The register is loaded only when a next page is received from the link partner. It is cleared each time the link goes down. Received Code Word Bit 15
14	Acknowledge	RO	0x0	0x0	Received Code Word Bit 14
13	Message Page	RO	0x0	0x0	Received Code Word Bit 13
12	Acknowledge2	RO	0x0	0x0	Received Code Word Bit 12
11	Toggle	RO	0x0	0x0	Received Code Word Bit 11
10:0	Message/ Unformatted Field	RO	0x000	0x000	Received Code Word Bit 10:0

Table 126: Extended Status Register
Page 1, Register 15

Bits	Field	Mode	HW Rst	SW Rst	Description
15	1000BASE-X Full-Duplex	RO	See Descr	See Descr	If Register 16_1.1:0 (MODE[1:0]) = 00 then this bit is 0, else this bit is 1. 1 = 1000BASE-X full-duplex capable 0 = Not 1000BASE-X full-duplex capable
14	1000BASE-X Half-Duplex	RO	See Descr	See Descr	If Register 16_1.1:0 (MODE[1:0]) = 00 then this bit is 0, else this bit is 1. 1 = 1000BASE-X half-duplex capable 0 = Not 1000BASE-X half-duplex capable
13	1000BASE-T Full-Duplex	RO	0x0	0x0	0 = Not 1000BASE-T full-duplex capable
12	1000BASE-T Half-Duplex	RO	0x0	0x0	0 = Not 1000BASE-T half-duplex capable
11:0	Reserved	RO	0x000	0x000	000000000000

Table 127: Fiber Specific Control Register 1
Page 1, Register 16

Bits	Field	Mode	HW Rst	SW Rst	Description
15:14	Fiber Transmit FIFO Depth	R/W	0x1	Retain	00 = ± 16 Bits 01 = ± 24 Bits 10 = ± 32 Bits 11 = ± 40 Bits
13	Block Carrier Extension Bit	R/W	0x0	Retain	Carrier extension and carrier extension with error are converted to idle symbols on the RXD only during full duplex mode. 1 = Enable Block Carrier Extension 0 = Disable Block Carrier Extension
12	SERDES Loop-back	R/W	0x0	0x0	Register 16_1.8 selects the line loopback path. 1 = Enable loopback from SERDES input to SERDES output 0 = Normal Operation
11	Assert CRS on Transmit	R/W	0x0	Retain	This bit has no effect in full-duplex. 1 = Assert on transmit 0 = Never assert on transmit
10	Force Link Good	R/W	0x0	Retain	If link is forced to be good, the link state machine is bypassed and the link is always up. 1 = Force link good 0 = Normal operation
9	Reserved	R/W	0x0	Retain	Set to 0.

Table 127: Fiber Specific Control Register 1 (Continued)
Page 1, Register 16

Bits	Field	Mode	HW Rst	SW Rst	Description
8	SERDES Loop-back Type	R/W	0x0	Retain	0 = Loopback Through PCS (Tx and Rx can be asynchronous) 1 = Loopback raw 10 bit data (Tx and Rx must be synchronous)
7:6	Enhanced SGMII	R/W	0x0	Update	00 = Do not pass flow control bits through SGMII Auto-Negotiation. SGMII (System mode) registers 4_1.9:7 are always 000. EEE bits are passed per IEEE. 01 = Pass flow control bits through SGMII Auto-Negotiation using 4_1.9:7. EEE bits are passed in alternate locations. 10 = Pass flow control bits through SGMII Auto-Negotiation using bits other than 4_1.9:7. EEE bits use 9:8. 11 = Reserved
5	Marvell Remote Fault Indication Enable	R/W	0x0	Retain	0 = Disable 1 = Enable, Remote Fault is indicated to link partner in less than 2 ms, only one bit of bit 5:4 can be set to 1
4	IEEE Remote Fault Indication Enable	R/W	0x0	Retain	0 = Disable 1 = Enable, Remote Fault is indicated to link partner after 20ms according to IEEE standard, only one bit of bit 5:4 can be set to 1
3	Reserved	R/W	0x1	Update	Reserved.
2	Interrupt Polarity	R/W	0x1	Retain	1 = INTn active low 0 = INTn active high
1:0	MODE[1:0]	RO	See Desc.	See Desc.	These bits reflects the mode as programmed in register of 20_18.2:0 00 = 100BASE-FX 01 = 1000BASE-X 10 = SGMII System mode 11 = SGMII Media mode

Table 128: Fiber Specific Status Register
Page 1, Register 17

Bits	Field	Mode	HW Rst	SW Rst	Description
15:14	Speed	RO	0x0	Retain	These status bits are valid only after resolved bit 17_1.11 = 1. The resolved bit is set when Auto-Negotiation is completed or Auto-Negotiation is disabled. In 100BASE-FX mode this bit is always 01. 11 = Reserved 10 = 1000 Mbps 01 = 100 Mbps 00 = 10 Mbps

Table 128: Fiber Specific Status Register (Continued)
Page 1, Register 17

Bits	Field	Mode	HW Rst	SW Rst	Description
13	Duplex	RO	0x0	Retain	This status bit is valid only after resolved bit 17_1.11 = 1. The resolved bit is set when Auto-Negotiation is completed or Auto-Negotiation is disabled. In 100BASE-FX mode this bit follows register 0_1.8. 1 = Full-duplex 0 = Half-duplex
12	Page Received	RO, LH	0x0	0x0	In 100BASE-FX mode this bit is always 0. 1 = Page received 0 = Page not received
11	Speed and Duplex Resolved	RO	0x0	0x0	When Auto-Negotiation is not enabled or in 100BASE-FX mode this bit is always 1. 1 = Resolved 0 = Not resolved If bit 26_1.5 is 1, then this bit will be 0.
10	Link (real time)	RO	0x0	0x0	1 = Link up 0 = Link down
9:8	Reserved	RO	Always 00	Always 00	Reserved.
7:6	Remote Fault Received	RO, LH	0x0	0x0	The mapping for this status is as follows: 00 = No Fault 01 = Link Failure detected at link partner 10 = Offline 11 = Auto-neg Error
5	Sync status	RO	0x0	0x0	1 = Sync 0 = No Sync
4	Fiber Energy Detect Status	RO	0x1	0x1	1 = No energy detected 0 = Energy Detected
3	Transmit Pause Enabled	RO	0x0	0x0	This is a reflection of the MAC pause resolution. This bit is for information purposes and is not used by the device. This status bit is valid only after resolved bit 17_1.11 = 1. The resolved bit is set when Auto-Negotiation is completed or Auto-Negotiation is disabled. In 100BASE-FX mode this bit is always 0. 1 = Transmit pause enabled 0 = Transmit pause disable
2	Receive Pause Enabled	RO	0x0	0x0	This is a reflection of the MAC pause resolution. This bit is for information purposes and is not used by the device. This status bit is valid only after resolved bit 17_1.11 = 1. The resolved bit is set when Auto-Negotiation is completed or Auto-Negotiation is disabled. In 100BASE-FX mode this bit is always 0. 1 = Receive pause enabled 0 = Receive pause disabled
1:0	Reserved	RO	Always 00	Always 00	Reserved.

Table 129: Fiber Interrupt Enable Register
Page 1, Register 18

Bits	Field	Mode	HW Rst	SW Rst	Description
15	Reserved	RO	Always 0	Always 0	Reserved.
14	Speed Changed Interrupt Enable	R/W	0x0	Retain	1 = Interrupt enable 0 = Interrupt disable
13	Duplex Changed Interrupt Enable	R/W	0x0	Retain	1 = Interrupt enable 0 = Interrupt disable
12	Page Received Interrupt Enable	R/W	0x0	Retain	1 = Interrupt enable 0 = Interrupt disable
11	Auto-Negotiation Completed Interrupt Enable	R/W	0x0	Retain	1 = Interrupt enable 0 = Interrupt disable
10	Link Status Changed Interrupt Enable	R/W	0x0	Retain	1 = Interrupt enable 0 = Interrupt disable
9	Symbol Error Interrupt Enable	R/W	0x0	Retain	1 = Interrupt enable 0 = Interrupt disable
8	False Carrier Interrupt Enable	R/W	0x0	Retain	1 = Interrupt enable 0 = Interrupt disable
7	Fiber FIFO Over/ Underflow Interrupt Enable	R/W	0x0	Retain	1 = Interrupt enable 0 = Interrupt disable
6	Reserved	RO	Always 0	Always 0	Reserved.
5	Remote Fault Receive Interrupt Enable	R/W	0x0	0x0	1 = Interrupt enable 0 = Interrupt disable
4	Fiber Energy Detect Interrupt Enable	R/W	0x0	Retain	1 = Interrupt enable 0 = Interrupt disable
3:0	Reserved	RO	Always 0000	Always 0000	0000

Table 130: Fiber Interrupt Status Register
Page 1, Register 19

Bits	Field	Mode	HW Rst	SW Rst	Description
15	Reserved	RO	Always 0	Always 0	0
14	Speed Changed	RO,LH	0x0	0x0	1 = Speed changed 0 = Speed not changed

Table 130: Fiber Interrupt Status Register (Continued)
Page 1, Register 19

Bits	Field	Mode	HW Rst	SW Rst	Description
13	Duplex Changed	RO,LH	0x0	0x0	1 = Duplex changed 0 = Duplex not changed
12	Page Received	RO,LH	0x0	0x0	1 = Page received 0 = Page not received
11	Auto-Negotiation Completed	RO,LH	0x0	0x0	1 = Auto-Negotiation completed 0 = Auto-Negotiation not completed
10	Link Status Changed	RO,LH	0x0	0x0	1 = Link status changed 0 = Link status not changed
9	Symbol Error	RO,LH	0x0	0x0	1 = Symbol error 0 = No symbol error
8	False Carrier	RO,LH	0x0	0x0	1 = False carrier 0 = No false carrier
7	Fiber FIFO Over/ Underflow	RO,LH	0x0	0x0	1 = Over/Underflow Error 0 = No FIFO Error
6	Reserved	RO	0x0	0x0	Reserved.
5	Remote Fault Receive Interrupt Enable	RO, LH	0x0	0x0	1 = Remote Fault received changed, read 1.17.7:6 for detail 0 = No change on remote fault received
4	Fiber Energy Detect Changed	RO,LH	0x0	0x0	1 = Energy Detect state changed 0 = No Energy Detect state change detected
3:0	Reserved	RO	Always 00000	Always 00000	00000

Table 131: Fiber Receive Error Counter Register
Page 1, Register 21

Bits	Field	Mode	HW Rst	SW Rst	Description
15:0	Receive Error Count	RO, LH	0x0000	Retain	Counter will peg at 0xFFFF and will not roll over. Both False carrier and symbol errors are reported.

Table 132: PRBS Control
Page 1, Register 23

Bits	Field	Mode	HW Rst	SW Rst	Description
15:8	Reserved	R/W	0x00	Retain	Set to 0s
7	Invert Checker Polarity	R/W	0x0	Retain	0 = Normal 1 = Invert

Table 132: PRBS Control
Page 1, Register 23

Bits	Field	Mode	HW Rst	SW Rst	Description
6	Invert Generator Polarity	R/W	0x0	Retain	0 = Normal 1 = Invert
5	PRBS Lock	R/W	0x0	Retain	0 = Counter Free Runs 1 = Do not start counting until PRBS locks first
4	Clear Counter	R/W, SC	0x0	0x0	0 = Normal 1 = Clear Counter
3:2	Pattern Select	R/W	0x0	Retain	00 = PRBS 7 01 = PRBS 23 10 = PRBS 31 11 = Generate 1010101010... pattern
1	PRBS Checker Enable	R/W	0x0	0x0	0 = Disable 1 = Enable
0	PRBS Generator Enable	R/W	0x0	0x0	0 = Disable 1 = Enable

Table 133: PRBS Error Counter LSB
Page 1, Register 24

Bits	Field	Mode	HW Rst	SW Rst	Description
15:0	PRBS Error Count LSB	RO	0x0000	Retain	A read to this register freezes register 25_1. Cleared only when register 23_1.4 is set to 1.

Table 134: PRBS Error Counter MSB
Page 1, Register 25

Bits	Field	Mode	HW Rst	SW Rst	Description
15:0	PRBS Error Count MSB	RO	0x0000	Retain	This register does not update unless register 24_1 is read first. Cleared only when register 23_1.4 is set to 1.

Table 135: Fiber Specific Control Register 2
Page 1, Register 26

Bits	Field	Mode	HW Rst	SW Rst	Description
15	Force INT	R/W	0x0	Retain	1 = Force INTn to assert 0 = Normal Operation
14	1000BASE-X Noise Filtering	R/W	0x0	Retain	1 = Enable 0 = Disable
13:10	Reserved	R/W	0x0	Retain	Must set to 0

Table 135: Fiber Specific Control Register 2 (Continued)
Page 1, Register 26

Bits	Field	Mode	HW Rst	SW Rst	Description
9	FEFI Enable	R/W	0x0	Retain	100BASE-FX FEFI 1 = Enable 0 = Disable
8:7	Reserved	R/W	0x0	Retain	Must set to 0
6	Serial Interface Auto-Negotiation bypass enable	R/W	0x1	Update	Changes to this bit are disruptive to the normal operation; hence, any changes to these registers must be followed by software reset to take effect. 1 = Bypass Allowed 0 = No Bypass Allowed
5	Serial Interface Auto-Negotiation bypass status	RO	0x0	0x0	1 = Serial interface link came up because bypass mode timer timed out and fiber Auto-Negotiation was bypassed. 0 = Serial interface link came up because regular fiber Auto-Negotiation completed. If this bit is 1, then bit 17_1.11 will be 0.
4	Reserved	R/W	0x0	Update	Must set to 0
3	Fiber Transmitter Disable	R/W	0x0	Retain	1 = Transmitter Disable 0 = Transmitter Enable
2:0	SGMII Output Amplitude	R/W	0x2	Retain	Differential voltage peak measured. See AC/DC section for valid VOD values. 000 = 14 mV 001 = 112 mV 010 = 210 mV 011 = 308 mV 100 = 406 mV 101 = 504 mV 110 = 602 mV 111 = 700 mV

Table 136: MAC Specific Control Register 1
Page 2, Register 16

Bits	Field	Mode	HW Rst	SW Rst	Description
15:14	Transmit FIFO Depth	R/W	01	Retain	00 = ± 16 Bits 01 = ± 24 Bits 10 = ± 32 Bits 11 = ± 40 Bits See 20_0.14 description. This bit enables a TX FIFO low latency mode- packet size is max 1518 bytes. 20_0.14 has priority over 16_2.15:14.
13	Reserved	R/W	0x0	Retain	

Table 136: MAC Specific Control Register 1 (Continued)
Page 2, Register 16

Bits	Field	Mode	HW Rst	SW Rst	Description
12	RCLK Frequency Select	R/W	0x0	Retain	Selects the frequency of RCLK (the SyncE clock that goes out of the CLK125 output pin) 0 = 25 MHz 1 = 125 MHz
11	RCLK Link Down Disable	R/W	0x0	Retain	Selects what the RCLK output is during link down and 10BT. 0 = RCLK is the 25 MHz XTAL1 clock 1 = RCLK is low
10	RCLK enable	R/W	0x1	Retain	1 = Enable RCLK 0 = Disable RCLK
9:7	Reserved	R/W	0x0	Retain	
6	Pass Odd Nibble Preambles	R/W	0x1	Update	0 = Pad odd nibble preambles in copper receive packets. 1 = Pass as is and do not pad odd nibble preambles in copper receive packets.
5:4	Reserved	R/W	0x0	Retain	
3	RGMII Interface Power Down	R/W	0x1	Update	Changes to this bit are disruptive to the normal operation; therefore, any changes to these registers must be followed by a software reset to take effect. This bit determines whether the RGMII RX_CLK powers down when Register 0.11, 16_0.2 are used to power down the device or when the PHY enters the energy detect state. 1 = Always power up 0 = Can power down
2	Disable CLK125	R/W	0x0	Retain	0 = Enable internally generated 125 MHz clock 1 = Disable internally generated 125 MHz clock
1	CLK125 Pin Output Disable	R/W	0x0	Retain	Disables the CLK125 output pin (tristates it) regardless of the enabled clock source. 1 = Stop the current clock selected from going out the CLK125 pin and tristate the output 0 = Allow the current clock selected to go out the CLK125 pin
0	CLK125 Pin Output Source Select	R/W	0x0	Retain	Selects the clock source for the CLK125 output pin. 1 = Enable RCLK to go out the CLK125 pin 0 = Enable the internally generated 125 MHz clock to go out the CLK125 pin.

Table 137: MAC Specific Interrupt Enable Register
Page 2, Register 18

Bits	Field	Mode	HW Rst	SW Rst	Description
15:8	Reserved	R/W	0x00	Retain	000000000

Table 137: MAC Specific Interrupt Enable Register
Page 2, Register 18

Bits	Field	Mode	HW Rst	SW Rst	Description
7	Copper FIFO Over/Underflow Interrupt Enable	R/W	0x0	Retain	1 = Interrupt enable 0 = Interrupt disable
6:4	Reserved	R/W	0x0	Retain	000
3	Copper FIFO Idle Inserted Interrupt Enable	R/W	0x0	Retain	1 = Interrupt enable 0 = Interrupt disable
2	Copper FIFO Idle Deleted Interrupt Enable	R/W	0x0	Retain	1 = Interrupt enable 0 = Interrupt disable
1:0	Reserved	R/W	0x0	Retain	00

Table 138: MAC Specific Status Register
Page 2, Register 19

Bits	Field	Mode	HW Rst	SW Rst	Description
15:8	Reserved	RO	Always 00	Always 00	00000000
7	Copper FIFO Over/Underflow	RO,LH	0x0	0x0	1 = Over/Underflow Error 0 = No FIFO Error
6:4	Reserved	RO	Always 0	Always 0	000
3	Copper FIFO Idle Inserted	RO,LH	0x0	0x0	1 = Idle Inserted 0 = No Idle Inserted
2	Copper FIFO Idle Deleted	RO,LH	0x0	0x0	1 = Idle Deleted 0 = Idle not Deleted
1:0	Reserved	RO	Always 0	Always 0	00

Table 139: Copper RX_ER Byte Capture
Page 2, Register 20

Bits	Field	Mode	HW Rst	SW Rst	Description
15	Capture Data Valid	RO	0x0	0x0	1 = Bits 14:0 Valid 0 = Bits 14:0 Invalid
14	Reserved	RO	0x0	0x0	0

Table 139: Copper RX_ER Byte Capture (Continued)
Page 2, Register 20

Bits	Field	Mode	HW Rst	SW Rst	Description
13:12	Byte Number	RO	0x0	0x0	00 = 4 bytes before RX_ER asserted 01 = 3 bytes before RX_ER asserted 10 = 2 bytes before RX_ER asserted 11 = 1 byte before RX_ER asserted The byte number increments after every read when register 20_2.15 is set to 1.
11	RX_ER	RO	0	0	RX Error. Normally this bit will be low since the capture is triggered by RX_ER being high. However it is possible to see an RX_ER high when the capture is re-enabled after reading the fourth byte and there happens to be a long sequence of RX_ER when the capture restarts.
10	RX_DV	RO	0	0	RX Data Valid
9	RX_ER	RO	0	0	RX Error. Normally this bit will be low since the capture is triggered by RX_ER being high. However it is possible to see an RX_ER high when the capture is re-enabled after reading the fourth byte and there happens to be a long sequence of RX_ER when the capture restarts.
8	RX_DV	RO	0	0	RX Data Valid
7:0	RXD[7:0]	RO	00	00	RX Data

Table 140: MAC Specific Control Register 2
Page 2, Register 21

Bits	Field	Mode	HW Rst	SW Rst	Description
15	Reserved	R/W	0x0	0x0	0
14	Copper Line Loopback	R/W	0x0	0x0	1 = Enable Loopback of MDI to MDI 0 = Normal Operation
13	Default MAC interface speed (LSB)	R/W	See Descr	Update	Changes to these bits are disruptive to the normal operation; therefore, any changes to these registers must be followed by software reset to take effect. Also, used for setting speed of MAC interface during MAC side loopback. Requires that customer set both these bits and force speed using register 0 to the same speed. MAC Interface Speed during Link down. This bit defaults to a 1 for 88EA1517 and when MII mode is used for either package, otherwise this bit defaults to 0. Bits 6,13 00 = 10 Mbps 01 = 100 Mbps 10 = 1000 Mbps
12:7	Reserved		0x20	0x20	Reserved.

Table 140: MAC Specific Control Register 2 (Continued)
Page 2, Register 21

Bits	Field	Mode	HW Rst	SW Rst	Description
6	Default MAC interface speed (MSB)	R/W	See Desc	Update	Changes to these bits are disruptive to the normal operation; therefore, any changes to these registers must be followed by software reset to take effect. Also, used for setting speed of MAC interface during MAC side loopback. Requires that customer set both these bits and force speed using register 0 to the same speed. MAC Interface Speed during Link down. This bit is a 0 always (a write of 1 has no effect) for 88EA1517 and when MII mode is used for either package, otherwise this bit defaults to 1 and may be written to 0. Bits 6, 13 00 = 10 Mbps 01 = 100 Mbps 10 = 1000 Mbps
5	RGMII Receive Timing Control	R/W	0x1	Update	Changes to these bits are disruptive to the normal operation; therefore, any changes to these registers must be followed by software reset to take effect. 1 = Receive clock transition when data stable 0 = Receive clock transition when data transitions
4	RGMII Transmit Timing Control	R/W	0x1	Update	Changes to these bits are disruptive to the normal operation; therefore, any changes to these registers must be followed by software reset to take effect. 1 = Transmit clock internally delayed 0 = Transmit clock not internally delayed
3	Block Carrier Extension Bit	R/W	0x0	Retain	1 = Enable Block Carrier Extension 0 = Disable Block Carrier Extension
2:0	Reserved	R/W	0x6	0x6	Reserved.

Table 141: RGMII Output Impedance Calibration Override
Page 2, Register 24

Bits	Field	Mode	HW Rst	SW Rst	Description
15	Restart Calibration	R/W, SC	0x0	Retain	Calibration will start once bit 15 is set to 1. 0 = Normal 1 = Restart
14	Calibration Complete	RO	0x0	Retain	Calibration is done once bit 14 becomes 1. 0 = Not done 1 = Done

Table 141: RGMII Output Impedance Calibration Override (Continued)
Page 2, Register 24

Bits	Field	Mode	HW Rst	SW Rst	Description
13	VDDO Level	R/W	See Descr.	Retain	VDDO level- must be programmed to indicate the VDDO supply voltage used when 1.8V is not used. The bit mapping is: 0 = 3.3V 1 = 2.5V If the CONFIG pin input values bit 1:0 are: 00, then VDDO Level = 3.3V 11, then VDDO Level = 3.3V 10, then VDDO Level = 2.5V 01, then VDDO Level = 2.5V **NOTE:** 3.3V is assumed initially until this value is changed.
12	1.8V VDDO Used	R/O	See Descr	Retain	This bit indicates whether VDDO = 1.8V is used or not. 1 = VDDO = 1.8V 0 = VDDO = 2.5V or 3.3V
11:8	PMOS Value	R/W	See Descr	Retain	0000 = All fingers off 1111 = All fingers on The automatic calibrated values are stored here after calibration completes. Once 24_2.6 is set to 1 the new calibration value is written into the I/O pad. The automatic calibrated value is lost.
7	Reserved	RW	0x0	Retain	Reserved.
6	Force PMOS/ NMOS	R/W	0x0	Retain	1 = Force value from 24_2.11:8 to PMOS, and 24_2.3:0 to NMOS. (Used for manual settings)
5:4	Reserved	R/O	0x0	Retain	Reserved.
3:0	NMOS value	R/W	See Descr	Retain	0000 = All fingers off 1111 = All fingers on The automatic calibrated values are stored here after calibration completes. Once 24_2.6 is set to 1 the new calibration value is written into the I/O pad. The automatic calibrated value is lost.

Table 142: RGMII Output Impedance Target
Page 2, Register 25

Bits	Field	Mode	HW Rst	SW Rst	Description
15:5	Reserved	R/W	0x000	Retain	Reserved.
4	Reserved	R/W	0x1	Retain	Must always be set to 1.
3	Reserved	R/W	0x0	Retain	Reserved.

Table 142: RGMII Output Impedance Target (Continued)
Page 2, Register 25

Bits	Field	Mode	HW Rst	SW Rst	Description
2:0	Calibration target	RW	0x3	Retain	000 = 78.8 Ohm 001 = 64.5 Ohm 010 = 54.6 Ohm 011 = 47.3 Ohm 100 = 41.7 Ohm 101 = 37.3 Ohm 110 = 33.8 Ohm 111 = 30.9 Ohm

Table 143: LED[2:0] Function Control Register Added Features
Page 3, Register 15

Bits	Field	Mode	HW Rst	SW Rst	Description
15:2	Reserved	R/W	0	Retain	Reserved.
1	RX SFD Detect Pulse Output Enable	R/W	0	Retain	1 = Enable RX datapath SFD detection pulse to go out LED[1] 0 = LED[1] outputs according to other logic
0	TX SFD Detect Pulse Output Enable	R/W	0	Retain	1 = Enable TX datapath SFD detection pulse to go out LED[0] 0 = LED[0] outputs according to other logic

Table 144: LED[2:0] Function Control Register
Page 3, Register 16

Bits	Field	Mode	HW Rst	SW Rst	Description
15:12	Reserved	R/W	0x1	Retain	
11:8	LED[2] Control	R/W	0x0	Retain	0000 = On - Link, Off - No Link 0001 = On - Link, Blink - Activity, Off - No Link 0010 = On- Full Duplex, Blink- Collision, Off- Half Duplex 0011 = On - Activity, Off - No Activity 0100 = Blink - Activity, Off - No Activity 0101 = On - Transmit, Off - No Transmit 0110 = On - 10/1000 Mbps Link, Off - Else 0111 = On - 10 Mbps Link, Off - Else 1000 = Force Off 1001 = Force On 1010 = Force Hi-Z 1011 = Force Blink 11xx = Reserved

Table 144: LED[2:0] Function Control Register (Continued)
Page 3, Register 16

Bits	Field	Mode	HW Rst	SW Rst	Description
7:4	LED[1] Control	R/W	0x1	Retain	If 16_3.3:2 is set to 11 then 16_3.7:4 has no effect 0000 = On- Receive, Off- No Receive 0001 = On - Link, Blink - Activity, Off - No Link 0010 = On - Link, Blink - Receive, Off - No Link 0011 = On - Activity, Off - No Activity 0100 = Blink - Activity, Off - No Activity 0101 = On- 100 Mbps Link/ Fiber Link 0110 = On - 100/1000 Mbps Link, Off - Else 0111 = On - 100 Mbps Link, Off - Else 1000 = Force Off 1001 = Force On 1010 = Force Hi-Z 1011 = Force Blink 11xx = Reserved
3:0	LED[0] Control	R/W	0xE	Retain	0000 = On - Link, Off - No Link 0001 = On - Link, Blink - Activity, Off - No Link 0010 = 3 blinks - 1000 Mbps 2 blinks - 100 Mbps 1 blink - 10 Mbps 0 blink - No Link 0011 = On - Activity, Off - No Activity 0100 = Blink - Activity, Off - No Activity 0101 = On - Transmit, Off - No Transmit 0110 = On - Copper Link, Off - Else 0111 = On - 1000 Mbps Link, Off - Else 1000 = Force Off 1001 = Force On 1010 = Force Hi-Z 1011 = Force Blink 1100 = MODE 1 (Dual LED mode) 1101 = MODE 2 (Dual LED mode) 1110 = MODE 3 (Dual LED mode) 1111 = MODE 4 (Dual LED mode)

Table 145: LED[2:0] Polarity Control Register
Page 3, Register 17

Bits	Field	Mode	HW Rst	SW Rst	Description
15:12	LED[1] mix percentage	R/W	0x4	Retain	When using 2 terminal bi-color LEDs the mixing percentage should not be set greater than 50%. 0000 = 0% 0001 = 12.5% ... 0111 = 87.5% 1000 = 100% 1001 to 1111 = Reserved

Table 145: LED[2:0] Polarity Control Register
Page 3, Register 17

Bits	Field	Mode	HW Rst	SW Rst	Description
11:8	LED[0] mix percentage	R/W	0x4	Retain	When using 2 terminal bi-color LEDs the mixing percentage should not be set greater than 50%. 0000 = 0% 0001 = 12.5% ..., 0111 = 87.5%, 1000 = 100% 1001 to 1111 = Reserved
7:6	Reserved	R/W	0x0	Retain	Reserved.
5:4	LED[2] Polarity	R/W	0x0	Retain	00 = On - drive LED[2] low, Off - drive LED[2] high 01 = On - drive LED[2] high, Off - drive LED[2] low 10 = On - drive LED[2] low, Off - tristate LED[2] 11 = On - drive LED[2] high, Off - tristate LED[2]
3:2	LED[1] Polarity	R/W	0x0	Retain	00 = On - drive LED[1] low, Off - drive LED[1] high 01 = On - drive LED[1] high, Off - drive LED[1] low 10 = On - drive LED[1] low, Off - tristate LED[1] 11 = On - drive LED[1] high, Off - tristate LED[1]
1:0	LED[0] Polarity	R/W	0x0	Retain	00 = On - drive LED[0] low, Off - drive LED[0] high 01 = On - drive LED[0] high, Off - drive LED[0] low 10 = On - drive LED[0] low, Off - tristate LED[0] 11 = On - drive LED[0] high, Off - tristate LED[0]

Table 146: LED Timer Control Register
Page 3, Register 18

Bits	Field	Mode	HW Rst	SW Rst	Description
15	Force INT	R/W	0x0	Retain	1 = Force INTn to assert 0 = Normal Operation
14:12	Pulse stretch duration	R/W	0x4	Retain	000 = No pulse stretching 001 = 21 ms to 42 ms 010 = 42 ms to 84 ms 011 = 84 ms to 170 ms 100 = 170 ms to 340 ms 101 = 340 ms to 670 ms 110 = 670 ms to 1.3s 111 = 1.3s to 2.7s
11	Interrupt Polarity	R/W	0x1	Retain	0 = INTn active high 1 = INTn active low
10:8	Blink Rate	R/W	0x1	Retain	000 = 42 ms 001 = 84 ms 010 = 170 ms 011 = 340 ms 100 = 670 ms 101 to 111 = Reserved
7	Interrupt Enable	R/W	0	Retain	Allows the interrupt output to be brought out on LED[2] for 88EA1512. For 88EA1517, this bit enables the interrupt output to be brought out on LED[1] if LED[1] or LED[2] functionality is selected by 19_3.12 1 = Interrupt is brought out LED[1] for 88EA1517/LED[2] otherwise 0 = Interrupt is not brought out
6:4	Reserved	R/W	0x0	Retain	000
3:2	Speed Off Pulse Period	R/W	0x1	Retain	00 = 84 ms 01 = 170 ms 10 = 340 ms 11 = 670 ms
1:0	Speed On Pulse Period	R/W	0x1	Retain	00 = 84 ms 01 = 170 ms 10 = 340 ms 11 = 670 ms

Table 147: PTP LED Function Control Register
Page 3, Register 19

Bits	Field	Mode	HW Rst	SW Rst	Description
15	LED[5] Status Select	R/W	0x0	Retain	This bit allows legacy LED[5] status to be brought out to LED[1]. 1 = Select LED[5] status for LED[1] 0 = Select non-LED[5] status for LED[1]
14	LED[4] Status Select	R/W	0x0	Retain	This bit allows legacy LED[4] status to be brought out to LED[2] for 88EA1512. For 88EA1517 LED[4] status may be brought out LED[1], but only if 19_3.12 = 1, which allows LED[2] status to be sent to LED[1] 1 = Select LED[4] status for LED[2] 0 = Select non-LED[4] status for LED[2]
13	Block PTP Activity	R/W	0x0	Retain	1 = Block PTP activity from lighting the LEDs 0 = Allow PTP activity to light the LEDs
12	88EA1517 LED[1] Select	R/W	0x0	Retain	This bit allows LED[2] status to be brought out to LED[1] only for the 88EA1517 package. 1 = Select LED[2] status for LED[1] only for the 88EA1517 package 0 = Select LED[1] status for LED[1]
11:10	LED[5] Polarity	R/W	00	Retain	00 = On - drive LED[5] low, Off - drive LED[5] high 01 = On - drive LED[5] high, Off - drive LED[5] low 10 = On - drive LED[5] low, Off - tristate LED[5] 11 = On - drive LED[5] high, Off - tristate LED[5]
9:8	LED[4] Polarity	R/W	00	Retain	00 = On - drive LED[4] low, Off - drive LED[4] high 01 = On - drive LED[4] high, Off - drive LED[4] low 10 = On - drive LED[4] low, Off - tristate LED[4] 11 = On - drive LED[4] high, Off - tristate LED[4]
7:4	LED[5] Control	R/W	0111	Retain	If 19_3.3:2 is set to 11 then 19_3.7:4 has no effect 0000 = On - Receive, Off - No Receive 0001 = On - Link, Blink - Activity, Off - No Link 0010 = On - Link, Blink - Receive, Off - No Link 0011 = On - Activity, Off - No Activity 0100 = Blink - Activity, Off - No Activity 0101 = On - Transmit, Off - No Transmit 0110 = On - Full-duplex, Off - Half-duplex 0111 = On - Full-duplex, Blink - Collision Off - Half-duplex 1000 = Force Off 1001 = Force On 1010 = Force Hi-Z 1011 = Force Blink 11xx = Reserved

Table 147: PTP LED Function Control Register (Continued)
Page 3, Register 19

Bits	Field	Mode	HW Rst	SW Rst	Description
3:0	LED[4] Control	R/W	0011	Retain	0000 = On - Receive, Off - No Receive 0001 = On - Link, Blink - Activity, Off - No Link 0010 = On - Link, Blink - Receive, Off - No Link 0011 = On - Activity, Off - No Activity 0100 = Blink - Activity, Off - No Activity 0101 = On - Transmit, Off - No Transmit 0110 = On - Full-duplex, Off - Half-duplex 0111 = On - Full-duplex, Blink - Collision Off - Half-duplex 1000 = Force Off 1001 = Force On 1010 = Force Hi-Z 1011 = Force Blink 1100 = MODE 1 (Dual LED mode) 1101 = MODE 2 (Dual LED mode) 1110 = MODE 3 (Dual LED mode) 1111 = MODE 4 (Dual LED mode)

Table 148: RGMII RX_ER Byte Capture
Page 4, Register 20

Bits	Field	Mode	HW Rst	SW Rst	Description
15	Capture Data Valid	RO	0	0	1 = Bits 14:0 Valid 0 = Bits 14:0 Invalid
14	Reserved	RO	0	0	Reserved.
13:12	Byte Number	RO	00	00	00 = 4 bytes before RX_ER asserted 01 = 3 bytes before RX_ER asserted 10 = 2 bytes before RX_ER asserted 11 = 1 byte before RX_ER asserted The byte number increments after every read when register 20_4.15 is set to 1.
11	RX_ER	RO	0	0	RX Error. Normally this bit will be low since the capture is triggered by RX_ER being high. However it is possible to see an RX_ER high when the capture is re-enabled after reading the fourth byte and there happens to be a long sequence of RX_ER when the capture restarts.
10	RX_DV	RO	0	0	RX Data Valid.
9	RX_ER	RO	0	0	RX Error. Normally this bit will be low since the capture is triggered by RX_ER being high. However it is possible to see an RX_ER high when the capture is re-enabled after reading the fourth byte if there happens to be a long sequence of RX_ER when the capture restarts.

Table 148: RGMII RX_ER Byte Capture (Continued)
Page 4, Register 20

Bits	Field	Mode	HW Rst	SW Rst	Description
8	RX_DV	RO	0	0	RX Data Valid
7:0	RXD[7:0]	RO	0x00	0x00	RX Data

Table 149: Advanced VCT TX to MDI[0] Rx Coupling
Page 5, Register 16

Bits	Field	Mode	HW Rst	SW Rst	Description
15	Reflected Polarity	RO	xx	Retain	1 = Positive reflection 0 = Negative reflection
14:8	Reflected Amplitude	RO	xx	Retain	0000000 = No reflection (0 mV) Each bit above represents an increase in 7.8125 mV. When 23_5.7 = 0 and 23_5.13:11 = 000 or 100 the reflected amplitude between the thresholds specified in registers 26_5, 27_5, 28_5.6:0, 26_7, 27_7, and 28_7.6:0 are reported as 0 mV. When 23_5.7 = 0 and 23_5.13:11 is not 000 or 100 the reflected amplitude between the thresholds specified in register 25_5 and 25_7 are reported as 0 mV. When 23_5.7 = 1 the actual offset or reflected amplitude is reported and the threshold specified in registers 25_5, 26_5, 27_5, 28_5, 25_7, 26_7, 27_7, and 28_7 are ignored. The amplitude value is valid only When 23_5.14 = 1. If bit 15:8 = 0x00 indicates that the test failed.
7:0	Distance	RO	xx	Retain	Distance of reflection. The distance value is valid only when 23_5.7 = 0 and 23_5.14 = 1.

Table 150: Advanced VCT TX to MDI[1] Rx Coupling
Page 5, Register 17

Bits	Field	Mode	HW Rst	SW Rst	Description
15	Reflected Polarity	RO	xx	Retain	1 = Positive Reflection 0 = Negative reflection
14:8	Reflected Amplitude	RO	xx	Retain	0000000 = No reflection (0 mV) each bit above increases 7.8125 mV. When 23_5.7 = 0 and 23_5.13:11 = 000 or 101 the reflected amplitude between the thresholds specified in registers 26_5, 27_5, 28_5.6:0, 26_7, 27_7, and 28_7.6:0 are reported as 0 mV. When 23_5.7 = 0 and 23_5.13:11 is not 000 or 101 the reflected amplitude between the thresholds specified in register 25_5 and 25_7 are reported as 0 mV. When 23_5.7 = 1 the actual offset or reflected amplitude is reported and the threshold specified in registers 25_5, 26_5, 27_5, 28_5, 25_7, 26_7, 27_7, and 28_7 are ignored. The amplitude value is valid only When 23_5.14 = 1. If bit 15:8 = 0x00 indicates that the test failed.
7:0	Distance	RO	xx	Retain	Distance of reflection. The distance value is valid only when 23_5.7 = 0 and 23_5.14 = 1.

Table 151: Advanced VCT TX to MDI[2] Rx Coupling
Page 5, Register 18

Bits	Field	Mode	HW Rst	SW Rst	Description
15	Reflected Polarity	RO	xx	Retain	1 = Positive Reflection 0 = Negative reflection
14:8	Reflected Amplitude	RO	xx	Retain	0000000 = No reflection (0 mV) each bit above increases 7.8125 mV. When 23_5.7 = 0 and 23_5.13:11 = 000 or 110 the reflected amplitude between the thresholds specified in registers 26_5, 27_5, 28_5.6:0, 26_7, 27_7, and 28_7.6:0 are reported as 0 mV. When 23_5.7 = 0 and 23_5.13:11 is not 000 or 110 the reflected amplitude between the thresholds specified in register 25_5 and 25_7 are reported as 0 mV. When 23_5.7 = 1 the actual offset or reflected amplitude is reported and the threshold specified in registers 25_5, 26_5, 27_5, 28_5, 25_7, 26_7, 27_7, and 28_7 are ignored. The amplitude value is valid only When 23_5.14 = 1. If bit 15:8 = 0x00 indicates that the test failed.
7:0	Distance	RO	xx	Retain	Distance of reflection. The distance value is valid only when 23_5.7 = 0 and 23_5.14 = 1.

Table 152: Advanced VCT TX to MDI[3] Rx Coupling
Page 5, Register 19

Bits	Field	Mode	HW Rst	SW Rst	Description
15	Reflected Polarity	RO	xx	Retain	1 = Positive Reflection 0 = Negative reflection
14:8	Reflected Amplitude	RO	xx	Retain	0000000 = No reflection (0 mV) each bit above increases 7.8125 mV. When 23_5.7 = 0 and 23_5.13:11 = 000 or 111 the reflected amplitude between the thresholds specified in registers 26_5, 27_5, 28_5.6:0, 26_7, 27_7, and 28_7.6:0 are reported as 0 mV. When 23_5.7 = 0 and 23_5.13:11 is not 000 or 111 the reflected amplitude between the thresholds specified in register 25_5 and 25_7 are reported as 0 mV. When 23_5.7 = 1 the actual offset or reflected amplitude is reported and the threshold specified in registers 25_5, 26_5, 27_5, 28_5, 25_7, 26_7, 27_7, and 28_7 are ignored. The amplitude value is valid only When 23_5.14 = 1. If bit 15:8 = 0x00 indicates that the test failed.
7:0	Distance	RO	xx	Retain	Distance of reflection. The distance value is valid only when 23_5.7 = 0 and 23_5.14 = 1.

Table 153: 1000BASE-T Pair Skew Register
Page 5, Register 20

Bits	Field	Mode	HW Rst	SW Rst	Description
15:12	Pair 7,8 (MDI[3]±)	RO	0x0	Retain	Skew = Bit value x 8 n s. Value is correct to within ± 8 ns. The contents of 20_5.15:0 are valid only if Register 21_5.6 = 1
11:8	Pair 4,5 (MDI[2]±)	RO	0x0	Retain	Skew = bit value x 8 ns. Value is correct to within ± 8 ns.
7:4	Pair 3,6 (MDI[1]±)	RO	0x0	Retain	Skew = bit value x 8 ns. Value is correct to within ± 8 ns.
3:0	Pair 1,2 (MDI[0]±)	RO	0x0	Retain	Skew = bit value x 8 ns. Value is correct to within ± 8 ns.

Table 154: 1000BASE-T Pair Swap and Polarity
Page 5, Register 21

Bits	Field	Mode	HW Rst	SW Rst	Description
15:7	Reserved	RO	0x000	Retain	Reserved.
6	Register 20_5 and 21_5 valid	RO	0x0	Retain	The contents of 21_5.5:0 and 20_5.15:0 are valid only if Register 21_5.6 = 1 1= Valid 0 = Invalid
5	C, D Crossover	RO	0x0	Retain	1 = Channel C received on MDI[2]± Channel D received on MDI[3]± 0 = Channel D received on MDI[2]± Channel C received on MDI[3]±
4	A, B Crossover	RO	0x0	Retain	1 = Channel A received on MDI[0]± Channel B received on MDI[1]± 0 = Channel B received on MDI[0]± Channel A received on MDI[1]±
3	Pair 7,8 (MDI[3]±) Polarity	RO	0x0	Retain	1 = Negative 0 = Positive
2	Pair 4,5 (MDI[2]±) Polarity		0x0	Retain	1 = Negative 0 = Positive
1	Pair 3,6 (MDI[1]±) Polarity	RO	0x0	Retain	1 = Negative 0 = Positive
0	Pair 1,2 (MDI[0]±) Polarity	RO	0x0	Retain	1 = Negative 0 = Positive

Table 155: Advance VCT Control
Page 5, Register 23

Bits	Field	Mode	HW Rst	SW Rst	Description
15	Enable Test	R/W, SC	0x0	Retain	0 = Disable test 1 = Enable test This bit will self clear when the test is completed
14	Test status	RO	0x0	Retain	0 = Test not started/in progress 1 = Test completed
13:11	Transmitter Channel Select	R/W	0x0	Retain	000 - Tx 0 => Rx 0, Tx 1 => Rx 1, Tx 2 => Rx 2, Tx 3 => Rx 3. 100 - Tx 0 => Rx 0, Tx 0 => Rx 1, Tx 0 => Rx 2, Tx 0 => Rx 3. 101 - Tx 1 => Rx 0, Tx 1 => Rx 1, Tx 1 => Rx 2, Tx 1 => Rx 3. 110 - Tx 2 => Rx 0, Tx 2 => Rx 1, Tx 2 => Rx 2, Tx 2 => Rx 3. 111 - Tx 3 => Rx 0, Tx 3 => Rx 1, Tx 3 => Rx 2, Tx 3 => Rx 3. 01x - reserved 0x1 - reserved
10:8	Number of Sample Averaged	R/W	6	Retain	0 = 2 samples 1 = 4 samples 2 = 8 samples 3 = 16 samples 4 = 32 samples 5 = 64 samples 6 = 128 samples 7 = 256 samples
7:6	Mode	R/W	0x0	Retain	00 = Maximum peak above threshold 01 = First or last peak above threshold. See register 28_5.13. 10 = Offset 11 = Sample point at distance set by 24_5.9:0
5:0	Peak Detection Hysteresis	R/W	0x03	Retain	0x00 = 0 mV, 0x01 = 7.81 mV,..., 0x3F = ± 492 mv

Table 156: Advanced VCT Sample Point Distance
Page 5, Register 24

Bits	Field	Mode	HW Rst	SW Rst	Description
15:10	Reserved	RO	0x00	Retain	Reserved.
9:0	Distance to measure/ Distance to start	R/W	0x000	Retain	When 23_5.7:6 = 11 the measurement is taken at this distance. (00 to 3FF) When 23_5.7:6 = 0x any distance below this distance is not considered (00 to FF). Bit 9:8 is ignored.

Table 157: Advanced VCT Cross Pair Positive Threshold
Page 5, Register 25

Bits	Field	Mode	HW Rst	SW Rst	Description
15	Reserved	RO	0x0	Retain	Reserved.
14:8	Cross Pair Positive Threshold > 30m	R/W	0x01	Retain	0x00 = 0 mV, 0x01 = 7.81 mV,..., 0x7F = 9.92 mV
7	Reserved	RO	0x0	Retain	Reserved.
6:0	Cross Pair Positive Threshold < 30m	R/W	0x04	Retain	0x00 = 0 mV, 0x01 = 7.81 mV,..., 0x7F = 992 mV

Table 158: Advanced VCT Same Pair Impedance Positive Threshold 0 and 1
Page 5, Register 26

Bits	Field	Mode	HW Rst	SW Rst	Description
15	Reserved	RO	0x0	Retain	Reserved.
14:8	Same-Pair Positive Threshold 10m - 50m	R/W	0x0F	Retain	0x00 = 0 mV, 0x01 = 7.81 mV,..., 0x7F = 9.92 mV
7	Reserved	RO	0x0	Retain	Reserved.
6:0	Same-Pair Positive Threshold < 10m	R/W	0x12	Retain	0x00 = 0 mV, 0x01 = 7.81 mV,..., 0x7F = 9.92 mV

Table 159: Advanced VCT Same Pair Impedance Positive Threshold 2 and 3
Page 5, Register 27

Bits	Field	Mode	HW Rst	SW Rst	Description
15	Reserved	RO	0x0	Retain	Reserved.
14:8	Same-Pair Positive Threshold 110m - 140m	R/W	0x0A	Retain	0x00 = 0 mV, 0x01 = 7.81 mV,..., 0x7F = 9.92 mv
7	Reserved	RO	0x0	Retain	Reserved.
6:0	Same-Pair Positive Threshold 50m - 110m	R/W	0x0C	Retain	0x00 = 0 mV, 0x01 = 7.81 mV,..., 0x7F = 9.92 mv

Table 160: Advanced VCT Same Pair Impedance Positive Threshold 4 and Transmit Pulse Control
Page 5, Register 28

Bits	Field	Mode	HW Rst	SW Rst	Description
15:14	Reserved	RO	0x0	Retain	0
13	First Peak/Last Peak Select	R/W	0x0	Retain	This register takes effect only if register 23_5.7:6 = 01. 0 = First Peak 1 = Last Peak
12	Break Link Prior to Measurement	R/W	0x0	Retain	1 = Do not wait 1.5s to break link before starting VCT 0 = Wait 1.5s to break link before starting VCT
11:10	Transmit Pulse Width	R/W	0x0	Retain	00 = Full pulse (128 ns) 01 = 3/4 pulse 10 = 1/2 pulse 11 = 1/4 pulse
9:8	Transmit Amplitude	R/W	0x0	Retain	00 = Full amplitude 01 = 3/4 amplitude 10 = 1/2 amplitude 11 = 1/4 amplitude

7	Distance Measurement Point	R/W	0x0	Retain	If 23_5.7:6 = 00 then
					0 = Measure distance when amplitude drops to 50% of peak amplitude
					1 = Measure distance at actual maximum amplitude
					If 23_5.7:6 = 01 then
					0 = Measure distance when amplitude drops below hysteresis
					1 = Measure distance at actual maximum amplitude
					If 23_5.7:6 = 1X then this bit is ignored.

6:0	Same-Pair Positive Threshold > 140m	R/W	0x06	Retain	0x00 = 0 mV, 0x01 = 7.81 mV,..., 0x7F = 992 mv

Table 161: Copper Port Packet Generation
Page 6, Register 16

Bits	Field	Mode	HW Rst	SW Rst	Description
15:8	Packet Burst	R/W	0x00	Retain	0x00 = Continuous 0x01 to 0xFF = Burst 1 to 255 packets
7	Packet Generator Transmit Trigger	R/W	0x0	Retain	This bit is only valid when all of the following are true: bit 6 =1 bit3 =1 bit15:8 is not equal to all 0s A read of this bit gives the following: 1: Packet generator transmit done 0: Packet generator is transmitting data When this bit is 1 a write of 0 will trigger the packet generator to transmit again. When this bit is 0 a write of 0 or 1 will have no effect.
6	Packet Generator Enable Self Clear Control	R/W	0x0	Retain	0 = Bit 3 will self clear after all packets are sent 1 = Bit 3 will stay high after all packets are sent
5	Reserved	R/W	0x0	Retain	Reserved
4	Enable CRC Checker	R/W	0x0	Retain	1 = Enable 0 = Disable
3	Enable Packet Generator	R/W	0x0	Retain	1 = Enable 0 = Disable
2	Payload of Packet to Transmit	R/W	0x0	Retain	0 = Pseudo-random 1 = 5A,A5,5A,A5,...
1	Length of Packet to Transmit	R/W	0x0	Retain	1 = 1518 bytes 0 = 64 bytes
0	Transmit an Errored Packet	R/W	0x0	Retain	1 = Tx packets with CRC errors & Symbol Error 0 = No error

Table 162: Copper Port CRC Counters
Page 6, Register 17

Bits	Field	Mode	HW Rst	SW Rst	Description
15:8	Packet Count	RO	0x00	Retain	0x00 = No packets received 0xFF = 256 packets received (max count). Bit 16_6.4 must be set to 1 in order for register to be valid.
7:0	CRC Error Count	RO	0x00	Retain	0x00 = No CRC errors detected in the packets received. 0xFF = 256 CRC errors detected in the packets received (max count). Bit 16_6.4 must be set to 1 in order for register to be valid.

Table 163: Checker Control
Page 6, Register 18

Bits	Field	Mode	HW Rst	SW Rst	Description
15:5	Reserved	R/W	0x000	Retain	Set to 0s
4	CRC Counter Reset	R/W, SC	0x0	Retain	1 = Reset This bit will self-clear after writing 1.
3	Enable Stub Test	R	0x0	Retain	1 = Enable stub test 0 = Normal Operation
2:0	Reserved	R/W	0x0	Retain	Reserved.

Table 164: Copper Port Packet Generation
Page 6, Register 19

Bits	Field	Mode	HW Rst	SW Rst	Description
15	Constant Packet Payload Enable	R/W	0	Retain	Enables a constant payload for packets generated by the packet generator. 1 = Enable constant packet payload 0 = Use other payload options
14	Constant Packet Payload Type	R/W	0	Retain	This bit selects the type of constant payload to use and is valid only when 19_6.15 = 1 1 = use all 1s packet payload 0 = use all 0s packet payload
13:8	Reserved	R/W	0	Retain	Reserved.
7:0	IPG Length	R/W	8'd12	Retain	The number in bit [7:0]+1 is the number of bytes for IPG

Table 165: General Control Register
Page 6, Register 20

Bits	Field	Mode	HW Rst	SW Rst	Description
15	100BT Low Latency J-only Detection	R/W	0	Update	Allows for fast detection of a 100BT packet for lower latency. 1 = Enable the 100BT copper receive PCS to only detect J 0 = Use the IEEE defined way of detecting JK
14:10	Reserved	R/W	0	Retain	Reserved.
9	PTP Power Down	R/W	1	Retain	1 = Power Down 0 = Power Up
8	PTP Reference Clock Source	R/W	0	Retain	1 = Use 125 MHz clock applied to the CONFIG pin after configuration is completed 0 = Use internal 125 MHz clock
7	PTP Input Source	R/W	0	Retain	1 = Use LED[1] pin for PTP input trigger pulse after configuration is completed 0 = Force input to 0
6	PTP Output Source	R/W	0	Retain	1 = Use LED[1] pin for PTP output trigger pulse after configuration is completed 0 = Use LED[1] for non-PTP functions
5	PTP 1PPS Output Enable	R/W	0	Retain	1 = Use LED[0] pin for PTP One Pulse Per Second (1PPS) output 0 = Use LED[0] for other functions
4:0	Reserved	R/W	0	Retain	Reserved.

Table 166: Late Collision Counters 1 & 2
Page 6, Register 23

Bits	Field	Mode	HW Rst	SW Rst	Description
15:8	Late Collision 97-128 bytes	RO, SC	0x00	Retain	This counter increments by 1 when the PHY is in half duplex and a start of packet is received while the 97th to 128th bytes of the packet are transmitted. The measurement is done at the internal GMII interface. The counter will not roll over and will clear on read.
7:0	Late Collision 65-96 bytes	RO, SC	0x00	Retain	This counter increments by 1 when the PHY is in half duplex and a start of packet is received while the 65th to 96th bytes of the packet are transmitted. The measurement is done at the internal GMII interface. The counter will not roll over and will clear on read.

Table 167: Late Collision Counters 3 & 4
Page 6, Register 24

Bits	Field	Mode	HW Rst	SW Rst	Description
15:8	Late Collision >192 bytes	RO, SC	0x00	Retain	This counter increments by 1 when the PHY is in half duplex and a start of packet is received after 192 bytes of the packet are transmitted. The measurement is done at the internal GMII interface. The counter will not roll over and will clear on read.
7:0	Late Collision 129-192 bytes	RO, SC	0x00	Retain	This counter increments by 1 when the PHY is in half duplex and a start of packet is received while the 129th to 192nd bytes of the packet are transmitted. The measurement is done at the internal GMII interface. The counter will not roll over and will clear on read.

Table 168: Late Collision Window Adjust/Link Disconnect
Page 6, Register 25

Bits	Field	Mode	HW Rst	SW Rst	Description
15:13	Reserved	R/W	0x0	Retain	Set to 0s
12:8	Late Collision Window Adjust	R/W	0x00	Retain	Number of bytes to advance in late collision window. 0 = Start at 64th byte 1 = Start at 63rd byte, etc.
7:0	Reserved	R/W	0x00	Retain	Set to 0s

Table 169: Misc Test
Page 6, Register 26

Bits	Field	Mode	HW Rst	SW Rst	Description
15	TX_TCLK Enable	R/W	0x0	0x0	The bit enables driving the transmit clock to the HSDACP/N pin. 1 = Enable 0 = Disable
14:13	Temperature Sensor Acceleration	R/W	0x0	Retain	00 = Sample once per second 01 = Sample once per 10ms 1x = Disable Polling
12:8	Temperature Threshold	R/W	0x19	Retain	Temperature in C = 5 x 26_6.4:0 - 25 i.e. for 100C the value is 11001
7	Temperature Sensor Interrupt Enable	R/W	0x0	Retain	1 = Interrupt Enable 0 = Interrupt Disable
6	Temperature Sensor Interrupt	RO, LH	0x0	Retain	1 = Temperature Reached Threshold 0 = Temperature Below Threshold

Table 169: Misc Test
Page 6, Register 26

Bits	Field	Mode	HW Rst	SW Rst	Description
5	Temperature Manual Control	R/W	0x0	Retain	Manual Control of temp_sense_en 1 = Temperature Acquire 0 = Temperature Read Set register 250_8.5:4 = 10 to use
4:0	Temperature Sensor	RO	xxxxx	xxxxx	Temperature is the 5MSBs of temperature value - Temp_val[5:1]

Table 170: Misc Test: Temperature Sensor Alternative Reading
Page 6, Register 27

Bits	Field	Mode	HW Rst	SW Rst	Description
15:13	Reserved	R/W	0x0	Retain	Reserved.
12:11	Temp Sensor: Number to average samples	R/W	2'b01	Retain	00: average over 2^9 samples 01: average over 2^11 samples 10: average over 2^13 samples 11: average over 2^15 samples
10:8	Temp Sensor: sampling rate	R/W	3'b100	Retain	Sampling rate 000: 28 us 001: 56 us 010: 168 us 011: 280 us 100: 816 us 101: 2.28 ms 110: 6.22 ms 111: 11.79 ms
7:0	Temperature Sensor Alternative reading	RO	xxxxx	Retain	Temperature in C = 1 x 27_6.7:0 - 25 i.e. for 100C the value is 0111_1101

Table 171: PHY Cable Diagnostics Pair 0 Length
Page 7, Register 16

Bits	Field	Mode	HW Rst	SW Rst	Description
15:0	Pair 0 Cable Length	RO	0x0000	Retain	Length to fault in meters or centimeters based on register 21_7.10.

Table 172: PHY Cable Diagnostics Pair 1 Length
Page 7, Register 17

Bits	Field	Mode	HW Rst	SW Rst	Description
15:0	Pair 1 Cable Length	RO	0x0000	Retain	Length to fault in meters or centimeters based on register 21_7.10.

Table 173: PHY Cable Diagnostics Pair 2 Length
Page 7, Register 18

Bits	Field	Mode	HW Rst	SW Rst	Description
15:0	Pair 2 Cable Length	RO	0x0000	Retain	Length to fault in meters or centimeters based on register 21_7.10.

Table 174: PHY Cable Diagnostics Pair 3 Length
Page 7, Register 19

Bits	Field	Mode	HW Rst	SW Rst	Description
15:0	Pair 3 Cable Length	RO	0x0000	Retain	Length to fault in meters or centimeters based on register 21_7.10.

Table 175: PHY Cable Diagnostics Results
Page 7, Register 20

Bits	Field	Mode	HW Rst	SW Rst	Description
15:12	Pair 3 Fault Code	RO	0x0	Retain	0000 = Invalid 0001 = Pair Ok 0010 = Pair Open 0011 = Same Pair Short 0100 = Cross Pair Short 1001 = Pair Busy else = Reserved
11:8	Pair 2 Fault Code	RO	0x0	Retain	0000 = Invalid 0001 = Pair Ok 0010 = Pair Open 0011 = Same Pair Short 0100 = Cross Pair Short 1001 = Pair Busy else = Reserved

Table 175: PHY Cable Diagnostics Results (Continued)
Page 7, Register 20

Bits	Field	Mode	HW Rst	SW Rst	Description
7:4	Pair 1 Fault Code	RO	0x0	Retain	0000 = Invalid 0001 = Pair Ok 0010 = Pair Open 0011 = Same Pair Short 0100 = Cross Pair Short 1001 = Pair Busy else = Reserved
3:0	Pair 0 Fault Code	RO	0x0	Retain	0000 = Invalid 0001 = Pair Ok 0010 = Pair Open 0011 = Same Pair Short 0100 = Cross Pair Short 1001 = Pair Busy else = Reserved

Table 176: PHY Cable Diagnostics Control
Page 7, Register 21

Bits	Field	Mode	HW Rst	SW Rst	Description
15	Run Immediately	R/W, SC	0	Retain	0 = No Action 1 = Run VCT Now
14	Run At Each Auto-Negotiation Cycle	R/W	1	Retain	0 = Do No Run At Auto-Negotiation Cycle 1 = Run At Auto-Negotiation Cycle
13	Disable Cross Pair Check	R/W	0	Retain	0 = Enable Cross Pair Check 1 = Disable Cross Pair Check
12	Run After Break Link	R/W, SC	0	0	0 = No Action 1 = Run VCT After Breaking Link
11	Cable Diagnos-tics Status	RO	0	Retain	0 = Complete 1 = In Progress
10	Cable Length Unit	R/W	0	Retain	0 = Centimeters 1 = Meters
9:8	DSP VCT Select	R/W	00	Retain	If 88EA1517 is used the default is 01 otherwise it is 00. 2'b00 = TDR+DSPVCT 2'b01 = TDR Only 2'b10 = DSP VCT Only 2'b11 = Reserved
7	DSP VCT Hold	R/W	0	Retain	0 = Normal 1 = Hold DSP VCT until this bit is released
6:0	Reserved	RO	0	0	Reserved.

Table 177: Advanced VCT Cross Pair Negative Threshold
Page 7, Register 25

Bits	Field	Mode	HW Rst	SW Rst	Description
15	Reserved	RO	0x0	Retain	Reserved.
14:8	Cross Pair Negative Threshold > 30m	R/W	0x01	Retain	0x00 = 0 mV, 0x01 = - 7.81 mV,..., 0x7F = - 992 mV
7	Reserved	RO	0x0	Retain	Reserved.
6:0	Cross Pair Negative Threshold < 30m	R/W	0x04	Retain	0x00 = 0 mV, 0x01 = - 7.81 mV,..., 0x7F = - 992 mV

Table 178: Advanced VCT Same Pair Impedance Negative Threshold 0 and 1
Page 7, Register 26

Bits	Field	Mode	HW Rst	SW Rst	Description
15	Reserved	RO	0x0	Retain	Reserved.
14:8	Same-Pair Negative Threshold 10m - 50m	R/W	0x0F	Retain	0x00 = 0 mV, 0x01 = - 7.81 mV,..., 0x7F = - 992 mV
7	Reserved	RO	0x0	Retain	Reserved.
6:0	Same-Pair Negative Threshold < 10m	R/W	0x12	Retain	0x00 = 0 mV, 0x01 = - 7.81 mV,..., 0x7F = - 992 mV

Table 179: Advanced VCT Same Pair Impedance Negative Threshold 2 and 3
Page 7, Register 27

Bits	Field	Mode	HW Rst	SW Rst	Description
15	Reserved	RO	0x0	Retain	0
14:8	Same-Pair Negative Threshold 110m - 140m	R/W	0x0A	Retain	0x00 = 0 mV, 0x01 = - 7.81 mV,..., 0x7F = - 992 mV
7	Reserved	RO	0x0	Retain	Reserved.
6:0	Same-Pair Negative Threshold 50m - 110m	R/W	0x0C	Retain	0x00 = 0 mV, 0x01 = - 7.81 mV,..., 0x7F = - 992 mV

Table 180: Advanced VCT Same Pair Impedance Negative Threshold 4
Page 7, Register 28

Bits	Field	Mode	HW Rst	SW Rst	Description
15:7	Reserved	RO	0x000	0x000	Reserved.
6:0	Same-Pair Negative Threshold > 140m	R/W	0x06	Retain	0x00 = 0 mV, 0x01 = - 7.81 mV,..., 0x7F = - 992 mV

Table 181: WOL Control
Page 17, Register 16

Bits	Field	Mode	HW Rst	SW Rst	Description
15	SRAM Packet Match Enable	R/W	0x0	Retain	Allows for matching received packets with SRAM packets to assert WOL detection logic. 1 = Enable SRAM packet matching for WOL detection 0 = Disable SRAM packet matching for WOL detection
14	Magic Packet Match Enable	R/W	0x0	Retain	Allows matching received packets with a Magic Packet (packet containing 6 bytes of FF followed by 16 instances of the destination address) to assert the WOL detection logic 1 = Enable Magic Packet matching for WOL detection 0 = Disable Magic Packet matching for WOL detection
13	Link Up Enable	R/W	0x0	Retain	Allows link up event to assert the WOL detection logic 1 = Enable link up event for WOL detection 0 = Disable link up event for WOL detection
12	Clear WOL Status	R/W, SC	0x0	Retain	1 = Clear status in 17_17 0 = Retain status in 17_17
11:9	Reserved	R/W	0x0	Retain	
8	10BT Low Power Mode Select	R/W	0x0	Retain	1 = 10BT Low Power mode enabled. The transmitter is powered down in between transmitted link pulses. 10BT packets cannot be transmitted when this mode is enabled. 0 = Normal Operation This feature may be enabled when WOL event detection is enabled during 10BT operation for lower power.
7	SRAM Packet Match Packet 7 Enable	R/W	0x0	Retain	1 = Enable received packet matching to SRAM's Packet 7 0 = Disable received packet matching to SRAM's Packet 7
6	SRAM Packet Match Packet 6 Enable	R/W	0x0	Retain	1 = Enable received packet matching to SRAM's Packet 6 0 = Disable received packet matching to SRAM's Packet 6
5	SRAM Packet Match Packet 5 Enable	R/W	0x0	Retain	1 = Enable received packet matching to SRAM's Packet 5 0 = Disable received packet matching to SRAM's Packet 5

Table 181: WOL Control (Continued)
Page 17, Register 16

Bits	Field	Mode	HW Rst	SW Rst	Description
4	SRAM Packet Match Packet 4 Enable	R/W	0x0	Retain	1 = Enable received packet matching to SRAM's Packet 4 0 = Disable received packet matching to SRAM's Packet 4
3	SRAM Packet Match Packet 3 Enable	R/W	0x0	Retain	1 = Enable received packet matching to SRAM's Packet 3 0 = Disable received packet matching to SRAM's Packet 3
2	SRAM Packet Match Packet 2 Enable	R/W	0x0	Retain	1 = Enable received packet matching to SRAM's Packet 2 0 = Disable received packet matching to SRAM's Packet 2
1	SRAM Packet Match Packet 1 Enable	R/W	0x0	Retain	1 = Enable received packet matching to SRAM's Packet 1 0 = Disable received packet matching to SRAM's Packet 1
0	SRAM Packet Match Packet 0 Enable	R/W	0x0	Retain	1 = Enable received packet matching to SRAM's Packet 0 0 = Disable received packet matching to SRAM's Packet 0

Table 182: WOL Status
Page 17, Register 17

Bits	Field	Mode	HW Rst	SW Rst	Description
15	SRAM Packet Match Detected	RO,LH	0x0	0x0	1 = At least 1 SRAM packet matched a received packet 0 = No SRAM packets have matched with a received packet
14	Magic Packet Match Detected	RO,LH	0x0	0x0	1 = At least 1 received packet contained the Magic Packet data 0 = No received packets contained the Magic Packet data
13	Link Change Detected	RO,LH	0x0	0x0	1 = Link status changed from down to up 0 = Link status did not change from down to up
12:8	Reserved	R/W	0x00	Retain	Reserved.
7	SRAM Packet Match With Packet 7 Detected	RO,LH	0x0	0x0	1 = At least 1 received packet matched SRAM's Packet 7 0 = No received packets matched SRAM's Packet 7
6	SRAM Packet Match With Packet 6 Detected	RO,LH	0x0	0x0	1 = At least 1 received packet matched SRAM's Packet 6 0 = No received packets matched SRAM's Packet 6
5	SRAM Packet Match With Packet 5 Detected	RO,LH	0x0	0x0	1 = At least 1 received packet matched SRAM's Packet 5 0 = No received packets matched SRAM's Packet 5

Table 182: WOL Status (Continued)
Page 17, Register 17

Bits	Field	Mode	HW Rst	SW Rst	Description
4	SRAM Packet Match With Packet 4 Detected	RO,LH	0x0	0x0	1 = At least 1 received packet matched SRAM's Packet 4 0 = No received packets matched SRAM's Packet 4
3	SRAM Packet Match With Packet 3 Detected	RO,LH	0x0	0x0	1 = At least 1 received packet matched SRAM's Packet 3 0 = No received packets matched SRAM's Packet 3
2	SRAM Packet Match With Packet 2 Detected	RO,LH	0x0	0x0	1 = At least 1 received packet matched SRAM's Packet 2 0 = No received packets matched SRAM's Packet 2
1	SRAM Packet Match With Packet 1 Detected	RO,LH	0x0	0x0	1 = At least 1 received packet matched SRAM's Packet 1 0 = No received packets matched SRAM's Packet 1
0	SRAM Packet Match With Packet 0 Detected	RO,LH	0x0	0x0	1 = At least 1 received packet matched SRAM's Packet 0 0 = No received packets matched SRAM's Packet 0

Table 183: SRAM Packets 7/6 Length
Page 17, Register 18

Bits	Field	Mode	HW Rst	SW Rst	Description
15:14	Reserved	R/W	0x0	Retain	Reserved.
13:7	SRAM Packet 7 Length	RW	111_1111	Retain	Number of bytes (including byte 0) that a received packet must have in order to match Packet 7. Note that this matching requirement is in addition to matching the bytes that need to be matched.
6:0	SRAM Packet 6 Length	RW	111_1111	Retain	Number of bytes (including byte 0) that a received packet must have in order to match Packet 6. Note that this matching requirement is in addition to matching the bytes that need to be matched.

Table 184: SRAM Packets 5/4 Length
Page 17, Register 19

Bits	Field	Mode	HW Rst	SW Rst	Description
15:14	Reserved	R/W	0x0	Retain	Reserved.
13:7	SRAM Packet 5 Length	RW	111_1111	Retain	Number of bytes (including byte 0) that a received packet must have in order to match Packet 5. Note that this matching requirement is in addition to matching the bytes that need to be matched.
6:0	SRAM Packet 4 Length	RW	111_1111	Retain	Number of bytes (including byte 0) that a received packet must have in order to match Packet 4. Note that this matching requirement is in addition to matching the bytes that need to be matched.

Table 185: SRAM Packets 3/2 Length
Page 17, Register 20

Bits	Field	Mode	HW Rst	SW Rst	Description
15:14	Reserved	R/W	0x0	Retain	Reserved.
13:7	SRAM Packet 3 Length	RW	111_1111	Retain	Number of bytes (including byte 0) that a received packet must have in order to match Packet 3. Note that this matching requirement is in addition to matching the bytes that need to be matched.
6:0	SRAM Packet 2 Length	RW	111_1111	Retain	Number of bytes (including byte 0) that a received packet must have in order to match Packet 2. Note that this matching requirement is in addition to matching the bytes that need to be matched.

Table 186: SRAM Packets 1/0 Length
Page 17, Register 21

Bits	Field	Mode	HW Rst	SW Rst	Description
15:14	Reserved	R/W	0x0	Retain	Reserved.
13:7	SRAM Packet 1 Length	RW	111_1111	Retain	Number of bytes (including byte 0) that a received packet must have in order to match Packet 1. Note that this matching requirement is in addition to matching the bytes that need to be matched.
6:0	SRAM Packet 0 Length	RW	111_1111	Retain	Number of bytes (including byte 0) that a received packet must have in order to match Packet 0. Note that this matching requirement is in addition to matching the bytes that need to be matched.

Table 187: Magic Packet Destination Address Word 2
Page 17, Register 23

Bits	Field	Mode	HW Rst	SW Rst	Description
15:0	Destination Address {7:0,15:8}	R/W	0x0000	Retain	This data is the upper 47:32 bits of the destination address that must be seen repeated 16 times in a received packet in order for it to be a Magic Packet. This must be programmed since the PHY has no other way of knowing what the Destination Address is.

Table 188: Magic Packet Destination Address Word 1
Page 17, Register 24

Bits	Field	Mode	HW Rst	SW Rst	Description
15:0	Destination Address {23:16,31:24}	R/W	0x0000	Retain	This data is the middle 31:16 bits of the destination address that must be seen repeated 16 times in a received packet in order for it to be a Magic Packet. This must be programmed since the PHY has no other way of knowing what the Destination Address is.

Table 189: Magic Packet Destination Address Word 0
Page 17, Register 25

Bits	Field	Mode	HW Rst	SW Rst	Description
15:0	Destination Address {39:32,47:40}	R/W	0x0000	Retain	This data is the lower 15:0 bits of the destination address that must be seen repeated 16 times in a received packet in order for it to be a Magic Packet. This must be programmed since the PHY has no other way of knowing what the Destination Address is.

Table 190: SRAM Byte Address Control
Page 17, Register 26

Bits	Field	Mode	HW Rst	SW Rst	Description
15:12	Reserved	R/W	0x0	Retain	
11	Read Enable	R/W	0x0	Retain	Enables reading the mask byte bit and data stored at SRAM Byte Address (indicated by 26_17.9:7) and SRAM Row Address (indicated by 26_17.6:0) and placed in 28_17.8:0. This completes DMA capability (writing is already advertised). 1 = read SRAM mask byte bit and data 0 = do not read SRAM mask byte bit and data Note that this bit must be set to 1 only after 26_17.9:0 have been written with the address of data to be read from the SRAM.
10	Write Enable	R/W, SC	0x0	Retain	When a 1 is written to this bit, Byte Data and Mask Byte Data are written to the location in the SRAM addressed by 26_17.9:0.
9:7	SRAM Byte Address	R/W	0x0	Retain	Indicates the byte address for a given SRAM row that should store Byte Data (27_17.7:0) from 000 (bits 7:0) up to 111 (bits 63:56). Mask Byte Data (27_17.8) is stored in the most significant byte location in the bit corresponding to the Packet number (bit 64 for Packet 0's mask bit up to bit 71 for Packet 7's mask bit). Note that this must be programmed before 26_17.10 is set to 1.
6:0	SRAM Row Address	R/W	0x00	Retain	Indicates the Row number of the 128 row X 72 bit SRAM used for the SRAM Packet Matching where Byte Data (27_17.7:0) is to be stored. Note that this must be programmed before 26_17.10 is set to 1.

Table 191: SRAM Byte Data Control
Page 17, Register 27

Bits	Field	Mode	HW Rst	SW Rst	Description
15:9	Reserved	R/W	0x00	Retain	Reserved.

Table 191: SRAM Byte Data Control
Page 17, Register 27

Bits	Field	Mode	HW Rst	SW Rst	Description
8	Mask Byte Data	R/W	0x0	Retain	Programmed with Byte Data (27_17.7:0). Indicates if Byte Data should be compared or not in the SRAM packet matching. 1 = compare Byte Data with the corresponding byte of a received packet (do not mask the comparison) 0 = do not compare Byte Data with the corresponding byte of a received packet (mask the comparison) Note that this must be programmed before 26_17.10 is set to 1.
7:0	Byte Data	R/W	0x00	Retain	Data to be stored in the SRAM at the address indicated by 26_17.9:0. Note that this must be programmed before 26_17.10 is set to 1.

Table 192: SRAM Read Control
Page 17, Register 28

Bits	Field	Mode	HW Rst	SW Rst	Description
15:9	Reserved	R/W	0x00	Retain	Reserved.
8:0	SRAM Read Data	R/W	0x000	Retain	WOL SRAM DMA Read Data where bit 8 is the byte mask bit and 7:0 are the byte data.

Table 193: EEE Buffer Control Register 1
Page 18, Register 0

Bits	Field	Mode	HW Rst	SW Rst	Description
15:8	IPG Length	R/W	0x0C	Retain	Minimum Number of IPGs in bytes
7:2	Reserved	R/W	0	Retain	Reserved.
1	10BT Time Select	R/W	0	Retain	1 = use 4_18.15:0 for 10BT exit and enter time 0 = use 1_18.7:0 & 2_18.7:0 for 10BT exit and enter time
0	EEE Buffer Enable	R/W	See Desc	Retain	This bit is 1 always (a write of 0 has no effect) for when the device is using MII mode and EEE is enabled, otherwise this bit defaults to 0 and may be written to 1. 1 = Enable EEE Buffer, and EEE Buffer will initiate LPI_I-DLE based on TX traffic 0 = Disable EEE Buffer, Device will pass LPI_IDLE in both directions

Table 194: EEE Buffer Control Register 2
Page 18, Register 1

Bits	Field	Mode	HW Rst	SW Rst	Description
15:8	Fast_exit_time	R/W	0x11	Retain	LPI exit timer when port speed is 1000 Mbps default is 17 us. Each increment in the register corre- sponds to 1 us.
7:0	Slow_exit_time	R/W	0x1E	Retain	LPI exit timer when port speed is 100 Mbps, unit is 1us LPI exit timer when port speed is 10 Mbps, unit is 6.8us (when 0_18.1 is 0) default is 30 us for 100Mbps, 204us for 10Mbps

Table 195: EEE Buffer Control Register 3
Page 18, Register 2

Bits	Field	Mode	HW Rst	SW Rst	Description
15:8	Fast_enter_time	R/W	0x11	Retain	LPI enter timer when port speed is 1000 Mbps default is 17us
7:0	Slow_enter_time	R/W	0x1E	Retain	LPI enter timer when port speed is 100 Mbps LPI enter timer when port speed is 10 Mbps, unit is 6.8 us (when 0_18.1 is 0) Default is 30 us for 100 Mbps, 204 us for 10 Mbps

Table 196: EEE Buffer Control Register 4
Page 18, Register 4

Bits	Field	Mode	HW Rst	SW Rst	Description
15:8	10bps_enter_time	R/W	0x0C	Retain	LPI enter timer when port speed is 10 Mbps, when 0_18.1 is 1. Default is 120 us
7:0	10bps_exit_time	R/W	0x0C	Retain	LPI exit timer when port speed is 10 Mbps, when 0_18.1 is 1. Default is 120 us

Table 197: Packet Generation
Page 18, Register 16

Bits	Field	Mode	HW Rst	SW Rst	Description
15:8	Packet Burst	R/W	0x00	Retain	0x00 = Continuous 0x01 to 0xFF = Burst 1 to 255 packets

Table 197: Packet Generation (Continued)
Page 18, Register 16

Bits	Field	Mode	HW Rst	SW Rst	Description
7:5	Enable Packet Generator	R/W, SC	0x0	Retain	000 = Normal Operation 010 = Generate Packets on Copper Interface 100 = Generate Packets on SGMII Interface 101 = Reserved 110 = Generate Packets on RGMII Interface 111 = Reserved else = Reserved
4	Packet Generator Transmit Trigger	R/W	0x0	Retain	This bit is only valid when all of the following are true: bit 7:5 are not equal to 000 bit3 =1 bit15:8 is not equal to all 0s A read of this bit gives the following: 1: Packet generator transmit done 0: Packet generator is transmitting data When this bit is 1 a write of 0 will trigger the packet generator to transmit again. When this bit is 0 a write of 0 or 1 will have no effect.
3	Packet Generator Enable Self Clear Control	R/W	0x0	Retain	0 = Bit 7:5 will self clear after all packets are sent 1 = Bit 7:5 will stay at the current value after all packets are sent
2	Payload of packet to transmit	R/W	0x0	Retain	0 = Pseudo-random 1 = 5A,A5,5A,A5,...
1	Length of packet to transmit	R/W	0x0	Retain	1 = 1518 bytes 0 = 64 bytes
0	Transmit an Errored packet	R/W	0x0	Retain	1 = Tx packets with CRC errors & Symbol Error 0 = No error

Table 198: CRC Counters
Page 18, Register 17

Bits	Field	Mode	HW Rst	SW Rst	Description
15:8	Packet Count	RO	0x00	Retain	0x00 = No packets received 0xFF = 256 packets received (max count). Bit 18_18.2:0. must not be all 0 in order for these bits to be valid.
7:0	CRC Error Count	RO	0x00	Retain	0x00 = No CRC errors detected in the packets received 0xFF = 256 CRC errors detected in the packets received (max count) Bit 18_18.2:0. must not be all 0 in order for these bits to be valid.

Table 199: Checker Control
Page 18, Register 18

Bits	Field	Mode	HW Rst	SW Rst	Description
15:5	Reserved	R/W	0x000	Retain	Set to 0s
4	CRC Counter Reset	R/W, SC	0x0	Retain	1 = Reset This bit will self-clear after writing 1.
3	Reserved	R/W	0x0	Retain	Reserved.
2:0	Enable CRC Checker	R/W	0x0	Retain	000 = Disable/reset CRC checker 010 = Check data from Copper Interface 100 = Check data from SGMII Interface 101 = Reserved 110 = Check data from RGMII Interface 111 = Reserved else = Reserved

Table 200: Packet Generation
Page 18, Register 19

Bits	Field	Mode	HW Rst	SW Rst	Description
15	Constant Packet Payload Enable	R/W	0	Retain	.Enables a constant payload for packets generated by the packet generator. 1 = Enable constant packet payload 0 = Use other payload options
14	Constant Packet Payload Type	R/W	0	Retain	This bit selects the type of constant payload to use and is valid only when 19_18.15 = 1 1 = use all 1s packet payload 0 = use all 0s packet payload
13:8	Reserved	R/W	0	Retain	Reserved
7:0	IPG Length	R/W	8'd12	Retain	The number in bit 7:0+1 is the number of bytes for IPG

Table 201: General Control Register 1
Page 18, Register 20

Bits	Field	Mode	HW Rst	SW Rst	Description
15	Reset	R/W, SC	0x0	SC	Mode Software Reset. Affects Page 6 and 18 Writing a 1 to this bit causes the main PHY state machines to be reset. When the reset operation is done, this bit is cleared to 0 automatically. The reset occurs immediately. 1 = PHY reset 0 = Normal operation
14:13	Reserved	R/W	0x0	Retain	Set to 0s.

Table 201: General Control Register 1
Page 18, Register 20

Bits	Field	Mode	HW Rst	SW Rst	Description
12:10	Reserved	R/W	0x0	Retain	Reserved for future use.
9:7	Reserved	R/W	0x0	Retain	Set to 0s.
6	100BASE-FX Option Select	R/W	0	Retain	This bit selects between using 100BASE-FX or using Auto Media Detect between Copper and 100BASE-FX for mode 011. The bit definition is as follows: 0 = RGMII (System mode) to 100BASE-FX 1 = RGMII (System mode) to Auto Media Detect Copper/ 100BASE-FX
5:4	Auto Media Detect Preferred Media	R/W	00	Retain	00 = Link on first media 01 = Copper Preferred 10 = Fiber Preferred 11 = Reserved
3	MII mode	RO		Retain	1 = MII mode 0 = RGMII mode
2:0	MODE[2:0]	R/W	See Descr.	Update	Changes to this bit are disruptive to the normal operation; therefore, any changes to these registers must be followed by a software reset to take effect. When RGMII is used, the mapping is: 000 = RGMII (System mode) to Copper 001 = SGMII (System mode) to Copper 010 = RGMII (System mode) to 1000BASE-X 011 = RGMII (System mode) to 100BASE-FX (20_18.6 = 0) 011 = RGMII (System mode) to Auto Media Detect Copper/100BASE-FX (20_18.6 = 1) 100 = RGMII (System mode) to SGMII (Media mode) 101 = Reserved 110 = RGMII (System mode) to Auto Media Detect Copper/SGMII (Media mode) 111 = RGMII (System mode) to Auto Media Detect Copper/1000BASE-X When MII is used, the mapping is: 000 = MII (System mode) to Copper 001 = SGMII (System mode) to Copper 010 = MII (System mode) to SGMII (Media mode) 011 = MII (System mode) to 100BASE-FX 100 = MII (System mode) to SGMII (Media mode) 101 = Reserved 110 = MII (System mode) to Copper 111 = MII (System mode) to Copper

Table 202: Link Disconnect count
Page 18, Register 25

Bits	Field	Mode	HW Rst	SW Rst	Description
15:8	Reserved	R/W	0x00	Retain	Set to 0s
7:0	Link Disconnect	RO, SC	0x00	Retain	This counter counts the number of times link status changed from up to down. The counter will not roll over and will clear on read.

Table 203: SERDES RX_ER Byte Capture
Page 18, Register 26

Bits	Field	Mode	HW Rst	SW Rst	Description
15	Capture Data Valid	RO	0x0	0x0	1 = Bits 14:0 Valid 0 = Bits 14:0 Invalid
14	Reserved	RO	0x0	0x0	Reserved.
13:12	Byte Number	RO	0x0	0x0	00 = 4 bytes before RX_ER asserted 01 = 3 bytes before RX_ER asserted 10 = 2 bytes before RX_ER asserted 11 = 1 byte before RX_ER asserted The byte number increments after every read when register 26_18.15 is set to 1.
11:10	Reserved	RO	0x0	0x0	Reserved.
9	RX_ER	RO	0x0	0x0	RX Error. Normally this bit will be low since the capture is triggered by RX_ER being high. However it is possible to see RX_ER high when the capture is re-enabled after reading the fourth byte and there happens to be a long sequence of RX_ER when the capture restarts.
8	RX_DV	RO	0x0	0x0	RX Data Valid
7:0	RXD[7:0]	RO	0x00	0x00	RX Data

3.3 PHY XMDIO Register Description

88EA1517/88EA1512 supports Clause 22 MMD extension registers to access Clause 45 MMD registers, using register 13 and 14, as specified in the IEEE Annex 22D.

Table 204: XMDIO MMD Control Register
Device 0 Register 13

Bits	Field	Mode	HW Rst	SW Rst	Description
15:14	Function	R/W	0	0	11 = Data, post increments on writes only 10 =Data, post increment on reads and writes 01= Data, no post increment 00 = Address
13:5	Reserved	RO	0	0	Reserved.
4:0	DEVAD	RO	0	0	Device Address.

Table 205: XMDIO MMD Address Data Register
Device 0 Register 14

Bits	Field	Mode	HW Rst	SW Rst	Description
15:0	Address Data	R/W	0	0	If 13.15:14 = 00, MMD DEVAD's address register. Otherwise, MMD DEVAD's data register as indicated by the contents of its address register

For example, to write single Clause 45 register, perform the following accesses:

1. To Register 13, write 00 (address) to bit 15:14 and the device address value to bit 4:0;
2. To Register 14, write the address value;
3. To Register 13, write 01 (Data, no post increment) to bit 15:14 and the same device address value to bit 4:0;
4. To Register 14, write the content of the MMD's selected register.

Step 1 and Step 2 can be skipped if the MMD's address register was previously configured.

For example, to read single Clause 45 register, perform the following accesses:

1. To Register 13, write 00 (address) to bit 15:14 and the device address value to bit 4:0;
2. To Register 14, write the address value;
3. To Register 13, write 01 (Data, no post increment) to bit 15:14 and the same device address value to bit 4:0;
4. From Register 14, read the content of the MMD's selected register.

Step 1 and Step 2 can be skipped if the MMD's address register was previously configured.

Table 206: 88EA1517/88EA1512 Register Map

Register Name	Register Address	Table and Page
PCS Control 1 Register	Device 3 Register 0	Table 207 on page 213
PCS Status 1 Register	Device 3 Register 1	Table 208 on page 213
PCS EEE Capability Register	Device 3 Register 20	Table 209 on page 213
PCS EEE Wake Error Counter	Device 3 Register 22	Table 210 on page 214
EEE Advertisement Register	Device 7 Register 60	Table 211 on page 214
EEE Link Partner Advertisement Register	Device 7 Register 61	Table 212 on page 215

Table 207: PCS Control 1 Register
Device 3 Register 0

Bits	Field	Mode	HW Rst	SW Rst	Description
15:11	Reserved	R	0	Retain	Set to 0s.
10	Clock Stoppable	R/W	0	Retain	1 = Clock stoppable during LPI 0 = Clock not stoppable
9:0	Reserved	R	0	Retain	Set to 0s.

Table 208: PCS Status 1 Register
Device 3 Register 1

Bits	Field	Mode	HW Rst	SW Rst	Description
15:12	Reserved	R	0	Retain	Set to 0s.
11	Tx LP idle received	RO/LH	0	Retain	1 = Tx PCS has received LP idle 0 = LP Idle not received
10	Rx LP idle received	RO/LH	0	Retain	1 = Rx PCS has received LP idle 0 = LP Idle not received
9	Tx LP idle indication	RO	0	Retain	1 = Tx PCS is currently receiving LP idle 0 = PCS is not currently receiving LP idle
8	Rx LP idle indication	RO	0	Retain	1 = Rx PCS is currently receiving LP idle 0 = PCS is not currently receiving LP idle
7:3	Reserved	R	0	Retain	Set to 0s.
2	PCS receive link status	R	0	Retain	1 = PCS receive link up 0 = PCS receive link down
1	Low-power ability	R	0	Retain	1 = PCS supports low-power mode 0 = PCS does not support low-power mode
0	Reserved	R	0	Retain	Set to 0.

Table 209: PCS EEE Capability Register
Device 3 Register 20

Bits	Field	Mode	HW Rst	SW Rst	Description
15:7	Reserved	R	0	Retain	Set to 0s.

**Table 209: PCS EEE Capability Register (Continued)
Device 3 Register 20**

Bits	Field	Mode	HW Rst	SW Rst	Description
6	10GBASE-KR EEE	R	0	Retain	1 = EEE is supported for 10GBASE-KR 0 = EEE is not supported for 10GBASE-KR
5	10GBASE-KX4 EEE	R	0	Retain	1 = EEE is supported for 10GBASE-KX4 0 = EEE is not supported for 10GBASE-KX4
4	1000BASE-KX EEE	R	0	Retain	1 = EEE is supported for 1000BASE-KX 0 = EEE is not supported for 1000BASE-KX
3	10GBASE-T EEE	R	0	Retain	1 = EEE is supported for 10GBASE-T 0 = EEE is not supported for 10GBASE-T
2	1000BASE-T EEE	R	1	Retain	1 = EEE is supported for 1000BASE-T 0 = EEE is not supported for 1000BASE-T
1	100BASE-TX EEE	R	1	Retain	1 = EEE is supported for 100BASE-TX 0 = EEE is not supported for 100BASE-TX
0	Reserved	R	0	Retain	Set to 0.

**Table 210: PCS EEE Wake Error Counter
Device 3 Register 22**

Bits	Field	Mode	HW Rst	SW Rst	Description
15:0	EEE Wake Error Counter	R, NR	0	Retain	This counter is incremented for each transition of lpi_wake_timer_done from FALSE to TRUE

**Table 211: EEE Advertisement Register
Device 7 Register 60**

Bits	Field	Mode	HW Rst	SW Rst	Description
15:7	Reserved	R	0	Retain	Set to 0s.
6	10GBASE-KR EEE	R	0	Retain	1 = EEE is supported for 10GBASE-KR 0 = EEE is not supported for 10GBASE-KR
5	10GBASE-KX4 EEE	R	0	Retain	1 = EEE is supported for 10GBASE-KX4 0 = EEE is not supported for 10GBASE-KX4
4	1000BASE-KX EEE	R	0	Retain	1 = EEE is supported for 1000BASE-KX 0 = EEE is not supported for 1000BASE-KX
3	10GBASE-T EEE	R	0	Retain	1 = EEE is supported for 10GBASE-T 0 = EEE is not supported for 10GBASE-T
2	1000BASE-T EEE	R/W	0	Retain	1 = EEE is supported for 1000BASE-T 0 = EEE is not supported for 1000BASE-T Changes to this bit are disruptive to the normal operation; therefore, any changes to these registers must be followed by a software reset to take effect.

Table 211: EEE Advertisement Register (Continued)
Device 7 Register 60

Bits	Field	Mode	HW Rst	SW Rst	Description
1	100BASE-TX EEE	R/W	0	Retain	1 = EEE is supported for 100BASE-TX 0 = EEE is not supported for 100BASE-TX Changes to this bit are disruptive to the normal operation; therefore, any changes to these registers must be followed by a software reset to take effect.
0	Reserved	R	0	Retain	Set to 0.

Table 212: EEE Link Partner Advertisement Register
Device 7 Register 61

Bits	Field	Mode	HW Rst	SW Rst	Description
15:7	Reserved	R	0	Retain	Set to 0s.
6	LP 10GBASE-KR EEE	R	0	Retain	1 = EEE is supported for 10GBASE-KR 0 = EEE is not supported for 10GBASE-KR
5	LP 10GBASE-KX4 EEE	R	0	Retain	1 = EEE is supported for 10GBASE-KX4 0 = EEE is not supported for 10GBASE-KX4
4	LP 1000BASE-KX	R	0	Retain	1 = EEE is supported for 1000BASE-KX 0 = EEE is not supported for 1000BASE-KX
3	LP 10GBASE-T EEE	R	0	Retain	1 = EEE is supported for 10GBASE-T 0 = EEE is not supported for 10GBASE-T
2	LP 1000BASE-T EEE	R	0	Retain	1 = EEE is supported for 1000BASE-T 0 = EEE is not supported for 1000BASE-T
1	LP 100BASE-TX EEE	R	0	Retain	1 = EEE is supported for 100BASE-TX 0 = EEE is not supported for 100BASE-TX
0	Reserved	R	0	Retain	Set to 0.

4 PTP Register Description

The 88EA1517/88EA1512 device's PTP Registers are accessible using the MDC and MDIO pins and support the IEEE Serial Management Interface (SMI – Clause 22) used for PHY devices.

4.1 Register Types

The PTP Registers in the 88EA1517/88EA1512 device are made up of one or more fields. The way in which each of these fields operate is defined by the field's Type. The function of each Type is described below.

Table 213: Register Types

Type	Description
LH	Register field with latching high function. If status is high, then the register is set to one and remains set until a read operation is performed through the management interface or a reset occurs.
LL	Register field with latching low function. If status is low, then the register is cleared to zero and remains zero until a read operation is performed through the management interface or a reset occurs.
RES	Reserved. All reserved bits are read as zero unless otherwise noted.
Retain	The register value is retained after software reset is executed.
RO	Read only.
ROC	Read only clear. After read, register field is cleared.
RW	Read and Write with initial value indicated.
RWC	Read/Write clear on read. All bits are readable and writable. After reset or after the register field is read, register field is cleared to zero.
SC	Self-Clear. Writing a one to this register causes the desired function to be immediately executed, then the register field is automatically cleared to zero when the function is complete.
Update	Value written to the register field doesn't take effect until soft reset is executed.
WO	Write only. Reads from this type of register field return undefined data.
NR	Non-Rollover Register

4.1.1 PTP Registers

4.1.1.1 PTP Port Registers

Table 214: PTP Port Config Register
Page 8, Register 0 - PTP Port

Bits	Field	Type	Description
15:12	TransSpec	RWS 0x1	PTP Transport Specific value. The Transport Specific bits present in PTP Common header are used to differentiate between IEEE1588, IEEE802.1AS etc. frames. This is to differentiate between various timing protocols running on either Layer2 or higher protocol layers. In addition to comparing the EtherType to determine that the incoming frame is a PTP frame, the TransSpec bits are compared to the incoming PTP common headers' Transport Specific bits. If there is a match then hardware logic time stamps the frames indicated by MsgTypeEn and optionally interrupts the CPU. If there is no match, and Transport Spec checking is enabled (see DisTSpecCheck bit below) the hardware will not perform any operations in the PTP core. For IEEE 1588 networks this is expected to be configured to a 0x0 and for IEEE 802.1AS networks this is expected to be configured to 0x1. The only valid TransSpec values for PTP hardware acceleration (PTP Port Page 8, Register 2) are 0x0 and 0x1.
11	DisTSpec Check	RWR	Disable Transport Specific Check. 0 = Enable checking for Transport Spec 1 = Disable checking for Transport Spec When this bit is cleared to a zero the Transport Spec part of the PTP Common header of incoming frames must match the configured TransSpec (above) in order for time stamping to occur (PTP hardware accelerated or not). This setting limits PTP time stamping to frames containing a TransSpec value that matches the value in the TransSpec register above. This allows time stamping to be limited to only IEEE 1588 or to only IEEE 802.1AS per their frame's Transport Specs (assuming their value is contained in the TransSpec register above). When this bit is set to a one the Transport Spec checking of the PTP frames is not performed before time stamping occurs. This setting allows PTP time stamping for all TransSpec values (although PTP hardware acceleration, PTP Port Page 8, Register 2, only works on IEEE 1588 and IEEE 802.1AS TransSpec values).
10:2	Reserved	RES	Reserved for future use.

Table 214: PTP Port Config Register (Continued)
Page 8, Register 0 - PTP Port

Bits	Field	Type	Description
1	DisTS Overwrite	RWR	Disable Time Stamp Counter Overwriting. 0 = Overwrite unread Time Stamps in the registers with new data 1 = Keep unread Time Stamps in the registers until read When this bit is cleared to a zero, PTPArr0Time, PTPArr1Time and PTPDepTime values get overwritten even though their corresponding valid bits (defined in PTP Port Status Data Structure below), are not cleared. When this bit is set to one, PTPArr0Time, PTPArr1Time and PTPDepTime values don't get overwritten with new time stamps until their corresponding valid bits (defined in PTP Port Status Data Structure below), are cleared.
0	DisPTP	RWS	Disable Precise Time Stamp logic. 0 = PTP logic on this port is enabled 1 = PTP logic on this port is disabled When PTP logic is disabled the hardware logic doesn't recognize or timestamp PTP frames. Even interrupt generation logic is disabled. This disable disables all modes of PTP on this port (including PTP hardware acceleration if enabled - PTP Port Page 8, Register 2). **NOTE:** PTP should not be enabled on half-duplex ports when ArrTSMode or HWAccel (PTP Port Page 8, Register 2) are non zero.

Table 215: PTP Port Config Register
Page 8, Register 1 - PTP Port

Bits	Field	Type	Description
15:14	Reserved	RES	Reserved for future use.
13:8	IPJump	RWR	Internet Protocol Jump added to ETJump below. Set this register to point to the start of the frame's IP Version byte (802.1Q tagged frames are automatically compensated by ETJump). This field specifies how many bytes to skip starting at the first byte of the frame's EtherType (i.e., where ETJump, below, left off) in order to jump to the beginning of the IPv4 or IPv6 headers in the frame. If an IPv4 or IPv6 version is found at this location of the frame, Layer 4 PTP processing occurs from there. This allows flexibility in the hardware to skip past the protocol chains that are specific to customer networks including MPLS etc. For example if ETJump is programmed to 0xC and IPJump is programmed to 0x16, this indicates to hardware to skip 0x22 bytes in order to get to the IP header. It can either be IPv4 or IPv6 header. **NOTE:** A value of 0x0 (default) is a special case that prevents further frame searching if ETJump did not find a match (i.e., a zero value in IPJump prevents the IPJump mechanism from searching further).
7:5	Reserved	RES	Reserved for future use.
4:0	ETJump	RWS 0xC	EtherType Jump points to the start of the frame's EtherType (assuming it is not 802.1Q tagged). This field specifies how many bytes to skip starting from the start of the MAC-DA of the frame in order to get to the first byte of the EtherType of the frame. Frames found with an 0x8100 EtherType (802.1Q tag) at this location are automatically searched 4 bytes further into the frame for the next EtherType to compare (this extension is done once). If the PTPEType value (PTP Global Page 14, Register 0) is found as the Ether Type in the frame, Layer 2 PTP processing occurs. If 0x0800 or 0x86DD is found the Layer 4 PTP processing occurs. If none of these values is found, PTP frame decoding is passed to the IPJump field above. This allows flexibility in the hardware to skip past the protocol chains that are specific to customer networks including DSA-Tag, IEEE802.1Q tag, Provider tag etc.

Table 216: PTP Port Config Register
Page 8, Register 2 – PTP Port

Bits	Field	Type	Description
15:8	ArrTSMode	RWR	Arrival Time Stamp Mode. This field is used to configure the Arrival Time Stamp mode as follows: 0x00 = Arrival Time Stamp frame modification is disabled. 0x01 = Add the PTPArr[x]Time associated with enabled PTP Event frames at the end of the frame increasing the frame's size by four bytes. 0x02 to 0x0F = Reserved for future use. 0x10 to 0xEF = Overwrite the PTPArr[x]Time associated with enabled PTP Event frames into the frame without increasing the frame's size. The location in the frame where the PTPArr[x]Time is placed is controlled by this register. It is placed ArrTSMode bytes past the start of the PTP Common Header. **For example, to place the time stamp in the four Reserved bytes in the PTP Common Header, set this register to a value of 0x10.** If the end of the frame is reached prior to the completion of this overwrite, the PTPArr[x]Time is placed at the end of the frame increasing the frame's size by enough bytes for it to fit. 0xF0 to 0xFF = Reserved for future use. Notes: All frames will be processed using the above settings unless the frame can be hardware accelerated via setting the HWAccel bit below to a one. Changing this register's value can only occur when PTP is idle. Added PTPArr[x]Time bytes that increase the frame's size are included in the MIB counters and policy is performed on the resulting frame (e.g., TCAM matching and frame size checking). This register must be zero if ExtHWAccel (below) is set to a one.
7	FilterAct	RWR	Filter LED Activity. 0 = LED Activity is activated for all frames 1 = LED Activity is not activated for most IEEE 802.1 frames This bit can filter all or most of the 802.1 Protocol frames from the Port's Activity LEDs. When this bit is set to a one all 802.1 protocol frames (those with a DA = 01:C2:80:00:00:0x) will be potentially filtered from the port's Activity LED as determined by the ArrLEDCtrl and DepLEDCtrl registers (PTP Port Page 8, Register 3). When this bit is cleared to a zero only the 802.1 gPTP protocol frames will be potentially filtered from the port's Activity LED.

Table 216: PTP Port Config Register (Continued)
Page 8, Register 2 – PTP Port

Bits	Field	Type	Description
6	HWAccel	RWR	Port PTP Hardware Acceleration enable. 0 = No, or only Ingress PTP hardware acceleration 1 = Ingress and Egress PTP hardware acceleration is enabled Setting this bit to a one enables PTP hardware acceleration on this port. PTP hardware acceleration will automatically occur in the selected PTPMode (PTP Global Page 14 Register 7) for the enabled PTP Domains once a Time Array (PTP Global Page 15 Register 2 to Page 15 Register 13) is configured and enabled. Even then PTP hardware acceleration will only occur for the Transport Specs enabled on this port (PTP Port Page 8 register 0). In this mode, any frame that cannot be hardware accelerated will be processed using the settings defined by ArrTSMode above (a type of fallback mode). Clearing this bit to a zero causes all frames to be processed using the settings defined by ArrTSMode above. Do not set this bit to a one if ExtHWAccel (below) is set to a one. Notes: PTP Hardware Acceleration requires that the Time Stamping Clock Period (TAI Page 12 Register 1) be 8000 pico seconds (8 ns) ± 100 PPM. If the PTP_EXTCLK (external clock) is used (TAI Page 12 Register 0) it must also be within this range before enabling PTP Hardware Acceleration. Changing this register's value can only occur when PTP is idle.
5	KeepSA	RWR	Keep Frame's SA. 0 = Place Port's SA into modified egressing PTP frames 1 = Keep the frame's SA even for modified egressing PTP frames Normally when the Data portion of a frame (the part of the frame between the EtherType and the CRC) is modified the address of the modifying entity is placed into the Source Address (SA) field of egressing frames. When this bit is cleared to zero this is what is done for modified PTP frames. When this bit is set to a one the SA portion of PTP frames is not modified.
4:2	Reserved	RES	Reserved for future use.
1	PTPDepInt En	RWR	Precise Time Protocol Port Departure Interrupt enable. 0 = Disable PTP Departure capture interrupts 1 = Enable PTP Departure capture interrupts This field enables the per-port interrupt for outgoing PTP frame from this port. When this bit is set to a one and this ports's PTPDepTimeValid bit (PTP Port Page 9 Register 0) is set to a one a PTP interrupt for this port will be indicted in PTP Global Register (Page 14 Register 8). Note that hardware logic only time stamps the PTP frames when configured to do so by MsgTypeEn field (see PTP Global Page 14 Register 0).

Table 216: PTP Port Config Register (Continued)
Page 8, Register 2 – PTP Port

Bits	Field	Type	Description
0	PTPArrInt En	RWR	Precise Time Protocol Port Arrival Interrupt enable. 0 = Disable PTP Arrival capture interrupts 1 = Enable PTP Arrival capture interrupts This field enabled the per-port interrupt for incoming PTP frames from this port. When this bit is set to a one and this port's PTPArr0TimeValid bit (PTP Port Page 8 Register 8) or its PTPArr1TimeValid bit (PTP Port Page 8 Register 12) is set to a one a PTP interrupt for this port will be indicted in PTP Global Register (Page 14 Register 8). Note that hardware logic only time stamps the PTP frames when configured to do so by MsgTypeEn field (see PTP Global Page 14 Register 0).

Table 217: PTP Port Config Register
Page 8, Register 3 – PTP Port

Bits	Field	Type	Description
15:8	ArrLED Ctrl	RWR	LED control for packets entering the device. When 0x0, if a received frame is classified as a PTP frame or an 802.1 protocol frame, the LED does not blink. But it blinks for every non-PTP or non-802.1 protocol frame. When 0x1, the LED blinks for every received frame classified as a PTP frame or an 802.1 protocol frame. It also blinks for every non-PTP frame or non-802.1 protocol frame. When 0xn, the LED blinks once for every n received frames classified as PTP or 802.1 protocol. It also blinks for every non-PTP frame or non-802.1 protocol frame. **NOTE:** This tracks all received PTP frames (even though the PTP core time stamps only the PTP event messages) and not just PTP frames that need time stamping or it tracks all received 802.1 protocol frames (any frame with a DA = 01:C2:80:00:00:0x). The FilterAct bit in PTP Port (Page 8 Register 2) controls which frames are tracked. This register affect the port's Activity LED only. It does not change how the frames progress through the switch.

Table 217: PTP Port Config Register
Page 8, Register 3 – PTP Port

Bits	Field	Type	Description
7:0	DepLED Ctrl	RWS 0x80	LED control for packets departing the device. When 0x0, if a transmitting frame is classified as a PTP frame or an 802.1 protocol frame, the LED does not blink. But it blinks for every non-PTP frame or non-802.1 protocol. When 0x1, the LED blinks for every transmitting frame classified as a PTP frame or an 802.1 protocol frame. It also blinks for every non-PTP frame or non-802.1 protocol frame. When 0xn, the LED blinks once for every n transmitting frames classified as PTP or 802.1 protocol. It also blinks for every non-PTP frame or non-802.1 protocol frame. **NOTE:** This tracks all transmitting PTP frames (even though the PTP core time stamps only the PTP event messages) and not just PTP frames that need time stamping or it tracks all transmitted 802.1 protocol frames (any frame with a DA = 01:C2:80:00:00:0x). The FilterAct bit in PTP Port (Page 8 Register 2) controls which frames are tracked. This register affect the port's Activity LED only. It does not change how the frames progress through the switch.

Table 218: PTP Port Status Register
Page 8, Register 8 – PTP Port

Bits	Field	Type	Description
15:3	Reserved	RES	Reserved for future use.
2:1	PTPArr0IntStatus	RWR	Precise Time Protocol Arrival Time 0 Interrupt Status. The PTP Arrival time 0 Interrupt bit gets set for a given port when an incoming PTP frame is time stamped in PTPArr0Time counter as long as that frame was not hardware accelerated (see HWAccel in PTP Port (Page 8 Register 2)). 0x0 = Normal i.e., none of the error conditions stated below are valid for this packet. 0x1 = If the PTPArr0Time counter with its associated valid and SequenceID got overwritten as there were more than one PTP frame that needed to use arrival0 counters arrived into the switch through this port before CPU cleared the corresponding valid and counter bits for the previous PTP frame(s). 0x2 = If the incoming frame couldn't be time stamped in hardware because the DisTSOverwrite was set to a 0x1 and PTPArr0TimeValid was 0x1 when this PTPFrame was processed in hardware logic. This can happen when there is more than one PTP frame that needs time stamping into arrival 0 counters arrives into the switch core before CPU clears the valid bits for the previous frame. 0x3 = Reserved Note that if the PTP frame gets discarded inside the switch for policy, CRC, queue congestion or any other reasons then one of the PTP arrival discard counters get updated (PTPNonTSArrDisCtr or PTPTSArrDisCtr). See the discard counter descriptions (PTP Port Page 9 Register 5) for further details.
0	PTPArr0 TimeValid	RWR	Precise Time Protocol Arrival 0 Time Valid. When the PTPArr0Time value is updated by hardware (which it won't do as long as the frame is hardware accelerated – see HWAccel in PTP Port (Page 8 Register 2)), this bit is set to a 0x1 validating the time counter. 0x0 = PTPArr0Time is not valid. 0x1 = PTPArr0Time is valid and PTPArr0IntStatus represents the status information for the PTPArr0Time counter. Note that this is set by hardware for the frames which are assured to reach the CPU. For frames with CRC error etc., this bit will not be set but either PTPNonTSArrCtr or PTPTSArrCtr is updated. Note that this valid bit needs to be cleared by software after reading the value and hardware doesn't provide any auto-clearing mechanisms. This is because hardware has no way to figure out if software is done reading all the relevant registers for a particular Time counter before clearing the valid bit.

Table 219: PTP Port Status Register
Page 8, Register 9 – PTP Port

Bits	Field	Type	Description
15:0	PTPArr0 Time [15:0]	RWR	Precise Time Protocol Arrival 0 Time counter bits 15 to 0 of a 32-bit register. This indicates the PTP Arrival 0 time stamp value that is captured by the PTP logic for a PTP frame that needs to be time stamped. The captured time stamp value is from a Global Timer counter running off of a switch internal clock. The value in this counter is validated by PTPArr0TimeValid bit and PTPArr0IntStatus indicates the status of the PTP frame through the device as described above. Note that maximum jitter associated with time stamping within the hardware is one TSClkPer amount. The upper 16-bits of this register are contained in the register below.

Table 220: PTP Port Status Register
Page 8, Register 10– PTP Port

Bits	Field	Type	Description
15:0	PTPArr0 Time [31:16]	RWR	Precise Time Protocol Arrival 0 Time counter bits 31 to 16 of a 32-bit register. This indicates the PTP Arrival 0 time stamp value that is captured by the PTP logic for a PTP frame that needs to be time stamped. The captured time stamp value is from a Global Timer counter running off of a switch internal clock. The value in this counter is validated by PTPArr0TimeValid bit and PTPArr0IntStatus indicates the status of the PTP frame through the device as described above. Note that maximum jitter associated with time stamping within the hardware is one TSClkPer amount. The lower 16-bits of this register are contained in the register above.

Table 221: PTP Port Status Register
Page 8, Register11 – PTP Port

Bits	Field	Type	Description
15:0	PTPArr0 SeqId	RWR	Precise Time Protocol Arrival 0 Sequence Identifier. This indicates the sequence identifier (extracted in hardware from incoming frames PTPCommonHeader) for the frame whose time stamp information has been captured by hardware logic in PTPArr0Time register.

Table 222: PTP Port Status Register
Page 8, Register 12 – PTP Port

Bits	Field	Type	Description
15:3	Reserved	RES	Reserved for future use.
2:1	PTPArr1IntStatus	RWR	Precise Time Protocol Arrival Time 1 Interrupt Status. The PTP Arrival time 1 Interrupt bit gets set for a given port when an incoming PTP frame is time stamped in PTPArr1Time counter as long as that frame was not hardware accelerated (see HWAccel in PTP Port (Page 8 Register 2)). 0x0 = Normal i.e., none of the error conditions stated below are valid for this packet. 0x1 = If the PTPArr1Time counter with its associated valid and SequenceID got overwritten as there were more than one PTP frame that needed to use arrival1 counters arrived into the switch through this port before CPU cleared the corresponding valid and counter bits for the previous PTP frame(s).0x2 = If the incoming frame couldn't be time stamped in hardware because the DisTSOverwrite was set to a 0x1 and PTPArr1TimeValid was 0x1 when this PTPFrame was processed in hardware logic. This can happen when there is more than one PTP frame that needs time stamping into arrival 1 counters arrives into the switch core before CPU clears the valid bits for the previous frame. 0x3 = Reserved Note that if the PTP frame gets discarded inside the switch for policy, CRC, queue congestion or any other reasons then one of the PTP arrival discard counters get updated (PTPNonTSArrDisCtr or PTPTSArrDisCtr). See the discard counter description (PTP Port Page 9 Register 5) for further details.

Table 222: PTP Port Status Register (Continued)
Page 8, Register 12 – PTP Port

Bits	Field	Type	Description
0	PTPArr1 TimeValid	RWR	Precise Time Protocol Arrival 1 Time Valid. When the PTPArr1Time value is updated by hardware (which it won't do as long as the frame is hardware accelerated – see HWAccel in PTP Port (Page 8 Register 2)), this bit is set to a 0x1 validating the time counter. 0x0 = PTPArr1Time is not valid. 0x1 = PTPArr1Time is valid and PTPArr1IntStatus represents the status information for the PTPArr1Time counter. Note that this is set by hardware for the frames which are assured to reach the CPU. For frames with CRC error etc., this bit will not be set but either PTPNonTSArrCtr or PTPTSArrCtr is updated. Note that this valid bit needs to be cleared by software after reading the value and hardware doesn't provide any auto-clearing mechanisms. This is because hardware has no way to figure out if software is done reading all the relevant registers for a particular Time counter before clearing the valid bit.

Table 223: PTP Port Status Register
Page 8, Register 13 – PTP Port

Bits	Field	Type	Description
15:0	PTPArr1 Time [15:0]	RWR	Precise Time Protocol Arrival 1 Time counter bits 15 to 0 of a 32-bit register. This indicates the PTP Arrival 1 time stamp value that is captured by the PTP logic for a PTP frame that needs to be time stamped. The captured time stamp value is from a Global Timer counter running off of a switch internal clock. The value in this counter is validated by PTPArr1TimeValid bit and PTPArr1IntStatus indicates the status of the PTP frame through the device as described above. Note that maximum jitter associated with time stamping within the hardware is one TSClkPer amount. The upper 16-bits of this register are contained in the register below.

Table 224: PTP Port Status Register
Page 8, Register 14 – PTP Port

Bits	Field	Type	Description
15:0	PTPArr1 Time [31:16]	RWR	Precise Time Protocol Arrival 1 Time counter bits 31 to 16 of a 32-bit register. This indicates the PTP Arrival 1 time stamp value that is captured by the PTP logic for a PTP frame that needs to be time stamped. The captured time stamp value is from a Global Timer counter running off of a switch internal clock. The value in this counter is validated by PTPArr1TimeValid bit and PTPArr1IntStatus indicates the status of the PTP frame through the device as described above. Note that maximum jitter associated with time stamping within the hardware is one TSClkPer amount. The lower 16-bits of this register are contained in the register above.

Table 225: PTP Port Status Register
Page 8, Register 15 – PTP Port

Bits	Field	Type	Description
15:0	PTPArr1 SeqId	RWR	Precise Time Protocol Arrival 1 Sequence Identifier. This indicates the sequence identifier (extracted in hardware from incoming frames PTPCommonHeader) for the frame whose time stamp information has been captured by hardware logic in PTPArr1Time register.

Table 226: PTP Port Status Register
Page 9, Register 0 – PTP Port

Bits	Field	Type	Description
15:3	Reserved	RES	Reserved for future use.
2:1	PTPDepIntStatus	RWR	Precise Time Protocol Departure Time Interrupt Status. The PTP Departure time Interrupt bit gets set for a given port when an incoming PTP frame is time stamped in PTPDepTime counter as long as that frame was not hardware accelerated (see HWAccel in PTP Port (Page 8 Register 2)). 0x0 = Normal i.e., none of the error conditions stated below are valid for this packet. 0x1 = If the PTPDepTime counter with its associated valid and SequenceID got overwritten as there were more than one PTP frame that needed to use departure counter departed out of the switch through this port before CPU cleared the corresponding valid and counter bits for the previous PTP frame(s). 0x2 = If the outgoing frame couldn't be time stamped in hardware because the DisTSOverwrite was set to a 0x1 and PTPDepTimeValid was 0x1 when this PTPFrame was processed in hardware logic. This can happen when there is more than one PTP frame that needs time stamping into departure counter leaves the switch core before CPU clears the valid bits for the previous frame. 0x3 = Reserved Note that if the PTP frame gets discarded inside the switch for CRC reasons then the PTP departure discard counter gets updated (PTPNonTSDepDisCtr or PTPTSDepDisCtr). See the discard counter description (PTP Port Page 9 Register 5) for further details.
0	PTPDep TimeValid	RWR	Precise Time Protocol Departure Time Valid. When the PTPDepTime value is updated by hardware (which it won't do as long as the frame is hardware accelerated – see HWAccel in PTP Port (Page 8 Register 2)), this bit is set to a 0x1 validating the time counter. 0x0 = PTPDepTime is not valid. 0x1 = PTPDepTime is valid and PTPDepIntStatus represents the status information for the PTPDepTime counter. Note that this is set by hardware for the frames which are assured to depart the port. For frames with CRC error etc., this bit will not be set but either PTPNonTSDepCtr or PTPTSDepCtr is updated. Note that this valid bit needs to be cleared by software after reading the value and hardware doesn't provide any auto-clearing mechanisms. This is because hardware has no way to figure out if software is done reading all the relevant registers for a particular Time counter before clearing the valid bit.

Table 227: PTP Port Status Register
Page 9, Register 1 – PTP Port

Bits	Field	Type	Description
15:0	PTPDep Time [15:0]	RWR	Precise Time Protocol Departure Time counter bits 15 to 0 of a 32-bit register. This indicates the PTP Departure time stamp value that is captured by the PTP logic for a PTP frame that needs to be time stamped. The captured time stamp value is from a Global Timer counter running off of a switch internal clock. The value in this counter is validated by PTPDepTimeValid bit and PTPDepIntStatus indicates the status of the PTP frame through the device described above. Note that maximum jitter associated with time stamping within the hardware is one TSClkPer amount. The upper 16-bits of this register are contained in the register below.

Table 228: PTP Port Status Register
Page 9, Register 2 – PTP Port

Bits	Field	Type	Description
15:0	PTPDep Time [31:16]	RWR	Precise Time Protocol Departure Time counter bits 31 to 16 of a 32-bit register. This indicates the PTP Departure time stamp value that is captured by the PTP logic for a PTP frame that needs to be time stamped. The captured time stamp value is from a Global Timer counter running off of a switch internal clock. The value in this counter is validated by PTPDepTimeValid bit and PTPDepIntStatus indicates the status of the PTP frame through the device described above. Note that maximum jitter associated with time stamping within the hardware is one TSClkPer amount. The lower 16-bits of this register are contained in the register above.

Table 229: PTP Port Status Register
Page 9, Register 3 – PTP Port

Bits	Field	Type	Description
15:0	PTPDep SeqId	RWR	Precise Time Protocol Departure Sequence Identifier. This indicates the sequence identifier (extracted in hardware from incoming frames PTPCommonHeader) for the frame whose time stamp information has been captured by hardware logic in PTPDepTime register.

Table 230: Ingress Mean Path Delay
Page 9, Register 12 – PTP Port

Bits	Field	Type	Description
15:0	Mean PathDelay	RWR	Ingress Mean Path Delay register. This indicates the cable delay between this port and its link partner in unsigned nano seconds. This is used in HWAccel PTP mode (PTP Port (Page 8 Register 2)).

Table 231: Ingress Path Delay Asymmetry
Page 9, Register 13 – PTP Port

Bits	Field	Type	Description
15	IPDA Sign	RWR	Ingress Path Delay Asymmetry Sign. 0 = The Ingress Path Delay Asymmetry number is added 1 = The Ingress Path Delay Asymmetry number is subtracted This indicates the sign of the asymmetry value (below) beyond the Mean Path Delay (PTP Port Page 9 Register 12) that needs to be adjusted for more accurate cable measurements.
14:0	Ingress PathDelay Asymmetry	RWR	Ingress Path Delay Asymmetry register. This indicates the asymmetry value beyond the Mean Path Delay (PTP Port Page 9 Register 12) that needs to be added for more accurate cable measurements. This register is in unsigned nano seconds.

Table 232: Egress Path Delay Asymmetry
Page 9, Register 14 – PTP Port

Bits	Field	Type	Description
15	EPDA Sign	RWR	Egress Path Delay Asymmetry Sign. 0 = The Egress Path Delay Asymmetry number is added 1 = The Egress Path Delay Asymmetry number is subtracted This indicates the sign of the asymmetry value (below) beyond the Mean Path Delay (PTP Port Page 9 Register 12) that needs to be adjusted for more accurate cable measurements.
14:0	Egress PathDelay Asymmetry	RWR	Egress Path Delay Asymmetry register. This indicates the asymmetry value beyond the Mean Path Delay (PTP Port Page 9 Register 12) needs to be subtracted for more accurate cable measurements. This register is in unsigned nano seconds.

4.1.1.2 PTP Global Registers

Table 233: PTP Global Config
Page 14, Register 0– PTP Global

Bits	Field	Type	Description
15:0	PTPEType	RWS to 0x88F7	Precise Time Protocol EtherType. All layer 2 PTP frames are recognized using a combination of a specific EtherType and MessageType values (part of the PTP Common Header). This field is used to identify the EtherType on these frames. The MsgTypeEn (specified below in Page 14 Register 1) qualifies the types of frames that the hardware needs to time stamp. For IEEE 802.1AS and IEEE1588 over Layer 2 Ethernet, the EtherType is expected to be programmed to 0x88F7.

Table 234: PTP Global Config
Page 14, Register 1– PTP Global

Bits	Field	Type	Description
15:0	MsgType En	RWR	Message Type Time Stamp Enable. MessageType is part of the PTP common header. There are PTP frames which need to be time stamped and some that don't need to be. This field identifies the PTP frame types that need to be time stamped for frames that are not hardware accelerated – see HWAccel in PTP Port Page 8 Register 2). The MessageType read from PTP frames is vectorized[1] and then used to access the appropriate bit in this register. If the selected bit is set to a one then that frame type will be time stamped both in ingress and egress. Else that frame type will not be time stamped. For example if MessageType field (in the PTP common header) with a value of 0x4 needs to be time stamped in hardware then MsgTypeEn[4] should be set to a one. Then the incoming PTP frames with a MessageTypefield of 0x4 will get time stamped into one of the Port's two available arrival capture registers (either PTPArr0Time or PTPArr1Time as identified by TSArrPtr[4] bit below). All outgoing PTP frames with the MessageType field of 0x4 will be timestamped into the Port's PTPDepTime capture register.

1. Vectorized term here refers to converting the hexadecimal MessageType field into a sixteen bit binary number.

Table 235: PTP Global Config
Page 14, Register 2 – PTP Global

Bits	Field	Type	Description
15:0	TSArrPtr	RWR	Time Stamp Arrival Time Capture Pointer. If the incoming PTP frame needs to be time stamped (based on MsgTypeEn), this field determines whether the hardware logic should use PTPArr0Time or PTPArr1Time for storing the arriving frames' time stamp information. This field corresponds to the sixteen combinations of the vectorized MessageType. For example if TSArrPtr[2] is set to a one it indicates to the hardware that if MsgTypeEn [2] is set to a one then PTP frames with MessageType = 0x2 will use PTPArr1Time counter for storing the incoming PTP frames' time stamp. On the contrary if TSArrPtr[2] is cleared to a zero that indicates to the hardware that if MsgTypeEn[2] is set to a one then PTP frames with MessageType = 0x2 will use PTPArr0Time counter for storing the incoming PTP frames' time stamp.

Table 236: PTP Global Config
Page 14, Register 7 – PTP Global

Bits	Field	Type	Description
15	Update	SC	Update Data. When this bit is set to a one the data written to bits 7:0 will be loaded into the PTP Global Config register referenced by the Pointer bits below. After the write has taken place this bit self clears to zero.
14:8	Pointer	RWR	Pointer to the desired octet of PTP Global Config. These bits select one of the possible PTP Global Config registers for both read and write operations. A write operation occurs if the Update bit is a one (the registers can be written to by writing this register in a single operation). Otherwise a read of the current Pointer occurs and the data found there is placed in the Data bits below (the desired register can be read by first writing to this register, with Update = 0, and then reading this register). The Pointer bits are used to access the Index registers as follows: 0x00 = PTP Mode Register 0x01 to 0x7F = Reserved for future use
7:0	Data	RWR	Octet Data of the PTP Global Config register referenced by the Pointer bits above.

The individual registers accessed by the PTP Global Config register are described below and are indicated with a light purple background heading color.

Table 237: PTP Mode, Index: 0x00

Bits	Field	Type	Description
7:5	Reserved	RES	Reserved for future use.
4	AltScheme	RWR	Alternate Scheme. This bit is used to define the values that are returned in the requestReceiptTimestamp field in IEEE 1588 pDelay Response messages and the correctionField & responseOriginTimestamp in pDelay Response Follow Up messages as follows: 0 = RequestReceiptTimestamp = t2, responseOriginTimestamp = t3 & the correctionField = 0 (fractional ns) 1 = RequestReceiptTimestamp = 0, responseOriginTimestamp = 0 & the correctionField = turn around time **NOTE:** This bit has no effect if the PTP frame is an IEEE 802.1AS frame (as indicated by the frame's TransSpec field).
3	GrandMstr	RWR	Grand Master Enable. This bit is used to enable hardware support for the ports on this device to act as if they are the PTP Grand Master as follows: 0 = Hardware accelerate with this device NOT being the Grand Master 1 = Hardware accelerate with this device being the Grand Master
2	OneStep	RWR	OneStep Enable. This bit is used to enable hardware One Step support for PTP frames in the mode selected by the PTPMode bits below as follows: 0 = Hardware accelerate using AutoFollowUp Two Step frame formats 1 = Hardware accelerate using One Step frame formats **NOTE:** This device does not support the receiving of One Step frames and the hardware conversion of these frames into Two Step when the PTPMode is End to End Transparent Clock.
1:0	PTPMode	RWR	PTP Mode. This register selects the PTP Mode of the device (for all ports) as follows: 0x0 = Boundary Clock 0x1 = Peer to Peer Transparent Clock 0x2 = End to End Transparent Clock 0x3 = Reserved for future use For these settings to have an effect, at least one Time Array (PTP Global Page 15 Register 0 - Page 15 Register 15) must be configured for the Domain that will be used. This mode will then take effect on the ports whose Hardware Acceleration is enabled (HWAccel is set to a one in PTP Port Page 8 Register 2).

Table 238: PTP Status
Page 14, Register 8 – PTP Global

Bits	Field	Type	Description
15	TrigGen Int	ROC	Trigger generate mode Interrupt. The TrigGenInt bit gets set by the TAI block when the TrigGenIntEn is set to a one (TAI Page 12 Register 0) and when the one shot pulse is generated (TrigMode is set to one in TAI Page 12 Register 0). It gets cleared by the reading of this register.
14	Event Int	RO	Event Capture Interrupt. This bit gets set by the TAI block when the EventIntEn is set to a one (TAI Page 12 Register 0) and when an EventReq is captured in the EventCap Register. It gets cleared by writing a zero to the EventCapValid register (TAI Page 12 Register 9). **NOTE:** This interrupt bit will also be set to a one, if enabled, whenever the EventCapValid bit (TAI Page 12 Register 9) is set to a one. This allows software to test its interrupt routine.
13:11	Reserved	RES	Reserved for future use.
10:0	PTPInt [10:0]	RO	These PTP Interrupt bits gets set for a given port when an incoming PTP frame is time stamped and PTPArrIntEn for that port is set to a one. Similarly the PTP Interrupt bits get set for a given port when an outgoing PTP frame is time stamped and PTPDepIntEn for that port is set to a one. The hardware logic sets this per port bit based on above criteria and gets cleared upon software reading and clearing the corresponding time counter valid bits that are valid for that port.

Table 239: ReadPlus Command Register
Page 14, Register 14 – PTP Global

Bits	Field	Type	Description
15	Read Plus Enable	R/W	Read Plus Enable 1 = Enable 0 = Disable
14:12	Reserved	Res	Reserved

**Table 239: ReadPlus Command Register (Continued)
Page 14, Register 14 – PTP Global**

Bits	Field	Type	Description
11:8	PTPReg	R/W	PTP Registers 0xE = Page 12 (TAI Registers, Event Capture Registers 0xF = Page 15 (PTP Global Time Array Registers) **NOTE:** These registers need to use ReadPlus command otherwise readback value is not correct Page 15 Register 0 to register 15 Page 12 Register 14 to register 15 Page 12 register 2 to register 3 Page 12 register 10 to register 11 Page 13 register 0 to register 1
7:5	Reserved	Res	Reserved.
4:0	PTPAddr	R/W	PTP Address This is the starting address of the PTP Registers to be read.

**Table 240: ReadPlus Data Register
Page 14, Register 15 – PTP Global**

Bits	Field	Type	Description
15:0	Read Plus Data	R/W	Read Plus Data This register is used to read out the ReadPlus Data. To read 32-bit wide PTP Registers, read from this register back-to-back.

Table 241: PTP Global Time Array
Page 15, Register 0 – PTP Global

Bits	Field	Type	Description
15:0	ToDLoadPt [15:0]	RWR	Time of Day Load Point Register bits 15 to 0 of a 32-bit register. The ToDLoadPt register is used in multiple ways, but its contents are always relative to the PTP Global Time (aka, H/W Time) found in TAI Global Page 12 Register 14 & Page 12 Register 15. This register is used as follows in the various ToD Operations: ToD Store All Registers – it is used to determine the instant in time that the selected Time Array is loaded. The load occurs at the instant the PTP Global Time (TAI Global Page 12 Register 14 & Page 12 Register 15) matches the contents of this register. ToD Capture – it is used to capture the instant in time that the Capture occurred. On each ToD Capture, the contents of this register will be loaded with the current value contained in the PTP Global Time (TAI Global Page 12 Register 14 & Page 12 Register 15). The upper 16-bits of this register are contained in the register below.

Table 242: TAI Global Time Array
Page 15, Register 1 – PTP Global

Bits	Field	Type	Description
15:0	ToDLoadPt [31:16]	RWR	Time of Day Load Point Register bits 31 to 16 of a 32-bit register. Value to be matched with global timer value – used to either load or capture the TOD. See the description above. The lower 16-bits of this register are contained in the register above.

Table 243: PTP Global Time Array
Page 15, Register 2– PTP Global

Bits	Field	Type	Description
15	ToDBusy	SC	Time of Day Busy. This bit must be set to a one to start a ToD operation (see ToD Op below). Only one ToD operation can be executing at one time so this bit must be zero before setting it to a one. When the requested ToD operation completes this bit will automatically be cleared to a zero.

Table 243: PTP Global Time Array (Continued)
Page 15, Register 2– PTP Global

Bits	Field	Type	Description
14:12	ToDOp	RWR	Time of Day Opcode. The following ToD Opcodes are supported and are applied to the selected Time Array as indicated below: 0x0 = Reserved for future use 0x1 = Reserved for future use 0x2 = Store Comp register only to selected TimeArray[1] 0x3 = Store All Registers to selected TimeArray @ ToDLoadPt[2] 0x4 = Capture selected TimeArray, Comp=Comp & ToDLoadPt=PTPGT[3] 0x5 = Reserved for future use 0x6 = Reserved for future use 10x7 = Reserved for future use **NOTE:** A Store ToDOp will start the timer if ClkValid below is set to a one.
11	Reserved	RES	Reserved for future use.
10:9	TimeArray [1:0]	RWR	Time Array. The above ToD operation is performed to/from the physical Time Array specified by this register. The device contains multiple independent Time Arrays where each Time Array consists of: 10 byte ToD time (PTP Global Page 15 Register 3 to Page 15 Register 7), 8 byte 1722 time (PTP Global Page 15 Register 8 to Page 15 Register 11), 4 byte Compensation (PTP Global Page 15 Register 12 to Page 15 Register 13), 1 byte Domain Number (bits 7:0 below), and a 1 bit Clock Valid (bit 8 below).
8	ClkValid	RWR	Clock Valid. When this bit is set to a one and then stored to a Time Array (by using ToDOp Store All Registers) the selected Time Array will start keeping time using the parameters loaded and that Time Array will be considered active, When this bit is cleared to a zero and then stored to a Time Array the selected Time Array will stop keeping time and it will be considered inactive.
7:0	Domain Number	RWR	Domain Number. This is the IEEE 1588 or IEEE 802.1ASbt frame Domain Number that is to be associated with the selected Time Array. It is used to select which Time Array is to be used when processing PTP frames in hardware. This Domain number will only be used on active Time Arrays (i.e., Time Arrays whose ClkValid bit above is set to a one).

1. Updating only the Comp (Compensation) register is needed when a Time Array is active but the PPM difference between the local crystal and the associated Time Array's Grand Master clock has changed and needs to be adjusted
2. Loading all the parameters to a Time Array at a specific ToD Load Pt time is the way to start a Time Array clock with the predetermined relationship between its associated Grand Master clock and the PTP Global Time (TAI Global Page 12 Register 14 & Page 12 Register 15).
3. OpCode 0x4 reads the selected Time Array's current clock values. The Tod Load Pt register is set to the current PTP Global Time value so the relationship between the two clock can be compared.

Table 244: PTP Global Time Array
Page 15, Register 3 – PTP Global

Bits	Field	Type	Description
15:0	ToD Nano [15:0]	RWR	Time Array Time of Day, Nano second portion, bits 15 to 0 of a 32-bit register. The five ToD registers (at PTP Global offsets Page 15 Register 3 to Page 15 Register 7) contain the 10 byte representation of time used in IEEE 1588 & IEEE 802.1AS frames. These registers are used to load this representation of time into the selected Time Array on ToD Store All Registers operations. They contain the selected Time Array's representation of this time after ToD Capture operations complete. The upper 16-bits of this register are contained in the register below.

Table 245: PTP Global Time Array
Page 15, Register 4 – PTP Global

Bits	Field	Type	Description
15:0	ToD Nano [31:16]	RWR	Time Array Time of Day, Nano second portion, bits 31 to 16 of a 32-bit register. See the description above. The lower 16-bits of this register are contained in the register above.

Table 246: PTP Global Time Array
Page 15, Register 5 – PTP Global

Bits	Field	Type	Description
15:0	ToD Sec [15:0]	RWR	Time Array Time of Day, Seconds portion, bits 15 to 0 of a 48-bit register. See the description above. The upper 32-bits of this register are contained in the registers below.

Table 247: PTP Global Time Array
Page 15, Register 6 – PTP Global

Bits	Field	Type	Description
15:0	ToD Sec [31:16]	RWR	Time Array Time of Day, Seconds portion, bits 31 to 16 of a 48-bit register. See the description above. The lower 16-bits of this register are contained in the register above. The upper 16-bits of this register are contained in the register below.

Table 248: PTP Global Time Array
Page 15, Register 7– PTP Global

Bits	Field	Type	Description
15:0	ToD Sec [47:32]	RWR	Time Array Time of Day, Seconds portion, bits 47 to 32 of a 48-bit register. See the description above. The lower 32-bits of this register are contained in the registers above.

Table 249: PTP Global Time Array
Page 15, Register 8– PTP Global

Bits	Field	Type	Description
15:0	1722 Nano [15:0]	RWR	Time Array 1722 Time of Day in Nano seconds, bits 15 to 0 of a 64-bit register. The four 1722 ToD registers (at PTP Global Page 15 Register 8 to Page 15 Register 11) contain an 8 byte representation of time used in IEEE 1722 frames (IEEE1722 uses only the lower 32-bits of this time. The 64-bit representation is used in PCI-e and it is a simple extension of the IEEE 1722 representation of time that wraps). These registers are used to load this representation of time into the selected Time Array on ToD Store All Registers operations. They contain the selected Time Array's representation of this time after ToD Capture operations complete. The upper 48-bits of this register are contained in the registers below.

Table 250: PTP Global Time Array
Page 15, Register 9 – PTP Global

Bits	Field	Type	Description
15:0	1722 Nano [31:16]	RWR	Time Array 1722 Time of Day in Nano seconds, bits 31 to 16 of a 64-bit register. See the description above. The lower 16-bits of this register are contained in the register above. The upper 32-bits of this register are contained in the registers below.

Table 251: PTP Global Time Array
Page 15, Register 10– PTP Global

Bits	Field	Type	Description
15:0	1722 Nano [47:32]	RWR	Time Array 1722 Time of Day in Nano seconds, bits 47 to 32 of a 64-bit register. See the description above. The lower 32-bits of this register are contained in the registers above. The upper 16-bits of this register are contained in the register below.

Table 252: PTP Global Time Array
Page 15, Register 11 – PTP Global

Bits	Field	Type	Description
15:0	1722 Nano [63:48]	RWR	Time Array 1722 Time of Day in Nano seconds, bits 63 to 48 of a 64-bit register. See the description above. The lower 48-bits of this register are contained in the registers above.

Table 253: PTP Global Time Array
Page 15, Register 12 – PTP Global

Bits	Field	Type	Description
15:0	ToD Comp [15:0]	RWR	Time Array Time of Day Compensation bits 15 to 0 of a 32-bit register. The two Time of Day Compensation registers (at PTP Global Page 15 Register 12 to Page 15 Register 13) are used to define the PPM difference between the local crystal clocking this PTP block and the PTP Grand Master device that this Time Array is tracking. Bits 30:0 of this register are used to define the difference between these two clocks in increments of 465.661 zeptoseconds (or 4.65661E-19 seconds which is less than a single attosecond. For reference, picoseconds is 10^{-12}, femptoseconds is 10^{-15}, attoseconds is 10^{-18} and zeptoseconds is 10^{-21}). Set this register to the amount of time needed to be added or subtracted to each local 125 MHz PTP clock period in order to make it match its associated Grand Master's rate (i.e., the PPM difference between the two). A difference of 1 PPM for a 125 MHz local PTP clock is 8 femtoseconds (8.0E-15 seconds) or a setting of 17,182 decimal (0x431E) in this register. The full range of this register (0x7FFF FFFF) results in 1.0 ns of compensation per local PTP Clock. Bit 31 of this register is used to indicate the direction of the difference: 0 = Local clock is slow (need to add the Comp to each cycle) 1 = Local clock is fast (need to subtract the Comp from each cycle) These registers are used to load the needed Compensation to the selected Time Array on ToD Store All Registers and on ToD Store only the Comp registers operations. They contain the selected Time Array's Compensation after a ToD Capture operation completes. Once a Time Array is set active, use the ToD Store only the Comp register operation to update any detected changes in the PPM difference between the local crystal and its associated Grand Master. The upper 16-bits of this register are contained in the register below.

Table 254: PTP Global Time Array
Page 15, Register 13– PTP Global

Bits	Field	Type	Description
15:0	ToD Comp [31:16]	RWR	Time Array Time of Day Compensation bits 31 to 16 of a 32-bit register. See the description above. The lower 16-bits of this register are contained in the register above.

Table 255: PTP Global Time Array
Page 15, Register 14– PTP Global

Bits	Field	Type	Description
15:12	1PPS Width	RWR	Pulse Width for the 1 Pulse Per Second on the Second signal. This register defines the pulse width of the selected 1 PPS signal (see 1 PPS Select bits below) in the units defined by the 1 PPS Width Range bits below. A value of 0x1 in this register selects 1 unit. A value of 0x0 selects 0 units (i.e., stops the clock).
11	Reserved	RES	Reserved for future use.
10:8	1PPS Width Range	RWR	Pulse Width Range for the 1 Pulse Per Second on the Second signal. This register selects the units of time used to define the 1PPS Width (above) as follows (each higher numbered selection generates units that are 8x larger than the previous lower numbered selection): 0x0 = 8 nSec units for a 125 MHz PTP clock 0x1 = 64 nSec units for a 125 MHz PTP clock 0x2 = 512 nSec units for a 125 MHz PTP clock 0x3 = 4,096 nSec units for a 125 MHz PTP clock 0x4 = 32.768 uSec units for a 125 MHz PTP clock 0x5 = 262.144 uSec units for a 125 MHz PTP clock 0x6 = 2.097 mSec units for a 125 MHz PTP clock 0x7 = 16.777 mSec units for a 125 MHz PTP clock The narrowest width is 8 nSec (by setting this register to 0x0 and the 1 PPS Width register to 0x1). The widest width is 251 mSec or just a bit over ¼ second (by setting this register to 0x7 and the 1 PPS Width register to 0xF).
7:4	Reserved	RES	Reserved for future use.
3	1PPS Phase	RWR	Phase of the 1 Pulse Per Second on the Second signal. When this bit is set to a one the leading edge of the 1 PPS signal (the edge that occurs at the exact start of each second) is the falling edge of the signal. When this bit is cleared to a zero the leading edge of the 1 PPS signal is the rising edge of the signal.
2	Reserved	RES	Reserved for future use.

Table 255: PTP Global Time Array (Continued)
Page 15, Register 14– PTP Global

Bits	Field	Type	Description
1:0	1PPS Select	RWR	Select for the 1 Pulse Per Second on the Second signal. This register selects the Time Array that is used to generate the 1 PPS signal. Any of the Time Arrays can be selected, but only one at a time can be selected. The selected Time Array will output the leading edge of the 1 PPS signal when its ToD Seconds (PTP Global Page 15 Register 5 to Page 15 Register 7) gets incremented. This increment occurs whenever the Time Array's ToD Nano Seconds (PTP Global Page 15 Register 13 & Page 15 Register 14) reaches 1,000,000 nSec such that this 1 PPS signal occurs at the exact start of each second. The selected 1 PPS signal is available on LED[0] pins.

4.1.1.3 PTP TAI Registers

Table 256: TAI Global Config
Page 12, Register 0 – TAI

Bits	Field	Type	Description
15	Event CapOv	RWR	Event Capture Overwrite. 0 = Capture & Hold first PTP Event 1 = Capture all PTP Events & Retain the last PTP Event When this bit is cleared to a zero it configures the hardware to capture a single event, i.e., take a snapshot of PTP Global Timer value at the first EventReq (see EventPhase below) and wait for software to read the EventCapRegister before capturing another event. This mode returns the data from the first EventReq. When this bit is set to a one it enables overwriting the EventCap registers (TAI Page 12 Register 9 to Page 12 Register 11) whenever an EventReq occurs (see EventPhase below). In this mode the hardware will overwrite the EventCapRegister even if the previously captured event register data has not been read by the software. This mode returns the data from the last EventReq.
14	EventCtr Start	RWR	Event Counter Start. 0 = Do not increment the Event Capture Counter 1 = Increment the Event Capture Counter on EventReq's When this bit is cleared to zero the EventCapCtr is not modified even when EventReq occurs (see EventPhase below). When this bit is set to a one it enables incrementing the EventCapCtr register (TAI offset Page 12 Register 9) whenever an EventReq occurs (see EventPhase below).
13	Event Phase	RWR	Event Phase. 0 = Event Requests occur on the rising edge of the PTP_EVREQ LED[1] pin 1 = Event Requests occur on the falling edge of the PTP_EVREQ LED[1] pin When this bit is cleared to a zero an EventReq occurs on the rising edge of the PTP_EVREQ input or on the leading edge of PTP_TRIG when internally sampled (see the CaptureTrig bit in TAI Page 12 Register 9). When this bit is set to a one an EventReq occurs on the falling edge of the PTP_EVREQ input or on the trailing edge of PTP_TRIG when internally sampled. When PTP_TRIG is selected to be internally captured (instead of using the PTP_EVREQ LED[1] pin – see Capture Trig in TAI Page 12 Register 9) this Event Phase is used to invert the value of the normal internal PTP_TRIG that is captured. When Event Phase = 0 the leading edge (or normal rising edge) of the internal PTP_TRIG is captured. When Event Phase = 1 the trailing edge (or normal falling edge) of the internal PTP_TRIG is captured.

Table 256: TAI Global Config (Continued)
Page 12, Register 0 – TAI

Bits	Field	Type	Description
12	TrigPhase	RWR	Trigger Phase. 0 = The PTP Trigger output is active high on the PTP_TRIG LED[1] pin 1 = The PTP Trigger output is active low on the PTP_TRIG LED[1] pin When this bit is cleared to a zero the active phase of the PTP_TRIG output to the LED[1] is normal active high. For example, the pulse mode of PTP_TRIG will be normally low with a high pulse and the 50% duty cycle's leading edge is the rising edge. When this bit is set to a one the active phase of the PTP_TRIG output to the LED[1] is inverted to be active low. For example, the pulse mode of PTP_TRIG will be normally high with a low pulse and the 50% duty cycle's leading edge is the falling edge. **NOTE:** This bit has no effect on the internal phase of PTP_TRIG or any other signal used in the internal blocks of the device.
11	Reserved	RES	Reserved for future use.
10	IRLClk Gen Req	RWS	Ingress Rate Limiter's Clock Generation Request/Enable. When this bit is set to a one, it enables a 50% duty cycle clock generation for the Ingress Rate Limiter's logic (IRL Clk) as configured by the IRLClkGenAmt, IRLClkComp (both ps & Sub ps) and IRLGenTime fields. On Reset, this IRL Clock defaults to a 3.125 uSec rate. This rate can be changed at any time by updating the IRLClkGenAmt &/or IRLClkComp (ps & Sub ps) registers.
9	TrigGen IntEn	RWR	Trigger Generator Interrupt Enable. 0 = Mask interrupts generated by the PTP Trigger logic 1 = Enable interrupts generated by the PTP Trigger logic When this bit is cleared to zero no interrupts are generated by the TrigGen logic. When this bit is set to a one the TAI block will generate an interrupt whenever a TrigGen pulse event has occurred. This interrupt will appear in the Trigger Mode Interrupt in PTP Global Page 14 Register 8.
8	EventCap IntEn	RWR	Event Capture Interrupt Enable. 0 = Mask interrupts generated by the PTP Event Capture logic 1 = Enable interrupts generated by the PTP Event Capture logic When this bit is cleared to zero no interrupts are generated by the EventCap logic. When this bit is set to a one the TAI block will generate an interrupt whenever an EventReq occurs. This interrupt will appear in the Event Capture Interrupt in PTP Global Page 14 Register 8.

Table 256: TAI Global Config (Continued)
Page 12, Register 0 – TAI

Bits	Field	Type	Description
7	TrigLock	SC	Trigger Lock. When this bit is set to a one the leading edge of PTP_TRIG (see TrigPhase above) will be adjusted to the value contained in the TrigGenTime register (TAI Page 13 Register 0 and Page 13 Register 1) if and only if the leading edge of PTP_TRIG occurs ± the number of PTP Clocks as defined in the TrigLockRange register below and PTP_TRIG is enabled (TrigGenReq, below, is set to a one) and the TrigGenTime register is non-zero. **NOTE:** The TrigLockRange, the TrigGenTime registers must be configured before this bit is set to a one. Once the TrigGenTime past in time, this bit will self clear (i.e., it will be active for only one possible correction per wrap around of the 32-bit Global Timer). This bit will clear if the Global time has passed even if a correction was not needed or done. When this bit clears the Lock correction amount, if any, will be registered in the Lock Correction fields for PTP_TRIG (TAI Page 13 Register 2).
6:4	TrigLock Range	RWR	Trigger Locking Range. These bits are used along with the TrigLock bit above. They determine the +/- error limit to adjust and re-center the leading edge of PTP_TRIG in PTP_CLK increments (8ns if using the internal clock) or PTP_EXTCLK increments (if using an external PTP clock).

Table 256: TAI Global Config (Continued)
Page 12, Register 0 – TAI

Bits	Field	Type	Description
3	Block Update	RWR	Block Update. 0 = Update the 50% duty cycle clock mode registers as written 1 = Update the 50% duty cycle clock mode registers as a block When the 50% duty cycle clock mode is enabled (see TrigMode below), the following registers are used to configure that mode: TrigGenAmt (TAI Page 12 Register 2 & Page 12 Register 3), TrigClkComp (TAI Page 12 Register 4) and TrigClkCompSubPs (TAI Page 12 Register 5). To compensate for PPM drift between this node's crystal and the Grand Master's crystal, these registers may need to be updated periodically. The same is true for the IRL clock which is configured using IRLClkGenAmt (TAI Page 12 Register 6), IRLClkComp (TAI Page 12 Register 7) and IRLClkCompSupPs (TAI offset Page 12 Register 8). Setting this register bit to a one allows the updated values in these register groups (TAI Page12 Register 2 to Page 12 Register 5 and TAI Page 12 Register 6 to Page 12 Register 8) to be presented to the hardware together at the same time and at a time when the hardware is not using these register values. This mode ensures smooth, glitch free, updates when the contents of more than one register needs to change during an update. The contents of the TRIG registers (TAI Page12 Register 2 to Page 12 Register 5) are presented to the hardware as a block whenever the TrigClkCompSubPs register is written to (at TAI Page 12 Register 5). The contents of the IRL_Clk registers (TAI Page 12 Register 6 to Page 12 Register 8) are presented to the hardware as a block whenever the IRLClkCompSubPs register is written to (at TAI Page 12 Register 8). This means that the software doesn't need to write all of these registers during an update. Only the registers that are changing need to be written to (it is assumed that the sub pico second register will need to be adjusted for each update, so writing to this register triggers the update).
2	MultiPTP Sync	RWR	Multiple PTP device sync mode. Used in systems where multiple PTP enabled devices' need to synchronize their PTP Global Time counters (TAI Page 12 Register 14 to Page 12 Register 15). 0 = Normal Event Request mode 1 = Enable Multiple PTP device sync mode When this bit is cleared to zero, the EventRequest interface operates normally (i.e. an EventReq transfers the value of the PTP Global Time[31:0] register to the EventCapTime[31:0] register based on the setting of the EventCapOv register above). When this bit is set to a one an EventReq (see EventPhase above) transfers the value in TrigGenAmt[31:0] (TAI Page 12 Register 2 to Page 12 Register 3) into the PTP Global Time[31:0] register (TAI Page 12 Register 14 to Page 12 Register 15). The EventCapTime[31:0] (TAI Page 12 Register 10 to Page 12 Register 11) is also updated at the same time with the previous value that the PTP Global Time[31:0] register contained prior to be updated.

Table 256: TAI Global Config (Continued)
Page 12, Register 0 – TAI

Bits	Field	Type	Description
1	TrigMode	RWR	Trigger Mode. 0 = 50% duty cycle clock mode 1 = Pulse (one-shot) mode When this bit is cleared to zero the 50% duty cycle clock mode is enabled. In this mode the value specified in the TrigGenAmt is used as the period for generating a 50% duty cycle clock on the PTP_TRIG signal. Note that the minimum clock period that can be generated on the PTP_TRIG signal is 4 times the TSClkPer amount. The frequency of this clock can be adjusted in ps increments (see TrigClkComp, TAI Page 12 Register 4) and it can be realigned to a specific time (see TrigLock bit above). The first leading edge of the 50% duty cycle clock will occur the first time the PTP Global Time (PTP TAI Page 12 Register E to Page 12 Register F) equals the value in the non-zero TrigGenTime register (PTP TAI Page 13 Register 0 to Page 13 Register 1) after TrigGenReq, below, is set to a one. This leading edge control occurs as long as the TrigGenTime register is non-zero. If it is zero the leading edge will occur when the TrigGenReq bit below is set to a one without regard to the PTP Global Time. The phase of the leading edge is controlled by the TrigPhase bit above. When this bit is set to a one, Pulse mode is enabled. This mode matches the PTP Global Timer (TAI Page 12 Register 14 to Page 12 Register 15) and the TrigGenAmt register (TAI Page 12 Register 2 to Page 12 Register 3) to generate a pulse at that time on the PTP_TRIG signal. The width of the pulse is specified by PulseWidth & PulseWitdhRange (TAI Page 12 Register 5). **NOTE:** The minimum pulse width that can be generated is one TSClkPer amount (TAI Page 12 Register 1) and the maximum pulse width is more than 30 million times the TSClkPer. The phase of the pulse is controlled by TrigPhase bit above.
0	TrigGen Req	RWR or SC	Trigger Generation Request/Enable. When this bit is set to a one, it enables a one-shot pulse or the 50% duty cycle clock generation on PTP_TRIG as previously configured by the TrigGenAmt, TrigMode, TrigClkComp (ps & Sub ps), PulseWidth, and TrigGenTime fields. If TrigMode (above) is set to a one (pulse mode) this bit will self clear after the pulse occurs (i.e., the trailing edge of the pulse as defined by the Pulse Width and Pulse Width Range registers (TAI Page 12 Register 5). If TrigMode is cleared to a zero the 50% duty cycle clock will continue running as long as this bit is set to a one.

Table 257: TAI Global Config
Page 12, Register 1 – TAI

Bits	Field	Type	Description
15:0	TSClkPer	RWS 0x1F40	Time Stamping Clock Period in pico seconds. This field specifies the clock period for the time stamping clock supplied to the PTP hardware. When this device is using the 125 MHz internally generated clock for the PTP hardware, the value of this register must be 0x1F40, or 8000 decimal which indicates a clock period of 8000 ps or 8 ns (or 125 MHz). When this device's PTP hardware is clocked by an external clock (using PTP_EXTCLK – see PtpExtClk in TAI Page 12 Register 8) this register must be set to the number of ps in that clock's period.

Table 258: TAI Global Config
Page 12, Register 2 – TAI

Bits	Field	Type	Description
15:0	TrigGen Amt [15:0]	RWR	Trigger Generation Amount bits 15 to 0 of a 32-bit register. This field specifies the PTP Time Application Interface trigger generation time amount. When TrigMode is set to one, the value specified in this field is compared with the PTP Global Timer (TAI Page 12 Register 9 to Page 12 Register 10) and when it matches the first time, a pulse is generated on PTP_TRIG whose width is controlled by PulseWidth (TAI Page 12 Register 5). In this mode there is an internal delay of three TSClkPer before the leading edge of the pulse will be seen on the PTP_TRIG output pin. When TrigMode is cleared to zero, the value in this field is used as a clock period in TSClkPer increments (TAI Page 12 Register 1) to generate an output clock on the PTP_TRIG signal (see TrigPhase in TAI Page 12 Register 0). In this mode the TrigClkComp amount (TAI Page 12 Register 4) and TrigClkCompSubPs (TAI Page 12 Register 5) gets accumulated once per TrigGenAmt cycle and when this accumulated value exceeds the value specified in TSClkPer, one TSClkPer amount gets added to or subtracted from the next trailing edge of PTP_TRIG clock output. **NOTE:** In 50% duty cycle clock mode the contents of this register can be presented to the hardware at the same time all the other 50% duty cycle registers are. See BlockUpdate in TAI Page 12 Register 0. The upper 16-bits of this register are contained in the register below.

Table 259: TAI Global Config
Page 12, Register 3 – TAI

Bits	Field	Type	Description
15:0	TrigGen Amt [31:16]	RWR	Trigger Generation Amount bits 31:16 of a 32-bit register. This field specifies the PTP Time Application Interface trigger generation time amount. See the description above. The lower 16-bits of this register are contained in the register above.

Table 260: TAI Global Config
Page 12, Register 4 – TAI

Bits	Field	Type	Description
15	TrigComp Dir	RWR	Trig Clock Compensation Direction. When the accumulated TrigClkComp amount (below) exceeds the value in TSClkPer (TAI Page 12 Register 1), one TSClkPer amount gets added to or subtracted from the next PTP_TRIG clock output. This bit determines which as follows: 0 = Add one TSClkPer to the next PTP_TRIG cycle 1 = Subtract one TSClkPer from the next PTP_TRIG cycle
14:0	TrigClk Comp	RWR	Trigger mode Clock Compensation Amount. This value is in pico seconds as an unsigned number. This field is used in 50% duty cycle clock mode only (when TrigMode is cleared to zero and TrigGenReq is set to one). This field specifies the remainder amount in ps for the clock that is being generated with a period specified by the TrigGenAmt (TAI Page 12 Register 2 & Page 12 Register 3). This field must be set as an absolute error number in ps (in other words it is a magnitude difference) regardless if the local clock is too fast or too slow compared to the reference clock. The direction of the clock compensation is configured in the CompDir bit above. In the 50% duty cycle clock mode this register gets accumulated once per TrigGenAmt cycle and when this accumulated value exceeds the value specified in TSClkPer (TAI Page 12 Register 1), one TSClkPer amount gets added to or subtracted from the next PTP_TRIG clock output. This requires that the absolute value of TrigClkComp must not exceed the size of the TSClkPer. If it does, the TSClkPer needs to be adjusted in size (either up or down) until the remainder that remains is less that the TSClkPer and that value gets put into this register. **NOTE:** In 50% duty cycle clock mode the contents of this register can be presented to the hardware at the same time all the other 50% duty cycle registers are. See BlockUpdate in TAI Page 12 Register 0.

Table 261: TAI Global Config
Page 12, Register 5 – TAI

Bits	Field	Type	Description
15:12	Pulse Width	RWS 0xF	Pulse width for PTP_TRIG. This pulse width is in units of TSClkPer (TAI Page 12 Register 1). This specifies the width of the pulse that gets generated on PTP_TRIG (see TrigPhase in TAI Page 12 Register 0) when the one shot pulse mode is selected (TrigMode is set to one and TrigGenReq is set to one). If the PTP_EXTCLK is not used (i.e., the internal clock is being used - see PtpExtClk in TAI Page 12 Register 8) the TSClkPer, or Pulse Width unit is 8ns (assuming the Pulse Width Range, below, is 0x0). **NOTE:** Setting this register to 0x0 will cause unpredictable results.
11	Reserved	RES	Reserved for future use.
10:8	Pulse Width Range	RWR	Pulse Width Range for the PTP_TRIG signal. This register selects the units of time used to define the Pulse Width (above) as follows (each higher numbered selection generates units that are 8x larger than the previous lower numbered selection): 0x0 = 8 nSec units for a 125 MHz PTP clock or 1 x TSClkPer 0x1 = 64 nSec units for a 125 MHz PTP clock or 8 x TSClkPer 0x2 = 512 nSec units for a 125 MHz PTP clock or 64 x TSClkPer 0x3 = 4,096 nSec units for a 125 MHz PTP clock or 512 x TSClkPer 0x4 = 32.768 uSec units for a 125 MHz PTP clock or 4,096 x TSClkPer 0x5 = 262.144 uSec units for a 125 MHz PTP clock or 32,768 x TSClkPer 0x6 = 2.097 mSec units for a 125 MHz PTP clock or 262,144 x TSClkPer 0x7 = 16.777 mSec units for a 125 MHz PTP clock or 2,097,152 x TSClkPer The narrowest width is 8 nSec (assuming a 125 MHz PTP clock) or one TSClkPer (by setting this register to 0x0 and the Pulse Width register, above, to 0x1). The widest width is 251 mSec (assuming a 125 MHz PTP clock) or just a bit over ¼ second (by setting this register to 0x7 and the Pulse Width register to 0xF). The maximum number is 31,457,280 TSClkPer.

Table 261: TAI Global Config
Page 12, Register 5 – TAI

Bits	Field	Type	Description
7:0	TrigClk Comp SubPs	RWR	Trigger mode Clock Compensation Amount. This value is in Sub Pico seconds as an unsigned number. This field is used in 50% duty cycle clock mode only (when TrigMode is cleared to zero and TrigGenReq is set to one). This field specifies the remainder amount in sub ps increments for the clock that is being generated with a period specified by the TrigGenAmt (TAI Page 12 Register 2 & Page 12 Register 3). This field must be set as an absolute error number in Sub ps (in other words it is a magnitude difference) regardless if the local clock is too fast or too slow compared to the reference clock. The direction of the clock compensation is configured in the CompDir bit above. Each unit in this register is 1/256th of a ps or approximately 4 femto seconds (actual number is 3.90625 femto seconds per unit). In the 50% duty cycle clock mode this register gets accumulated once per TrigGenAmt cycle and when this accumulated value exceeds on ps, one ps gets added to the accumulated Trig Clock Compensation (TAI Page 12 Register 4). **NOTE:** In 50% duty cycle clock mode the contents of this register can be presented to the hardware at the same time all the other 50% duty cycle registers are. The writing to this register is used to transfer this data as a block. See BlockUpdate in TAI Page 12 Register 0.

Table 262: TAI Global Status
Page 12, Register 9 – TAI

Bits	Field	Type	Description
15	Reserved	RES	Reserved for future use.
14	Capture Trig	RWR	Capture Trig. 0 = Capture PTP_EVREQ pin events 1 = Capture PTP_TRIG internal events When this bit is cleared to a zero the Event Capture register looks at events on the PTP_EVREQ pin. When this bit is set to a one the Event Capture register looks at events from the waveform generated by PTP_TRIG. This allows observing the rising or falling edge of the PTP_TRIG (the EventPhase register is still active, PTP TAI Page 12 Register 0) without the need of using pins. This is used to ensure the edges have not drifted over time so they can be re-aligned if needed.
13:10	Reserved	RES	Reserved for future use.
9	EventCap Err	RWR	Event Capture Error. This bit gets set by the hardware logic when an EventReq has occurred (see EventPhase in TAI Page 12 Register 0) where the EventCapValid bit, below, is already set to a one and the EventCapOv bit (TAI Page 12 Register 0) is cleared to a zero. This condition could happen if the EventReqs are occurring faster than the local CPU can process them (and clear the EventCapValid bit before the next EventReq). Some number of missed EventReq can be seen in the EventCapCtr, below, if its enabled.
8	EventCap Valid	RWR	Event Capture Valid. This bit is set to a one whenever the EventCap (Event Capture – TAI Page 12 Register 10 & Page 12 Register 11) register contains the time of a captured event. Software needs to clear this bit to a zero to enable the EventCap Register to be able to acquire a subsequent event if the EventCapOv (Event Capture Override – TAI Page 12 Register 0) is not enabled. Clearing this bit to a zero also clears the EventInt (Event Capture Interrupt – PTP Global Page 14 Register 8).
7:0	EventCap Ctr	RWR	Event Capture Counter. This field is incremented once by each EventReq (see EventPhase in PTP TAI Page 12 Register 0) as long as EventCtrStart (PTP TAI Page 12 Register 0) is set to one. This counter wraps around and can be cleared by writing zeros to it.

Table 263: TAI Global Status
Page 12, Register 10 – TAI

Bits	Field	Type	Description
15:0	EventCap Register [15:0]	RWR	Event Capture Register bits 15 to 0 of a 32-bit register. This register captures the value of the PTP Global Timer (TAI Page 12 Register 14 & Page 12 Register 15) when an EventReq (see EventPhase in TAI Page 12 Register 0) has occurred. If the EventCapOv (TAI Page 12 Register 0) is set to a one, then this register indicates the time captured for the last event. When EventCapErr is set to a one, the contents in this register no longer represent the time of the first event. Note that the maximum jitter for the EventCapRegister time amount with respect to the EventReq on a LED[1] pin is one TSClkPer amount. Note that the minimum EventReq input signal high or low width has to be equal to or greater than 1.5 times the TSClkPer amount. Note that in order for hardware to capture the EventReq on the LED[1] input signal, the minimum gap between two consecutive events has to be 150 plus 5 times the TSClkPer amount. The upper 16-bits of this register are contained in the register below.

Table 264: TAI Global Status
Page 12, Register 11 – TAI

Bits	Field	Type	Description
15:0	EventCap Register [31:16]	RWR	Event Capture Register bits 31 to 16 of a 32-bit register. This register captures the value of the PTP Global Timer (TAI Page 12 Register 14 & Page 12 Register 15) when an EventReq (see EventPhase in TAI Page 12 Register 0) has occurred. See the description above. The lower 16-bits of this register are contained in the register above.

Table 265: TAI Global Status
Page 12, Register 14– TAI

Bits	Field	Type	Description
15:0	PTPGlobalTime [15:0]	RO	Precise Time Protocol Global Timer bits 15 to 0 of a 32-bit register. This indicates the global timer value that is running off of the free running switch core clock. This counter wraps around in hardware. To support synchronization of PTP Global Time between multiple devices in a system, this register gets updated with the value specified in TrigGenAmt when MultiPTPSync is set to a one (TAI Page 12 Register 0) and an EventReq occurs (see EventPhase in TAI Page 12 Register 0). The upper 16-bits of this register are contained in the register below.

Table 266: TAI Global Status
Page 12, Register 15 – TAI

Bits	Field	Type	Description
15:0	PTPGlobalTime [31:16]	RO	Precise Time Protocol Global Timer bits 31 to 16 of a 32-bit register. This indicates the global timer value that is running off of the free running switch core clock. This counter wraps around in hardware. See the description above. The lower 16-bits of this register are contained in the register above.

Table 267: TAI Global Config
Page 13, Register 0 – TAI

Bits	Field	Type	Description
15:0	TrigGen Time [15:0]	RWR	Trigger Generation Time bits 15 to 0 of a 32-bit register. This field specifies the PTP Global Time (TAI Page 12 Register 14 & Page 12 Register 15) where the 1st leading edge of PTP_TRIG will occur (with a three TSClkPer latency) when PTP Trig is in the continuous square wave mode (i.e, when TrigMode is 0x0, offset 0x00 above) as long as this register's value is non-zero. If its value is zero, the 1st leading edge of PTP_TRIG will occur when TrigGenReg is set to a one (TAI Page 12 Register 0). This register is also used to for re-locking the leading edge of the square wave (see TrigLock in TAI Page 12 Register 0). The upper 16-bits of this register are contained in the register below.

Table 268: TAI Global Config
Page 13, Register 1 – TAI

Bits	Field	Type	Description
15:0	TrigGen Time [31:16]	RWR	Trigger Generation Time bits 31 to 16 of a 32-bit register. See the description above. The lower 16-bits of this register are contained in the register above.

Table 269: TAI Global Config
Page 13, Register 2 – TAI

Bits	Field	Type	Description
15:5	Reserved	RES	Reserved for future use
4	LockCorr Valid	RO	Trig Lock Correction Valid. 0 = Trig Lock Correction is not valid or did not occur 1 = Trig Lock Correction is valid & did occur When a Trigger Lock is enabled (TAI Page 12 Register 0) this bit is cleared to zero. When the Trigger Lock completes this bit will be set to a one if and only if a Trigger Lock occurred for PTP_TRIG. In this case the Lock Correction value below will show the amount of adjustment that was made (if any).

Table 269: TAI Global Config (Continued)
Page 13, Register 2 – TAI

Bits	Field	Type	Description
3:0	Lock Correction	RO	Trig Lock Correction amount. When the TrigLock bit is set to a one (TAI Page 12 Register 0) enabling a potential clock adjustment, these bits are cleared to zero. When the TrigLock bit is cleared to zero (indicating that the requested clock adjustment is now past in time) these bits will reflect the magnitude and direction that was applied to the PTP_TRIG leading edge of the generated clock. A value of zero means no adjustment was necessary. If bit 3 is a one then the leading edge of the clock was moved n number of clocks earlier in time where n is shown in bits 2:0. If bit 3 is a zero then the leading edge of the clock was moved n number of clocks later in time where n is shown in bits 2:0.

Table 270: TAI Global Config
Page 13, Register 14 – TAI

Bits	Field	Type	Description
15	Reserved	RES	Reserved for future use.
14	PtpExtClk	RWR	PTP External Clock select. 0 = Use internal clock for the PTP core 1 = Use external clock for the PTP core When this bit is cleared to a zero the PTP core gets its clock from an internal 125 MHz clock based on the device's XTAL_IN input. When this bits is set to a one the PTP core gets its clock from the device's PTP_EXTCLK pin. **NOTE:** Do not select the PTP_EXTCLK pin unless the pin (CONFIG pin) has a clock and CONFIG pins is configured to be the PTP_EXTCLK.
13:0	Reserved	RES	Reserved for future use.

5 Electrical Specifications

5.1 Absolute Maximum Ratings

Table 271: Absolute Maximum Ratings

Stresses above those listed in Absolute Maximum Ratings may cause permanent device failure. Functionality at or above these limits is not implied. Exposure to absolute maximum ratings for extended periods may affect device reliability.

Symbol	Parameter	Min	Typ	Max	Units
V_{DDA}	Power Supply Voltage on AVDD18 with respect to VSS	-0.5		2.5	V
V_{DDAC}	Power Supply Voltage on AVDDC18 with respect to VSS	-0.5		2.5	V
V_{DDAR}	Power Supply Voltage on AVDD33 with respect to VSS	-0.5		3.6	V
V_{DD}	Power Supply Voltage on DVDD with respect to VSS	-0.5		1.5	V
V_{DDO}	Power Supply Voltage on VDDO with respect to VSS	-0.5		3.6	V
V_{PIN}	Voltage applied to any digital input pin	-0.5		3.6V or VDDO + 0.7 whichever is less	V
$T_{STORAGE}$	Storage temperature	-55		+125[1]	°C

1. 125 °C is only used as bake temperature for not more than 24 hours. Long term storage (e.g weeks or longer) should be kept at 85 °C or lower.

5.2 Recommended Operating Conditions

Table 272: Recommended Operating Conditions

Symbol	Parameter	Condition	Min	Typ	Max	Units
V_{DDA}[1,2]	AVDD18 Supply	For AVDD18	1.71	1.8	1.995	V
V_{DDAC}[1,2]	AVDDC18 Supply	For AVDDC18	1.71	1.8	1.995	V
V_{DDAR}[1]	AVDD33 Supply	For AVDD33	3.14	3.3	3.46	V
V_{DD}[1,2]	DVDD Supply	For DVDD	0.95	1.0	1.05	V
V_{DDO}[1]	VDDO Supply (88EA1517/88EA1512)	For VDDO at 1.8V	1.71	1.8	1.995	V
		For VDDO at 2.5V	2.38	2.5	2.62	V
		For VDDO at 3.3V	3.14	3.3	3.46	V
RSET	Internal bias reference	Resistor connected to V_{SS}		4990 ± 1% Tolerance		Ω
T_A	Ambient operating temperature	Automotive Grade	-40		+105	°C
T_J	Maximum junction temperature				125[3]	°C

1. Maximum noise allowed on supplies is 50 mV peak-peak.
2. The recommended operating conditions mentioned here for AVDD18, AVDDC18 and DVDD are applicable only when using external supplies. They are not applicable when using internal regulators.
3. Refer to white paper on TJ Thermal Calculations for more information.

5.3 Internal Regulator Specifications

Table 273: Internal Regulator Specifications

Symbol	Parameter	Conditions	Min	Typ	Max	Units
VDDA	AVDD18 Supply	For AVDD18	1.61		1.8	V
VDDAC	AVDDC18 Supply	For AVDDC18	1.61		1.8	V
VDD	DVDD Supply	For DVDD	0.90		1.05	V
VREG_IN	Regulator Supply Input	For REG_IN	3.14	3.3	3.46	V

5.4 Package Thermal Information

5.4.1 Thermal Conditions for 88EA1517 40-pin, QFN Package

Table 274: Thermal Conditions for 88EA1517 40-pin, QFN Package

Symbol	Parameter	Condition	Min	Typ	Max	Units
θ_{JA}	Thermal resistance[1] - junction to ambient for the device 40-Pin, QFN package $\theta_{JA} = (T_J - T_A)/ P$ P = Total power dissipation	JEDEC 3 in. x 4.5 in. 4-layer PCB with no air flow		37.5		°C/W
		JEDEC 3 in. x 4.5 in. 4-layer PCB with 1 meter/sec air flow		33		°C/W
		JEDEC 3 in. x 4.5 in. 4-layer PCB with 2 meter/sec air flow		31.2		°C/W
		JEDEC 3 in. x 4.5 in. 4-layer PCB with 3 meter/sec air flow		30.2		°C/W
ψ_{JT}	Thermal characteristic parameter[a] - junction to top center of the device 40-Pin, QFN package $\psi_{JT} = (T_J-T_{top})/P$. P = Total power dissipation T_{top}: Temperature on the top center of the package.	JEDEC 3 in. x 4.5 in. 4-layer PCB with no air flow		0.9		°C/W
		JEDEC 3 in. x 4.5 in. 4-layer PCB with 1 meter/sec air flow		1.57		°C/W
		JEDEC 3 in. x 4.5 in. 4-layer PCB with 2 meter/sec air flow		2.24		°C/W
		JEDEC 3 in. x 4.5 in. 4-layer PCB with 3 meter/sec air flow		2.48		°C/W
θ_{JC}	Thermal resistance[1] - junction to case for the device 40-Pin, QFN package $\theta_{JC} = (T_J - T_C)/ P_{top}$ P_{top} = Power dissipation from the top of the package	JEDEC with no air flow		20.7		°C/W
θ_{JB}	Thermal resistance[1] - junction to board for the device 40-Pin, QFN package $\theta_{JB} = (T_J - T_B)/ P_{bottom}$ P_{bottom} = Power dissipation from the bottom of the package to the PCB surface.	JEDEC with no air flow		24		°C/W

1. Refer to white paper on TJ Thermal Calculations for more information.

5.4.2 Thermal Conditions for 88EA1512 56-pin QFN Package

Table 275: Thermal Conditions for 88EA1512 56-pin QFN Package

Symbol	Parameter	Condition	Min	Typ	Max	Units
θ_{JA}	Thermal resistance[1] - junction to ambient for the device 56-Pin QFN package $\theta_{JA} = (T_J - T_A)/ P$ P = Total power dissipation	JEDEC 3 in. x 4.5 in. 4-layer PCB with no air flow		33.1		°C/W
		JEDEC 3 in. x 4.5 in. 4-layer PCB with 1 meter/sec air flow		28.7		°C/W
		JEDEC 3 in. x 4.5 in. 4-layer PCB with 2 meter/sec air flow		27.6		°C/W
		JEDEC 3 in. x 4.5 in. 4-layer PCB with 3 meter/sec air flow		26.7		°C/W
ψ_{JT}	Thermal characteristic parameter[a] - junction to top center of the device 56-Pin QFN package $\psi_{JT} = (T_J-T_{top})/P$. P = Total power dissipation $T_{top:}$ Temperature on the top center of the package.	JEDEC 3 in. x 4.5 in. 4-layer PCB with no air flow		0.54		°C/W
		JEDEC 3 in. x 4.5 in. 4-layer PCB with 1 meter/sec air flow		0.92		°C/W
		JEDEC 3 in. x 4.5 in. 4-layer PCB with 2 meter/sec air flow		1.17		°C/W
		JEDEC 3 in. x 4.5 in. 4-layer PCB with 3 meter/sec air flow		1.31		°C/W
θ_{JC}	Thermal resistance[1] - junction to case for the device 56-Pin QFN package $\theta_{JC} = T_J - T_C)/ P_{top}$ P_{top} = Power dissipation from the top of the package	JEDEC with no air flow		17.8		°C/W
θ_{JB}	Thermal resistance[1] - junction to board for the device 56-Pin QFN package $\theta_{JB} = (T_J - T_B)/ P_{bottom}$ P_{bottom} = Power dissipation from the bottom of the package to the PCB surface.	JEDEC with no air flow		20.7		°C/W

1. Refer to white paper on TJ Thermal Calculations for more information.

5.5 88EA1517 Current Consumption

5.5.1 Current Consumption when using External Regulators

Table 276: Current Consumption AVDD18 + AVDDC18

(Over full range of values listed in the Recommended Operating Conditions unless otherwise specified)

Symbol	Parameter	Pins	Condition	Min	Typ	Max	Units
I_{AVDD}	1.8V Power to analog core	AVDD18, AVDDC18	RGMII over 100BASE-TX with traffic		30		mA
			RGMII over 10BASE-T with traffic		20		mA
			RGMII over 100BASE-TX with EEE		30		mA
			RGMII over 10BASE-Te		20		mA
			Energy Detect		13		mA
			IEEE Power Down		6		mA

Table 277: Current Consumption AVDD33

(Over full range of values listed in the Recommended Operating Conditions unless otherwise specified)

Symbol	Parameter	Pins	Condition	Min	Typ	Max	Units
I_{AVDDR}	Analog 3.3V supply	AVDD33	RGMII over 100BASE-TX with traffic		17		mA
			RGMII over 10BASE-T with traffic		33		mA
			RGMII over 100BASE-TX with EEE		2		mA
			RGMII over 10BASE-Te		25		mA
			Energy Detect		3		mA
			IEEE Power Down		3		mA

Table 278: Current Consumption DVDD

(Over full range of values listed in the Recommended Operating Conditions unless otherwise specified)

Symbol	Parameter	Pins	Condition	Min	Typ	Max	Units
I_{VDD}	1.0V Power to digital core	DVDD	RGMII over 100BASE-TX with traffic		18		mA
			RGMII over 10BASE-T with traffic		8		mA
			RGMII over 100BASE-TX with EEE		10		mA
			RGMII over 10BASE-Te		8		mA
			Energy Detect		7		mA
			IEEE Power Down		7		mA

Table 279: Current Consumption VDDO

(Over full range of values listed in the Recommended Operating Conditions unless otherwise specified)

Symbol	Parameter	Pins	Condition		Min	Typ	Max	Units
I_{VDDO}	Power to the digital I/Os	VDDO	RGMII over 100BASE-TX with traffic	VDDO = 3.3V		14		mA
				VDDO = 2.5V		6		mA
				VDDO = 1.8V		8		mA
			RGMII over 10BASE-T with traffic	VDDO = 3.3V		9		mA
				VDDO = 2.5V		7		mA
				VDDO = 1.8V		6		mA
			RGMII over 100BASE-TX with EEE	VDDO = 3.3V		6		mA
				VDDO = 2.5V		6		mA
				VDDO = 1.8V		6		mA
			RGMII to 10BASE-Te	VDDO = 3.3V		8		mA
				VDDO = 2.5V		7		mA
				VDDO = 1.8V		6		mA
			Energy Detect	VDDO = 3.3V		9		mA
				VDDO = 2.5V		7		mA
				VDDO = 1.8V		6		mA
			IEEE Power Down	VDDO = 3.3V		9		mA
				VDDO = 2.5V		7		mA
				VDDO = 1.8V		6		mA

5.5.2 Current Consumption when using Internal Regulators

Table 280: Current Consumption REG_IN

(Over full range of values listed in the Recommended Operating Conditions unless otherwise specified)

Symbol	Parameter	Pins	Condition	Min	Typ	Max	Units
I_{REG_IN}	3.3V Internal Regulator Supply	REG_IN	RGMII over 100BASE-TX with traffic		32		mA
			RGMII over 10BASE-T with traffic		22		mA
			RGMII over 100BASE-TX with EEE		30		mA
			RGMII over 10BASE-Te		22		mA
			Energy Detect		25		mA
			IEEE Power Down		13		mA

NOTE: When using internal regulators, AVDD18, AVDDC18, and DVDD are supplied internally using 3.3V REG_IN regulator supply. VDDO and AVDD33 still need to be supplied externally.

5.6 88EA1512 Current Consumption

5.6.1 Current Consumption when using External Regulators

Table 281: Current Consumption AVDD18 + AVDDC18

(Over full range of values listed in the Recommended Operating Conditions unless otherwise specified)

Symbol	Parameter	Pins	Condition	Min	Typ	Max	Units
I_{AVDD}	1.8V Power to analog core	AVDD18, AVDDC18	RGMII over 1000BASE-T with traffic		67		mA
			RGMII over 100BASE-TX with traffic		30		mA
			RGMII over 10BASE-T with traffic		20		mA
			RGMII over 1000BASE-T with EEE		35		mA
			RGMII over 100BASE-TX with EEE		30		mA
			RGMII over 10BASE-Te		20		mA
			SGMII over 1000BASE-T with traffic		88		mA
			SGMII over 100BASE-TX with traffic		51		mA
			SGMII over 10BASE-T with traffic		41		mA
			SGMII over 1000BASE-T with EEE		40		mA
			SGMII over 100BASE-TX with EEE		32		mA
			SGMII over 10BASE-Te		38		mA
			RGMII over 1000BASE-X		25		mA
			RGMII over SGMII at 1000 Mbps		25		mA
			RGMII over SGMII at 100 Mbps		25		mA
			RGMII over SGMII at 10 Mbps		25		mA
			Energy Detect		10		mA
			IEEE Power Down		4		mA

Table 282: Current Consumption AVDD33

(Over full range of values listed in the Recommended Operating Conditions unless otherwise specified)

Symbol	Parameter	Pins	Condition	Min	Typ	Max	Units
I_{AVDDR}	Analog 3.3V supply	AVDD33	RGMII over 1000BASE-T with traffic		58		mA
			RGMII over 100BASE-TX with traffic		17		mA
			RGMII over 10BASE-T with traffic		33		mA
			RGMII over 1000BASE-T with EEE		7		mA
			RGMII over 100BASE-TX with EEE		2		mA
			RGMII over 10BASE-Te		25		mA
			SGMII over 1000BASE-T with traffic		58		mA
			SGMII over 100BASE-TX with traffic		16		mA
			SGMII over 10BASE-T with traffic		32		mA
			SGMII over 1000BASE-T with EEE		7		mA
			SGMII over 100BASE-TX with EEE		4		mA
			SGMII over 10BASE-Te		25		mA
			RGMII over 1000BASE-X		0		mA
			RGMII over SGMII at 1000 Mbps		0		mA
			RGMII over SGMII at 100 Mbps		0		mA
			RGMII over SGMII at 10 Mbps		0		mA
			Energy Detect		2		mA
			IEEE Power Down		1		mA

Table 283: Current Consumption DVDD

(Over full range of values listed in the Recommended Operating Conditions unless otherwise specified)

Symbol	Parameter	Pins	Condition	Min	Typ	Max	Units
I_{VDD}	1.0V Power to digital core	DVDD	RGMII over 1000BASE-T with traffic		87		mA
			RGMII over 100BASE-TX with traffic		18		mA
			RGMII over 10BASE-T with traffic		8		mA
			RGMII over 1000BASE-T with EEE		14		mA
			RGMII over 100BASE-TX with EEE		10		mA
			RGMII over 10BASE-Te		8		mA
			SGMII over 1000BASE-T with traffic		90		mA
			SGMII over 100BASE-TX with traffic		21		mA
			SGMII over 10BASE-T with traffic		11		mA
			SGMII over 1000BASE-T with EEE		15		mA
			SGMII over 100BASE-TX with EEE		13		mA
			SGMII over 10BASE-Te		11		mA
			RGMII over 1000BASE-X		14		mA
			RGMII over SGMII at 1000 Mbps		14		mA
			RGMII over SGMII at 100 Mbps		11		mA
			RGMII over SGMII at 10 Mbps		10		mA
			Energy Detect		7		mA
			IEEE Power Down		7		mA

Table 284: Current Consumption VDDO

(Over full range of values listed in the Recommended Operating Conditions unless otherwise specified)

Symbol	Parameter	Pins	Condition		Min	Typ	Max	Units
I_{VDDO}	Power to the digital I/Os	VDDO	RGMII over 1000BASE-T with traffic	VDDO = 3.3V		45		mA
				VDDO = 2.5V		36		mA
				VDDO = 1.8V		27		mA
			RGMII over 100BASE-TX with traffic	VDDO = 3.3V		14		mA
				VDDO = 2.5V		10		mA
				VDDO = 1.8V		8		mA
			RGMII over 10BASE-T with traffic	VDDO = 3.3V		9		mA
				VDDO = 2.5V		7		mA
				VDDO = 1.8V		6		mA
			RGMII over 1000BASE-T with EEE[1]	VDDO = 3.3V		7		mA
				VDDO = 2.5V		6		mA
				VDDO = 1.8V		6		mA
			RGMII over 100BASE-TX with EEE	VDDO = 3.3V		6		mA
				VDDO = 2.5V		6		mA
				VDDO = 1.8V		6		mA
			RGMII to 10BASE-Te	VDDO = 3.3V		8		mA
				VDDO = 2.5V		7		mA
				VDDO = 1.8V		6		mA
			SGMII over 1000BASE-T with traffic	VDDO = 3.3V		6		mA
				VDDO = 2.5V		6		mA
				VDDO = 1.8V		6		mA
			SGMII over 100BASE-TX with traffic	VDDO = 3.3V		6		mA
				VDDO = 2.5V		6		mA
				VDDO = 1.8V		6		mA
			SGMII over 10BASE-T with traffic	VDDO = 3.3V		6		mA
				VDDO = 2.5V		6		mA
				VDDO = 1.8V		6		mA

Table 284: Current Consumption VDDO (Continued)

(Over full range of values listed in the Recommended Operating Conditions unless otherwise specified)

Symbol	Parameter	Pins	Condition		Min	Typ	Max	Units
I_{VDDO} (contd.)	Power to the digital I/Os	VDDO	SGMII over 1000BASE-T with EEE	VDDO = 3.3V		6		mA
				VDDO = 2.5V		6		mA
				VDDO = 1.8V		6		mA
			SGMII over 100BASE-TX with EEE	VDDO = 3.3V		6		mA
				VDDO = 2.5V		6		mA
				VDDO = 1.8V		6		mA
			SGMII to 10BASE-Te	VDDO = 3.3V		6		mA
				VDDO = 2.5V		6		mA
				VDDO = 1.8V		6		mA
			RGMII over 1000BASE-X	VDDO = 3.3V		45		mA
				VDDO = 2.5V		36		mA
				VDDO = 1.8V		27		mA
			RGMII over SGMII at 1000 Mbps	VDDO = 3.3V		45		mA
				VDDO = 2.5V		36		mA
				VDDO = 1.8V		27		mA
			RGMII over SGMII at 100 Mbps	VDDO = 3.3V		14		mA
				VDDO = 2.5V		10		mA
				VDDO = 1.8V		8		mA
			RGMII over SGMII at 10 Mbps	VDDO = 3.3V		9		mA
				VDDO = 2.5V		7		mA
				VDDO = 1.8V		6		mA
			Energy Detect	VDDO = 3.3V		6		mA
				VDDO = 2.5V		6		mA
				VDDO = 1.8V		6		mA
			IEEE Power Down	VDDO = 3.3V		6		mA
				VDDO = 2.5V		6		mA
				VDDO = 1.8V		6		mA

1. VDDO EEE current consumption is measured with clock-stoppable bit enabled, wherein the RX_CLK is powered down after 9 clock cycles after entering EEE LPI state using Device 3 Register 0 bit 10.

5.6.2 Current Consumption when using Internal Regulators

Table 285: Current Consumption REG_IN

(Over full range of values listed in the Recommended Operating Conditions unless otherwise specified)

Symbol	Parameter	Pins	Condition	Min	Typ	Max	Units
I_{REG_IN}	3.3V Internal Regulator Supply	REG_IN	RGMII over 1000BASE-T with traffic		80		mA
			RGMII over 100BASE-TX with traffic		32		mA
			RGMII over 10BASE-T with traffic		22		mA
			RGMII over 1000BASE-T with EEE		35		mA
			RGMII over 100BASE-TX with EEE		30		mA
			RGMII over 10BASE-Te		22		mA
			SGMII over 1000BASE-T with traffic		92		mA
			SGMII over 100BASE-TX with traffic		53		mA
			SGMII over 10BASE-T with traffic		43		mA
			SGMII over 1000BASE-T with EEE		40		mA
			SGMII over 100BASE-TX with EEE		32		mA
			SGMII over 10BASE-Te		43		mA
			RGMII over 1000BASE-X		28		mA
			RGMII over SGMII at 1000 Mbps		28		mA
			RGMII over SGMII at 100 Mbps		27		mA
			RGMII over SGMII at 10 Mbps		26		mA
			Energy Detect		25		mA
			IEEE Power Down		13		mA

NOTE: When using internal regulators, AVDD18, AVDDC18, and DVDD are supplied internally using 3.3V REG_IN regulator supply. VDDO and AVDD33 still need to be supplied externally.

5.7 DC Operating Conditions

5.7.1 Digital Pins

Table 286: Digital Pins

(Over full range of values listed in the Recommended Operating Conditions unless otherwise specified)

Symbol	Parameter	Pins[1]	Condition	Min	Typ	Max	Units
VIH	Input high voltage	All digital inputs	VDDO = 3.3V	2.0			V
			VDDO = 2.5V	1.75			V
			VDDO = 1.8V	1.26		VDDO+ 0.6V	V
VIL	Input low voltage	All digital inputs	VDDO = 3.3V			0.8	V
			VDDO = 2.5V			0.75	V
			VDDO = 1.8V	-0.3		0.54	V
VOH	High level output voltage	All digital outputs		VDDO - 0.4V			V
VOL	Low level output voltage	All digital outputs				0.4	V
I_{ILK}	Input leakage current					10	uA
C_{IN}	Input capacitance	All pins				5	pF

1. VDDO supplies the CLK125, MDC, MDIO, RESETn, LED[2:0], CONFIG, TX_CLK, TX_CTRL, TXD[3:0], RX_CLK, RX_CTRL, and RXD[3:0].

5.7.2 LED Pins

Table 287: LED Pins

Symbol	Parameter	Pins	Condition	Min	Typ	Max	Units
VOH	High level output voltage	All LED outputs	IOH = -8 mA	VDDO - 0.4V			V
VOL	Low level output voltage	All LED outputs	IOL = 8 mA			0.4	V
I_{MAX}	Total maximum current per port	All LED pins					mA
I_{ILK}	Input leakage current	All LED pins				10	uA
CIN	Input capacitance	All LED pins				5	pF

5.7.3 IEEE DC Transceiver Parameters

Table 288: IEEE DC Transceiver Parameters

IEEE tests are typically based on template and cannot simply be specified by a number. For an exact description of the template and the test conditions, refer to the IEEE specifications.
-10BASE-T IEEE 802.3 Clause 14
-100BASE-TX ANSI X3.263-1995

(Over full range of values listed in the Recommended Operating Conditions unless otherwise specified)

Symbol	Parameter	Pins	Condition	Min	Typ	Max	Units
V_{ODIFF}	Absolute peak differential output voltage	MDIP/N[1:0]	10BASE-T no cable	2.2	2.5	2.8	V
		MDIP/N[1:0]	10BASE-T cable model	585[1]			mV
		MDIP/N[1:0]	100BASE-TX mode	0.950	1.0	1.050	V
		MDIP/N[3:0]	1000BASE-T[2]	0.67	0.75	0.82	V
	Overshoot[2]	MDIP/N[:0]	100BASE-TX mode	0		5%	V
	Amplitude Symmetry (positive/ negative)	MDIP/N[1:0]	100BASE-TX mode	0.98x		1.02x	V+/V-
V_{IDIFF}	Peak Differential Input Voltage	MDIP/N[:0]	10BASE-T mode	585[3]			mV
	Signal Detect Assertion	MDIP/N[1:0]	100BASE-TX mode	1000	460[4]		mV peak-peak
	Signal Detect De-assertion	MDIP/N[1:0]	100BASE-TX mode	200	360[5]		mV peak-peak

1. IEEE 802.3 Clause 14, Figure 14.9 shows the template for the "far end" wave form. This template allows as little as 495 mV peak differential voltage at the far end receiver.
2. IEEE 802.3ab Figure 40 -19 points A&B.
3. The input test is actually a template test ; IEEE 802.3 Clause 14, Figure 14.17 shows the template for the receive wave form.
4. The ANSI TP-PMD specification requires that any received signal with peak-to-peak differential amplitude greater than 1000 mV should turn on signal detect (internal signal in 100BASE-TX mode). The device will accept signals typically with 460 mV peak-to-peak differential amplitude.
5. The ANSI-PMD specification requires that any received signal with peak-to-peak differential amplitude less than 200 mV should de-assert signal detect (internal signal in 100BASE-TX mode). The Alaska® Quad will reject signals typically with peak-to-peak differential amplitude less than 360 mV.

5.7.4 SGMII Interface (88EA1512 only)

SGMII specification is a de-facto standard proposed by Cisco. It is available at the Cisco website ftp://ftp-eng.cisco/smii/sgmii.pdf. It uses a modified LVDS specification based on the IEEE standard 1596.3. Refer to that standard for the exact definition of the terminology used in the following table. The device adds flexibility by allowing programmable output voltage swing and supply voltage option.

5.7.4.1 Transmitter DC Characteristics

Table 289: Transmitter DC Characteristics

Symbol	Parameter[1]	Min	Typ	Max	Units
V_{OH}	Output Voltage High			1600	mV
V_{OL}	Output Voltage Low	700			mV
V_{RING}	Output Ringing			10	%
$\|V_{OD}\|$[2]	Output Voltage Swing (differential, peak)	Programmable - see Table 290.			mV peak
V_{OS}	Output Offset Voltage (also called Common mode voltage)	Variable - see 5.7.4.2 for details.			mV
R_O	Output Impedance (single-ended) (50 ohm termination)	40		60	Ωs
Delta R_O	Mismatch in a pair			10	%
Delta V_{OD}	Change in V_{OD} between 0 and 1			25	mV
Delta V_{OS}	Change in V_{OS} between 0 and 1			25	mV
I_{S+}, I_{S-}	Output current on short to VSS			40	mA
I_{S+-}	Output current when S_OUT+ and S_OUT- are shorted			12	mA
I_{X+}, I_{X-}	Power off leakage current			10	mA

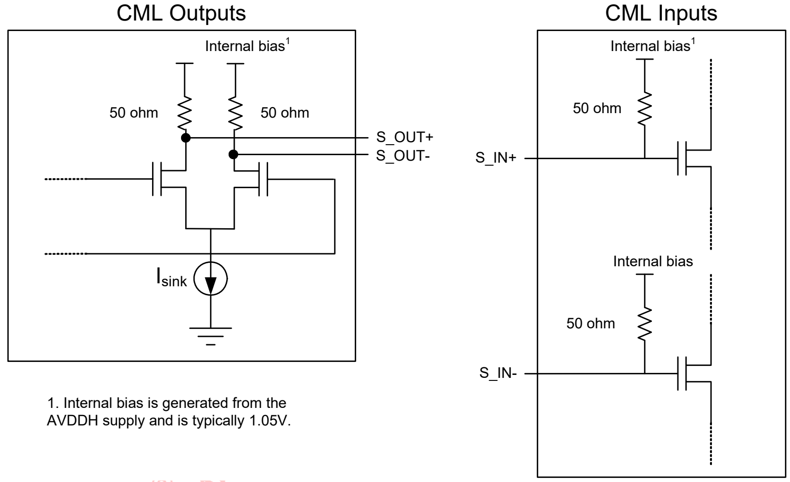
1. Parameters are measured with outputs AC connected with 100 ohm differential load.
2. Output amplitude is programmable by writing to Register 26_1.2:0.

Table 290: Programming SGMII Output Amplitude

Register 26_1 Bits	Field	Description
2:0	SGMII/Fiber Output Amplitude[1]	Differential voltage peak measured. Note that internal bias minus the differential peak voltage must be greater than 700 mV. 000 = 14 mV 001 = 112 mV 010 = 210 mV 011 = 308 mV 100 = 406 mV 101 = 504 mV 110 = 602 mV 111 = 700 mV

1. Cisco SGMII specification limits are |VOD| = 150 mV - 400 mV peak differential.

Figure 41: CML I/Os



5.7.4.2 Common Mode Voltage (Voffset) Calculations

There are four different main configurations for the SGMII/Fiber interface connections. These are:

- DC connection to an LVDS receiver
- AC connection to an LVDS receiver
- DC connection to an CML receiver
- AC connection to an CML receiver

If AC coupling or DC coupling to an LVDS receiver is used, the DC output levels are determined by the following:

- Internal bias. See Figure 41 for details. (If AVDD18 is used to generate the internal bias, the internal bias value will typically be 1.05V.)
- The output voltage swing is programmed by Register 26_1.2:0 (see Table 290).
- Voffset (i.e., common mode voltage) = internal bias - single-ended peak-peak voltage swing. See Figure 42 for details.

If DC coupling is used with a CML receiver, then the DC levels will be determined by a combination of the MACs output structure and the input structure shown in the CML Inputs diagram in Figure 43. Assuming the same MAC CML voltage levels and structure, the common mode output levels will be determined by:

- Voffset (i.e., common mode voltage) = internal bias - single-ended peak-peak voltage swing/2. See Figure 43 for details.
- If DC coupling is used, the output voltage DC levels are determined by the AC coupling considerations above, plus the I/O buffer structure of the MAC.

Figure 42: AC connections (CML or LVDS receiver) or DC connection LVDS receiver

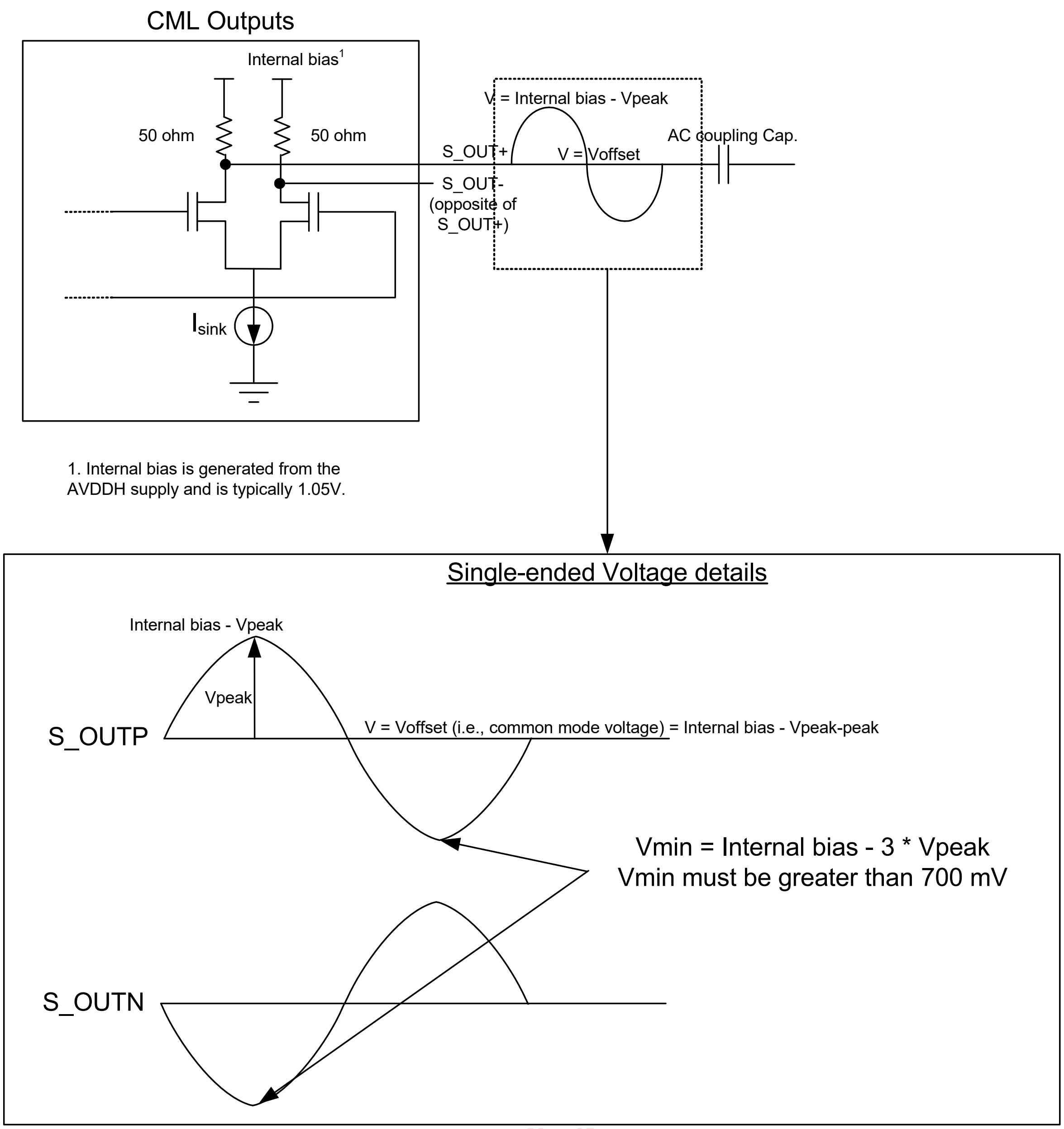
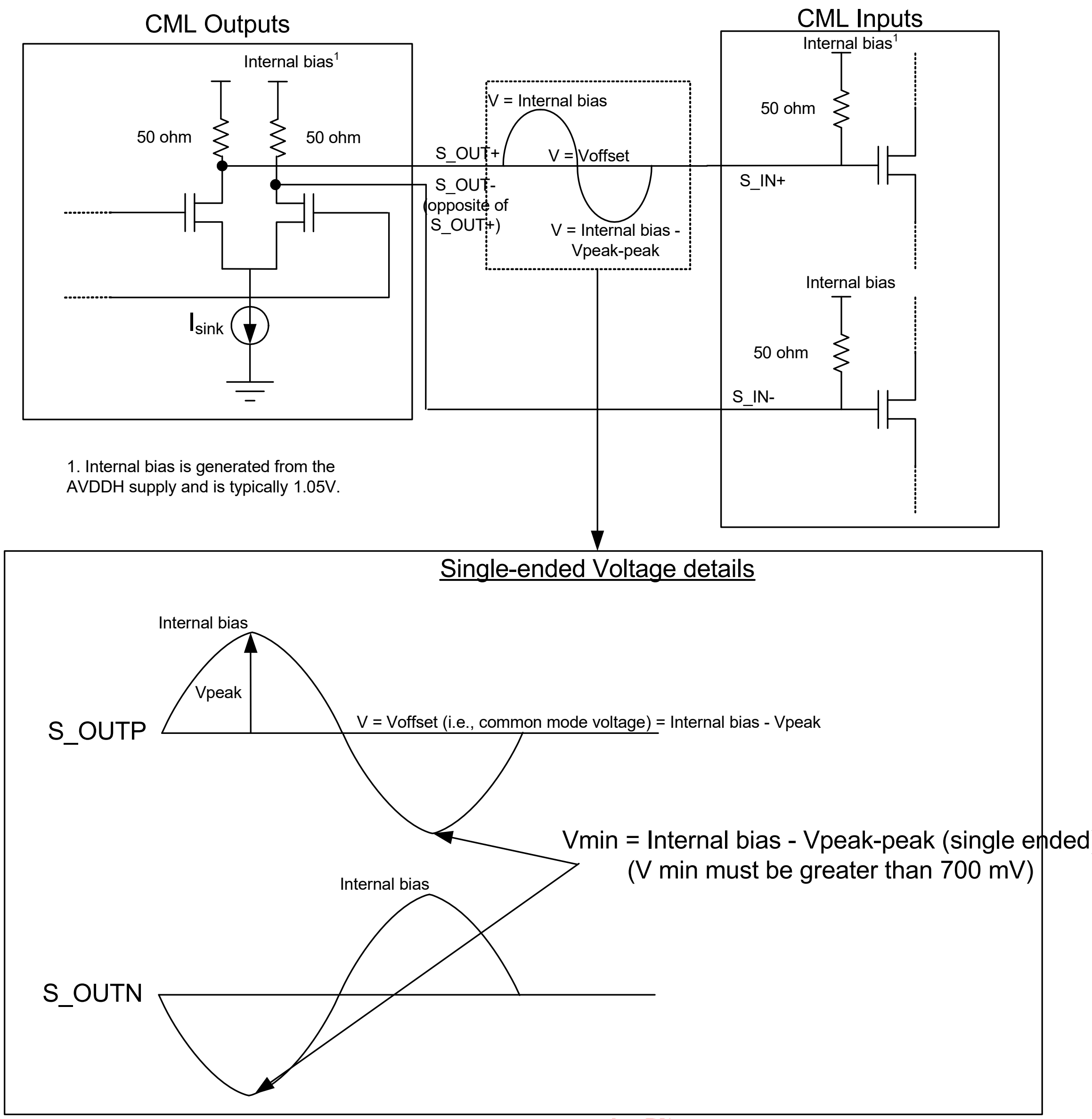


Figure 43: DC connection to a CML receiver



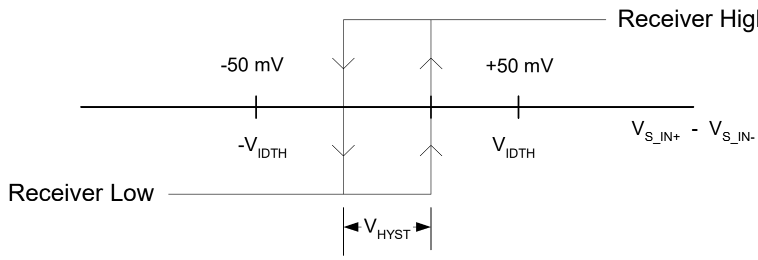
1. Internal bias is generated from the AVDDH supply and is typically 1.05V.

5.7.4.3 Receiver DC Characteristics

Table 291: Receiver DC Characteristics

Symbol	Parameter	Min	Typ	Max	Units
V_I	Input Voltage range a or b	675		1725	mV
V_{IDTH}	Input Differential Threshold	50		50	mV
V_{HYST}	Input Differential Hysteresis	25			mV
R_{IN}	Receiver 100 W Differential Input Impedance	80		120	W

Figure 44: Input Differential Hysteresis



5.8 AC Electrical Specifications

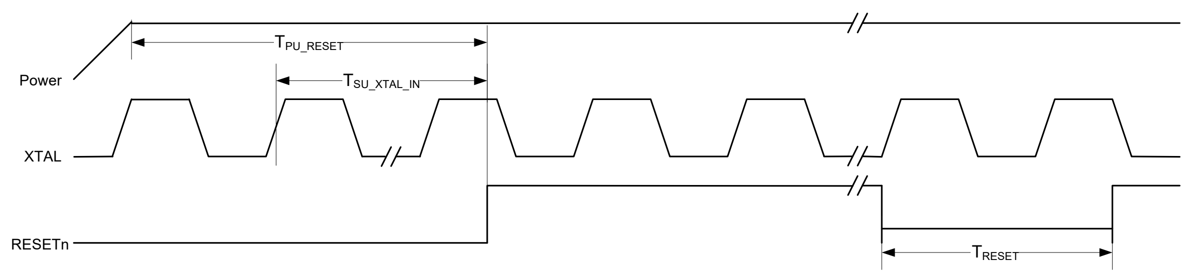
5.8.1 Reset Timing

Table 292: Reset Timing

(Over Full range of values listed in the Recommended Operating Conditions unless otherwise specified)

Symbol	Parameter	Condition	Min	Typ	Max	Units
T_{PU_RESET}	Valid power to RESETn de-asserted		10			ms
$T_{SU\_XTAL\_IN}$	Number of valid XTAL_IN cycles prior to RESETn de-asserted		10			clks
T_{RESET}	Minimum reset pulse width during normal operation		10			ms
T_{RESET_MDIO}	Minimum wait time from RESET de-assertion to first MDIO access		50			ms

Figure 45: Reset Timing



5.8.2 XTAL_IN/XTAL_OUT Timing

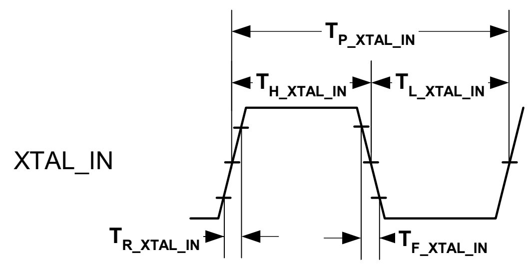
Table 293: XTAL_IN/XTAL_OUT Timing[1]

(Over full range of values listed in the Recommended Operating Conditions unless otherwise specified)

Symbol	Parameter	Condition	Min	Typ	Max	Units
$T_{P\_XTAL\_IN}$	XTAL_IN Period		40 -50 ppm	40	40 +50 ppm	ns
$T_{H\_XTAL\_IN}$	XTAL_IN High time		13	20	27	ns
$T_{L\_XTAL\_IN}$	XTAL_IN Low time		13	20	27	ns
$T_{R\_XTAL\_IN}$	XTAL_IN Rise	10% to 90%	-	3.0	-	ns
$T_{F\_XTAL\_IN}$	XTAL_IN Fall	90% to 10%	-	3.0	-	ns
$T_{J\_XTAL\_IN}$	XTAL_IN total jitter[2]		-	-	200	ps[3]
XTAL_ESR	Crystal ESR[4]		-	30	50	Ω

1. If the crystal option is used, ensure that the frequency is 25 MHz ± 50 ppm. Capacitors must be chosen carefully - see application note supplied by the crystal vendor.
2. PLL generated clocks are not recommended as input to XTAL_IN since they can have excessive jitter. Zero delay buffers are also not recommended for the same reason.
3. 12 kHz to 20 MHz rms jitter on XTAL_IN = 4 ps (RGMII to Copper), 3 ps (SGMII to Copper), 2 ps (RGMII to 1000BX/SGMII/100FX).
4. See “How to use Crystals as Clock Sources” application note for details.

Figure 46: XTAL_IN/XTAL_OUT Timing



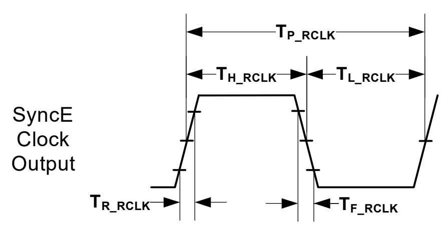
5.8.3 SyncE Recovered Clock Output Timing (88EA1512 only)

Table 294: SyncE Recovered Clock Output Timing (88EA1512 only)

(Over full range of values listed in the Recommended Operating Conditions unless otherwise specified)

Symbol	Parameter	Condition	Min	Typ	Max	Units
T_{P_RCLK}	Period	125 MHz		8		ns
		25 MHz		40		ns
T_{H_RCLK}	High Time	125 MHz		4		ns
		25 MHz		20		ns
T_{L_RCLK}	Low Time	125 MHz		4		ns
		25 MHz		20		ns
T_{R_RCLK}	Rise Time	125 MHz		0.9		ns
		25 MHz		0.9		ns
T_{F_RCLK}	Fall Time	125 MHz		0.5		ns
		25 MHz		0.5		ns
T_{J_RCLK}	Total Jitter	125 MHz			650	ps
		25 MHz			650	ps
T	Duty Cycle	125 MHz	45	50	55	%
		25 MHz	45	50	55	%

Figure 47: SyncE Clock Output Timing

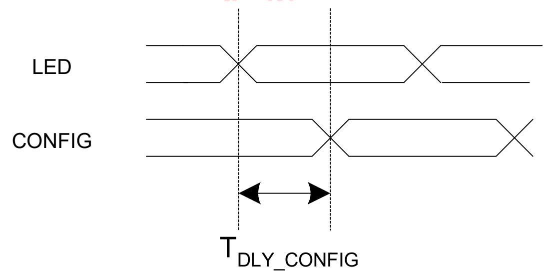


5.8.4 LED to CONFIG Timing

Table 295: LED to CONFIG Timing

Symbol	Parameter	Condition	Min	Typ	Max	Units
T_{DLY_CONFIG}	LED to CONFIG Delay		0		25	ns

Figure 48: LED to CONFIG Timing



5.9 SGMII Interface Timing (88EA1512 only)

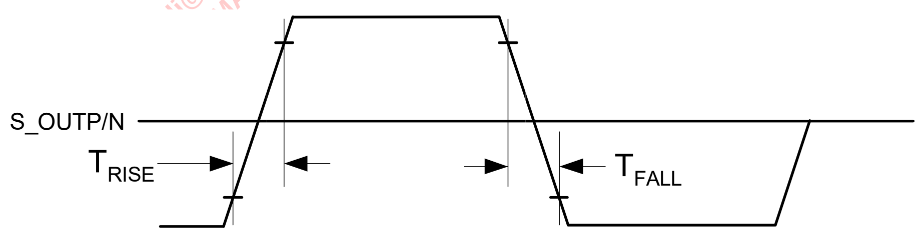
5.9.1 SGMII Output AC Characteristics

Table 296: SGMII Output AC Characteristics

Symbol	Parameter	Min	Typ	Max	Units
T_{FALL}	V_{OD} Fall time (20% - 80%)	100		200	ps
T_{RISE}	V_{OD} Rise time (20% - 80%)	100		200	ps
T_{SKEW1}[1]	Skew between two members of a differential pair			20	ps
$T_{OutputJitter}$	Total Output Jitter Tolerance (Deterministic + 14*rms Random)		127		ps

1. Skew measured at 50% of the transition.

Figure 49: Serial Interface Rise and Fall Times



5.9.2 SGMII Input AC Characteristics

Table 297: SGMII Input AC Characteristics

Symbol	Parameter	Min	Typ	Max	Units
$T_{InputJitter}$	Total Input Jitter Tolerance (Deterministic + 14*rms Random)			599	ps

5.10 PTP Interface Timing

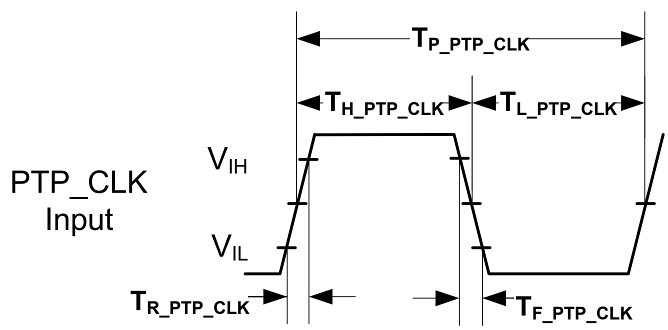
5.10.1 PTP Clock Input Timing

Table 298: PTP Clock Input Timing

(Over full range of values listed in the Recommended Operating Conditions unless otherwise specified)

Symbol	Parameter	Condition	Min	Typ	Max	Units
$T_{P\_PTP\_CLK}$	PTP Clock Period	125 MHz	8 -50 ppm	8	8 +50 ppm	ns
$T_{H\_PTP\_CLK}$	PTP Clock High time	125 MHz	2.8	4	5.2	ns
$T_{L\_PTP\_CLK}$	PTP Clock Low time	125 MHz	2.8	4	5.2	ns
$T_{R\_PTP\_CLK}$	PTP Clock Rise	V_{IL}(max) to V_{IH}(min) - 125 MHz		0.6		ns
$T_{F\_PTP\_CLK}$	PTP Clock Fall	V_{IH}(min) to V_{IL}(max) - 125 MHz		0.6		ns
$T_{J\_PTP\_CLK}$	PTP Clock jitter total				400	ps

Figure 50: PTP Clock Input Timing



5.10.2 PTP Event Request Input AC Timing (LED[1])

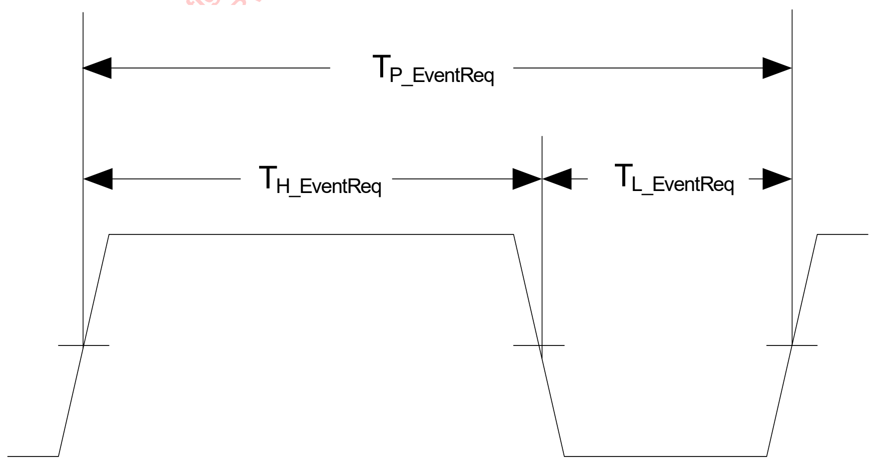
Table 299: PTP Event Request Input AC Timing (LED[1])

(Over full range of values listed in the Recommended Operating Conditions unless otherwise specified)

Symbol	Parameter	Condition	Min	Typ	Max	Units
$T_{P\_EventReq}$	Minimum gap between two PTP Event Request		280 ns + 5*TSClk-Per			ns, TSClk-Per[1]
$T_{H\_EventReq}$	PTP Event Request Pulse Width		1.5 * TSClkPer			TSClkPer
$T_{L\_EventReq}$	PTP Event Request Low Time		1.5 * TSClkPer			TSClkPer

1. TSClkPer is specified by bit[15:0] of the TAI Global Configuration Register 1 – Page 12, Register 1. The default value is 0x1F40 (8ns)

Figure 51: PTP Event Request Input Timing



5.10.3 PTP Trigger Generate Output AC Timing (LED[1])

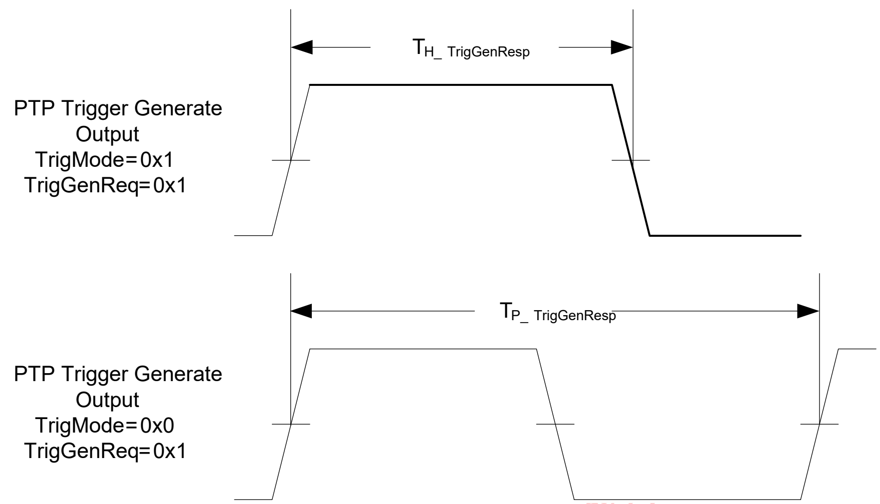
Table 300: PTP Trigger Generate Output AC Timing (LED[1])

(Over full range of values listed in the Recommended Operating Conditions unless otherwise specified)

Symbol	Parameter	Condition	Min	Typ	Max	Units
$T_{H\_}$ TrigGenResp	PTP Trigger Generate Output Pulse Width[1]		1*TSClkPer	PulseWidth[2]	15*TSClk Per	TSClkPer[3]
$T_{P\_}$ TrigGenResp	PTP Trigger Generate Output Clock Period[4]		2*TSClkPer	TrigGenAMT[5]		TSClkPer ns
$T_{P\_}$ TrigGenResp_DC	PTP Trigger Generate Output Clock Period Duty Cycle		40		60	%

1. The PTP Trigger Generate Output is used to generate a pulse when TrigMode is set to ‘1’ and TrigGenReq is set to ‘1’
2. The Pulse Width is specified by bit[15:12] of the TAI Global Configuration Register 5 – Page 12, Register 5 in increments of TSClkPer
3. TSClkPer is specified by bit[15:0] of the TAI Global Configuration Register 1 – Page 12, Register 1. The default value is 0x1F40 (8ns)
4. The PTP Trigger Generate Output is used to generate a periodic clock when TrigMode is set to ‘0’ and TrigGenReq is set to ‘1’.
5. The Output Period is specified by bit[15:0] of the TAI Global Configuration Register 2 – Page 12, Register 2 and bit[15:0] of the TAI Global Configuration Register 3 – Page 12, Register 3 in increments of TSClkPer. If the TrigGenAmt is not set in TSClkPer increments, the TrigClkComp, which is specified by bit[15:0] of the TAI Global Configuration Register 4 – Page 12, Register 4, gets accumulated internally and when this accumulated value exceeds the TSClkPer, a TSCLKPer amount will be added to the clock output.

Figure 52: PTP Trigger Generate Output Timing



5.11 RGMII Interface Timing

5.11.1 RGMII AC Characteristics

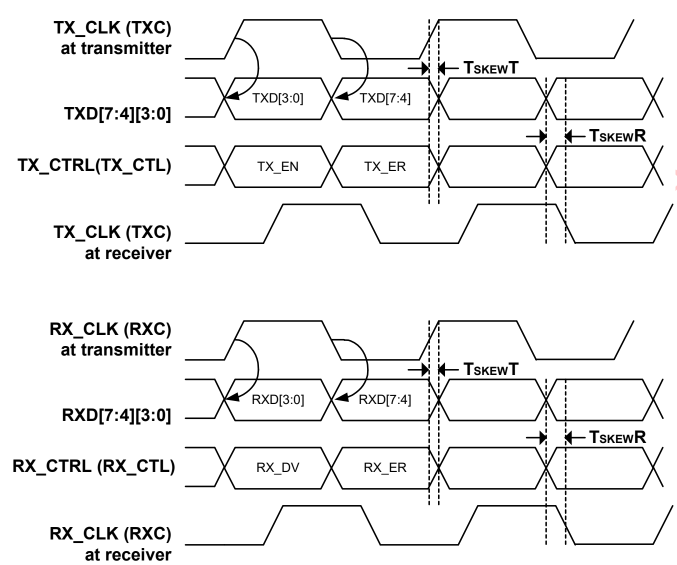
Table 301: RGMII AC Characteristics

(This table is copied from the RGMII Specification. See Application Note “RGMII Timing Modes” for details of how to convert the timings in this table to the four timing modes discussed in Section 5.11.2 "RGMII Delay Timing for different RGMII Modes" on page 289).

Symbol	Parameter	Min	Typ	Max	Units
TskewT	Data to Clock output Skew (at transmitter)	-500	0	500	ps
TskewR	Data to Clock input Skew (at receiver)	1.0	-	2.8	ns
T_{CYCLE}	Clock Cycle Duration[1]	7.2	8.0	8.8	ns
$T_{CYCLE\_HIGH1000}$	High Time for 1000BASE-T [2]	3.6	4.0	4.4	ns
$T_{CYCLE\_HIGH100}$	High Time for 100BASE-T[2]	16	20	24	ns
T_{CYCLE_HIGH10}	High Time for 10BASE-T[2]	160	200	240	ns
T_{RISE}/T_{FALL}	Rise/Fall Time (20-80%)			0.75	ns

1. For 10 Mbps and 100 Mbps, Tcycle will scale to 400 ns ± 40 ns and 40 ns ± 4ns respectively.
2. Duty cycle may be stretched/shrunk during speed changes or while transitioning to a received packet’s clock domain as long as minimum duty cycle is not violated and stretching occurs for no more than three TCYCLE of the lowest speed transitioned between.

Figure 53: RGMII Multiplexing and Timing



This figure is copied from the RGMII Specification. See Application Note “RGMII Timing Modes” for details of how to convert the timings in this table to the four timing modes discussed in Section 5.11.2 "RGMII Delay Timing for different RGMII Modes" on page 289

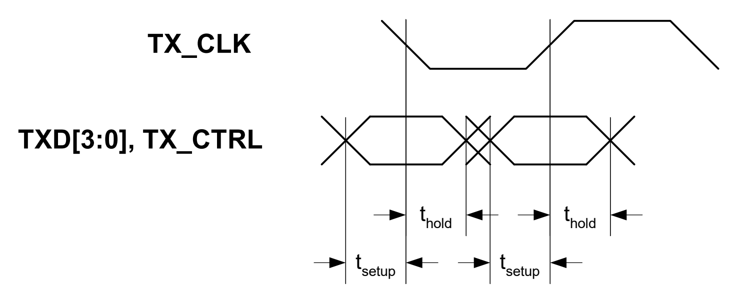
5.11.2 RGMII Delay Timing for different RGMII Modes

5.11.2.1 PHY Input - TX_CLK Delay when Register 21_2.4 = 0

Table 302: PHY Input - TX_CLK Delay when Register 21_2.4 = 0
(Over full range of values listed in the Recommended Operating Conditions unless otherwise specified)

Symbol	Parameter	Min	Typ	Max	Units
t_{setup}	Register 21_2.4 = 0	1.0			ns
t_{hold}		0.8			ns

Figure 54: TX_CLK Delay Timing - Register 21_2.4 = 0

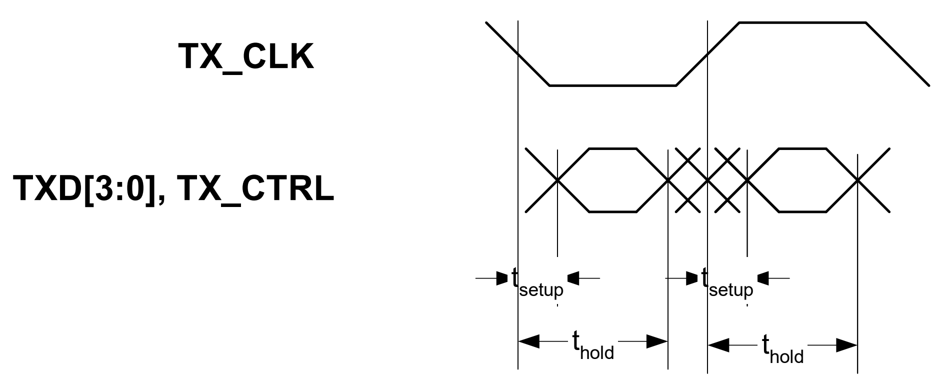


5.11.2.2 PHY Input - TX_CLK Delay when Register 21_2.4 = 1

Table 303: PHY Input - TX_CLK Delay when Register 21_2.4 = 1
(Over full range of values listed in the Recommended Operating Conditions unless otherwise specified)

Symbol	Parameter	Min	Typ	Max	Units
t_{setup}	Register 21_2.4 = 1 (add delay)	-0.9			ns
t_{hold}		2.7			ns

Figure 55: TX_CLK Delay Timing - Register 21_2.4 = 1 (add delay)



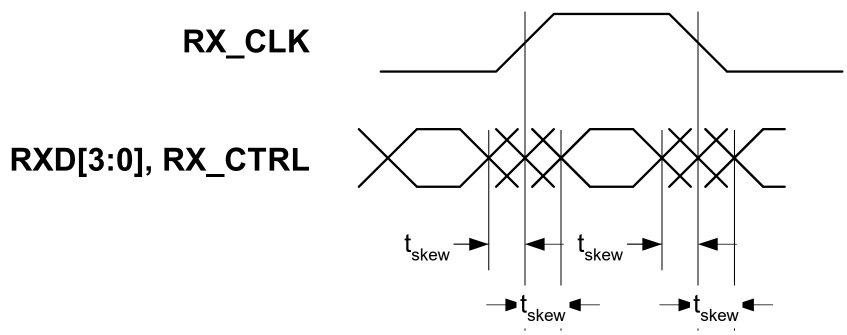
5.11.2.3 PHY Output - RX_CLK Delay

Table 304: PHY Output - RX_CLK Delay

(Over full range of values listed in the Recommended Operating Conditions unless otherwise specified)

Symbol	Parameter	Min	Typ	Max	Units
t_{skew}	Register 21_2.5 = 0	- 0.5		0.5	ns

Figure 56: RGMII RX_CLK Delay Timing - Register 21_2.5 = 0



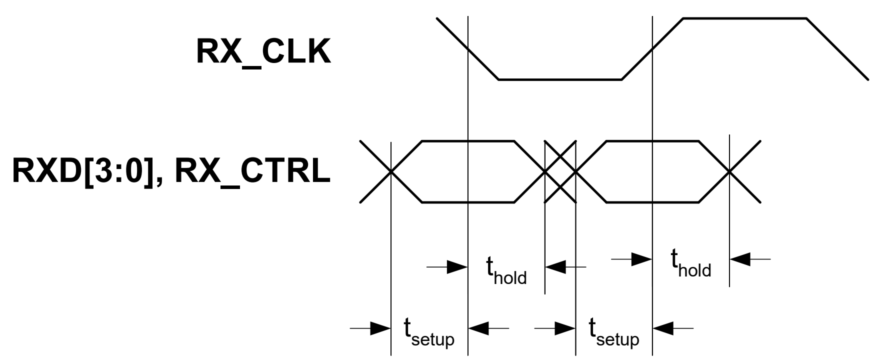
5.11.2.4 PHY Output - RX_CLK Delay

Table 305: PHY Output - RX_CLK Delay

(Over full range of values listed in the Recommended Operating Conditions unless otherwise specified)

Symbol	Parameter	Min	Typ	Max	Units
t_{setup}	Register 21_2.5 = 1 (add delay)	1.2			ns
t_{hold}		1.2			ns

Figure 57: RGMII RX_CLK Delay Timing - Register 21_2.5 = 1 (add delay)



5.12 MDC/MDIO Timing

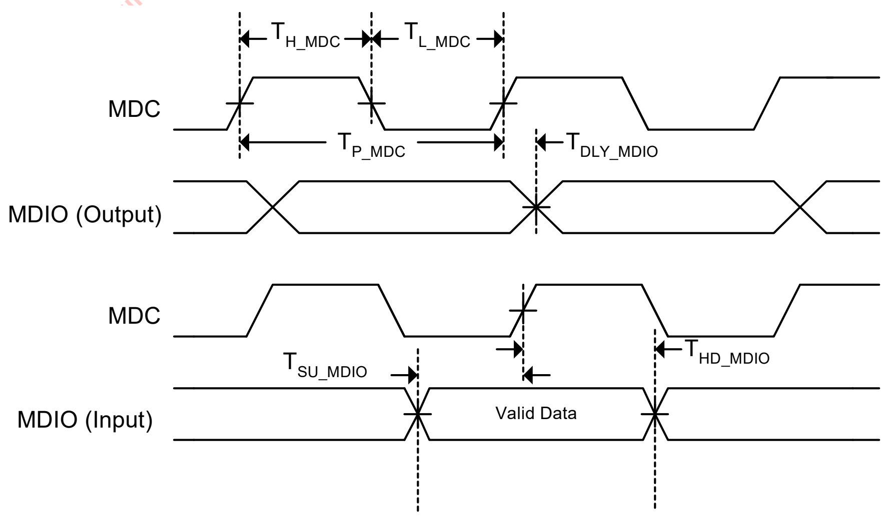
Table 306: MDC/MDIO Timing

(Over full range of values listed in the Recommended Operating Conditions unless otherwise specified)

Symbol	Parameter	Condition	Min	Typ	Max	Units
T_{DLY_MDIO}	MDC to MDIO (Output) Delay Time		0		20	ns
T_{SU_MDIO}	MDIO (Input) to MDC Setup Time		10			ns
T_{HD_MDIO}	MDIO (Input) to MDC Hold Time		10			ns
T_{P_MDC}	MDC Period		83.3			ns[1]
T_{H_MDC}	MDC High		30			ns
T_{L_MDC}	MDC Low		30			ns

1. Maximum frequency = 12 MHz.

Figure 58: MDC/MDIO Timing



5.13 IEEE AC Transceiver Parameters

Table 307: IEEE AC Transceiver Parameters

IEEE tests are typically based on templates and cannot simply be specified by number. For an exact description of the templates and the test conditions, refer to the IEEE specifications:
-10BASE-T IEEE 802.3 Clause 14-2000
-100BASE-TX ANSI X3.263-1995
-1000BASE-T IEEE 802.3ab Clause 40 Section 40.6.1.2 Figure 40-26 shows the template waveforms for transmitter electrical specifications.

(Over full range of values listed in the Recommended Operating Conditions unless otherwise specified)

Symbol	Parameter	Pins	Condition	Min	Typ	Max	Units
T_{RISE}	Rise time	MDIP/N[1:0]	100BASE-TX	3.0	4.0	5.0	ns
T_{FALL}	Fall Time	MDIP/N[1:0]	100BASE-TX	3.0	4.0	5.0	ns
$T_{RISE/}$ T_{FALL} Symmetry		MDIP/N[1:0]	100BASE-TX	0		0.5	ns
DCD	Duty Cycle Distortion	MDIP/N[1:0]	100BASE-TX	0		0.5[1]	ns, peak-peak
Transmit Jitter		MDIP/N[1:0]	100BASE-TX	0		1.4	ns, peak-peak

1. ANSI X3.263-1995 Figure 9-3

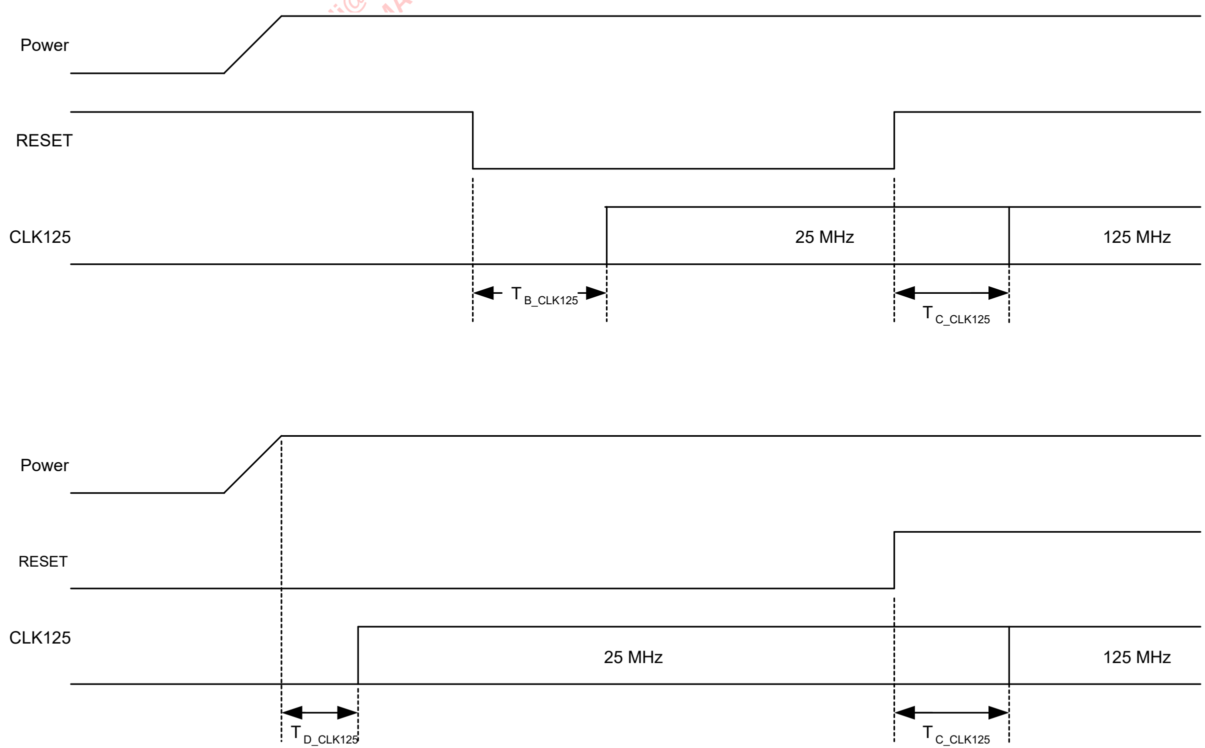
5.14 CLK125 (88EA1512 only)

Table 308: CLK125 (88EA1512 only)

(Over full range of values listed in the Recommended Operating Conditions unless otherwise specified)

Symbol	Parameter	Condition	Min	Typ	Max	Units
T_{J_CLK125}	CLK125 Total Jitter			500		ps
T_{B_CLK125}	RESETn Assert to 25 MHz					ns
T_{C_CLK125}	RESETn De-assert to 125 MHz					µs
T_{D_CLK125}	Power ON to 25 MHz					ns
Duty	Duty Cycle		40	50	60	%

Figure 59: Transition Timing



5.15 Latency Timing

5.15.1 RGMII to 10/100/1000BASE-T Transmit Latency Timing

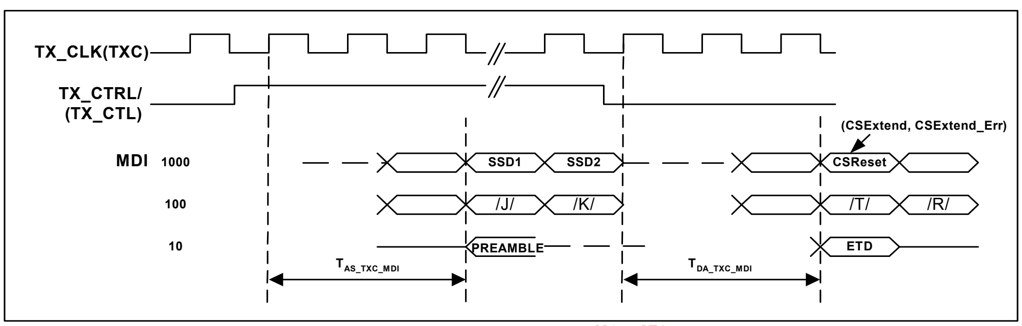
Table 309: RGMII to 10/100/1000BASE-T Transmit Latency Timing

(Over full range of values listed in the Recommended Operating Conditions unless otherwise specified)

Symbol	Parameter	Condition	Min	Typ	Max	Units
$T_{AS\_TXC\_MDI\_1000}$[1, 2]	1000BASE-T TX_CTRL Asserted to MDI SSD1		85		91	ns
$T_{DA\_TXC\_MDI\_1000}$[1,2,]	1000BASE-T TX_CTRL De-asserted to MDI CSReset, CSExtend, CSExtend_Err		85			ns
$T_{AS\_TXC\_MDI\_100}$	100BASE-TX TX_CTRL Asserted to /J/		120		152	ns
$T_{DA\_TXC\_MDI\_100}$	100BASE-TX TX_CTRL De-asserted to /T/		120		152	ns
$T_{AS\_TXC\_MDI\_10}$	10BASE-T TX_CTRL Asserted to Preamble					µs
$T_{DA\_TXC\_MDI\_10}$	10BASE-T TX_CTRL De-asserted to ETD					µs

1. Assumes 20_0.14 is set to 1 enabling TX FIFO Low Latency mode. See Section 2.7 for details. which is the minimum latency.
2. Minimum and maximum values on end of packet assume zero frequency drift between the received signal on MDI and TX_CLK. The worst case variation will be outside these limits if there is a frequency difference.

Figure 60: RGMII to 10/100/1000BASE-T Transmit Latency Timing



5.15.2 10/100/1000BASE-T to RGMII Receive Latency Timing

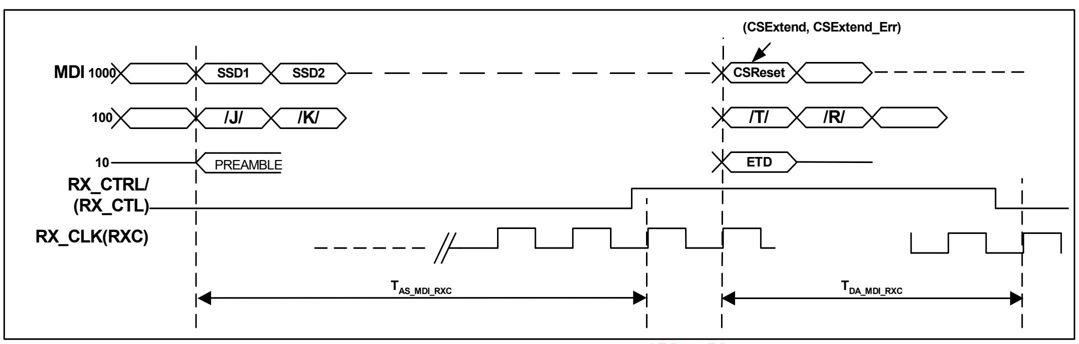
Table 310: 10/100/1000BASE-T to RGMII Receive Latency Timing

(Over full range of values listed in the Recommended Operating Conditions unless otherwise specified)

Symbol	Parameter	Condition	Min	Typ	Max	Units
$T_{AS\_MDI\_RXC\_1000}$	1000BASE-T MDI start of Packet to RX_CTRL Asserted			224		ns
$T_{DA\_MDI\_RXC\_1000}$	1000BASE-T MDI CSReset, CSExtend, CSExtend_Err to RX_CTRL De-asserted			224		ns
$T_{AS\_MDI\_RXC\_100}$	100BASE-TX MDI start of Packet to RX_CTRL Asserted			204		ns
$T_{DA\_MDI\_RXC\_100}$	100BASE-TX MDI /T/ to RX_CTRL De-asserted			204		ns
$T_{AS\_MDI\_RXC\_10}$[1]	10BASE-T MDI start of Packet to RX_CTRL Asserted					µs
$T_{DA\_MDI\_RXC\_10}$[1]	10BASE-T MDI ETD to RX_CTRL De-asserted					µs

1. Actual values depend on number of bits in preamble and number of dribble bits, since nibbles on MII are aligned to start of frame delimiter and dribble bits are truncated.

Figure 61: 10/100/1000BASE-T to RGMII Receive Latency Timing



5.15.3 RGMII to 1000BASE-X Transmit Latency Timing (88EA1512 only)

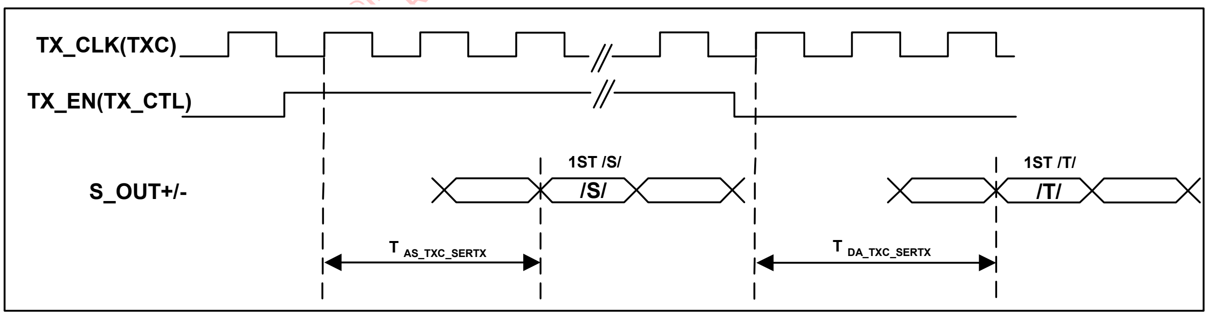
Table 311: RGMII to 1000BASE-X Transmit Latency Timing (88EA1512 only)

(Over full range of values listed in the Recommended Operating Conditions unless otherwise specified)

Symbol	Parameter	Condition	Min	Typ	Max	Units
$T_{AS\_TXC\_SERTX}$[1]	1000BASE-X TX_CTL Asserted to S_OUT± Start of Packet /S/					ns
$T_{DA\_TXC\_SERTX}$[1,2]	1000BASE-X TX_CTL De-asserted to S_OUT± /T/					ns

1. Assumes Register 16_1.13:12 is set to 00, which is the minimum latency. Each increase in setting adds 8 ns of latency.
2. Minimum and maximum values on end of packet assume zero frequency drift between the transmitted signal on S_OUT± and TX_-CLK. The worst case variation will be outside these limits, if there is a frequency difference.

Figure 62: RGMII to 1000BASE-X Transmit Latency Timing



5.15.4 1000BASE-X to RGMII Receive Latency Timing (88EA1512 only)

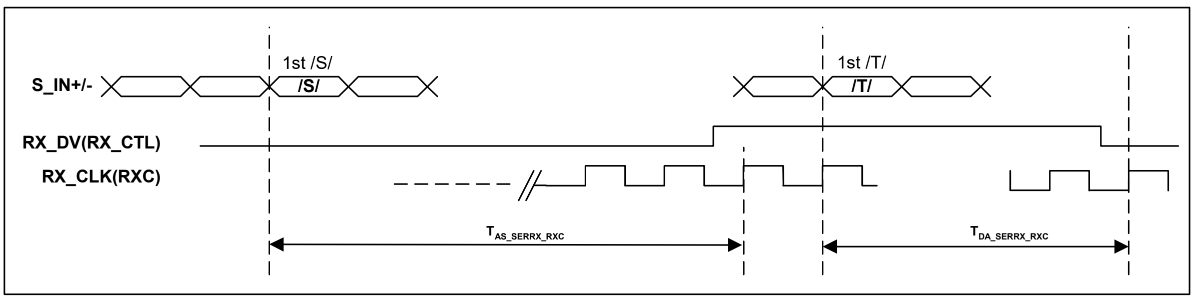
Table 312: 1000BASE-X to RGMII Receive Latency Timing (88EA1512 only)

(Over full range of values listed in the Recommended Operating Conditions unless otherwise specified)

Symbol	Parameter	Condition	Min	Typ	Max	Units
$T_{AS\_SERRX\_RXC}$[1]	S_IN± Start of Packet /S/ to RX_CTL Asserted					ns
$T_{DA\_SERRX\_RXC}$[1]	S_IN± /T/ to RX_CTL De-asserted					ns

1. Minimum and maximum values on end of packet assume zero frequency drift between the received signal on S_IN± and RX_CLK. The worst case variation will be outside these limits, if there is a frequency difference.

Figure 63: 1000BASE-X to RGMII Receive Latency Timing



5.15.5 RGMII to 100BASE-FX Transmit Latency Timing (88EA1512 only)

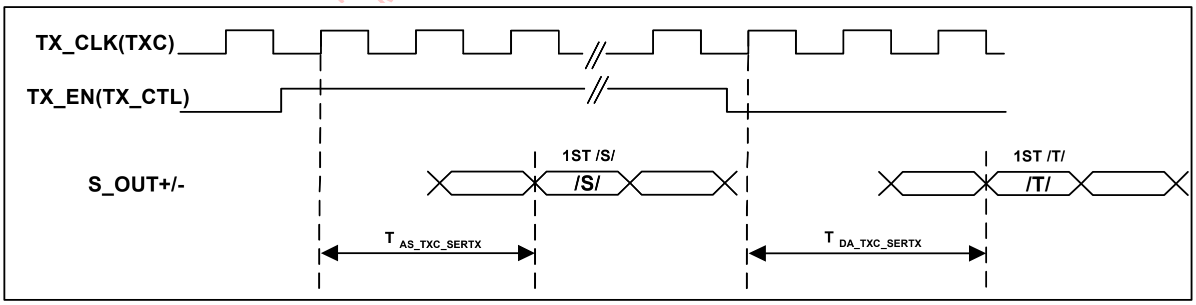
Table 313: RGMII to 100BASE-FX Transmit Latency Timing (88EA1512 only)

(Over full range of values listed in the Recommended Operating Conditions unless otherwise specified)

Symbol	Parameter	Condition	Min	Typ	Max	Units
$T_{AS\_TXC\_SERTX}$[1]	1000BASE-X TX_CTL Asserted to S_OUT± Start of Packet /S/					ns
$T_{DA\_TXC\_SERTX}$[1,2]	1000BASE-X TX_CTL De-asserted to S_OUT± /T/					ns

1. Assumes Register 16_1.13:12 is set to 00, which is the minimum latency. Each increase in setting adds 8 ns of latency.
2. Minimum and maximum values on end of packet assume zero frequency drift between the transmitted signal on S_OUT± and TX_-CLK. The worst case variation will be outside these limits, if there is a frequency difference.

Figure 64: RGMII to 100BASE-FX Transmit Latency Timing



5.15.6 100BASE-FX to RGMII Receive Latency Timing (88EA1512 only)

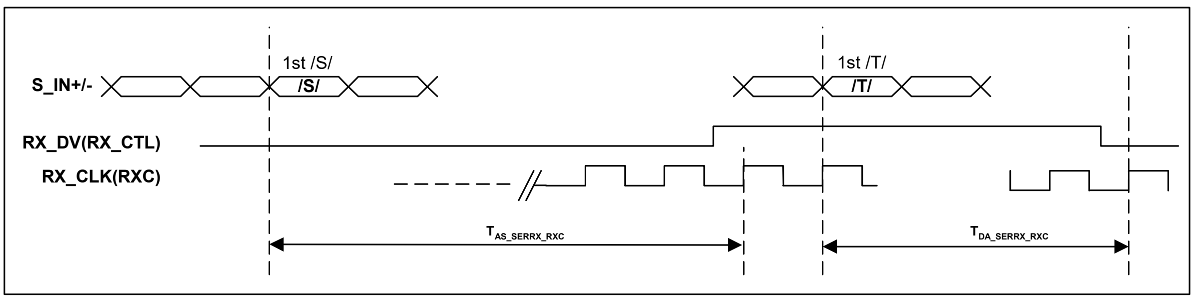
Table 314: 100BASE-FX to RGMII Receive Latency Timing (88EA1512 only)

(Over full range of values listed in the Recommended Operating Conditions unless otherwise specified)

Symbol	Parameter	Condition	Min	Typ	Max	Units
$T_{AS\_SERRX\_RXC}$ [1]	S_IN± Start of Packet /S/ to RX_CTL Asserted					ns
$T_{DA\_SERRX\_RXC}$ [1]	S_IN± /T/ to RX_CTL De-asserted					ns

1. Minimum and maximum values on end of packet assume zero frequency drift between the received signal on S_IN± and RX_CLK. The worst case variation will be outside these limits, if there is a frequency difference.

Figure 65: 100BASE-FX to RGMII Receive Latency Timing



5.15.7 10/100/1000BASE-T to SGMII Latency Timing (88EA1512 only)

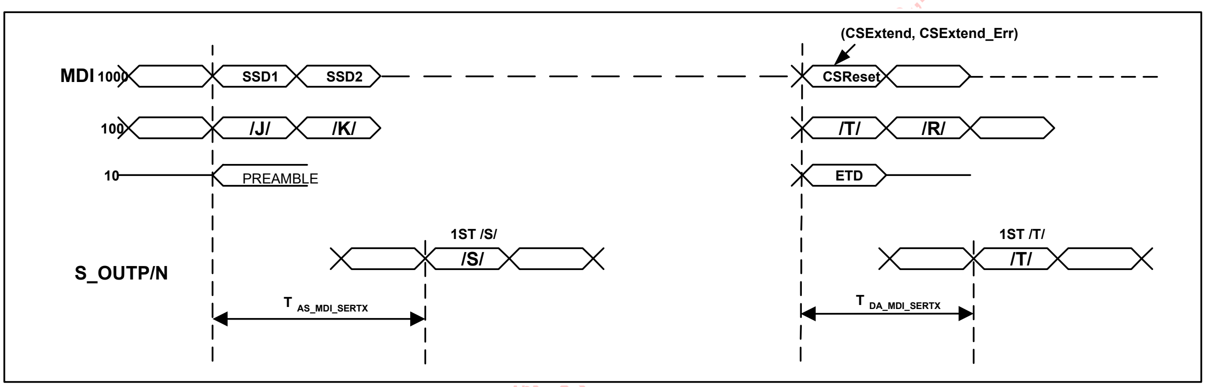
Table 315: 10/100/1000BASE-T to SGMII Latency Timing (88EA1512 only)

(Over full range of values listed in the Recommended Operating Conditions unless otherwise specified)

Symbol	Parameter	Condition	Min	Typ	Max	Units
$T_{AS\_MDI\_SERTX\_1000}$[1,2]	MDI SSD1 to S_OUTP/N Start of Packet		309		333	ns
$T_{DA\_MDI\_SERTX\_1000}$[1,2,3]	MDI CSReset, CSExtend, CSExtend_Err to S_OUTP/N /T/					ns
$T_{AS\_MDI\_SERTX\_100}$[2]	MDI /J/ to S_OUTP/N Start of Packet		457		505	ns
$T_{DA\_MDI\_SERTX\_100}$[2,3]	MDI /T/ to S_OUTP/N /T/					ns
$T_{AS\_MDI\_SERTX\_10}$[2,4]	MDI Preamble to S_OUTP/N Start of Packet		2877		3485	us
$T_{DA\_MDI\_SERTX\_10}$[2,3,4]	MDI ETD to S_OUTP/N /T/					us

1. In 1000BASE-T the signals on the 4 MDI pairs arrive at different times because of the skew introduced by the cable. All timing on MDIP/N[3:0] is referenced from the latest arriving signal.
2. Assumes Register 16_1.15:14 is set to 00, which is the minimum latency. Each increase in setting adds 8 ns of latency 1000 Mbps, 40 ns in 100 Mbps, and 400 ns in 10 Mbps.
3. Minimum and maximum values on end of packet assume zero frequency drift between the received signal on MDI and S_OUTP/N. The worst case variation will be outside these limits if there is a frequency difference.
4. Actual values depend on number of bits in preamble and number of dribble bits, since nibbles on MII are aligned to start of frame delimiter and dribble bits are truncated.

Figure 66: 10/100/1000BASE-T to SGMII Latency Timing



5.15.8 SGMII to 10/100/1000BASE-T Latency Timing (88EA1512 only)

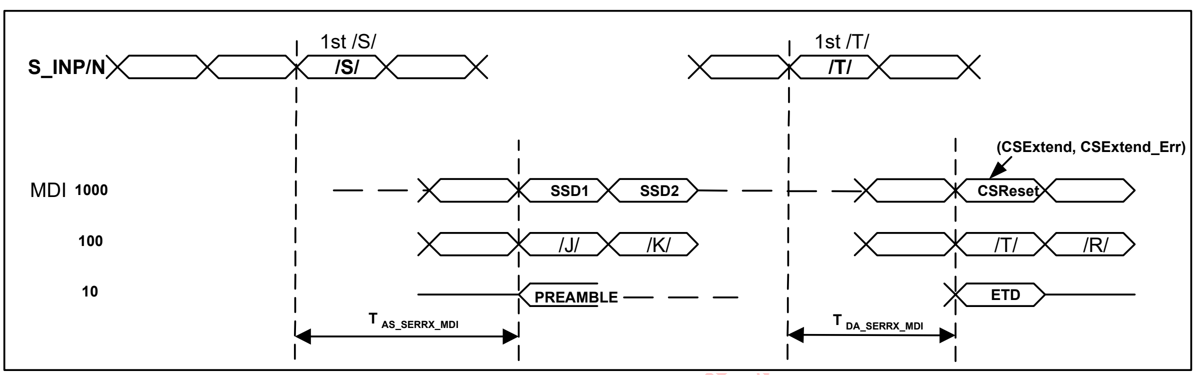
Table 316: SGMII to 10/100/1000BASE-T Latency Timing (88EA1512 only)

(Over full range of values listed in the Recommended Operating Conditions unless otherwise specified)

Symbol	Parameter	Condition	Min	Typ	Max	Units
$T_{AS\_SERRX\_MDI\_1000}$[1]	S_INP/N Start of Packet / S/ to MDI SSD1		170		186	ns
$T_{DA\_SERRX\_MDI\_1000}$[1,2]	S_INP/N /T/ to MDI CSReset, CSExtend, CSExtend_Err					ns
$T_{AS\_SERRX\_MDI\_100}$[1]	S_INP/N Start of Packet / S/ to MDI /J/		306		386	ns
$T_{DA\_SERRX\_MDI\_100}$[1,2]	S_INP/N /T/ to MDI /T/					ns
$T_{AS\_SERRX\_MDI\_10}$[1]	S_INP/N Start of Packet / S/ to MDI Preamble		2337		3137	us
$T_{DA\_SERRX\_MDI\_10}$[1,2]	S_INP/N /T/ to MDI ETD					us

1. Assumes register 16_2.15:14 is set to 00, which is the minimum latency. Each increase in setting adds 8 ns of latency in 1000 Mbps, 40 ns in 100 Mbps, and 400 ns in 10 Mbps.
2. Minimum and maximum values on end of packet assume zero frequency drift between the transmitted signal on MDI and the received signal on S_INP/N. The worst case variation will be outside these limits, if there is a frequency difference.

Figure 67: SGMII to 10/100/1000BASE-T Latency Timing



6 Package Mechanical Dimensions

6.1 40-Pin QFN Package

Figure 68: 88EA1517 40-pin QFN Package Mechanical Drawings

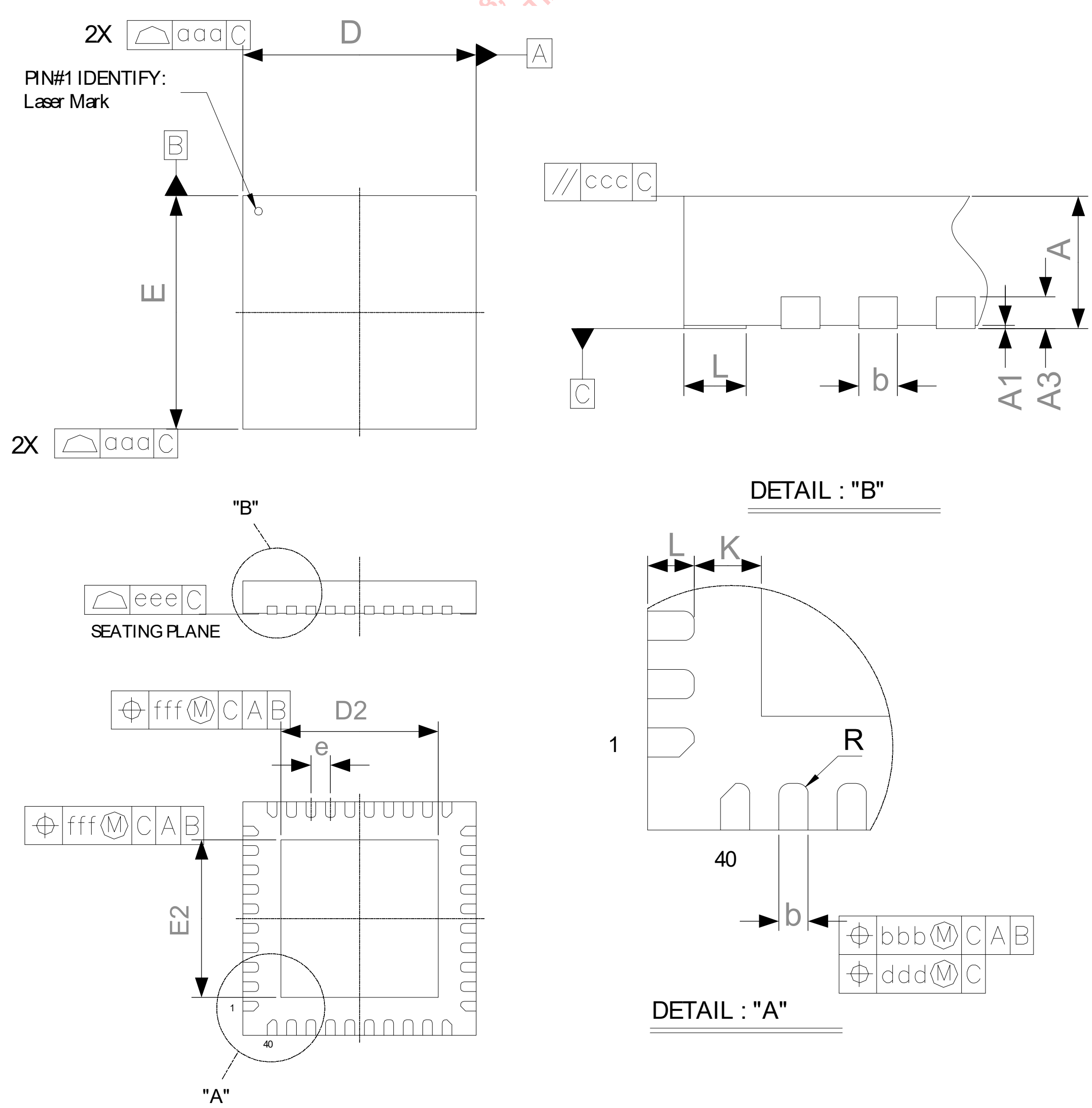


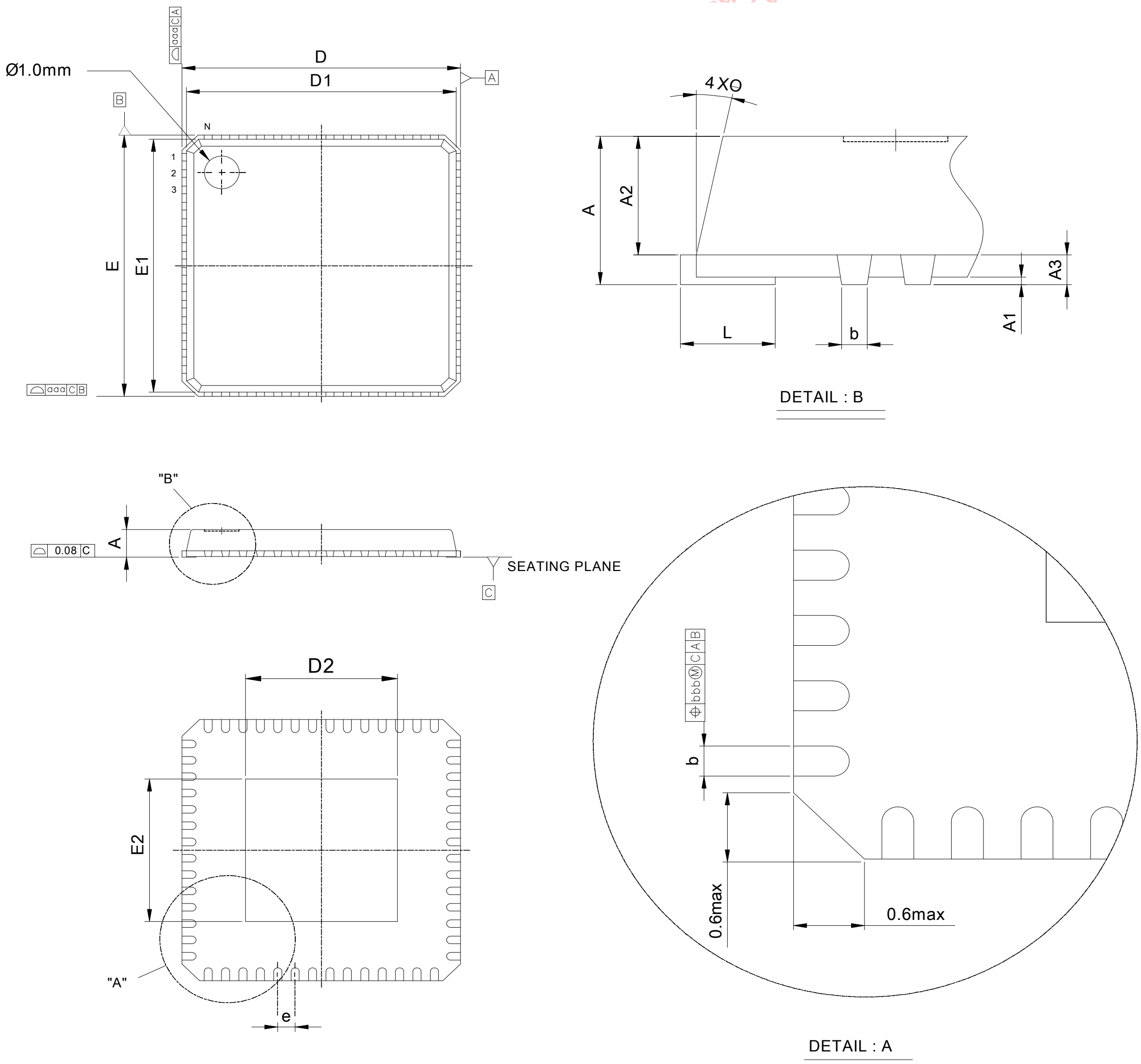
Table 317: 40-Pin QFN Mechanical Dimensions

Dimensions in mm			
Symbol	**MIN**	**NOM**	**MAX**
A	0.80	0.85	0.90
A1	0.00	0.02	0.05
A3		0.20 REF	
b	0.18	0.25	0.30
D/E	5.90	6.00	6.10
e		0.50 BSC	
L	0.30	0.40	0.50
K	0.20	--	--
P	0.09	--	--
aaa		0.15	
bbb		0.10	
ccc		0.10	
ddd		0.05	
eee		0.08	
fff		0.10	

Die Pad Size	
Symbol	**Dimension in mm**
D_2	4.04 ± 0.15
E_2	4.04 ± 0.15

6.2 56-Pin QFN Package

Figure 69: 88EA1512 56-pin QFN Package Mechanical Drawings



(All dimensions in mm.)

Table 318: 56-Pin QFN Mechanical Dimensions

Dimensions in mm			
Symbol	**MIN**	**NOM**	**MAX**
A	0.80	0.85	1.00
A1	0.00	0.02	0.05
A2	--	0.65	1.00
A3	0.20 REF		
b	0.18	0.23	0.30
D		8.00 BSC	
D1		7.75 BSC	
E		8.00 BSC	
E1		7.75 BSC	
e		0.50 BSC	
L	0.30	0.40	0.50
θ	0°	--	12°
aaa	--	--	0.15
bbb	--	--	0.10
chamfer	--	--	0.60

Die Pad Size	
Symbol	**Dimension in mm**
D_2	4.37 ± 0.2
E_2	4.37 ± 0.2

7 Ordering Part Number/Package Marking

7.1 Ordering Part Number

Figure 70 shows the ordering part numbering scheme for the device. Refer to the relevant release notes on the Marvell® extranet for the latest revision and complete part ordering information.

Figure 70: Ordering Part Number

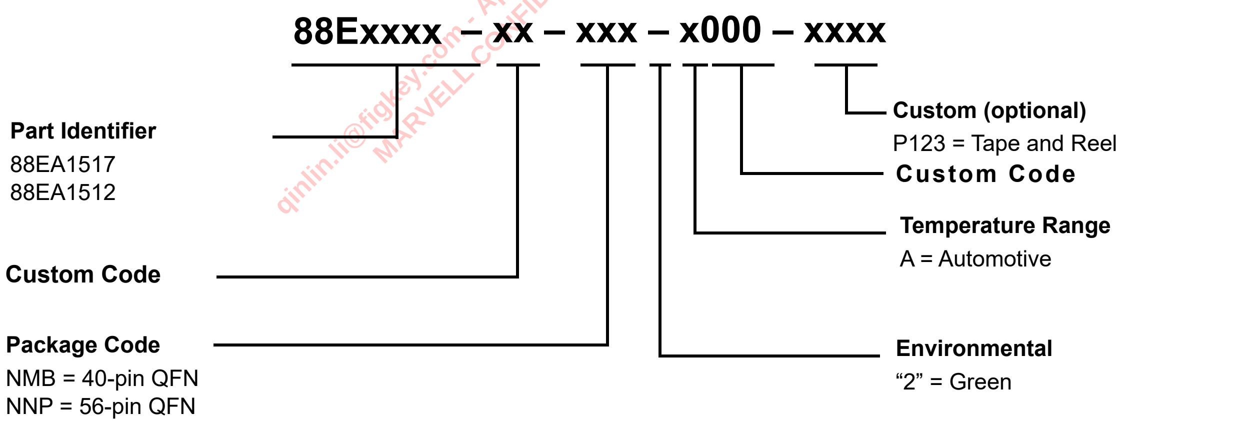


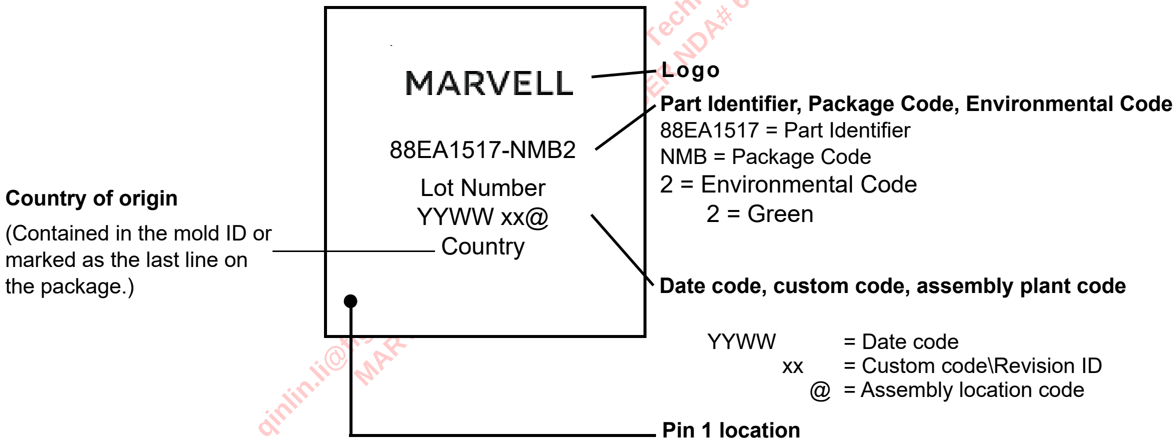
Table 319: Part Ordering Options

Package Type	Part Order Number
88EA1517 40-pin QFN - Green - Automotive	88EA1517xx-NMB2A000
88EA1517 40-pin QFN - Green - Automotive - Tape and Reel	88EA1517xx-NMB2A000-P123
88EA1512 56-pin QFN - Green - Automotive	88EA1512xx-NNP2A000
88EA1512 56-pin QFN - Green - Automotive - Tape and Reel	88EA1512xx-NNP2A000-P123

7.1.1 Package Marking

Figure 71 is an example of the package marking and pin 1 location for the 88EA1517 40-pin QFN Automotive Green compliant package.

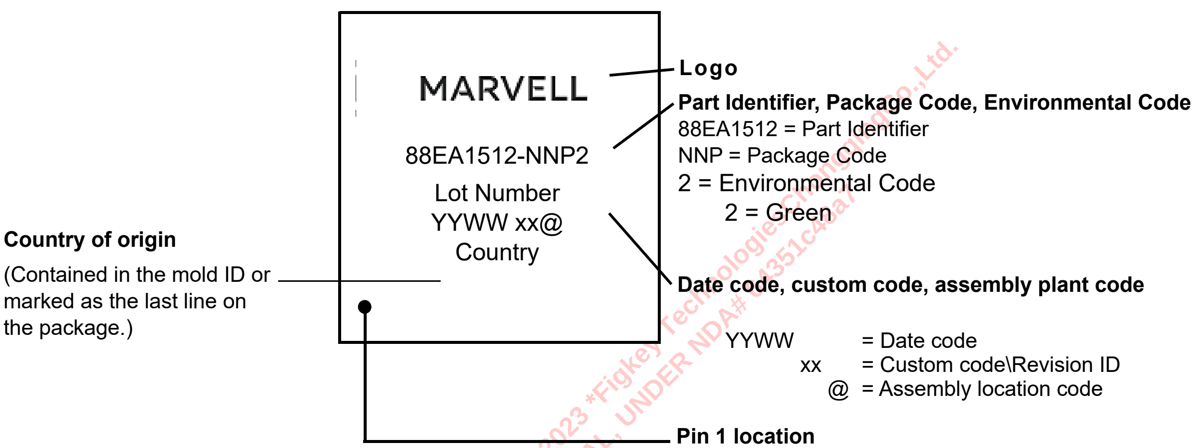
Figure 71: 88EA1517 40-pin QFN Automotive Green Compliant Package Marking and Pin 1 Location



Note: The above example is not drawn to scale. Location of markings is approximate.

Figure 72 is an example of the package marking and pin 1 location for the 88EA1512 56-pin QFN Automotive Green compliant package.

Figure 72: 88EA1512 56-pin QFN Automotive Green Compliant Package Marking and Pin 1 Location



Note: The above example is not drawn to scale. Location of markings is approximate.

A Revision History

Table 320: Revision History

Revision	Date	Section	Detail
Rev. B	12/16/2020	All applicable	Corporate update. Rebranding/logo updates throughout for all package markings.
		Cover	Updated Disclaimer
	12/16/2020	Datasheet release.	
Rev. A	04/24/2020	Features	“Automotive grade” to “Automotive grade 2”
		PHY Functional Description	Section 2.26, Two-wire Serial Interface - First paragraph - corrected 88EA1512 TWSI address details, and added 88EA1517 device details. Section 2.26.1, Bus Operation - corrected TBD cross-reference to Section 2.9. Section 2.26.2, Read and Write Operations - updated Figure 26 - Figure 30 addresses to {3'b100, PhyAddr.[3:0]} Section 2.31.2, MAC Interface Calibration Register Definitions - removed references to E1510Q device. Section 2.32.1, 88EA1517 Hardware Configuration - corrected Table 75 heading from 88EA1512 to 88EA1517. Section 2.32.2, Software Configuration - Management Interface - updated section to differentiate 88EA1512 device and 88EA1517 device details.
		General Register Description	Table 141, RGMII Output Impedance Calibration Override - bit 13, replaced “For either package if” with “If”. Table 165, General Control Register - exposed bits 9:5 Table 201, General Control Register 1 - bits 2:0 - removed reference to E1510Q.
		Package Mechanical Dimensions	Table 317, 40-Pin QFN Mechanical Dimensions D2 & E2 reduced from 4.05 to 4.04 mm
		Ordering Part Number/Package Marking	Updated Ordering Part Number to Automotive. Added Tape and Reel ordering options
	12/03/2019	All Applicable	Document status indicator removed
			Cosmetic enhancements
			Added Revision History

Table 320: Revision History (Continued)

Revision	Date	Section	Detail
Rev. A (cont.)	12/03/2019 (cont.)	Cover	Updated Disclaimer
		Electrical Specifications	Added 88EA1517 Current Consumption Values
	04/24/2020	Datasheet release.	
Rev. --	05/23/2016	Datasheet release.	
Rev.0.07	02/09/2016	Draft revision to customers.	

qinlin.li@figkey.com - Apr 18, 2023 *Figkey TechnologiesChongqingCo.,Ltd.
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